NATURE'S HEALING CENTER dba.

THE FLOWER SHOP

PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd Ste. 1 and Ste. 2, Phoenix, AZ 85008

PROJECT NUMBER: 3378.002.00

CODE COMPLIANCE NARRATIVE

THE FLOWER SHOP IS AN EXISTING ONE-STORY CANNABIS DISPENSARY MERCANTILE FACILITY LOCATED AT 3155 E. MCDOWELL ROAD IN PHOENIX ARIZONA. RENOVATIONS WILL OCCUR IN BOTH SUITES; SUITE 01 AND SUITE 02.

SUITE 01
NORTH PORTION OF BUILDING; ~1,932 S.F. ORIGINALLY PERMITTED AS FUTURE M OCCUPANCY
REMODELED TO ENCOMPASS A NEW CANNABIS DISPENSARY WHICH WILL INCLUDE A NEW RETAIL / DISPLAY AREA, UNISEX RESTROOMS, LOUNGE AREA – EMPLOYEE BREAK ROOM AND AN ADDITIONAL FULLFILMENT AND STORAGE AREA. DEMOLITION OF EXISTING PARTITIONS, REMOVAL AND UPDATE OF INTERIOR FLOOR AND WALL FINISHES, LIGHT FIXTURES, NEW PARTITIONS, DISPLAYS AND PLUMBING ASSOCIATED WITH NEW UNISEX BATHROOM AND BREAK ROOM. NEW INTERIOR ROOF ACCESS HATCH WILL BE PROVIDED.

SUITE 02
SOUTH PORTION OF BUILDING; ~1,967 S.F. EXISTING M OCCUPANCY – EXISTING CANNABIS DISPENSARY. AREA WILL BE REMODELED TO ENCOMPASS NEW STOCK ROOMS THAT WILL HAVE THE SAME INVENTORY AS THE MAIN RETAIL AREA. DEMOLITION OF EXISTING PARTITIONS, REMOVAL OF EXISTING DISPLAY CASES AND REMOVAL OF EXISTING CEILING AND LIGHT FIXTURES.

ONE OF THE REQUIRED EXITS FOR THE NORTHERN PORTION OF THE BUILDING WILL BE THROUGH THE SOUTHERN STOCK ROOM. IN ACCORDANCE WITH IBC SECTION 1016.2 MEANS OF EGRESS ARE NOT PROHIBITED THROUGH STOCK ROOMS IN GROUP M OCCUPANCIES.

NO EXTERIOR ELEVATION MODIFICATION WILL NECESSARY UNDER THIS PERMIT.

THE EXISTING BUILDING CONSTRUCTION TYPE IS VB AND NON-SPRINKLERED. THE BUILDING IS MIXED USE, NON-SEPARATED. RETAIL, AND ACCESSORY STORAGE WILL BE CLASSIFIED AS GROUP M OCCUPANCY. BREAK WILL BE CLASSIFIED AS GROUP M OCCUPANCY, AND STORAGE WILL BE CLASSIFIED AS ACCESSORY GROUP S-1 . GROUP M AND S OCCUPANCIES ARE NOT REQUIRED TO BE SEPARATED PER BUILDING CODE.

PER CITY OF PHOENIX AMENDMENTS TO THE INTERNATIONAL PLUMBING CODE, SEPARATE PLUMBING FACILITIES AND DRINKING FOUNTAINS ARE NOT REQUIRED IN MERCANTILE OCCUPANCIES IN WHICH THE MAXIMUM OCCUPANT LOAD IS 50 OR FEWER.

DESIGN AND CONSTRUCTION FOR ACCESSIBILITY WILL BE IN ACCORDANCE WITH THE 2010 ADA STANDARDS FOR ACCESSIBLE DESIGN, AND ICC/ANSI A117.1-2009 ACCESSIBLE AND USEABLE BUILDINGS AND FACILITIES.

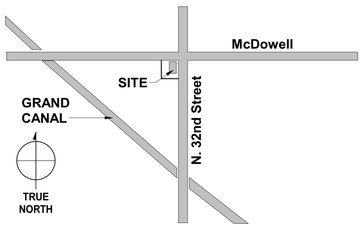
ZA CASE: ZA-480-21
ZA CASE: ZA-64-16

DOCUMENT PACKAGE

ISSUED: 10/13/2022

PACKAGE: PERMIT PACKAGE INTERIOR PHASE

LOCATION MAP



KEY PLAN



PROJECT TEAM

OWNER INFORMATION
ZONE LAND HOLDING LLC
3219 E Camelback Rd Suite 425
Phoenix, AZ 85018

TENANT REPRESENTATIVE
The Flower Shop
2575 E. Camelback Rd, Suite 400
Phoenix, AZ 85016
Phone: 480.532.8982

Chris Anaradian, Development Manager
Email: chrisa@theflowershopaz.com

ARCHITECT
RSP Architects, Ltd.
502 S. College Ave, Suite 203
Phoenix, AZ 85281
Phone: 480.889.2000

Jason Ploszaj, Principal
Phone: 480.889.2046
Email: jazon.ploszaj@rsparch.com

Shane Lucero, Project Architect/Manager
Phone: 480.889.2014
Email: shane.lucero@rsparch.com

CIVIL ENGINEER
KBELL ENGINEERING
1355 N 86th Place
Mesa, Arizona 85207
Phone: 602.980.8246

Kelly Bell, P.E.
Phone: 602.980.8246
Email: KBELL@KBELLENG.COM

LANDSCAPE CONSULTANT
TRUEFORM landscape architecture studio
2009 N 7th Street Suite F
Phoenix, AZ 85006
Phone: 480.382.4244

Roger Socha
Phone: 480.518.2053
Email: roger.socha@trueformlas.com

CONTRACTOR
Wespac
9440 N. 26th St Ste 100
Phoenix, AZ 85028
Phone: 602.956.1323

John Wasemiller
Phone: 602.343.0227
Email: jwasemiller@wespac.com

STRUCTURAL ENGINEER
MEYER BORGMAN JOHNSON
116 E. University Drive Ste. 500
Tempe, AZ 85281
Phone: 480.747.6721

Kurt Kindermann, PE
Phone: 602.327.2503
Email: kkindermann@mbjeng.com

MECH/ELEC ENGINEERS
IMEG (MSA ENGINEERING CONSULTANTS)
7878 North 16th Street Ste 140
Phoenix, AZ 85020
Phone: 602.371.1333 Fax: 602.371-0675

Brett Casperson
Phone: 602-371-1333
Email: brett.c.casperson@imegcorp.com

CODE SUMMARY

PROJECT NAME:
THE FLOWER SHOP

PROJECT DESCRIPTION:

This project consists of renovating an existing cannabis dispensary located at the southwest corner of E McDowell Rd and N 32nd St. Project includes renovating the north storage into retail space and converting the south existing retail into shell space.

APPLICABLE CODES:
The Phoenix Building Construction Code consists of the City of Phoenix Administrative Provisions, the following codes and standards, and city of Phoenix Amendments:

2018 International Building Code w/ Amendments
2018 International Energy Conservation Code w/ Amendments
2018 International Mechanical Code w/ Amendments
2018 International Plumbing Code w/ Amendments
2017 National Electrical Code w/ Amendments
2003 International Fire Code w/ Amendments
2010 ADA Standards for Accessible Design

OCCUPANCY:
Existing: Mercantile (M)
Proposed: Retail M
Storage (Accessory) M
Office B

TYPE OF CONSTRUCTION: VB (Non-Sprinklered)

HEIGHT AND NUMBER OF STORIES:
Allowable Height: 40'
Allowable Stories: 1

Actual Height: 15' - 6"
Actual Stories: 1

ALLOWABLE FLOOR AREA:
Allowable Floor Area: 9,000 SF

Actual Floor Area: 3,679 NSF

FIRE RESISTIVE REQUIREMENTS:
Type V construction is that type of construction in which the structural elements, exterior wall and interior walls are of any materials permitted by IBC code. (Section 602.5)

DEFERRED SUBMITTALS:

EXITING REQUIREMENTS:
Occupant Load Factors (OLF): Mercantile: 60 GSF
Per Table 1004.1.2 Mercantile Storage: 300 GSF

Occupant Load:

OCCUPANT LOAD CALCULATION				
FUNCTION OF SPACE	OCCUPANCY	AREA	OLF	DESIGN OCCUPANT LOAD
RETAIL	M	1476 SF	60 GSF	25
STORAGE (ACCESSORY)	M	379 SF	300 GSF	2
SUPPORT	M	406 SF	60 GSF	9
STORAGE (ACCESSORY)	M	1418 SF	300 GSF	5
TOTAL		3,674 SF		41

EXIT ACCESS:

Required number of exits or exit access doorways:
Per Section 1006

		Required	Provided
This Project:	Group B, M:	2	3

Exit Access Travel Distance: Group M (Without Sprinkler): 200 FT

Common Path of Egress Travel: Group M (Without Sprinkler), OL > 30: 75 FT

Dead Ends: Group M (Without Sprinkler): 20 FT

MEANS OF EGRESS SIZING:

Requirements:
Other- 1005.3.2, Occupant Load x 0.2"

Stairways: (NOT APPLICABLE)

Other components (Doors):

		Required	Provided
Group B, M	41 x 0.2" = 8.2" occ (Min. 32" per Door)=	2	3

MINIMUM PLUMBING FACILITIES:
Minimum Number of Fixtures per Table 2902.1

OCCUPANTS: 41					
OCCUPANCY MERCANTILE	WATERCLOSET MALE	WATERCLOSET FEMALE	LAVATORIES MALE	LAVATORIES FEMALE	DRINKING FOUNTAINS	OTHER
RATIO:	1 PER 500		1 PER 750		1 PER 1000	
REQUIRED:	.08		.05		.04	1 SERVICE SINK
PROVIDED:	1		1		0	1 SERVICE SINK

Note:
1. Per City of Phoenix Amendment to 2018 International Plumbing Code (IPC), Section 403.2, separate facilities shall not be required in Mercantile Occupancies in which the Maximum Occupant Load is 50 or fewer.
2. Drinking Fountains are not required for an Occupant Load of 50 or fewer per City of Phoenix Amendment to 2018 International Plumbing Code (IPC) Section 410.2

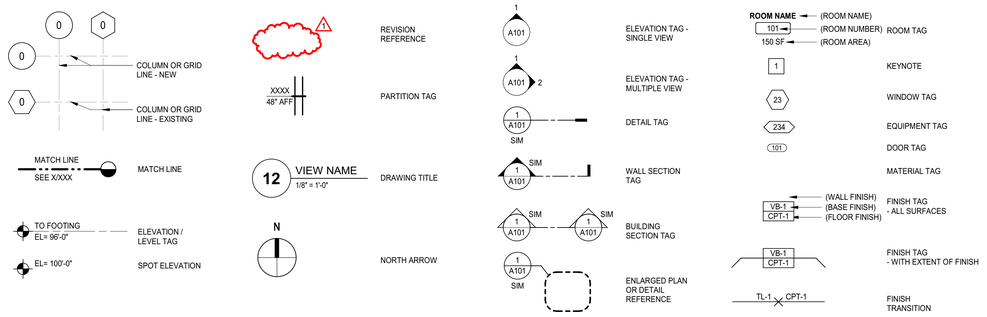
SHEET INDEX

SHEET NO.	DRAWINGS NAME	INTERIOR PHASE	ISSUES

ISSUE KEY:
A ISSUED (WITH SIGNATURE)
B REVISED AND RE-ISSUED (WITH SIGNATURE)
C NOT FOR CONSTRUCTION - REFERENCE ONLY
D RE-ISSUED FROM A PRIOR BID PACK - NO REVISIONS

SHEET NO.	NAME	INTERIOR PHASE
00 GENERAL		
G000	COVER SHEET RETAIL	A
G001	SPECIFICATIONS & ABBREVIATIONS	A
G002	SPECIFICATIONS	A
G003	SPECIFICATIONS	A
G010	MOUNTING HEIGHTS AND DETAILS	A
G101	FIRE AND LIFE SAFETY PLANS	A
10 ARCHITECTURAL SITE		
A000	OVERALL SITE LAYOUT PLAN	A
30 ARCHITECTURAL DEMO		
D101	DEMO PLANS	A
D102	ROOF PLAN DEMOLITION	A
40 ARCHITECTURAL		
A101	FLOOR PLAN & REFLECTED CEILING PLAN	A
A201	ROOF PLAN	A
A310	SCHEDULES AND DOOR DETAILS	A
A320	PARTITION TYPES AND DETAILS	A
A901	BUILDING SECTIONS, INTERIOR ELEVATIONS & ENLARGED PLANS	A
A940	ARCHITECTURAL MILLWORK	A
60 INTERIORS		
I101	FIRST FLOOR FINISH PLAN	A
70 STRUCTURAL		
S001	LEGEND SHEET	A
S002	GENERAL STRUCTURAL NOTES	A
S003	STRUCTURAL SPECIAL INSPECTIONS	A
S201	FOUNDATION PLAN	A
S202	ROOF FRAMING PLAN	A
S501	FOUNTIATION DETAILS	A
S701	FRAMING DETAILS	A
75 MECHANICAL		
M000	MECHANICAL COVERSHEET	A
M001	MECHANICAL SPECIFICATIONS	A
M100	MECHANCIAL PLANS	A
M101	MECHANICAL ROOF PLANS	A
M400	MECHANICAL DIAGRAMS	A
M500	MECHANICAL SCHEDULES	A
80 PLUMBING		
P000	PLUMBING COVER SHEET	A
P001	PLUMBING SPECIFICATIONS	A
P100	PLUMBING DEMO PLANS	A
P101	PLUMBING PLANS	A
P400	PLUMBING DIAGRAMS	A
P500	PLUMBING SCHEDULES	A
85 ELECTRICAL		
E000	ELECTRICAL COVER SHEET	A
E001	ELECTRICAL SPECIFICATIONS	A
E100	ELECTRICAL DEMO PLANS	A
E101	ELECTRICAL PLANS	A
E102	ELECTRICAL ROOF PLANS	A
E301	SINGLE LINE DIAGRAM AND CALCULATIONS	A

DRAWING SYMBOLS



City of Phoenix

PLANNING & DEVELOPMENT DEPARTMENT

This set of plans has been reviewed by the City of Phoenix Planning & Development Department for compliance with applicable City codes and ordinances and shall be kept at the construction site. Such review shall not prevent the City from requiring correction of errors in the plans where such errors are subsequently found to be in violation of any code, law, ordinance, health, safety, or other design issues.

SITE - Brandon Zurick 602-262-7404

IBC - Gilbert Tellez 602-534-2768

STRUC - Manny Bahraini

NEC - Romabelle Ramos 602-495-7437

IMC-UPC - Thomas Fledderjohn 602-534-7651

RSP Architects
502 South College Avenue Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO.
35861
MICHAEL L. KRAGER
ARIZONA, U.S.A.
13OCT2022
Signed

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd
Ste. 1 and Ste. 2,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	JJZ/DM
Checked By	SL
Date	10/13/2022

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Sheet Issues / Revisions

No.	Date	Description

COVER SHEET RETAIL

G000

DIVISION 1 - GENERAL REQUIREMENTS

SECTION 01 00 01 - GENERAL CONDITIONS

1.01 GENERAL

A. The "General Conditions of the Contract for Construction", AIA Document A107, 1997 Edition, are hereby made a part of these Contract Documents to the same extent as if bound herein. Owner will provide Supplementary General Conditions as necessary to customize the General Conditions to this specific Project.

B. The term Owner refers to: **THE FLOWER SHOP**

C The term Project Manager refers to: **SHANE LUCERO**

D The term Architect refers to: RSP Architects.

E. The term Engineer refers to **IMEG CORP.**

SECTION 01 00 02 - BASIC REQUIREMENTS

1.01 GENERAL

A. The following abbreviated specifications are a listing of products/systems only, with their respective accessories. "General" and "Execution" language typically associated with the CSI 3-part specification format has been omitted. Quality control, installation procedures and tolerances normally associated with each trade will, however, be expected.

1.02 DEFINITIONS

A. Approved: Where used in conjunction with Architect's response to submittals, requests, applications, inquiries, reports and claims by Contractor, the meaning of the term "approved" will be held to limitations of Architects' responsibilities and duties as specified in General and Supplementary Conditions. In no case will "approval" by Architect be interpreted as a release of Contractor from responsibilities to fulfill requirements of Contract Documents.

B. By Owner: Items that will be ordered, paid for and shipped to Project by Owner. Contractor to receive, unload, unpack or uncrate, protect, move into place, install and connect these items as specified or indicated in the Contract Documents.

C. Directed, Requested, Etc.: Where not otherwise explained, terms such as "directed", "requested", "authorized", "selected", "required", "accepted", and "permitted", mean "directed by Project Manager", etc. However, no such implied meaning will be interpreted to extend Project Manager's responsibility into Contractor's area of construction and responsibility.

D. Furnish: Supply and deliver to Project site, ready for unloading, unpacking, assembly, installation, etc. , as applicable, except as otherwise defined in greater detail.

E. Indicated: Cross-reference to graphics, notes or schedules on Drawings, to other paragraphs or schedules in specifications, and to similar means of recording requirements in Contract Documents. Where terms such as "shown", "noted", "scheduled", and "specified" are used in lieu of "indicated", it is for the purpose of helping reader locate cross-reference, and no limitation of location is intended except as specifically noted.

F. Install: Operations at Project site including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protecting, cleaning, and similar operations, as applicable in each instance, except as defined in greater detail.

G. Installer: The entity (person or firm) engaged by Contractor, Subcontractor or Sub-subcontractor for performance of a particular unit of work at Project site, including installation, erection, application and similar required operations. Installers are required to be experienced in operations they are engaged to perform.

H. Not In Contract (N.I.C.): Products not in the Contract, but which may require provisions in the construction for future installation by Others.

I. Project Site: The space available for the performance of the work, either exclusively to the Contractor or in conjunction with others performing work as part of the project.

J. Provide: Furnish and install, including without limitation, labor, materials, equipment, transportation, services, and other items required to complete referenced tasks.

K. Product: Materials, systems, and equipment incorporated in or to be incorporated in the Project.

L. Replace and Provide New: Remove existing items, furnish and install new items in their place.

M. Reuse: Remove items and store, reinstall stored items as indicated.

N. Work: Construction and services required by Contract Documents, whether completed of partially completed, and includes labor, materials, equipment and services provided or to be provided by Contractor to fulfill the Contractor's obligations. The Work may constitute the whole or a part of the Project.

1.03 CONTRACTOR USE OF PREMISES

A. Contractor shall verify with the Building Manager the locations of base building utilities and services. Contractor shall coordinate scheduling and use of elevators and loading docks for deliveries, handling of materials, equipment and debris with the Building Manager and the Project Manager.

B. Cooperate with Project Manager in scheduling, executing Work and Site usage. Confine operations and storage of material to areas assigned by Project Manager.

C. The existing building will be occupied and open for business during construction. Coordinate use of premises under direction of the Building Manager and the Project Manager. Conduct operations to permit public access to existing building. Maintain public walks, driveways, and entrances in safe condition, free of equipment, material and debris.

D. Existing exit stairs and all fire-protection systems must remain accessible and operable during selective demolition and/or construction and shall be protected from physical damage. Notify Building Manager of damaged or disabled systems and repair or replace damaged systems immediately.

E. Cleaning During Construction: Control accumulation of waste materials and rubbish on a continuous or daily basis. Comply with requirements of NFPA 241 for removal of combustible waste material and debris. Clean interior areas prior to start of finish work, maintain areas free of dust and other contaminants during finishing operations.

1.04 OVERLAPPING AND CONFLICTING REQUIREMENTS

A. Contractor shall visit the site and familiarize himself with all existing conditions and verify all dimensions and conditions of the Drawings. Notify the Project Manager and Architect regarding discrepancies before proceeding with the work affected.

B. Where compliance with two or more standards or requirements is indicated, and where overlapping requirements establish different or conflicting levels of quality, the most stringent, (and generally the most expensive), requirement is intended, and will be enforced unless written approval is granted otherwise by the Architect.

C. During Construction: Refer to Project Manager for resolution conflicting requirements and uncertainties as to which level of quality is more stringent, and receive written clarification from Project Manager and/or Architect before proceeding with questioned work.

1.05 CUTTING AND PATCHING

A. Remove, cut, fit, and/or patch old work as required to join new Work to existing attached and/or adjacent construction

B. Remove and replace existing construction to remain which is loosened, cracked, or otherwise damaged or defaced as a result of Work and is unsuitable for use intended

C. Patch existing defective or incomplete surfaces caused or exposed by work to match adjacent materials. Perform patching using skilled mechanics experienced in type of Work involved. Conform to standards of Contract Documents where applicable; where not specified, conform to highest standards of the trade.

D. Inspect existing conditions, including elements subject to damage or movement during cutting and patching. Beginning of cutting or patching means acceptance of existing conditions affecting performance of work

E. Provide protective coverings and enclosures necessary to prevent damage to existing construction to remain.

F. Salvage and material not specifically designated otherwise shall become property of Contractor, and shall be removed promptly from Site. Remove trash, debris and other waste material promptly from Site.

SECTION 01 10 00 - ENVIRONMENTAL GOALS

1.01 General:

A. The following environmental goals have been established for the project. These goals are general in nature; refer to specific specification sections for more detailed requirements. Notify Architect if conflicts arise between performance of the work and environmental goals. This specification is not intended to limit alternative means of achieving these goals. Suggestions and input from the Contractor(s) for implementing these goals are encouraged.

1. Use resources efficiently. The construction, operation, and ultimately reuse or removal of the building should be optimized to eliminate unnecessary use of energy, water, and raw materials. As such, the project has been designed to:
a. Support occupant efforts to recycle and reuse materials.
b. Be consistent with expected current and future uses.
c. Utilize materials and construction methods that require a minimum of energy, water, and raw materials.
d. Be reusable and or recyclable in whole or in component.
2. Provide a healthy and productive indoor environment to the extent that the building can affect their experience, occupants should be enlivened, relaxed, and stimulated when they are in the building. To this end, the project has been designed to:
a. Utilize daylight as a primary source of illumination, while minimizing glare and extreme contrast.
b. Provide optimal amounts of clean, fresh air to occupied spaces.
c. Release a minimum of unhealthy pollutants from building components, fixtures, and furnishings.
d. Evacuate effectively any pollutants generated by occupant activities in the building.
e. Provide views.

3. Enhance, restore, and protect the global environment. The project is designed to minimize releases of pollutants that affect the global environment and to avoid the need for materials that are polluting to manufacture. These measures include:
a. Minimizing the release of carbon dioxide (CO2) from fuels burned on site or fuels burned off site to supply electricity to the building
b. Avoiding the release of ozone-depleting compounds, such as HCFCs from refrigerants or foam insulation materials
c. Selection of building materials that, based on an overall life-cycle assessment, are the least damaging to the environment in their extraction, processing, use, and ultimate disposal

B. Material selection and construction process goals: In support of the overall goals listed above, the following goals have been identified for the material selection and construction process. These are also general in nature and should be considered as a guideline in the event of a conflict, any specific requirements in technical specification sections take precedence. Tradeoffs and compromises are inevitable in the pursuit of these goals, but everyone working on the project is expected to take all reasonable actions to further these goals. Notify Architect of any performance requirements that conflict unnecessarily with these goals.
1. Use resources efficiently:
a. Reuse existing buildings and materials.
b. Select materials that use resources efficiently.
c. Use construction practices that achieve the most efficient use of resources and materials.
d. Recycle or reuse job-site waste.
e. Select recycled-content materials.
f. Select materials that can be recycled.
2. Avoid scarce, irreplaceable, or endangered resources:
a. Select materials from abundant, well-managed resources.
b. Select materials that are replaceable, renewable, or can be replenished.
c. Select materials that minimize damage to natural habitats.
3. Use durable materials:
a. Select materials with the longest usable life.
b. Select materials that can be reused.
c. Select materials with the least burdensome maintenance requirements.
4. Create spaces that are healthy for occupants:
a. Select low-toxic products and materials.
b. Select materials without toxic maintenance requirements.
c. Specify mechanical equipment that will provide fresh air and will not trap water or pollutants.
5. Use energy efficiently:
a. Select materials with low embodied energy.
b. Select materials that save energy during building operations.
c. Select products and equipment that save energy during building operations.
6. Use water efficiently:
a. Use construction practices that achieve the most efficient use of water.
b. Select water-conserving appliances and equipment.
7. Select materials that generate the least amount of pollution.

SECTION 01 33 00 - SUBMITTALS

1.01 SUBMITTAL PROCEDURES

A. Deliver submittals to Architect addressed as follows:

RSP Architects
502 South College Ave.
Phoenix, AZ 85281

Attention: **SHANE LUCERO**
Electronic copies may be e-mailed to **shane.lucero@rsparch.com**

B. Submittal Schedule:

Key: PD = Product Data
SA = Samples
SC = Schedule
SD = Shop Drawing
Section Product/System Submittals, but not limited to:

06410 Custom Casework SA SD
08400 Aluminum Framing Systems SD SA
08800 Glazing PD SA
09655 Linoleum Flooring SA
09900 Painting SA

C. Submittals received from sources other than through Contractor's office, and which do not have Contractor's executed review and approved marking, will be returned "without action".

D. Do not allow submittals without appropriate final "action" markings by Architect to be used in connection with the Work.

E. Shop Drawings:

1. Submit in the form of bound .pdf's.

F. Product Data:

1. Submit .pdf copy to be retained by the Architect. Submit to the Contractor, one .pdf copy for Project Record Documents, and Operation and Maintenance Manuals.

G. Samples:

1. Submit the number specified in respective Specification section; one will be retained by Architect. Reviewed samples which may be used in the Work are indicated in the Specification Section.

2. Submit full range of manufacturers' standard colors, textures, and patterns for Architect's selection, where required by individual Specification Sections.

H. Contractor Review:

1. Review submittals prior to transmittal; determine and verify field measurements, field construction criteria, manufacturer's catalog numbers, and conformance of submittal with requirements of Contract Documents.

2. Apply Contractor's stamp, signed or initialed, and dated, indicating review, verification of products, field dimensions and field construction criteria, and coordination of information with requirements of Work and Contract Documents. Notify Architect in writing at time of submittal of any deviations from requirements of Contract Documents.

3. Do not fabricate products or begin work which requires submittals until return of submittal with Architect's "approval".

I. Architect Review:

1. Architect will review shop drawings, product data, and samples and will endeavor to return submittals within ten days from receipt.

2. Submittals will be reviewed with the following actions:
a. "Approved": Indicates that Submittal is in general conformance with the design intent of the Contract Documents. Contractor shall not be relieved of responsibility for deviations from requirements of the Contract Documents by the Architect's approval of Submittal.
b. "Approved as Noted": Indicates Submittal is approved as defined above, subject to noted corrections.
c. "Revise and Resubmit": Indicates unspecified material, insufficient information, unapproved system or similar inadequacies. Additional information must be revised and resubmitted before Submittal will be approved.

J. Re-submittals:

1. Submittals not "approved" by Architect will be returned to the Contractor for further action.

2. Make re-submittals under procedures specified for initial submittals; identify changes made since previous submittal.

1.02 PRODUCT OPTIONS AND SUBSTITUTIONS

A. Products specified by naming several manufacturers and, where applicable, sample products: Provide products of named manufacturers meeting specifications; no options, no substitutions allowed.
B. Products specified by naming only one manufacturer: No options, no substitutions allowed.

C. Substitutions will not be considered when indicated on Shop Drawings or Product Data submittals without separate formal request, when requested directly by subcontractor or supplier, or when acceptance will require substantial revision of Contract Documents

D. Project Manager and Architect will determine acceptability of any and all substitutions. Substitute products shall not be ordered or installed without written acceptance.

SECTION 01 35 16 - ALTERATION PROJECT PROCEDURES

1.01 SYSTEM DESCRIPTION

A. Items indicated to "remain" shall be left in their present condition.

B. Items indicated to be "repaired" shall be restored, not replaced, following the general guidelines this Section, and any more specific guidelines specified in Sections governing work of the respective materials.

C. Items indicated to be "replaced" shall follow the more specific guidelines specified in Sections governing work of the respective materials.

D. Contractor shall notify Architect if items indicated to "remain" or be "repaired", appear to need "replacement". Beginning replacement work without prior approval may result in loss of compensation for additional work.

1.02 PRODUCTS FOR PATCHING AND EXTENDING WORK

A. New Materials: As specified in individual Sections.

B. Match existing products and work for patching and extending work.

C. Determine type and quality of existing products by inspection and any necessary testing, and workmanship by use of existing as a standard. Presence of a product, finish, or type of work, requires that patching, extending, or matching shall be performed as necessary to make Work complete and consistent with specifications.

1.03 INSPECTION

A. Verify that demolition is complete, and areas are ready for installation of new work.

B. Beginning of restoration work means acceptance of existing conditions.

1.04 PREPARATION

A. Cut, move, or remove items as necessary for access to alterations and renovations work; replace and restore at completion.

B. Remove unsuitable material not marked for salvage, such as rotted wood, rusted metals, and deteriorated masonry and concrete; replace materials as specified for finished work.

C. Remove debris and abandoned items from area and from concealed spaces.

D. Prepare surfaces and remove surface finishes to provide for proper installation of new work and new finishes.

E. Close openings in exterior surfaces to protect existing work and salvage items from weather and extremes of temperature and humidity. Insulate ductwork and piping to prevent condensation in exposed areas.

1.05 INSTALLATION

A. Coordinate work of alterations and renovations to expedite completion sequentially and to accommodate Owner occupancy.

B. Project rooms shall be complete in all respects including operational mechanical and electrical systems.

C. Remove, cut, and patch work in a manner to minimize damage and to provide means of restoring products and finishes to specified condition.

D. Refinish visible existing surfaces to remain in renovated rooms and spaces, to specified condition for each material, with a neat transition to adjacent new finishes.

E. In addition to specified replacement of equipment and fixtures, restore existing plumbing, heating, ventilation, air conditioning and electrical systems.

F. Install products as specified in individual Sections.

1.06 TRANSITIONS

A. Where new work abuts or aligns with existing, make a smooth and even transition. Patched work shall match existing adjacent work in texture and appearance.

B. When finished surfaces are cut so that a smooth transition with new work is not possible, terminate existing surface along a straight line at a natural line of division and make recommendation to Architect.

1.07 ADJUSTMENTS

A. Where removal of partitions results in adjacent spaces becoming one, rework floors, walls, and ceilings to a smooth plane without breaks, steps, or bulkheads. Where a change of plane of 1/8 inch or more occurs, submit recommendation for providing a smooth transition for Architect review.

B. Trim existing doors as necessary to clear new floor finishes; refinish trimmed areas.

C. Fit work at penetrations of surfaces.

1.08 REPAIR OF DAMAGED SURFACES

A. Patch or replace portions of existing surfaces which are damaged, lifted, discolored, or showing other imperfections.

B. Repair substrate prior to patching finish.

1.09 FINISHES

A. Finish surfaces as specified in individual Sections.

B. Finish patches to produce uniform finish and texture over entire area. When finish cannot be matched, refinish entire surface to nearest intersections.

1.10 CLEANING

A. Clean daily the Owner-occupied areas of work in which Contractor is working or traversing.

SECTION 01 40 00 – QUALITY CONTROL

1.01 GENERAL

A. Maintain quality control over suppliers, manufacturers, products, services, site conditions, and workmanship, to produce work of specified quality.

B. Workmanship:
1. Comply with industry standards and manufacturer's instructions, except when more restrictive tolerances or specified requirements indicate more rigid standards or more precise workmanship.
2. Perform work by persons qualified to produce workmanship of specified quality.
3. Secure products in place with positive anchorage devices designed and sized to withstand reasonable stresses, vibration and racking.

C. Installer's Examination of Conditions:
1. Installer of each unit of work shall examine substrate to receive work, and conditions under which work will be performed, and report in writing to Architect unsatisfactory conditions. Do not proceed with work until unsatisfactory conditions have been corrected in a manner acceptable to Installer.

D. Manufacturers' Instructions:
1. Comply with instructions in full detail, including each step in sequence, except where more stringent requirements are indicated in Contract Documents. Should instructions conflict with Contract Documents, request clarification from Architect before proceeding.

SECTION 01 70 00 – PROJECT CLOSEOUT

1.01 FINAL COMPLETION CLOSEOUT PROCEDURES

A. Complete items on Contractor's Final Completion punch list.

B. Remove temporary facilities, services, surplus materials, rubbish and similar elements.

C. Change over door locks and other provisions for Contractor's access to Owner's property, if applicable.

D. Prior to inspection for Final Completion, perform Final Cleaning under provisions of this Section.

E. When Contractor considers Work is complete, submit the following:
1. Written Certificate of Final Completion, stating:
a. Contract Documents have been reviewed.
b. Work has been inspected for compliance with Contract Documents.
c. Work has been completed in accordance with Contract Documents, and deficiencies listed with Certificate of Substantial Completion have been corrected.
d. Equipment and systems have been tested, adjusted and balanced, and are fully operational.
e. Operation of systems has been demonstrated to Owner's personnel.
f. Work is complete and ready for final inspection.
2. Project Record Documents under provisions of this Section.
3. Operation and Maintenance Data under provisions of this Section.
4. Spare Parts, Attic Stock and Maintenance Materials under provisions of this Section.
5. Contractor's Affidavit of Payment of Debts and Claims (AIA G706); proving satisfactorily to Owner that taxes, fees and similar obligations of Contractor have been paid.
6. Contractor's Affidavit of Release of Liens (AIA G706A); providing evidence of payment and release of liens in accordance with Conditions of the Contract.
7. Contractor's Lien Waiver in full amount of Contract Sum.
8. Lien waivers from all subcontractors, sub- subcontractors, and major suppliers to Project legally entitled to file a lien in excess of one thousand dollars ($1000.00) arising out of Work of this Contract and covered by payments, totaling full amount of Contract Sum.
9. Certificates of Insurance for Products and Completed Operations in accordance with Conditions of the Contract.
10. Application for final payment in accordance with provisions of Conditions of the Contract.
11. Statement of Adjustment of Accounts to Contract Sum indicating:
a. Original Contract Sum.
b. Previous change orders.
c. Changes under allowances.
d. Changes under unit prices.
e. Deductions for uncorrected work.
f. Penalties and bonuses.
g. Deductions for re-inspection fees.
h. Other adjustments to Contract Sum.
i. Total Contract Sum as adjusted.
j. Previous payments.
k. Sum remaining due.

F. Should Architect find Work incomplete or defective:
1. Architect will notify Contractor and Owner in writing listing observed deficiencies.
2. Contractor shall remedy deficiencies and send a second certification of final completion.
3. Architect will re-inspect.

G. When Architect finds work is complete, Architect will:
1. Consider closeout submittals.

H. Final Cleaning:
1. Execute cleaning prior to inspection for Final Completion of the Work.
2. Use only materials and methods recommended by manufacturer of material being cleaned.
3. In addition to removal of debris and cleaning specified in other sections, clean interior and exterior exposed-to-view surfaces.
4. Remove temporary protection and labels not required to remain.
5. Clean finishes free of dust, stains, films and other foreign substances.
6. Vacuum clean carpeted surfaces. Clean and damp mop, resilient and hard-surfaced floors.
7. Clean plumbing fixtures to a sanitary condition.
8. Clean surfaces of equipment; remove excess lubrication. Clean permanent filters of ventilating equipment and replace disposable filters when units have been operated during construction.
9. Clean light fixtures and lamps.
10. Clean interior glass and frame surfaces of new and existing interior and exterior glazing systems.
11. Remove waste, debris, and surplus materials from site. Clean grounds; remove stains, spills, and foreign substances from paved areas and sweep clean. Rake clean other exterior surfaces.

I. Project Record Documents:
1. Maintain Record Documents in a clean, dry and legible condition. Do not use Record Documents for construction purposes. Label each document "PROJECT RECORD" in neat, large, printed letters.
2. Record information concurrently with construction progress. Do not conceal any work until required information is recorded.
3. At Contract closeout, deliver Record Documents and samples, signed by Contractor, to Architect under provisions of this Section.

J. Operation and Maintenance Data:
1. Submit two sets prior to final inspection, bound in 8-1/2 x 11 inch three-ring side binders with durable plastic covers.
2. Include the following:
a. Appropriate design criteria.
b. List of equipment.
c. Operating instructions.
d. Maintenance instructions, equipment.
e. Maintenance instructions, finishes.
f. Shop drawings and product data.
g. Warranties.
3. Instruction of Owner Personnel: Before final inspection, instruct Owner's designated personnel in operation, adjustment, and maintenance of products, equipment, and systems, at agreed upon times. For equipment requiring seasonal operation, perform instructions for other seasons within six months.

K. Warranties and Bonds:
1. Provide original, notarized copies to Owner, with duplicate to Architect. Execute Contractor's submittals and assemble documents executed by subcontractors, suppliers, and manufacturers. Provide table of contents and assemble in binder with durable plastic cover.
2. Any warranty shall not relieve rights available to Owner under common law or other reserved rights.
3. Provide individual product and system warranties as defined in specification Sections, which may extend beyond Complete Project Warranty period.

DIVISION 2 - EXISTING CONDITIONS

SECTION 02 41 19 - SELECTIVE DEMOLITION

Part 1 -General
1.1. Comply with ANSI A10.6 - "American National Standard Safety Requirements for Demolition."
1.2. Erect and maintain temporary bracing, shoring, lights, barricades, signs and other means to protect the public, workers and other persons; finishes and improvements to remain; and adjoining property from damage from demolition Work; in accordance with applicable regulatory requirements.

Part 2 -Products
2.1. Materials forming portions of the permanent structure designated for demolition shall become the contractor's property; remove and dispose of such materials unless otherwise indicated or specified.
2.2. Carefully disconnect and remove items to be salvaged, store and protect items removed and indicated for reuse in the Work.
2.3. Removal of items not indicated for removal on the Drawings shall be approved by Owner prior to removal.

Part 3 -Execution
3.1. Prior to starting demolition, inspect and report observable defects and structural weaknesses of construction designated for demolition, of adjacent structures and of other improvements to remain.
3.2. Perform demolition in accordance with ANSI A10.6, the accepted demolition plan and applicable regulatory requirements. If in doubt whether to remove an item, obtain written approval prior to proceeding.
3.3. Remove materials carefully, providing for neat and structurally sound junctions between existing and new materials.
3.4. Protect existing structures, facilities, and equipment from damage. Items damaged because of demolition operations shall be repaired or replaced at no increase in Contract Sum.
3.5. Perform demolition to provide the least interference and most protection to existing facilities and improvements to remain.
3.6. Clean remaining surfaces of loose particles and dust.
3.7. Remove deactivated plumbing piping, mechanical equipment and ducts, and electrical conduit, including fastenings, connections and other related appurtenances and accessories which would otherwise be exposed in the completed Work or interfere with construction operations. Cap deactivated piping systems at points of cutoff.
3.8. Regularly remove debris from the site so that its presence will not delay the progress of the Work. Clean surfaces on which new materials will be applied, removing adhesives and other adhering materials, as necessary to furnish acceptable substrates for new materials.

DIVISION 3 – CONCRETE

SECTION 03 01 35 - CONCRETE REPAIR WORK

1.01 FLOOR UNDERLAYMENT AND FLOOR TRANSITION MATERIAL

A. Materials:
1. Trowel-Applied Floor Transition Material:
a. Compress. Strength: 800 psi -1day; 4500psi - 28days..
b. Flexural Strength: 1360 psi -28days.

B. Troweled Products & Acceptable Manufacturers:
1. "Thin Patch", L & M Construction Chemicals, Inc., Omaha, NE. , note that Thin Patch has 1-1/2" max. depth per layer.
2. "Set Latex Cement, and Acryl-Set Liquid Polymer", by Master Builders, Cleveland, OH, note that Set Latex Cement has 1/4" max. depth per layer .
3. "Sonopatch 100", Sonneborn Building Products, Minneapolis, MN, note that Sonopatch 100 has 3/4" max. depth per layer.
4. "Thoro Underlayment Trowel Grade", by Thoro System Products, Centerville, IN, note that Thoro Underlayment

C. Troweled Applications that have a feathered edge condition (may) require primer to be used for entire area covered with floor transition material. Primer for Troweled Products & Acceptable Manufacturers:
1. "Everbond", by L & M Construction Chemicals, Inc., Omaha, NE.
2. No primer required by Master Builders.
3. "Sonoprep", by Sonneborn Building Products, Minneapolis, MN.
4. No primer required by Thoro.

D. Floor Sealer/Hardener (at trowel-applied locations that are permanently exposed without another floor finish over them) provide the following sealer/hardener:
1. "Seal Hard", L & M Construction Chemicals, Inc., Omaha, NE.
2. "Kure-N-Harden ",Sonneborn Building Products, Minneapolis, MN.
3. "Ashford Formula ", Curecrete Chemical Company, Inc.

DIVISION 4 – MASONRY

Not Used

DIVISION 5 – METAL

SECTION 05 50 00 - METAL FABRICATIONS

1.01 MATERIALS

A. Steel Sections: ASTM A36.

B. Sheet Steel: ASTM A570.

C. Bolts, Nuts, and Washers: ASTM A307.

D. Welding Materials: AWS D1.1; type required for materials being welded.

E. Primer: Fabricator's standard gray primer, 2.0 mils dry film thickness.

1.02 FABRICATION - GENERAL

A. Verify dimensions on site prior to shop fabrication.

B. Fabricate items with joints tightly fitted and secured.

C. Fit and shop assemble in largest practical sections, for delivery to site.

D. Supply components required for anchorage of metal fabrications. Fabricate anchorage and related components of same material and finish as metal fabrication, except where specifically noted otherwise.

E. Collect metal cutoffs and scrap; place in designated area for recycling.

1.03 FINISHING - GENERAL

A. Clean surfaces of rust, scale, grease, and foreign matter prior to finishing.

B. Galvanized Steel: Shop applied cold galvanized after fabrication. Apply in conformance with manufacturer's directions for intended use.

C. Shop Primed Steel: Shop prime steel items not indicated to be galvanized. Remove loose mill scale and rust from structural steel, and remove mill scale and rust from ornamental metal. Thoroughly clean to remove oil, grease, dirt and other foreign material, and apply one shop coat of rust-inhibitive primer containing at least 50% rust-inhibitive pigments. Carefully apply paint to provide smooth and even surface. Minimum dry film thickness 2.0 mils.

D. Stainless Steel: Directional Satin Finish: No. 4.

E. Do not prime surfaces in direct contact bond with concrete or where field welding is required.

F. Prime Paint: Minimum dry film thickness 2.0 mils.

1.04 METAL FABRICATION SCHEDULE

A. Shop fabricated ferrous metal items, galvanized or prime painted, including but not limited to the following:
1. Support framing for overhead door.
2. Miscellaneous clip angels, plates and other miscellaneous steel frames and supports.
3. Fasteners necessary for connection of metal fabrications.
4. Stainless steel base.

ABBREVIATIONS

A

AB	- ANCHOR BOLT
AC	- AIR CONDITIONING
ACC	- ACCESSIBLE
ACOUS	- ACOUSTICAL
ACT	- ACOUSTICAL CEILING TILE
AD	- AREA DRAIN
ADD	- ADDITIONAL
ADJ	- ADJUSTABLE
AFF	- ABOVE FINISH FLOOR
AGGR	- AGGREGATE
ALT	- ALTERNATE
ALUM	- ALUMINUM
ANCH	- ANCHOR
ANNO	- ANODIZED
APC	- ACOUSTICAL PANEL CEILING
APPD	- APPROVED
APPROX	- APPROXIMATE
ARCH	- ARCHITECTURAL
AUTO	- AUTOMATIC
AV	- AUDIO VISUAL

B

BEJ	- BRICK EXPANSION JOINT
BD	- BOARD
BLDG	- BUILDING
BLK	- BLOCK
BM	- BEAM
BOT	- BOTTOM
BRK	- BRICK
BSMT	- BASEMENT
BT	- BOLT
BUR	- BUILT-UP ROOFING

C

CAB	- CABINET
CAT	- CATEGORY
CB	- CATCH BASIN
CBD	- CHALK BOARD
CBU	- CEMENTITIOUS BACKER UNIT
CEM	- CEMENT
CER	- CERAMIC
CG	- CORNER GUARD
CH	- CHILLER
CHAN	- CHANNEL
CI	- CAST IRON
CIP	- CAST-IN-PLACE
CJ	- CONTROL JOINT / CONSTRUCTION JOINT
CL	- CENTER LINE
CLG	- CEILING
CLR	- CLEAR
CMU	- CONCRETE MASONRY UNIT
CNTR	- COUNTER
CO	- CLEANOUT
COL	- COLUMN
COMPART	- COMPARTMENT
CONC	- CONCRETE
COND	- CONDITION
CONN	- CONNECTION
CONT	- CONTINUOUS
CONTR	- CONTRACTOR
COORD	- COORDINATE
CORR	- CORRIDOR
CT	- CERAMIC TILE / COOLING TOWER
CTR	- CENTER
CTSK	- COUNTER SUNK
CW	- COLD WATER (PIPING)

D

D	- DEEP, DEPTH
DBL	- DOUBLE
DEG	- DEGREE
DEMO	- DEMOLITION
DEPT	- DEPARTMENT
DET	- DETAIL
DF	- DRINKING FOUNTAIN
DIA	- DIAMETER
DIFF	- DIFFUSER
DIM	- DIMENSION
DISP	- DISPENSER
DMPF	- DAMPPROOFING
DMT	- DEMOUNTABLE
DN	- DOWN
DO	- DOOR OPENING
DP	- DIMENSION POINT
DPTN	- DEMOUNTABLE PARTITION
DR	- DOOR
DRN	- DRAIN
DS	- DOWNSPOUT
DW	- DISHWASHER
DWG	- DRAWING
DWR	- DRAWER

E

(E)	- EXISTING
E	- EAST
EA	- EACH
EB	- EXPANSION BOLT
EIFS	- EXTERIOR INSULATION AND FINISH SYSTEM
EFS	- EXTERIOR FINISH SYSTEM
EJ	- EXPANSION JOINT
EL	- ELEVATION
ELEC	- ELECTRICAL
ELEV	- ELEVATOR
EMERG	- EMERGENCY
ENCLO	- ENCLOSURE
EP	- ELECTRICAL PANELBOARD
EQ	- EQUAL
EQUIP	- EQUIPMENT
ESCAL	- ESCALATOR
EWC	- ELECTRICAL WATER COOLER
EXH	- EXHAUST
EXP	- EXPANSION
EXIST	- EXISTING
EXT	- EXTERIOR

F

FA	- FIRE ALARM
FB	- FACE BRICK
FC	- FACE
FD	- FLOOR DRAIN
FDC	- FIRE DEPARTMENT CONNECTION
FND	- FOUNDATION
FE	- FIRE EXTINGUISHER
FEC	- FIRE EXTINGUISHER CABINET
FF&E	- FURNITURE, FINISHES & EQUIPMENT
FFEL	- FINISH FLOOR ELEVATION
FH	- FLAT HEAD
FHC	- FIRE HOSE CABINET
FIN	- FINISH
FIXT	- FIXTURE
FL	- FLOOR
FLASH	- FLASHING
FLUOR	- FLUORESCENT
FO	- FACE OF
FP	- FIRE PROTECTION
FPG	- FIREPROOFING
FR	- FRAME
FT	- FEET
FRTW	- FIRE RETARDANT TREATED WOOD
FTG	- FOOTING
FURN	- FURNITURE
FURR	- FURRING
FWC	- FABRIC WALLCOVERING
FWP	- FABRIC WRAPPED PANEL

G

GA	- GAUGE / GAGE
GALV	- GALVANIZED
GB	- GRAB BAR
GC	- GENERAL CONTRACT(OR)
GRC	- GLASS FIBER REINFORCED CONCRETE
GRG	- GLASS FIBER REINFORCED GYPSUM
GL	- GLASS
GR	- GRADE
GRND	- GROUND
GYP BD	- GYPSUM BOARD

H

H	- HIGH / HEIGHT
HB	- HOSE BIB
HC	- HOLLOW CORE
HCP	- HANDICAPPED
HDW	- HARDWARE
HDWD	- HARDWOOD
HS	- HEAT STRENGTHENED (GLASS)
HM	- HOLLOW METAL (STEEL FRAME)
HNDRL	- HANDRAIL
HO	- HOLD-OPEN
HORIZ	- HORIZONTAL
HPT	- HIGH POINT
HRC	- HOSE REEL CABINET
HR	- HOUR
HT	- HEIGHT
HVAC	- HEATING, VENTILATION, AIR CONDITIONING
HW	- HOT WATER
HYDR	- HYDRAULIC

I

ID	- INSIDE DIAMETER
IN	- INCH
INCAND	- INCANDESCENT
INCL	- INCLUDED / INCLUDING
INFO	- INFORMATION
INSUL	- INSULATION
INT	- INTERIOR
INTERM	- INTERMEDIATE
INV	- INVERT
IPS	- INTERNATIONAL PIPE STANDARD
IRMA	- INVERTED ROOF MEMBRANE ASSEMBLY

J

JAN	- JANITOR
JC	- JANITOR'S CLOSET
JST	- JOIST
JT	- JOINT

K

K	- KIP (1000 LBF)
KPL	- KICK PLATE
KG	- KILOGRAM
KIT	- KITCHEN
KO	- KNOCKOUT

L

L	- LONG OR LITER (METRIC DOCS)
LAB	- LABORATORY
LDG	- LANDING
LAM	- LAMINATE / LAMINATION
LAV	- LAVATORY
LB	- POUND
LF	- LINEAR FOOT
LKR	- LOCKER
LLH	- LONG LEG HORIZONTAL
LLV	- LONG LEG VERTICAL
LN	- LINE
LT	- LIGHT
LPT	- LOW POINT

M

M	- METER
MACH	- MACHINE
MAINT	- MAINTENANCE
MAS	- MASONRY
MATL	- MATERIAL
MAX	- MAXIMUM
MB	- MACHINE BOLT
MBL	- MARBLE
MDF	- MEDIUM DENSITY FIBERBOARD
MEP	- MECHANICAL, ELECTRICAL, PLUMBING
MDO	- MEDIUM DENSITY OVERLAY PLYWOOD
MECH	- MECHANICAL
MEMB	- MEMBRANE
MEZZ	- MEZZANINE
MFR	- MANUFACTURER
MH	- MANHOLE
MIN	- MINIMUM
MIR	- MIRROR
MISC	- MISCELLANEOUS
MM	- MILLIMETER
MO	- MASONRY OPENING
MS	- MACHINE SCREW
MTD	- MOUNTED
MTG	- MOUNTING
MTL	- METAL
MULL	- MULLION

N

N	- NORTH
NA	- NOT APPLICABLE
NC	- NOISE CRITERIA
NIC	- NOT IN CONTRACT
NO	- NUMBER
NOM	- NOMINAL
NTS	- NOT TO SCALE

O

OA	- OUTSIDE AIR
OC	- ON CENTER
OCEW	- ON CENTER EACH WAY
OD	- OUTSIDE DIAMETER/DIMENSION
OFCI	- OWNER FURNISHED, CONTR INSTALLED
OFOI	- OWNER FURNISHED, OWNER INSTALLED
OFF	- OFFICE
OH	- OVER HEAD
OPH	- OPPOSITE HAND
OPNG	- OPENING
OPP	- OPPOSITE
ORD	- OVERFLOW ROOF DRAIN
OTD	- OPEN TO DECK
OUTS	- OUTSIDE

P

P	- PAINT
PAV	- PAVING
PATD	- PAPER TOWEL DISPENSER
PBD	- PARTICLEBOARD
PC	- PRECAST CONCRETE
PDF	- POWER DRIV

DIVISION 6 - WOOD AND PLASTICS

SECTION 06 10 00 - ROUGH CARPENTRY

1.01 MATERIALS

A. Lumber: PS 20; graded in accordance with established Grading rules; of following species and grades:
1. Non-Structural Light Framing (less than 2 in thick): Douglas Fir (WCLIB or WWPA) or Southern Pine kiln dried (SPIB); moisture content 19% maximum at time of dressing "S-DRY", or 15% maximum "MC-15" or "K-D"; graded as follows:
a. General framing: Standard and better, or Stud grade.
b. Blocking and nailers: Up to 2 x 10 in. Utility grade and better; over 2 x 10 in. Construction grade and better.
c. General utility purposes: Economy grade.

B. Equipment Backing Panels: Plywood, APA A-C, Interior – Exposure 1, Group 2, fire-retardant treated; 5/8 in. min. thickness. Product shall have no added urea-formaldehyde.

C. Nails, Spikes and Staples: Galvanized for exterior locations, high humidity locations and treated wood; plain finish for other interior locations; size and type to suit application and in accordance with manufacturer's recommendations.

D. Bolts, Nuts, Washers, Lags, Pins and Screws: Medium carbon steel; sized to suit application and in accordance with manufacturer's recommendations; galvanized for exterior locations, high humidity locations and treated wood; plain finish for other interior locations.

E. Fasteners: Toggle bolt type for anchorage to hollow masonry. Expansion shield and lag bolt type for anchorage to solid masonry or concrete. Bolts or power activated type for anchorage to steel.

1.02 ACCESSORIES

A. Dust and Vapor Barrier: reinforced flame retardant polyethylene sheets, 6 mil minimum thickness.

B. Polypropylene Tape: Flame retardant self-adhering type, 2" wide.

1.04 BLOCKING

A. Fasten wood blocking to framing with fasteners capable of withstanding loads to be applied to blocking. Install blocking for support of items as required.

B. Install continuous pieces of longest possible lengths, cut to fit and fully bearing on framing.

1.05 DUST PROOF ENCLOSURES

A. Cover wood studs with reinforced flame retardant polyethylene sheets. Locate joints in polyethylene sheets over studs, and horizontal blocking and attach to studs and blocking with staples at 6" on center. Lap joints 6" and seal with tape. Seal perimeter of dust proof enclosures to adjacent existing construction with tape. Maintain dust proof enclosure in sound condition during construction and remove when no longer required.

1.06 ROUGH CARPENTRY SCHEDULE

A. Fire Retardant treatment of wood members where required.

B. Dust-proof enclosures.

SECTION 06 41 00 - CUSTOM CASEWORK

1.01 WOOD MATERIALS

A. Softwood Lumber: PS 20; graded in accordance with AWI; maximum moisture content of 11 percent; species and grade as follows:

Item	Species	Cut
Cabinet Frame Concealed	Pine	No.2

B. Hardwood Lumber (**W-X**): Graded in accordance with AWI; maximum moisture content of 6 percent. Provide species, grade and cut as indicated In Finish Schedule on the Drawings.

1.02 SHEET MATERIALS

A. Wood Particle board: Per AWI standard, composed of wood chips, made with high density board with waterproof resin binders; of grade to suit application. All products to be formaldehyde free.

B. Hardwood Plywood: AWI Grade AA face veneers with type of glue recommended for application, and the following:
1. Core: Veneer.
2. Face Species of Veneer: As indicated in Finish Schedule on the Drawings.
3. Finish: Walnut Satin to match Architect's sample.
4. Wood veneer to match Architect's approved sample.
5. Veneer-Faced Panel Products (Hardwood Plywood): HPVA HP-1,made with adhesive containing no urea formaldehyde.

1.03 LAMINATE MATERIALS

A. Plastic Laminate: NEMA LD 3, GP-50 for horizontal surfaces and GP-28 for vertical surfaces.

B. Laminate Backing Sheet: Backing grade, undecorated plastic laminate.

C. Colors/Finish: (**PLAM-X**): See Finish Schedule for plastic laminate material types, colors, finishes and locations as indicated and scheduled.

1.05 PLASTIC LAMINATE CLAD CABINETS (CASEWORK)

A. Fabricate plastic laminate finished casework using "Flush Overlay" style, "Custom" grade, as defined by AWS.

B. Shop assemble casework for delivery to site in units easily handled and to permit passage through building openings.

C. Fit doors and exposed edges with matching plastic laminate edging. Use full length pieces only.

D. Fit shelves with matching plastic edging. Use full length pieces only.

E. Door and Drawer Fronts: 3/4" thick.

F. When necessary to cut and fit on site, provide materials with ample allowance for cutting. Provide trim for scribing and site cutting.

G. Apply plastic laminate finish in full uninterrupted sheets consistent with manufactured sizes. Make corners and joints hairline. Slightly bevel arises.

H. Cabinet Liner: Melamine plastic laminate as supplied by manufacturer of plastic laminate. Provide Melamine on all exposed interior surfaces.

I. Cap exposed plastic laminate edges with material of same finish and pattern.

J. Apply laminate backing sheet to reverse side of plastic laminate finished surfaces.

K. Provide cutouts for plumbing fixtures, inserts, appliances, outlet boxes, and other fixtures and fittings. Verify locations of cutouts from on-site dimensions. Seal contact surfaces of cut edges.

1.07 SOLID SURFACING MATERIAL

A, Solid Surfacing Material (SS-X): One piece monolithic solid material, methyl methacrylate binder. Homogeneous material, not coated, laminated or of composite construction.
1. See Finish Schedule on the Drawings for Solid Surfacing material types, colors and finishes and locations as indicated and scheduled.

B. Fabrication:
1. Fabricate components by mold to achieve required shape and configuration.
2. Finish exposed surfaces smooth and polish to a satin sheen.
3. Radius corners and edges, unless indicated otherwise, eased edge detail.

1.08 HARDWARE
A. Cabinet Hardware Standards: ANSI A 156. 9, Type 2 Institutional quality; US26 brushed chrome finish; unless otherwise noted.

B. Drawer and Door Pulls: See Finish Schedule

C. Drawer Slides: Full extension, cold rolled steel, zinc plated, captive polymer ball bearing rollers with 100 lb. minimum load rating, positive in-stop, out-stop and out-keeper.
1. Accuride Model 3832.

D. File Drawer Slides: Full extension, cold rolled steel, zinc plated, captive polymer ball bearing rollers with 110 lb. minimum load rating, positive in-stop, out-stop and out-keeper.
1. Accuride Model 4037.

E. Magnetic Catches:
1. Pull Type: Knape and Voyt, #916.
2. Touch Type: Amerock #CM 32301 41; Lamp Sugatsune ML 80

F. Concealed Hinges: Concealed casework type, self-closing from 10 deg.; Stanley, #1510 Series.

G. Countertop Connectors: Concealed type; Knape and Voyt, #516.

H. Adjustable Shelf Standards and Supports: Steel, nickel plated, 5/8" wide x 3/16" deep; Knape and Voyt Mfg. Co., #255NP standards with #256 supports.

I. Locks: All cabinet drawers and doors where indicated, provide Best standard pin type (five pins)tumbler locks. All locks to have the same keyway.

J. Grommets: Doug Mockett and Company, minimum 3" diameter with cover.
1. Color: As selected by Architect.

DIVISION 7 - THERMAL AND MOISTURE PROTECTION

SECTION 07 01 50.19 - PREPERATION FOR RE-ROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Infill Materials
2. Partial roof tear-off

1.2 QUALITY ASSURANCE

A. Installer Qualifications: Installer of new membrane roofing system, approved by warrantor of existing roofing system to work on existing roofing.

1.3 PROJECT CONDITIONS.

A. Owner will occupy portions of building immediately below reroofing area. Conduct reroofing so Owner's operations will not be disrupted.

1.4 WARRANTY

A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during reroofing, by methods and with materials so as not to void existing roofing system warranty. Notify warrantor before proceeding.
1. Notify warrantor of existing roofing system on completion of reroofing and obtain documentation verifying that existing roofing system has been inspected and warranty remains in effect.

PART 2 - PRODUCTS

2.1 INFILL MATERIALS

A. Use infill materials matching existing membrane roofing system materials unless otherwise indicated.

PART 3 - EXECUTION

3.1 PREPERATION

A. Protect existing membrane roofing system that is indicated to be reroofed.

B. Coordinate with Owner to shut down air-intake equipment in the vicinity of the work. Cover air-intake louvers before proceeding with reroofing work that could affect indoor air quality or activate smoke detectors in ductwork.

C. During removal operations, have sufficient and suitable materials on-site to facilitate rapid installation of temporary protection in the event of unexpected rain.

3.2 INFILL MATERIALS INSTALLATION

A. Immediately after removal of selected portions of existing membrane roofing system, and inspection and repair, if needed, of deck, fill in the tear-off areas to match existing membrane roofing system construction.

3.3 DISPOSAL

A. Collect demolished materials and place in containers. Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site.

B. Transport and legally dispose of demolished materials off Owner's property.

SECTION 07 20 00 - INSULATION

1.01 INSULATION MATERIALS

A. INSUL-1: Glass Fiber Blanket/Batt Insulation (Unfaced); ASTM C665, Type I, preformed glass fiber blanket without facers. Minimum 20% total recycled content for glass cullet.
1. CertainTeed Corporation, Unfaced Building Insulation.
2. Johns Manville; Thermal-Shield Unfaced Insulation Blankets.
3. Owens-Corning Fiberglass Corp., Unfaced Glass Fiber Insulation.

B. INSUL-2: Sound Insulation Batts; ASTM C665, Type I and ASTM E136, semi-rigid fiberglass insulation, 6-1/4" minimum thickness. Minimum 20% total recycled content for glass cullet.
1. Owens-Corning Corp., Sonobatts.
2. Comparable units from other acceptable manufacturers that have been tested for air erosion and can be used in return air plenums.

C. Install sound attenuation insulation in partitions and above suspended ceiling according to manufacturer's instructions as indicated on the drawings.

SECTION 07 92 00 - JOINT SEALERS

1.01 SEALANT MATERIALS

A. SLNT-1: Self-leveling, two component, non-staining, non-bleeding polyurethane sealant with movement capability of + 25 percent and conforming to ASTM C920, Type M, Grade P, Class 25. Color: As selected by the Architect.
1. Mameco International, "Vulkem 245".
2. Pecora Corporation, "Urexpan NR-200".
3. Sonneborn Building Products, Sonolac.
4. Tremco, "THC-900".

B. SLNT-2: One part, non-sag, skinning, paintable acrylic latex sealant capable of + 7.5 percent minimum movement and conforming to ASTM C834. Color: As selected by Architect.
1. Pecora Corporation, AC-20.
2. Sonneborn Building Products, Sonolac.
3. Tremco, Tremflex 834.

C. SLNT-3: Silicone Sanitary Sealant; ASTM C920, Type S, Grade NS; mildew resistant. Color: Clear.
1. Dow Corning Corporation, "786 Mildew Resistant Silicone Sealant".
2. General Electric Company, "GE Silicone Sanitary 1700 Sealant".
3. Sonneborn Building Products, "Omniplus".

D. SLNT-5: Acoustical Sealant; Non-hardening, non-skinning acoustical sealant for concealed joints.
1. Tremco, Acoustical Sealant.
2. United States Gypsum, Acoustical Sealant.

1.02 PRIMERS

A. Non-staining type, as recommended by sealant manufacturer to suit application.

1.03 JOINT BACKING

A. Backer Rod: Provide pre-formed, compressible, non-staining, closed cell polyethylene or open cell expanded polyurethane foam rod compatible with the sealant used and acceptable by the sealant manufacturer for the specific application.

B. Provide round rod, size as recommended by manufacturer based on width and depth of joint and condition located.

C. Do not use joint backing impregnated with oil, tar, or asphalt.

1.04 ACCESSORIES

A. Joint Cleaner: Non-corrosive and non-staining type, recommended by sealant manufacturer; compatible with joint forming materials.

B. Bond Breaker: Pressure sensitive tape recommended by sealant manufacturer to suit application.

1.05 SEALANT SCHEDULE

Type	Location
SLNT-1	Joints in floor tilework.
SLNT-2	Miscellaneous interior joints subject to moderate movement and joints to be painted.
SLNT-3	Joints surrounding plastic laminate counter tops, fixtures and equipment requiring sanitary sealant.
SLNT-5	Joints at sound-proofed partitions.

SECTION 07 72 00 - ROOF ACCESSOREIS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Inbludes:
1. Roof hatches.

PART 2 - PRODUCTS

2.1 ROOF HATCHES

A. Roof Hatches: Metal roof-hatch units with lids and insulated single-walled curbs, welded or mechanically fastened and seals corner joints, contuous lid-to-curb counterflashing and weathertight perimeter gasketing and integrally formed desk-mounted flange at perimeter bottom.
B. Type and Size: Single leaf lid, 30 by 36 inches.
C. Hatch Material: Zinc-coated (galvanized) steel sheet
1. Thickness: Manufacturere's standard thickenss for hatch size indicated.
2. Finish: Backed enamel or power coated.
3. Color: As selected by Architect from manufacturer's full range.

D. Construction:
1. Insualtion: 2-inch (50mm) thick, polyisocyanurate board.
a. R-Value: 12.0 according to ASTM C1363.
2. Nailer: Factory-installed wood nailer continuous around hatch perimeter
3. Hatch Lid: Opaque, insulated, and double walled, with manufacturer's standard metal liner of same material and finish as outer metal lid.
4. Fabricate curbs to minimum height of 12 inhses (305 mm) above roofing surface unless otherwise indicted

E. Hardware: Spring operators, hold-open arm, galvanished steel spring latch with turn hanges, galvanized steel butt- or pintle-type hinge systesm, and padlock hasps inside and outside.

F. Safety Railing System: Roof-hatch manufacturer's standard system including rails, clamps, fasteners, safety barrier at railing opening, and accessories required for a complete installation; attahed to roof hatch and complying with 29 CFR 1910.23 requirements and authorities having jursdiction.
1. Height: 42 inches (1060 mm) above finished roof deck
2. Post and Rails: Galvanized-steel pipe, 1-1/4 inches (31 mm) in diameter to galvanized-steel tube, 1-5/8 inches (41 mm) in diameter.
3. Flas Bar: Galvanized steel, 2 inches (50 mm) high by 3/8 inch (9mm) thick.
4. Maximum Opening Size: System constructed to prevent passage of a sphere 21 inches (533 mm) in diameter.
5. Self-Latching Gate: Fabricated of same materials and rail spacing as safety railing system. Provide manufacturer's standard hinges and self-latching mechanism.
6. Post and Rail Tops and Ends: Weather resistant, closed or plugged with prefabricated end fittings.
7. Provide weep holes or another means to drain entrapped water in hollow sections of handrails and railing members.
8. Fabricate joints exposed to weather to be watertight.
9. Fasteners: Manufacturer's standard, finished to match railing sytem.
10. Finish: Manufacturer's standard.

G. Ladder-Assist Post: Roof-hatch manufacturer's standard device for attachement to roof-access ladder.
1. Operation: Post locks in place of full extension; release mechanism returns post to closed position.
2. Height: 42 inches (1060 mm) above finished roof deck.
3. Material: Steel tube.
4. Post: 1-5/8 inch (41 mm) diameter pipe.
5. Finish: Manufacturer's standard baked enamal or powder coat.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, to verify actual locations, dimensions, and other conditions affecting performance of the work.
B. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
C. Verify dimensions of roof openings for roof accessories.
D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Roof-Hatch Installation:
1. Install roof hatch so top surface of hatch curb is level.
a. Attach curbs to wood nailer on toof deck with wood screws.
2. Anchor counter-flashing to wood nailer with lag screws and lead washers.
3. Verify that roof hatch operates properly. Clean, lubricate, and adjust operating mechanism and hardware.
4. Attached safety railing system to roof-hatch curb.
5. Attach ladder-assist post accoring to manufacturer's written instructions.

3.3 REPAIR AND CLEANING

A. Clean exposed surfaces according to manufacturer's written instructions.
B. Clean off excess sealants.

END OF SECTION 07 72 00

SECTION 07 92 00 - JOINT SEALERS

1.01 VOC Content of Interior Sealants:

A. Provide sealants and sealant primers for use inside the weatherproofing system that comply with the following limits for VOC content when calculated according to 40 CFR 59, Part 59, Subpart D (EPA Method 24):
1. Architectural Sealants: 250 g/L.
2. Sealant Primers for Nonporous Substrates: 250 g/L.
3. Sealant Primers for Porous Substrates: 775 g/L.

1.02 SEALANT MATERIALS

A. SLNT-1: Self-leveling, two component, non-staining, non-bleeding polyurethane sealant with movement capability of + 25 percent and conforming to ASTM C920, Type M, Grade P, Class 25. Color: As selected by the Architect.
1. Mameco International, "Vulkem 245".
2. Pecora Corporation, "Urexpan NR-200".
3. Sonneborn Building Products, "Sonolastic SL2".
4. Tremco, "THC-900".

B. SLNT-2: One part, non-sag, skinning, paintable acrylic latex sealant capable of + 7.5 percent minimum movement and conforming to ASTM C834. Color: As selected by Architect.
1. Pecora Corporation, AC-20.
2. Sonneborn Building Products, Sonolac.
3. Tremco, Tremflex 834.

C. SLNT-3: Silicone Sanitary Sealant; ASTM C920, Type S, Grade NS; mildew resistant. Color: Clear.
1. Dow Corning Corporation, "786 Mildew Resistant Silicone Sealant".
2. General Electric Company, "GE Silicone Sanitary 1700 Sealant".
3. Sonneborn Building Products, "Omniplus".

D. SLNT-5: Acoustical Sealant; Non-hardening, non-skinning acoustical sealant for concealed joints.
1. Tremco, Acoustical Sealant.
2. United States Gypsum, Acoustical Sealant.

1.03 PRIMERS

A. Non-staining type, as recommended by sealant manufacturer to suit application.

1.04 JOINT BACKING

A. Backer Rod: Provide pre-formed, compressible, non-staining, closed cell polyethylene or open cell expanded polyurethane foam rod compatible with the sealant used and acceptable by the sealant manufacturer for the specific application.

B. Provide round rod, size as recommended by manufacturer based on width and depth of joint and condition located.

C. Do not use joint backing impregnated with oil, tar, or asphalt.

1.05 ACCESSORIES

A. Joint Cleaner: Non-corrosive and non-staining type, recommended by sealant manufacturer; compatible with joint forming materials.

B. Bond Breaker: Pressure sensitive tape recommended by sealant manufacturer to suit application.

1.06 SEALANT SCHEDULE

Type	Location
SLNT-1	Joints in floor tilework.
SLNT-2	Miscellaneous interior joints subject to moderate movement and joints to be painted.
SLNT-3	Joints surrounding plastic laminate counter tops, fixtures and equipment requiring sanitary sealant.
SLNT-5	Joints at sound-proofed partitions.

DIVISION 8 – DOORS AND WINDOWS

SECTION 08 11 13 - STEEL DOOR FRAMES

1.01 DOOR FRAMES/SIDELITES

A. Provide frames of required throat depth for wall thickness.

B. Attach fire rated label to each frame and door unit, where required.

1.02 ACCESSORIES

A. Rubber Silencers: Resilient rubber.

B. Glazing Stops: Construct removable glazing stops of 16 gauge steel. Factory fit and install stops. Form neat hairline joints. Install stops using screws not over 16" on center with at least two screws per piece.

1.03 FABRICATION

A. Fabricate frames with hardware reinforcement plates welded in place. Comply with SDI-100 for minimum hardware reinforcing requirements. Provide mortar guard boxes.

B. Prepare frame for silencers. Provide three single rubber silencers on strike jamb for single doors, and two single silencers on frame head at double doors.

C. Provide fixed glazing stops on secure side of interior frames.

D. Provide floor clips for frames. Provide jamb anchors spaced not over 30" on center for frames. Furnish temporary spreader bars and bracing.

E. Thoroughly clean, phosphate treat, and apply one coat of rust inhibitive primer containing at least 50% rust inhibitive pigments. Prime frames under removable glazing stops and prime back side of removable stops.

SECTION 08 41 13 - ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Aluminum-framed storefront systems.
2. Aluminum-framed entrance door systems.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.
1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

B. Shop Drawings: For aluminum-framed entrances and storefronts. Include plans, elevations, sections, full-size details, and attachments to other work.

C. Samples for Verification: For each type of exposed finish required, in manufacturer's standard sizes.

D. Delegated-Design Submittal: For aluminum-framed entrances and storefronts including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: For aluminum-framed entrances and storefronts to include in maintenance manuals.

B. Maintenance Data for Structural Sealant: For structural-sealant-glazed storefront to include in maintenance manuals. Include ASTM C1401 recommendations for post-installation-phase quality-control program.

1.4 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of aluminum-framed entrances and storefronts that do not comply with requirements or that fail in materials or workmanship within specified warranty period.
1. Warranty Period: Five years from date of Substantial Completion.

B. Special Finish Warranty, Factory-Applied Finishes: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of factory-applied finishes or anodized finishes within specified warranty period.
1. Deterioration Includes, but is not limited to, the following:
a. Color fading more than 5 Delta E units when tested according to ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested according to ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
2. Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain all components of aluminum-framed entrance and storefront system, including framing and accessories, from single manufacturer.

B. Comply with performance requirements specified, as determined by testing of aluminum framed entrances and storefronts representing those indicated for this Project without failure due to defective manufacture, fabrication, installation, or other defects in construction.

2.2 PERFORMANCE REQUIREMENTS

A. General Performance: Comply with performance requirements specified, as determined by testing of aluminum-framed entrances and storefronts representing those indicated for this Project without failure due to defective manufacture, fabrication, installation, or other defects in construction.

B. Deflection of Framing Members Supporting Glass: At design wind load, as follows:
1. Deflection Normal to Wall Plane: Limited to 1/175 of clear span or ¾ inch, whichever is smaller.

C. Structural: Test according to ASTM E330/E330M as follows:
1. When tested at positive and negative wind-load design pressures, storefront assemblies, including entrance doors, do not evidence deflection exceeding specified limits.
2. When tested at 150 percent of positive and negative wind-load design pressures, storefront assemblies, including entrance doors and anchorage, do not evidence material failures, structural distress, or permanent deformation of main framing members exceeding 0.2 percent of span.
3. Test Durations: As required by design wind velocity, but not less than 10 seconds.

D. Water Penetration under Static Pressure: Test according to ASTM E331 as follows:
1. No evidence of water penetration through fixed glazing and framing areas, including entrance doors, when tested according to a minimum static-air-pressure differential of 20 percent of positive wind-load design pressure, but not less than 15 lbf/sq. ft. (720 Pa).

E. Water Penetration under Dynamic Pressure: Test according to AAMA 501.1 as follows:
1. No evidence of water penetration through fixed glazing and framing areas when tested at dynamic pressure equal to 20 percent of positive wind-load design pressure, but not less than 15 lbf/sq. ft. (720 Pa).

F. Seismic Performance: Aluminum-framed entrances and storefronts shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

G. Energy Performance: Certified and labeled by manufacturer for energy performance as follows:
a. Fixed Glazing and Framing Areas: Air leakage for the system of not more than 0.06 cfm/sq. ft. (0.30 L/s per sq. m) at a static-air-pressure differential of [1.57 lbf/sq. ft. (75 Pa) when tested according to ASTM E283.
b. Entrance Doors: Air leakage of not more than 1.0 cfm/sq. ft. (5.08 L/s per sq. m) at a static-air-pressure differential of 1.57 lbf/sq. ft. (75 Pa).
2. Condensation Resistance: Design the wall and its components to not develop any visible interior condensation on framing members or glazing.

H. Thermal Movements: Allow for thermal movements resulting from ambient and surface temperature changes.

2.3 STOREFRONT SYSTEMS

A. Framing Members: Manufacturer's extruded- or formed-aluminum framing members of thickness required and reinforced as required to support imposed loads.
1. Exterior Framing Construction: Thermally broken.
2. Glazing System: Retained mechanically with gaskets on four sides.
3. Glazing Plane: Front.
4. Finish: Dark Bronze anodic finish.
5. Fabrication Method: Field-fabricated stick system.
6. Aluminum: Alloy and temper recommended by manufacturer for type of use and finish indicated.
7. Steel Reinforcement: As required by manufacturer.

B. Backer Plates: Manufacturer's standard, continuous backer plates for framing members, if not integral, where framing abuts adjacent construction.

C. Brackets and Reinforcements: Manufacturer's standard high-strength aluminum with nonstaining, nonferrous shims for aligning system components.

2.4 ENTRANCE DOOR SYSTEMS

A. Entrance Doors: Manufacturer's standard glazed entrance doors for manual-swing operation.
1. Door Construction: 1-3/4-inch (44.5-mm) overall thickness, with minimum 0.125-inch- (3.2-mm-), with minimum 0.125-inch- (3.2-mm-)] thick, extruded-aluminum tubular rail and stile members. Mechanically fasten corners with reinforcing brackets that are deeply penetrated and fillet welded or that incorporate concealed tie rods.
a. Thermal Construction: High-performance plastic connectors separate aluminum members exposed to the exterior from members exposed to the interior.
2. Door Design: Medium stile; 3-1/2-inch (88.9-mm) nominal width.
3. Glazing Stops and Gaskets: Manufacturer's standard.

2.5 ENTRANCE DOOR HARDWARE

A. General: Provide entrance door hardware for each entrance door, to comply with requirements in this Section. Hardware not specified in this Section is Specified in Section 087100 – Door Hardware.

1. Opening-Force Requirements:
a. Egress Doors: Not more than 15 lbf (67 N) to release the latch and not more than 30 lbf (133 N) to set the door in motion.

B. Strikes: Provide strike with black-plastic dust box for each latch or lock bolt; fabricated for aluminum framing.

C. Operating Trim: BHMA A156.6.

D. Weather Stripping: Manufacturer's standard replaceable components.
1. Compression Type: Made of ASTM D2000 molded neoprene or ASTM D2287 molded PVC.

E. Weather Sweeps: Manufacturer's standard exterior-door bottom sweep with concealed fasteners on mounting strip.

F. Thresholds: BHMA A156.21 raised thresholds beveled with a slope of not more than 1:2, with maximum height of 1/2 inch (12.7 mm).

2.6 GLAZING

A. Glazing: Comply with Section 088000 "Glazing."

B. Glazing Gaskets: Manufacturer's standard.

C. Glazing Sealants: As recommended by manufacturer.
1. Sealant shall have a VOC content of 250 g/L or less.

2.7 MATERIALS

A. Aluminum: Alloy and temper recommended by manufacturer for type of use and finish indicated.

B. Sheet and Plate: ASTM B209 (ASTM B209M).

C. Extruded Bars, Rods, Profiles, and Tubes: ASTM B221 (ASTM B221M).

D. Structural Profiles: ASTM B308/B308M.

E. Steel Reinforcement:
1. Structural Shapes, Plates, and Bars: ASTM A36/A36M.
2. Cold-Rolled Sheet and Strip: ASTM A1008/A1008M.
3. Hot-Rolled Sheet and Strip: ASTM A1011/A1011M.

F. Steel Reinforcement Primer: Manufacturer's standard zinc-rich, corrosion-resistant primer complying with SSPC-PS Guide No. 12.00; applied immediately after surface preparation and pretreatment. Select surface preparation methods according to recommendations in SSPC-SP COM, and prepare surfaces according to applicable SSPC standard.

2.8 ACCESSORIES

A. Fasteners and Accessories: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding fasteners and accessories compatible with adjacent materials.

B. Anchors: Three-way adjustable anchors with minimum adjustment of 1 inch that accommodate fabrication and installation tolerances in material and finish compatible with adjoining materials and recommended by manufacturer.

2.9 FABRICATION
A. Form or extrude aluminum shapes before finishing.

B. Weld in concealed locations to greatest extent possible to minimize distortion or discoloration of finish. Remove weld spatter and welding oxides from exposed surfaces by descaling or grinding.

C. Fabricate components that, when assembled, have the following characteristics:
1. Profiles that are sharp, straight, and free of defects or deformations.
2. Accurately fitted joints with ends coped or mitered.
3. Physical and thermal isolation of glazing from framing members.
4. Accommodations for thermal and mechanical movements of glazing and framing to maintain required glazing edge clearances.
5. Fasteners, anchors, and connection devices that are concealed from view to greatest extent possible.

D. Entrance Door Frames: Reinforce as required to support loads imposed by door operation and for installing entrance door hardware.
1. At interior and exterior doors, provide compression weather stripping at fixed stops.

E. Entrance Doors: Reinforce doors as required for installing entrance door hardware.
1. At pairs of exterior doors, provide sliding-type weather stripping retained in adjustable strip and mortised into door edge.
2. At exterior doors, provide weather sweeps applied to door bottoms.

F. Entrance Door Hardware Installation: Factory install entrance door hardware to the greatest extent possible. Cut, drill, and tap for factory-installed entrance door hardware before applying finishes.

G. After fabrication, clearly mark components to identify their locations in Project according to Shop Drawings.

2.10 ALUMINUM FINISHES

A. Color Anodic Finish: AAMA 611, AA-M12C22A42/A44, Class I, 0.018 mm or thicker.
1. Color: Dark bronze.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

A. Comply with manufacturer's written instructions.

B. Do not install damaged components.

C. Fit joints to produce hairline joints free of burrs and distortion.

D. Rigidly secure nonmovement joints.

E. Install anchors with separators and isolators to prevent metal corrosion and electrolytic deterioration and to prevent impeding movement of moving joints.

F. Seal perimeter and other joints watertight unless otherwise indicated.

G. Metal Protection:
1. Where aluminum is in contact with dissimilar metals, protect against galvanic action by painting contact surfaces with materials recommended by manufacturer for this purpose or by installing nonconductive spacers.
2. Where aluminum is in contact with concrete or masonry, protect against corrosion by painting contact surfaces with bituminous paint.

H. Set continuous sill members and flashing in full sealant bed, as specified in Section 079200 "Joint Sealants," to produce weathertight installation.

I. Install joint filler behind sealant as recommended by sealant manufacturer.

J. Install components plumb and true in alignment with established lines and grades.

3.3 INSTALLATION OF ALUMINUM-FRAMED ENTRANCE DOORS

A. Install entrance doors to produce smooth operation and tight fit at contact points.
1. Exterior Doors: Install to produce weathertight enclosure and tight fit at weather stripping.

Field-Installed Entrance Door Hardware: Install surface-mounted entrance door hardware according to entrance door hardware manufacturers' written instructions using concealed fasteners to greatest extent possible.

SECTION 08 71 00 - HARDWARE

1.01 SOURCE LIMITATIONS

A. Obtain each type and variety of aluminum, steel and wood door hardware from the same manufacturer, unless otherwise indicated.

B. Existing door hardware to be reused to the greatest extent possible.

1.02 GENERAL

A. Field Measurements: Field verify existing conditions, swings and functions for each opening which is part of the required work prior to submittal of Shop Drawings and ordering of hardware,. Clearly indicate on Shop Drawings dimension, installation and hardware items related to these openings.

B. Provide environmentally-sound practices in the manufacturing and shipping of finish hardware items. Provide Product Data for Credit EQ 4.1 & 4.2 – Recycled Content, documentation indicating that products incorporate recycled material.

1.03 ELECTRICAL

A. Electrical boxes in walls, electrical service, conductors and final connections provided under Division 16.

1.05 BUTT HINGES

A. Approved Manufacturers for bidding.
1. Ives (IVE). http://us.allegion.com/brands/ives/Pages/default.aspx
2. Bommer. www.bommerhinge.com .

B. Product notes and applications
1. All out swinging lockable doors shall have NRP hinges.
2. Width of hinges shall be sufficient to clear trim and wall conditions as shown on the drawings.
3. Size: 4 ½" x 4 ½" for doors up to 3'-0" in width and 4 ea. 4 ½" x 4 ½" for doors over 3'-4" in width, Exterior Doors and Vestibule Doors. Provide heavy weight hinges (.180) at exterior, vestibule and doors over 3'-4" and high traffic doors.

1.06 LOCKS AND LATCHES

A. Schlage (SCH): http://us.allegion.com/brands/schlage/Pages/default.aspx

B. Magnetic Locks and Accessories: Schlage (SCE)

http://us.allegion.com/Products/system_components/system_accessories/Pages/default.aspx?id=1
C. Install custom strikes to match existing frame preparations when new lock is scheduled to be installed in existing frame. Existing frames to be modified to accept new custom strikes.

1.07 CYLINDERS AND KEYING

A. Keying Schedule: Provide keying schedule per owner keying instructions.

B. Keys:
1. Master Keys: 4 cut keys per Masterkey symbol.
2. Change Keys: 2 cut keys per lock or cylinder.
3. Control Keys: 4 total.

C. Identification:
1. Emboss exclusive FACILITY CODE on all cut keys and key blanks.
2. Emboss DO NOT DUPLICATE on all cut keys and key blanks.

D. Bitting Transcript:
1. Provide factory generated bitting transcript on all factory master keyed products supplied.

1.08 PUSHES AND PULLS

A. Ives (IVE) http://us.allegion.com/brands/ives/Pages/default.aspx

B. Trimco www.trimcobbwmfg.com .

1.09 EXIT DEVICES

A. Von Duprin (VON) http://us.allegion.com/brands/von_duprin/Pages/default.aspx

1.10 DÉCOR EXIT DEVICES

A. CRLaurence (CRL) www.crlaurence.com

1.11 DOOR CLOSERS

A. LCN (LCN): http://us.allegion.com/brands/lcn/Pages/default.aspx - No Substitution.
B. Surface mounted LCN 4011/ 4111 series with closer arm as required. Closer bodies to have three separate control valves (including back check). All closers to be equipped with size adjustment (1 - 6).

C. Product notes and applications:
1. Closers shall have non-ferrous covers; heavy duty forged steel arms, and separate valves for adjusting back check, closing and latching cycles and adjustable spring to provide sizes 1 through 6.
2. Provide non-sized closers, adjustable to meet maximum opening force requirements of ADA.
3. Provide drop plates, brackets, or adapters for arms as required to suit details.
4. Mount closers on room side of corridor doors and inside of exterior doors. Where possible install closers on door for optimum aesthetics.
5. Provide forged heavy duty parallel arms. Non-hold open types.

1.12 OVERHEAD STOPS AND HOLDERS

A. Acceptable manufacturers and respective catalog numbers:
1. HD Concealed, HD Surface Mt. AND Std Surface Mt: Glynn-Johnson (GLY): www.glynnjohnson.ingersollrand.com

B. Product notes and applications:
1. Install overhead stops where conditions limit the use of wall stops and floor stops would be a tripping hazard.
2. Use special template closers to allow offset arms for surface applied stops.

1.13 AUTOMATIC FLUSH BOLTS AND COORDINATORS

A. Ives (IVE) http://us.allegion.com/brands/ives/Pages/default.aspx .

B. Trimco www.trimcobbwmfg.com .

1.14 MANUAL FLUSH BOLTS

A. Ives (IVE) http://us.allegion.com/brands/ives/Pages/default.aspx .

B. Trimco www.trimcobbwmfg.com .

1.15 GASKETING: WEATHERSTRIPPING, SMOKE SEALS, THRESHOLDS AND DOOR BOTTOMS

A. Zero International, Inc: www.zerointernational.com.

B. Reese www.reeseusa.com

C. Gasketing:
1. Weatherstrip shall be type as listed in hardware sets.
2. Smoke seals shall be type as listed in hardware sets.
3. Thresholds shall be type as detailed on plans or as listed in hardware sets.
4. Sweeps shall be type as listed in hardware sets.

1.16 PROTECTION PLATES

A. Ives (IVE) http://us.allegion.com/brands/ives/Pages/default.aspx .

B. Trimco www.trimcobbwmfg.com .

1.17 WALL STOPS AND HOLDERS

A. Acceptable manufacturers and respective catalog numbers:
1. Concave Wall Stop: McKinney WSG2 series, Rockwood 409
2. Stop/Holder: McKinney ADH01, Rockwood 490

B. Product notes and applications:
1. Reinforce behind all walls to ensure stop cannot be pushed through wall.

1.18 MISCELLANEOUS

A. Silencers: Supply three each at jambs of single doors and two

1.02 GYPSUM BOARD MATERIALS

A. Standard Gypsum Board: ASTM C36; 5/8" thick unless indicated otherwise, maximum permissible length; ends square cut, tapered and rounded edges.
1. Georgia-Pacific, G-P Gypsum Board (RE).
2. Gold Bond Building Products, Sta-Smooth Gypsum Wallboard.
3. United States Gypsum, Sheetrock SW Edge Gypsum Panels.

B. Moisture Resistant Gypsum Board: ASTM C630; 5/8" thick unless indicated otherwise, maximum permissible length; ends square cut, tapered edges.
1. Gold Bond Building Products, MR Board.
2. United States Gypsum, Sheetrock WR Gypsum Panels.

C. Glass Mat Water-Resistant Gypsum Backing Panel: ASTM C1178; 1/2" thick min unless indicated otherwise, maximum permissible length; ends square cut, tapered edges.
1. Georgia-Pacific; DensShield
2. United States Gypsum, Fiberock Aqua-tough Tile Backerboard Panels.

D. Sag-resistant Gypsum Ceiling Board: Complying with ASTM C 1396 (Section 12); 1/2" thick; ends square cut, tapered edges. Subject to compliance with requirements, provide one of the following:
1. Certainteed: 'ProRoc Interior Ceiling'.
2. G-P Gypsum: 'ToughRock CD Ceiling Board'.
3. Lafarge "Sagcheck Brand Sag-Resistant Drywall".
4. National Gypsum Co.: "Gold Bond Brand High-Strength Ceiling Board".
5. United States Gypsum Co.: 'Sheetrock Brand Interior Ceiling Board".

1.03 JOINT MATERIALS

A. Regular and Fire Rated Interior Gypsum Board Joint Material: Low VOC (20g/L max) Ready mixed, drying type, vinyl based joint material conforming to ASTM C475.
1. Georgia-Pacific, G-P Bedding Compound with G-P Tape, G-P Topping Compound.
2. Gold Bond Building Products, Sta-Smooth Joint Compound with Q-W Tape, Topping Compound.
3. United States Gypsum, Ready-Mixed Joint Compound-Tapping with Perf-A-Tape, Ready-Mixed Joint Compound-Topping.

B. Moisture Resistant Gypsum Board Joint Material: Factory prepackaged, job mixed, chemical hardening type joint material conforming to ASTM C475.
1. Georgia-Pacific, G-P Speed Set 90 with G-P Tape.
2. Gold Bond Building Products, Sta-Smooth 90 Joint Compound with Q-W Tape.
3. United States Gypsum, Sheetrock Brand Durabond 90 Joint Compound with Perf-A-Tape.

1.04 ACCESSORIES

A. Corner Beads: ASTM C1047, Cornerbead, galvanized steel corner bead with 1-1/4" perforated legs.
1. Clinch-On Products, Cornerbead.
2. Gold Bond Building Products, Wall Board Corner Bead.
3. United States Gypsum, Dur-A-Bead No. 103.

B. Control Joints:
1. Gold Bond Building Products, E-Z Strip.
2. United States Gypsum, Control Joint No. 93.

C. Edge Trim: ASTM C1047, LC Bead, galvanized steel "J"-shaped channel.
1. Clinch-On Products, Inc., U-Bead.
2. Gold Bond Building Products, No. 100 Wallboard Casing.
3. United States Gypsum, 200-A Metal Trim.

D. Laminating Adhesive: Screw attach. Adhesive attachment is not allowed.

E. Screws: ASTM C1002.

F. Reinforcing Tape: Paper only, fiberglass tape not permitted.

G. Partition clip: UPC- Partition Clip as manufactured by Armstrong.

H. Sound Attenuation Insulation: Specified in Section 07 20 0.

I. Acoustical Sealant: Specified in Section 07 92 00.

J. Ceiling Angle Trim:
1. Trim-Tex – 7/8" Mud on ceiling bead (Speed Bead).

1.05 GYPSUM CEILING SUSPENSION SYSTEM INSTALLATION

A. Install Gypsum Ceiling Suspension System in accordance with ASTM C636

1.06 ACOUSTICAL ACCESSORIES INSTALLATION

A. Place acoustical insulation in partitions fight within spaces, around cut openings, behind and around electrical and mechanical items within or behind partitions, and fight to items passing through partitions.

B. Install acoustical sealant within partitions in accordance with manufacturer's instructions.

1.07 CONTROL JOINTS

A. Install control joints as follows and in locations approved by Architect for visual effect:
1. Not to exceed 24'-0" in either direction for ceilings and not to exceed 30'-0" on center for partitions.
2. Where gypsum board abuts any dissimilar wall or ceiling assembly and where gypsum board construction changes within the same plane.
3. Where control joint occurs in continuous wall surface.
4. Where gypsum board construction abuts an exterior wall.
5. Where wall of different fire ratings abut or align in the same plane.
6. Make joint 1/4" wide with supports noncontinuous over joint.

1.08 DRYWALL FINISHING

A. Levels of finishing shall comply with those established by reference in GA-214 except as follows:

1. All walls to be level 5 gypsum board finish, apply joint compound combination specified for level 4 plus thin, uniform skim coat of joint compound over entire surface. Use joint compound specified for the finish (third coat) or a product specially formulated for this purpose and acceptable to gypsum board manufacturer. Produce surfaces free of tool marks and ridges ready for decoration of type indicated.
2. For level 4 gypsum board finish, embed tape in joint compound and apply three separate coats of joint compound over joints, angles, fastener heads, and accessories. Touch up and sand between coats and after last coat as needed to produce a surface free of visual defects and ready for decoration.
3. Level 3: For finished spaces scheduled for wall coverings classified as Grade III.
4. Level 2: For wallboard scheduled to receive ceramic tile, and in exposed spaces not painted.
5. Level 1: For concealed spaces, except where additional finishing is required to achieve fire-resistance rating, sound rating and where acting as an air or smoke barrier, meet Level 2 finishing requirements.
6. Level 0: Not permitted.

1.09 QUALITY ASSURANCE

A. Construct gypsum drywall partition and ceiling systems according to gypsum board manufacturer's current printed specifications.

B. Provide products of one manufacturer for each major product type, such as gypsum board or steel studs

C. Requirements of Regulatory Agencies:
1. Where a fire resistance rating is required, construct to achieve required fire resistance rating according to applicable UL listed assemblies.
2. For lateral support of partitions and suspended ceilings required for seismic bracing, construct the Work as specified herein and to meet applicable codes, ordinances, and regulations.
a. Provide lateral bracing for partitions not extending to structure above. Brace with opposing splayed hanger wires, minimum 12 gauge, at 8'-0" on center maximum.

D. Firestopping: Coordinate gypsum board Work with firestopping work at control joints, expansion joints and other penetrations through fire rated walls, smoke barrier walls, fire rated ceilings, and structural floors. Construct openings for firestopping assemblies to size indicated in the applicable UL systems

1.10 STUD GAGE/LIMITING HEIGHT SCHEDULE

A. Install framing in accordance with ASTM C754 and stud manufacturer's printed installation instructions where more stringent requirements are specified.

B. Metal stud spacing: 16" on center, unless otherwise indicated.

C. Isolate framing system from transfer of structural loading to system, both horizontally and vertically. Provide slip or cushioned type joint to attain lateral support and avoid axial loading.
1. Use deep-leg deflection track and firestop track where indicated.
2. Do not bridge building expansion joints with support system. Frame both sides of joint.

D. Door Opening Framing: Install double studs at door frame jambs. Install stud tracks on each side of opening, at frame head height, and between studs and adjacent studs.

E. Blocking: Bolt or screw wood blocking to studs. Wood blocking is required for support of plumbing fixtures, toilet partitions, wall cabinets, toilet accessories, items of finish hardware including wall mounted door stops, and other locations as indicated.

F. Coordinate installation of bucks, anchors, blocking, electrical and mechanical work placed in or behind partition framing.

SECTION 09 30 00 - TILING

1.01 GENERAL

A. Tile systems shall meet the requirements of ANSI A137.1 for slip resistance: Minimum .42 dynamic static coefficient of friction.

B. Factory Blending: For tile exhibiting color variations within ranges, blend tile in factory and package so tile units taken from one package show same range in colors as those taken from other packages and match approved Samples.
C. Mounting: For factory-mounted tile, provide back- or edge-mounted tile assemblies as standard with manufacturer unless otherwise indicated.
D. Factory-Applied Temporary Protective Coating: Where indicated, protect exposed surfaces by pre-coating with continuous film of petroleum paraffin wax, applied hot. Do not coat unexposed tile surfaces.

E. ANSI Standards for Tile Installation Materials: Provide materials complying with ANSI A108.02, ANSI standards referenced in other Part 2 articles, ANSI standards referenced by TCA installation methods specified in tile installation schedules, and other requirements specified.

1.02 TILE

A. Floor Tile: TL-1(R) & TL-2(R): as indicated in Finish Schedule on the Drawings.

1.03 SETTING MATERIALS

A. Floor Tile and Wall Tile (Wet Areas): Latex-Modified Portland Cement Mortar (Thin Set): Comply with requirements of ANSI A118.4.
1. For wall applications, provide mortar that complies with ANSI A118.4 requirements for non-sagging mortar in addition to the other requirements.
2. Subject to compliance with requirements, provide one of the following:
a. Bostik Construction Products: 'Title-Mate with 425 Multi-Purpose Acrylic Latex Admixture.'
b. C-Cure Corp.: 'MultiCure 905.'
c. Custom Building Products: 'Multiset Modified Thinset Mortar'.
d. Laticrete International:'254 Platinum,'or '211 Powder and 4237 Latex Additive.'
e. Mapei Corp.: 'Keraborid Dry-set Mortar with Keralastic Polymer Additive.'
f. TEC Specialty Construction Brands, Inc.; 'Full Set Plus with XtraFlex acrylic latex additive.'

B. Wall Tile (Dry Areas): Organic Adhesive: Comply with requirements of ANSI A136.1, Type I, with a VOC content of 65 g/L or less.
1. Bostick Construction Products, HDM 7000 Mastic.
2. Custom Building Products: 'ReliaBond Ceramic Tile Adhesive'.
3. Laticrete International, Latamastic.
4. Mapei Corporation: 'KER 903 Organic Adhesive.'
5. TEC Specialty Construction Brands, Inc.; 'Double Duty'.

1.04 GROUT

A. Floors: Sanded Polymer-Modified Tile Grout: Comply with requirements of ANSI A118.7.
1. Sanded products: Subject to compliance with requirements provide one of the following:
a. Bostik Construction Products: 'Hydroment Ceramic Tile Grout with 425 Multipurpose Acrylic Latex Admixture.'
b. Laticrete International: '1500 Sanded Grout with 1776 Grout Enhancer.'
c. Mapei Corp.: 'Keracolor S.'
d. TEC Specialty Construction Brands, Inc.: 'AccuColor Premium Sanded Grout with TEC Acrylic Grout Additive.'

B. Walls: Un-Sanded Polymer-Modified Tile Grout: Comply with requirements of ANSI A118.7.
1. Un-sanded products: Subject to compliance with requirements provide one of the following:
a. Bostik Construction Products: 'Hydroment Dry Tile Grout with 425 Multi Purpose Acrylic Latex Admixture.'
b. Custom Building Products: 'Dry Tile Grout with Acrylic Mortar Admix.'
c. Laticrete International: '1600 Unsanded Grout with 1776 Grout Enhancer.'
d. Mapei Corp.: 'Keracolor U.'
e. TEC Specialty Construction Brands, Inc.; 'AccuColor Premium Unsanded Grout with TEC Acrylic Grout Additive.'

1.05 WATERPROOF MEMBRANE

A. Elevated Tiled Floors: Chlorinated Polyethylene Sheet Waterproofing] Membrane: Non-plasticized, chlorinated polyethylene faced on both sides with non-woven polyester fabric; 0.030-inch nominal thickness. Subject to compliance with requirements, provide products similar to 'Nobleseal TS Waterproofing' as manufactured by The Noble Company.

1.06 TRANSITIONS

A. Metal Edge Strips: Angle or L-shape, height to match tile and setting-bed thickness. Profiles as indicated in Finish Schedule on Drawings.

1.07 INSTALLATION METHODS

A. Floors:
1. Slab-On-Grade Concrete: Thin-set; TCA F122
2. Elevated Concrete Slabs (Non-Wet Areas): Thin-set; TCA F122
3. Elevated Concrete Slabs (Wet Areas: Thin-set mortar on waterproof membrane; TCA F122

B. Walls:
1.Dry Areas: Organic adhesive on gypsum board; TCA W242
2.Wet Areas: Thin-set mortar on cementitious backer units or fiber cement underlayment; TCA W244

SECTION 09 51 00 - ACOUSTICAL CEILINGS

1.01 SUSPENDED ACOUSTICAL CEILINGS - GENERAL

A. All ceiling grid and acoustical tile units required for patching or replacement to match Building Standard. General Contractor to verify with Architect.

1.02 NEW SUSPENSION SYSTEM MATERIALS

A. Suspension System: ASTM C635, intermediate duty, non-fire rated exposed T, double web, 15/16" wide face, electro-galvanized commercial quality cold rolled steel, pre-painted white satin finish.
1. Armstrong World Industries, Prelude XL, Exposed Tee System.
2. USG Interiors, DX, Exposed Tee System.
3. Or approved substituted product of another manufacturer.

B. Accessories: Stabilizer bars, clips, splices edge moldings, hold down clips required for suspended grid system.

C. Support Channels and Hangers: Galvanized steel; size and type to suit application, to rigidly secure acoustic ceiling system including integral mechanical and electrical components with maximum deflection of 1/360.

1.03 NEW ACOUSTIC UNITS

A.ACT-1(R): As indicated in Finish Schedule on the Drawings.

1.04 PERIMETER TRIM

A. Trim Channel: Aluminum extrusions product by the suspension grid system manufacturer designed to attach to the suspension grid and provide positive mechanical lock with no visible fasteners. Factory-finished to match grid. Depth as indicated on drawings.
1. Armstrong World Industries, Axiom Classic Trim.
2. USG Interiors, Compasso Trim.
3. Or approved substituted product of another manufacturer.

1.05 EXTRA MATERIALS

A. Provide 5 percent of total installed for each acoustical panel type used. Deliver acoustical panels in manufacturer's original packaging, properly labeled

SECTION 09 54 26 – LINEAR WOOD CEILINGS

1.01 LINEAR WOOD CEILING PANELS AND SUSPENSION SYSTEM - GENERAL

A. Furnish and install as indicated in Finish Schedule and on the Drawings.

1.02 SUSPENSION SYSTEM MATERIALS

A. Suspension System: ASTM C635, heavy duty, non-fire rated T-Grid, double web, 15/16" wide face, electro-galvanized commercial quality cold rolled steel.

B. Accessories: Stabilizer bars, clips, splices edge moldings, required for suspended grid system.

C. Support Channels and Hangers: Galvanized steel; size and type to suit application, to rigidly secure linear wood ceiling system including integral mechanical and electrical components with maximum deflection of 1/360.

1.03 EXTRA MATERIALS

A. Provide 5 percent of total installed. Deliver acoustical panels in manufacturer's original packaging, properly labeled

SECTION 09 65 00 - RESILIENT FLOORING

1.01 RESILIENT FLOORING

A. Vinyl Composite Flooring [VCT-1(R)]: As indicated in the Finish Schedule on Drawings.

1.02 RESILIENT BASE MATERIALS

A. Base: FS SS-W-40, Type II vinyl; top set, 1/8" thick. Provide premolded endstops and external corners. Color: As scheduled on Drawings.
1. [VB-1(R)]: 4" high, coved base.

1.03 ACCESSORIES

A. Primers and Adhesives: Waterproof; of types recommended by resilient flooring manufacturer, low (50g/L max) or no VOC-emitting type, for specific flooring material and substrate.

B. Edge Strips/Transitions: Metal type, height as required for flooring thickness; as indicated in Finish Schedule on Drawings.

C. Sub.-floor Filler: Premixed cementitious paste. Tamms Industries Co., "Thin Patch"

D. Cleaning/Maintenance: Flooring manufacturer's recommended products for commercial applications.

SECTION 09 90 00 - PAINTING

1.01 MATERIALS

A. Provide best grade and quality materials and products manufactured by acceptable manufacturers.

B. Where a particular coating formula or custom mix is specified or indicated on Drawings, provide material from indicated manufacturer, unless other acceptable manufacturer's materials result in approved samples.

C. Except as otherwise specified, base coats and finish coats on a specified surface shall be compatible products of the same manufacturer. Provide field-applied materials that are compatible with factory-applied primers.

D. Other miscellaneous materials shall be pure, of highest commercial quality, and shall have identifying labels on containers.

E. All materials shall meet the 2006 VOC regulations of the Ozone Transport Commission (OTC).
Coating – Category CARB-OTC
Flat /Primer 100
Non-Flat 150

1.02 QUALITY ASSURANCE

A. Paint Coordination: Provide finish coats which are compatible with prime paints used. Review other Sections of these Specifications in which prime paints are to be provided to ensure compatibility of total coatings system for various substrates. Upon request from other trades, furnish information on characteristics of finish materials proposed for use, to ensure compatible prime coats are used. Provide barrier coats over incompatible primers or remove and re-prime as required. Notify the Architect, in writing, of anticipated problems using specified coating systems with substrates primed by others.

B. No claims as to the suitability of materials specified, or the applicator's inability to produce first class finishes with these materials will be considered unless such claims are made in writing prior to the start of the painting work.

C. Single-Source Responsibility: Provide primers and undercoat paint produced by the same manufacturer as the finish coats. Use only thinners approved by the paint manufacturer, and use only within recommended limits.

1.03 SUBMITTALS

A. Material and Equipment Submittals: Submit the following under provisions of Section 01 33 00:
1. Samples: Submit two 6" x 6" samples illustrating range of colors and textures available for each surface finishing product scheduled, for selection and final approval.
2. Manufacturer's application instructions.
3. Certification by the manufacturer that products supplied comply with local regulations controlling use of volatile organic compounds (VOCs).

1.04 EXTRA MATERIALS

A. Provide a one gallon container of each color and surface texture to Owner.

B. Label each container with color, texture, and room locations, in addition to the manufacturer's label. Store in Owner's designated storage area.

1.05 PRIMERS

A. Interior Primer/Sealer:
1. Glidden Professional, Lifemaster No VOC Interior Primer 9116-1200.
2. Hirshfield's, PRO-Z VOC Free Primer 1200
3. Moore, Moorcraft Latex Primer Sealer 273.
4. Pratt and Lambert, Vinyl Acrylic Wall Primer.
5. Sherwin Williams, Pro Mar 200 Zero VOC Interior Latex Wall Primer, B28W200.

B. Rust Inhibiting Primer:
1. Devoe Coatings, Devflex 4020PF Direct to Metal Primer & Flat Finish (VOC 91 g/l)
2. Hirshfield's, Acrylic DTM Primer 9250
3. (VOC < 250 g/l)
4. Moore, IronClad Retardo Rust Inhibitive Paint 163(VOC 375 g/l)
5. Sherwin Williams, KemKromik Metal Primer B50Universal Primer, B66-310 Series (99g/L VOCs)

1.06 FINISH PAINTS

A. Interior Eggshell Latex Enamel:
1. Glidden Professional, Ultra-Hide No VOC Interior Eggshell 1411 Series.
2. Hirshfield's, PRO-Z VOC Free Eggshell Latex Enamel 3590.
3. Moore, Moorecraft Latex Eggshell 274.
4. Pratt and Lambert, Aqua-Satin.
5. Sherwin-Williams, ProMar 200 Zero VOC Interior Latex Eg-Shel, B20-2600 Series.

B. Interior Flat Latex:
1. Glidden Professional, Ultra-Hide No VOC Interior Flat 1209 Series (VOC 0 g/l).
2. Hirshfield's, PRO-Z VOC Free Flat 3490 (VOC 0 g/l)
3. Benjamin Moore, Moorcraft Vinyl Latex Flat 275.
4. Pratt and Lambert, Vapex Flat Wall Finish.
5. Sherwin Williams, Pro Mar 200 Zero VOC Interior Latex Flat. (0 g/L VOCs)

C. Interior Semi-Gloss Latex Enamel:
1.Glidden Professional, Ultra-Hide No VOC Interior Semi-Gloss 1415 Series (VOC 0 g/l).
2.Hirshfield's, Contractor Select Semi-Gloss Interior Latex Enamel 2590 (VOC < 150 g/l)
3.Benjamin Moore, Moorcraft Latex Semi-gloss 276.
4.Pratt and Lambert, Accolade Interior Semi-Gloss Enamel.
5.Sherwin Williams, Pro Mar 200 Zero VOC Interior Latex Semi-Gloss Enamel, B31-2600 Series. (0 g/L VOCs)

D. Dry Fallout Coating:
1.Glidden Professional Waterborne Interior Flat Dry Fall 1280-1200 (VOC 50 g/l).
2.Hirshfield's, Prowall 2000 Interior Flat Dry Fog 1405 (VOC < 50 g/l).
3. Benjamin Moore, Sweep Up Airless Spray Alkyd Finish 347 Flat.
4. Pratt and Lambert, Pro Hide Dry Fall Out Spray Coating Satin.
5. Pittsburg Paints, Speedhide Super Tech Dry-Fog.
6. Sherwin-Williams, Low VOC Waterborne Acrylic Dry Fall, B42 Series

1.07 MISCELLANEOUS MATERIALS

A. Accessory Materials: Materials not specifically indicated but required to achieve the finishes specified, of commercial quality.

1.08 MIXING

A. Mix and thin materials according to manufacturer's instructions without adulteration. Use premixed paints insofar as possible. Thin only as permitted by manufacturer's label instructions, and avoid unnecessary thinning.

1.09 APPLICATION

A. Apply products in accordance with manufacturer's instructions or more stringent requirements specified herein. Roller applied coatings – where allowed - should pay special attention to the manufacture's recommended roller cover for each paint type.

B. During progress of Work maintain premises free of unnecessary accumulation of tools, equipment, surplus materials, and debris.

C. Match approved samples for color, texture, and coverage. Remove, refinish, or repaint work not in compliance with specified requirements, until such requirements are met.

1.10 PAINTING SCHEDULES - GENERAL

A. Painting schedules outline the primary paint systems required. For substrates and conditions not scheduled, provide paint systems recommended by paint manufacturers. Submit proposed paint material product data for approval.

B. The number of coats listed are field-applied coats. Provide all coats even if surfaces have been factory primed. Factory-applied one-coat combination primer-finishes shall be considered a factory prime coat only.

C. Match approved samples for color, texture, and coverage. Remove, refinish, or repaint work not in compliance with specified requirements, until such requirements are met.

D. At the completion of work of other trades, touch-up and restore damaged and defaced painted surfaces from wall corner to wall corner.
1. Costs for this touch-up painting shall be charged to the trade responsible for such damage, and at no additional cost to the Owner.
2. Notify the Contractor immediately upon such discovery.

1.11 INTERIOR PAINT SYSTEM SCHEDULE

A. IPS-1: Gypsum Board as indicated on Room Finish Schedule:
1 coat Interior Primer/Sealer
2 coats Interior Eggshell Latex Enamel.

B. IPS-2: Gypsum Board Ceilings as indicated on Room Finish Schedule:
2 coats Interior Flat Latex Enamel.

C. IPS-3: Hollow Metal Door Frames as indicated on Room Finish Schedule:
1 coat Rust Inhibiting Primer (if unprimed)
2 coats Interior Semi-gloss Latex Enamel.

D. IPS-4: Exposed ceiling structure as indicated on Room Finish Schedule:
1 coat Sprayed Dry Fallout Coating.

1.12 COLOR SCHEDULE

A. Color/Finish [PT-X(R)]: See Finish Plan for paint types and colors locations as indicated and scheduled.

1.13 SUBMITTALS

A. Material and Equipment Submittals: Submit under provisions of Section 01 33 00:
1. Product Data: Submit product data for each door type.
2. Samples: Submit 4" x 4" sample for final color approval.

SECTION 09 94 10 – ARCHITECTURAL FINISH FILMS

1.01 FINISH FILM

A. Vinyl Film [**WC-(R)**]: As indicated in Finish Schedule on the Drawings.

B. Substrate Primer and Sealer: As recommended by finish film manufacturer

1.02 SUBMITTALS

A. Material and Equipment Submittals: Submit the following under provisions of Section 01 30 00:
1. Manufacturer's installation instructions.
2. Samples: 7-inch by 9-inch sample of each finish film material.

1.03 EXTRA MATERIALS

A. Extra Materials: Provide 25 lineal feet of each type and color of finish film.

B. Package and label each roll by destination room number; store in Owner's designated storage area as directed.

SECTION 09 95 50 - WALL COVERINGS

1.01 WALLCOVERING

A. Vinyl Wall Graphic [WC-1(R)]: As indicated in Finish Schedule on the Drawings

B. Accessories:
1. Adhesive: Low (50g/L max) or no VOC-emitting type, mildew resistant, non-staining type recommended by wall covering manufacturer to suit application to substrate.
2. Substrate Primer and Sealer: Alkyd enamel, or type recommended by wall covering manufacturer

1.02 SUBMITTALS

A. Material and Equipment Submittals: Submit the following under provisions of Section 01 30 00:
1. Manufacturer's installation instructions.
2. Samples: 7-inch by 9-inch sample of each wall covering material.

1.03 EXTRA MATERIALS

A. Extra Materials: Provide 25 lineal feet of each type and color of wall covering.

B. Package and label each roll by destination room number; store in Owner's designated storage area as directed.

SECTION 10 28 00 – TOILET ACCESSORIES

1.01 ACCESSORIES

A. Toilet Accessories as indicated on the Drawings.

1.02 SUBMITTALS

A. Material and Equipment Submittals: Submit the following under provisions of Section 01 33 00:
1. Product Data: Submit data illustrating each accessory at large scale and show installation method.
2. Installation Instructions.

B. Closeout Submittals: Submit the following under provisions of Section 01 70 00:
1. Guarantee: Manufacturers 15 year guarantee against silver spoilage on mirrors.
2. Keys: Submit keys for each locked accessory item.

END OF SPECIFICATIONS

RSP

RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099 fax
Tempe, AZ 85281-3714 www.rsparch.com

Consultants

REGISTERED ARCHITECT
STATE OF ARIZONA
35561
MICHAEL L. KRAGER
08/23/2022
Signed
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	JJZ/DM
Checked By	SL
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description
	7/29/22	PERMIT PACKAGE

SPECIFICATIONS

G003

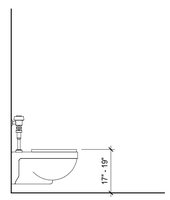
City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

GENERAL MOUNTING HEIGHT NOTES

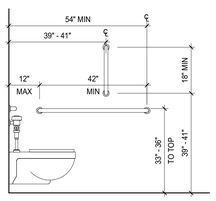
1. IT IS THE INTENT OF THE DESIGN THAT ALL ITEMS SHOWN MOUNTED AT TYPICAL HEIGHTS BE ACCESSIBLE TO PERSONS WITH DISABILITIES.
2. THE PURPOSE OF THIS SHEET IS TO ILLUSTRATE TYPICAL MOUNTING HEIGHTS AND, WHERE APPLICABLE, TYPICAL MINIMUM OR MAXIMUM CLEARANCES AND/OR TYPICAL MOUNTIN CONFIGURATIONS FOR A VARIETY OF ITEMS. CAUTION: THIS SHEET MAY ILLUSTRATE ITEMS OR CONFIGURATIONS WHICH DO NOT OCCUR AS PART OF THE WORK OF THIS PROJECT. REFER TO THE PLANS, ELEVATIONS, SECTIONS, DETAILS, AND SCHEDULES TO DETERMINE WHICH ITEMS AND CONFIGURATIONS APPLY TO THE WORK OF THIS PROJECT.

ICC / ANSI A117.1-2009 - MOUNTING DIAGRAM

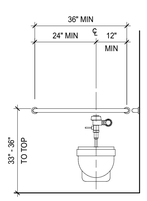
1/2" = 1'-0"



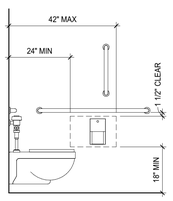
ANSI A117.1 FIGURE 604.4



ANSI A117.1 FIGURE 604.5.1



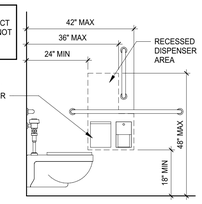
ANSI A117.1 FIGURE 604.5.2



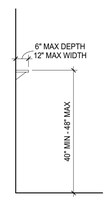
ANSI A117.1 FIGURE 604.7(a), PROTRUDING DISPENSER BELOW GRAB BAR

NOTE:
SANITARY PRODUCT DISPOSAL DOES NOT OCCUR IN MENS RESTROOMS

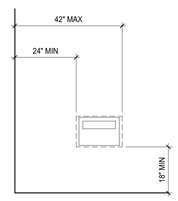
KEEP WASTE RECEPTACLE IN LINE WITH TOILET TISSUE DISPENSER



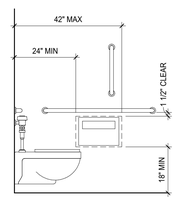
ANSI A117.1 FIGURE 604.7(c), RECESSED DISPENSER



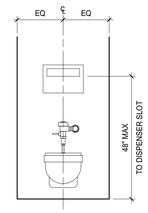
ANSI A117.1 FIGURE 604.8



ACCESSIBLE STALL



AMBULATORY STALL

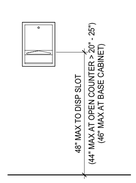


STANDARD TOILET STALL

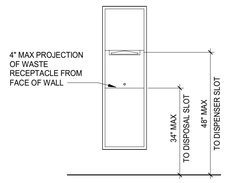
A	TOILET SEAT HEIGHT	B	GRAB BAR HEIGHT	C	REAR WALL GRAB BAR HEIGHT	D	TOILET TISSUE DISPENSER	E	SANITARY PRODUCT DISPOSAL	F	SHELF	G1	SANITARY SEAT COVER DISPENSER	G2	SANITARY SEAT COVER DISPENSER	G3	SANITARY SEAT COVER DISPENSER



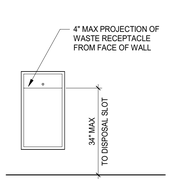
ANSI A117.1 SECTION 308, 309



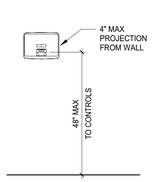
ANSI A117.1 SECTION 308



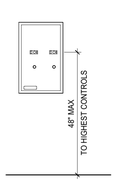
ANSI A117.1 SECTION 308, 604.3.3



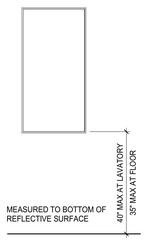
ANSI A117.1 SECTION 308, 604.3.3



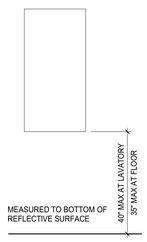
ANSI A117.1 SECTION 308, 604.3.3



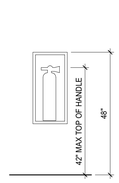
ANSI A117.1 SECTION 308, 604.3.3



ANSI A117.1 SECTION 603.3



ANSI A117.1 SECTION 603.3



MFC 906.9.2

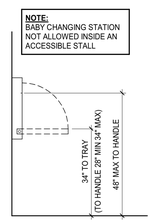


ANSI A117.1 SECTION 604.8

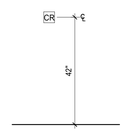


H1	SOAP DISPENSER	J1	PAPER TOWEL DISPENSER	J2	COMBO PAPER / WASTE	J3	WASTE RECEPTACLE	K	HAND DRYER	L	SANITARY PRODUCT DISPENSER	M1	FRAMED MIRROR	M2	UNFRAMED MIRROR	N	FIRE EXTINGUISHER CABINET	P1	COAT HOOK	P2	COAT HOOK

NOTE:
BABY CHANGING STATION NOT ALLOWED INSIDE AN ACCESSIBLE STALL



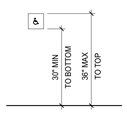
ANSI A117.1 SECTION 603.5, 902.4



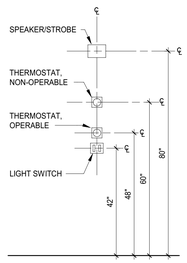
ANSI A117.1 SECTION 308



ANSI A117.1 SECTION 704



ANSI A117.1 FIGURE 404.3.5



Q	BABY CHANGING STATION	R	CARD READER	S	TELEPHONE	T	DOOR CONTROL SWITCHES	U	TYPICAL DEVICE MOUNTING

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO.
35061
MICHAEL L. KRAGER
Date Signed
8 AUG 2022
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	JJZ/DM
Checked By	SL
Date	08/23/2022

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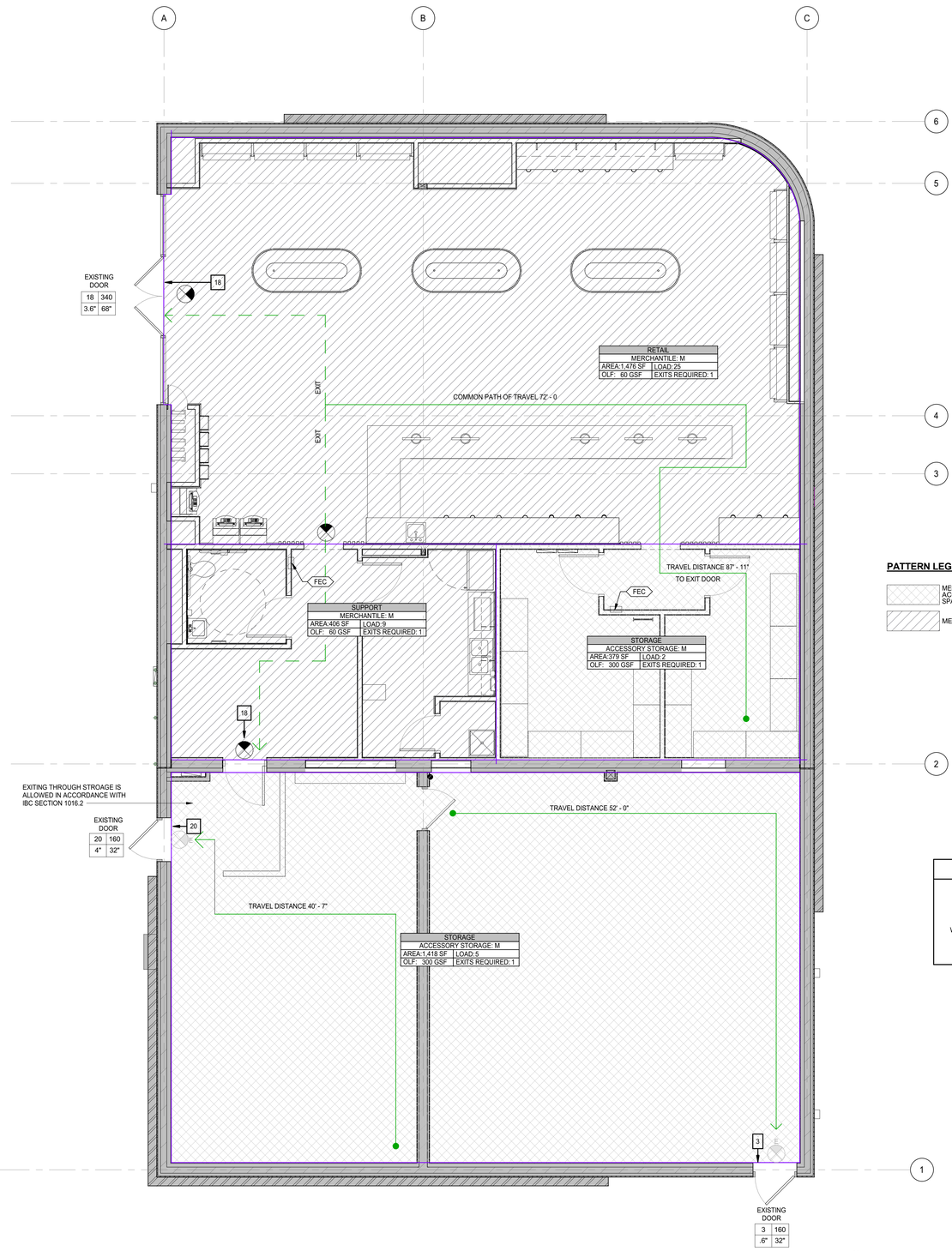
Sheet Issues / Revisions

No.	Date	Description
	7/29/22	PERMIT PACKAGE

MOUNTING HEIGHTS AND DETAILS

G010

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22



1 01 - FIRST FLOOR LIFE SAFETY PLAN
1/4" = 1'-0"

PATTERN LEGEND

- MERCHANTILE GROUP M: ACCESSORY STORAGE SPACE PER 311.1
- MERCHANTILE GROUP M

GRAPHIC LEGEND

REQUIRED CAPACITY | ACTUAL CAPACITY
150 | 160
CALCULATED EXIT WIDTH REQUIRED (IN.) 30" | 32" EXIT WIDTH PROVIDED (IN.)

CODE LEGEND

NOTES:

A. SEE G000 FOR CODE SUMMARY.

LEGEND:

- SMOKE PARTITION (NON-RATED)
- SMOKE PARTITION (RATED)
- 1-HOUR RATED PARTITION
- 2-HOUR RATED PARTITION
- 3-HOUR RATED PARTITION

RSP Architects
502 South College Avenue Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT CERTIFICATE NO. 35561 MICHAEL L. KRAGER Date Signed 13OCT2022 ARIZONA, U.S.A.

Project For

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Ste. 1 and Ste. 2,
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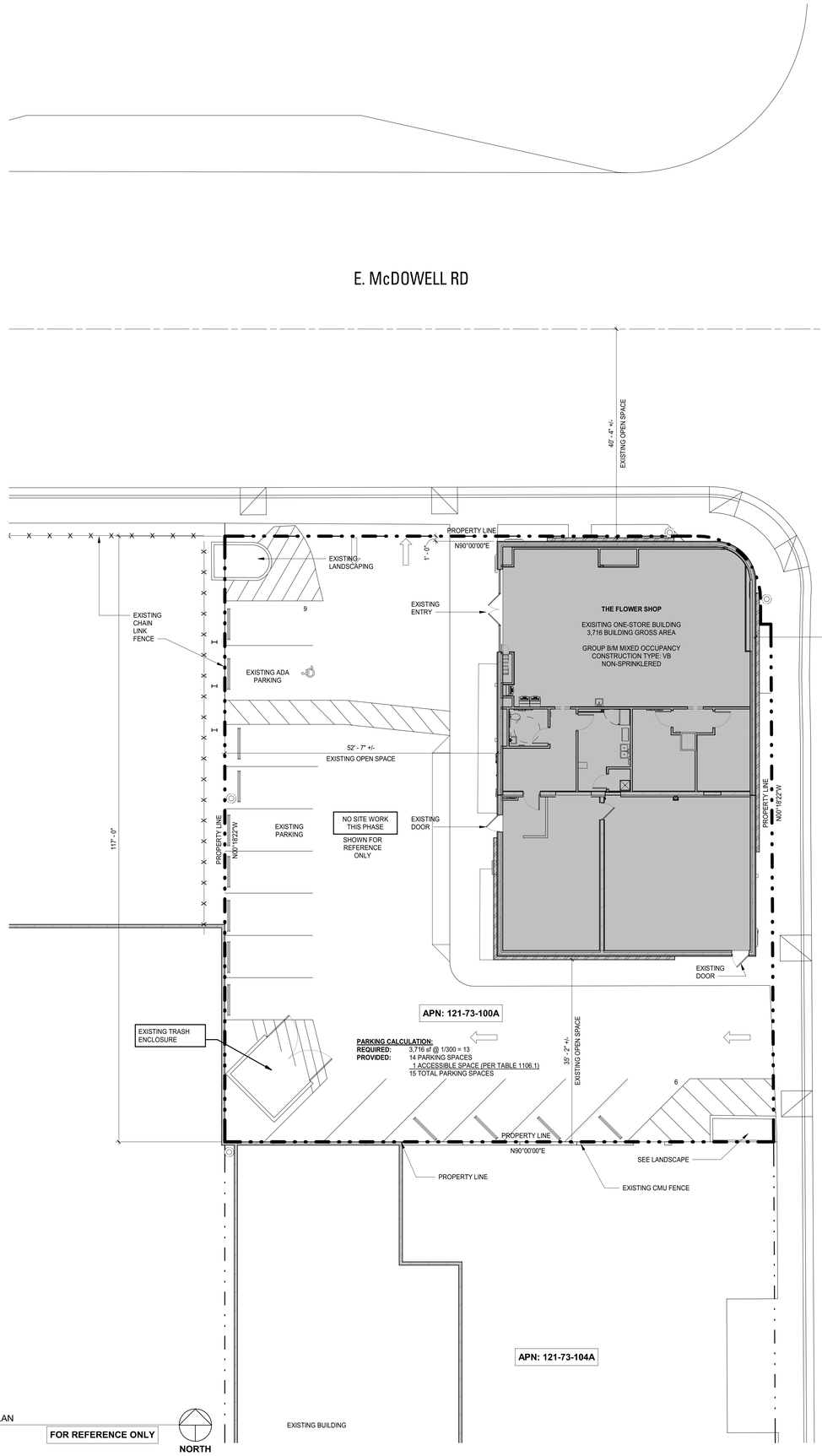
FIRE AND LIFE SAFETY PLANS

G101

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

SHEET NOTES

1. INFORMATION PROVIDED ON SITE PLAN IS FROM AVAILABLE RECORD DRAWINGS AND/OR VISUAL FIELD SURVEYS. THE CONTRACTOR SHALL VERIFY ACTUAL EXISTING CONDITIONS AT THE SITE PRIOR TO CONSTRUCTION AND NOTIFY THE ARCHITECT OF ANY DISCREPANCIES.



1 OVERALL SITE PLAN
1" = 10'-0"

FOR REFERENCE ONLY

NORTH

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO.
35981
MICHAEL L. KRAGER
13 OCT 2022
RENEWAL DATE
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

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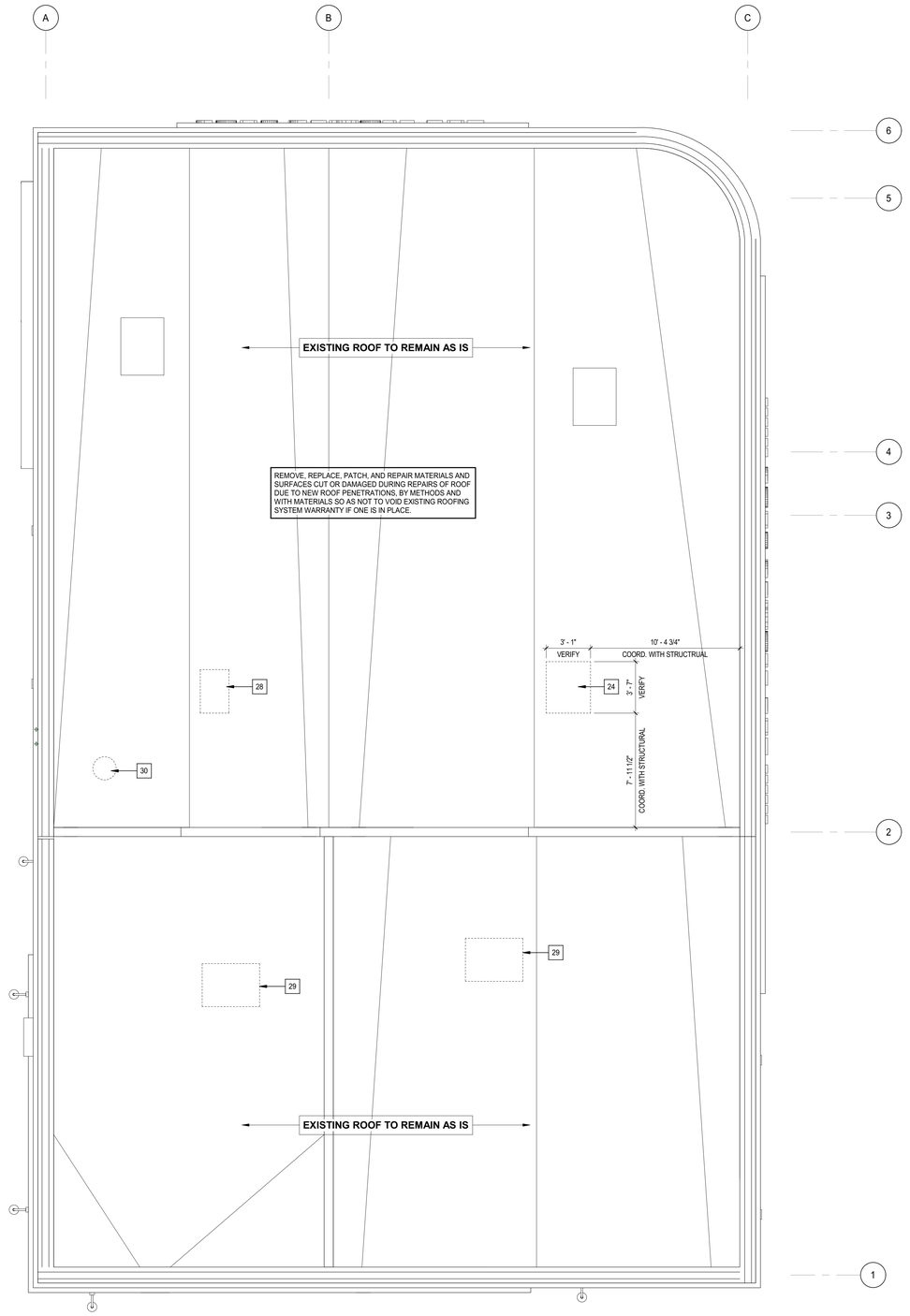
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No.	Date	Description
	7/29/22	PERMIT PACKAGE

OVERALL SITE LAYOUT PLAN

A000



1 ROOF PLAN DEMOLITION PLAN
1/4" = 1'-0"

GENERAL NOTES - DEMOLITION

A. SEE SHEETS G001-G003 FOR SPECIFICATIONS.

B. SEE ENGINEERING DRAWINGS FOR DUCTWORK, DIFFUSER, PLENUM BOX, ETC. DEMOLITION AND/OR PROTECTION. COORDINATE WITH MECHANICAL ENGINEER.

C. DEMOLITION IS NOT NECESSARILY LIMITED TO WHAT IS SHOWN ON DRAWINGS. THE INTENT IS TO INDICATE THE GENERAL SCOPE OF DEMOLITION REQUIRED TO COMPLETE THE WORK IN ACCORDANCE WITH THE CONTRACT DRAWINGS.

D. FIRE SAFETY MUST BE MAINTAINED FOR ALL PERSONNEL WORKING ON THE FLOOR. ALL STAIRS, ALARMS, ETC. MUST REMAIN ACCESSIBLE AND OPERABLE AT ALL TIMES. IMMEDIATELY NOTIFY BUILDING SECURITY AND BUILDING MANAGER OF DAMAGED OR DISABLED SYSTEMS AND REPAIR OR REPLACE DAMAGED SYSTEMS IMMEDIATELY.

E. ALL FIRE SAFETY EQUIPMENT (SMOKE DETECTORS, PUBLIC ADDRESS SPEAKERS, FIRE ALARM BOXES, ETC.) AND THE ASSOCIATED CONDUIT AND WIRING SYSTEM SHALL BE PROTECTED FROM ANY PHYSICAL DAMAGE DURING THE DEMOLITION AND SUBSEQUENT CONSTRUCTION, SHALL BE ACCOMPLISHED AS A FIRST PRIORITY, AND PER THE PLANS. NOTIFY BUILDING SECURITY AND BUILDING OWNER OF DAMAGED OR DISABLED SYSTEMS AND REPAIR OR REPLACE DAMAGED SYSTEMS IMMEDIATELY.

F. FURNISH ALL LABOR AND MATERIALS/ EQUIPMENT REQUIRED TO COMPLETE DEMOLITION AND REMOVAL OF ALL ITEMS AS INDICATED.

G. IF ANY QUESTIONS ARISE AS TO THE REMOVAL OF ANY MATERIAL, CLARIFY THE POINT IN QUESTION WITH THE BUILDING OWNER BEFORE PROCEEDING. IMMEDIATELY STOP WORK IF HAZARDOUS MATERIALS ARE FOUND AND CONTACT THE BUILDING OWNER.

H. REMOVAL OF ANY EQUIPMENT, CABLING SWITCHES, AND CONDUIT PERTAINING TO DATA/COMMUNICATIONS REPRESENTATIVE AS REQUIRED TO PREVENT NEW CONSTRUCTION DELAYS.

I. REMOVE ALL EXISTING FLOOR & WALL MOUNTED RECEPTACLES, JUNCTION BOXES, THERMOSTATS AND PIPING THROUGHOUT AREA OF WORK, UNLESS SCHEDULED TO REMAIN. REFER TO POWER AND DATA PLAN(S) WHERE FLOOR POWER/TELEPHONE RECEPTACLES ARE TO REMAIN. WHERE RECEPTACLES HAVE BEEN REMOVED, SEAL CORE DRILL HOLE WITH APPROVED FIRE STOPPING MATERIAL AND COVER IT WITH STEEL PLATE FLUSH WITH FLOOR. REPAIR FLOOR SLAB WHERE DAMAGED AS REQUIRED.

J. PROVIDE STRICT CONTROL OF JOB CLEANING AND PREVENT DUST AND DEBRIS FROM EMANATING FROM DEMOLITION/ CONSTRUCTION AREA.

K. AT COMPLETION OF DEMOLITION WORK, THE CONSTRUCTION AREA(S) SHALL BE LEFT IN "BROOM CLEAN" CONDITION. ALL DEBRIS AND MISCELLANEOUS MATERIAL SHALL BE REMOVED.

L. GRIND DOWN IMPERFECTIONS IN FLOOR SLAB AND PREPARE TO RECEIVE NEW FLOORING.

M. CONTRACTOR SHALL BE RESPONSIBLE FOR PATCHING AND/ OR REPAIRING ANY DAMAGE CAUSED BY HIM OR HIS SUBCONTRACTORS TO EXISTING CONSTRUCTION NOT IN SCOPE. REFINISH TO MATCH EXISTING ADJACENT FINISH, OR AS NOTED HEREIN.

N. DEMOLITION OF NETWORK/ COMMUNICATIONS CABLING IS SELECTIVE. CONTACT FACILITY/BUILDING TELECOMMUNICATIONS SPECIALIST FOR INFORMATION.

O. REMOVE ALL ABANDONED CABLING AND CONDUIT FROM ABOVE CEILING PRIOR TO FINAL CEILING CHANGES. COORDINATE WITH FACILITY/ BUILDING TELECOMMUNICATIONS SPECIALIST AND BUILDING ENGINEER TO IDENTIFY ITEMS TO REMAIN. SEE NOTES ABOVE.

P. REMOVE THROUGHOUT AREA OF WORK U.N.O., ALL LAYERS OF EXISTING FLOOR FINISH TO TOP OF SLAB. SCRAPE CONCRETE SLAB CLEAN. REMOVE ALL GLUE AND IMPERFECTIONS. FLASH PATCH CONCRETE FLOOR AS REQUIRED FOR NEW FINISH FLOOR INSTALLATION, LEVEL OF SLAB NOT EXCEED 1/8" RISE OVER 10'-0" RUN.

Q. FOR AREAS NOT IN DEMOLITION SCOPE OF WORK, PROTECT AS REQUIRED, ALL SURFACES, EQUIPMENT, FIXTURES AND HARDWARE DURING DEMOLITION AND/OR CONSTRUCTION.

R. SEE A101 FOR NEW REFLECTED CEILING PLAN.

S. REMOVE EXISTING INTERIOR GYPSUM BOARD THAT OCCURS ALONG THE PERIMETER OF SPACE WHERE STUD FRAMING OCCURS. PROVIDE NEW GYPSUM BOARD FROM FLOOR TO UNDERSIDE OF EXISTING FLOOR/ROOF DECK (EXCEPT WHERE NOTED.)

T. REMOVE EXISTING FURRING FROM EXISTING INTERIOR WALLS AT MASONRY CONDITIONS AND REPLACE WITH NEW METAL STUDS AS NOTED PER PLANS AND 5/8" GYPSUM BOARD WITH BATT INSULATION.

KEYNOTES - DEMOLITION PLAN

NOTE: ALL KEYNOTES MAY NOT BE USED ON THIS SHEET

NO.	DESCRIPTION
1	REMOVE PORTION OF EXISTING INTERIOR PARTITION AS SHOWN
3	SALVAGE EXISTING REFRIGERATOR FOR RE-INSTALLATION IN THIS PROJECT
4	DEMOLISH EXISTING CABINETRY AND COUNTERTOP
5	REMOVE EXISTING DOOR. SALVAGE TO OWNER'S STOCK, AT OWNER'S DISCRETION
6	REMOVE EXISTING FIRE EXTINGUISHER
7	REMOVE EXISTING SINK AND CAP PLUMBING BACK TO SOURCE
8	REMOVE EXISTING SOAP DISPENSER, PAPER TOWEL DISPENSER, AND MIRROR. SALVAGE FOR OWNER'S STOCK.
9	REMOVE EXISTING MILLWORK UPON COMPLETION OF NEW SUITE
10	ATM TO BE RELOCATED INTO NEW SPACE. COORDINATE WITH OWNER.
11	REMOVE EXISTING PLUMBING FIXTURE AND CAP AS REQUIRED. SEE MECHANICAL
12	REMOVE EXISTING WATER HEATER AND RELOCATE. SEE MECHANICAL
13	REMOVE EXISTING ELECTRICAL PANEL AND RELOCATE. SEE ELECTRICAL.
14	DEMOLISH EXISTING MASONRY WALL AND FURRING. NON-BEARING MASONRY. PATCH AND REPAIR CONCRETE AS NECESSARY.
15	DEMOLISH EXISTING INTERIOR WINDOW
16	DEMOLISH EXISTING COUNTER
17	EXISTING SAFE TO BE SALVAGED AND REINSTALLED - SEE FURNITURE PLAN
19	EXISTING UNDERCOUNTER REFRIGERATORS TO BE SALVAGED AND RELOCATED - SEE FLOOR PLAN
20	DEMOLISH EXISTING STRUCTURE INCLUDING ROOF AND SLAB ON GRADE SOUTH OF MASONRY WALL. PROTECT EXISTING STRUCTURE TO REMAIN.
21	REMOVE PORTIN OF METAL DECORATIVE WALL AS REQUIRED FOR NEW OPENING.
22	REMOVE EXISTING ROOF TOP MECH UNIT AND ASSOCIATED DUCTWORK, PIPING AND ELECTRICAL.
23	REMOVE EXISTING STOREFRONT SYSTEM
24	DEMO A PORTION OF EXISING ROOF AS REQUIRED FOR NEW ROOF HATCH; REFER TO STRUCTRUAL FOR ADDITIONAL INFORMATION
26	DEMO EXISTING GYPSUM FURRING. TYPICAL ALL EXISTING WALLS.
27	REMOVE EXISTING FLOORING, GRIND AND PREP FOR NEW FLOORING.
28	DEMO A PORTION OF EXISTING ROOF AS REQUIRED FOR NEW ROOF TOP A/C UNIT. REFER TO MECHANICAL AND STRUCTURAL FOR ADDITIONAL INFORMATION
29	REMOVE EXISING ROOF UNIT; REFER TO MECHANICAL FOR ADDITIONAL INFORMATION
30	DEMO PORTION OF EXISTING ROOF AS REQUIRED FOR NEW EXHAUST FAN; REFER TO MECHANICAL FOR ADDITIONAL INFORMATION

DEMOLITION LEGEND

EXISTING PARTITION TO REMAIN
EXISTING PARTITION TO BE REMOVED
NOT IN SCOPE
EXISTING DOOR AND FRAME TO REMAIN
EXISTING DOOR AND FRAME TO BE REMOVED

RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099 fax
Tempe, AZ 85281-3714 www.rsparch.com

Consultants

REGISTERED ARCHITECT 35061 MICHAEL L. KRAGER 8/30/2022 ARIZONA, U.S.A.

Project For

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3155 E. McDowell Rd,
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Project No.	3378.002.00
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Checked By	Checker
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ROOF PLAN DEMOLITION

D102

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

CEILING PLAN KEY

2X2 LIGHT FIXTURE	WALL SCONCE
RECESSED CAN FIXTURE	DECORATIVE FIXTURE
1X4 LINEAR PENDANT	PENDANT CAN

KEYNOTES - CEILING PLAN

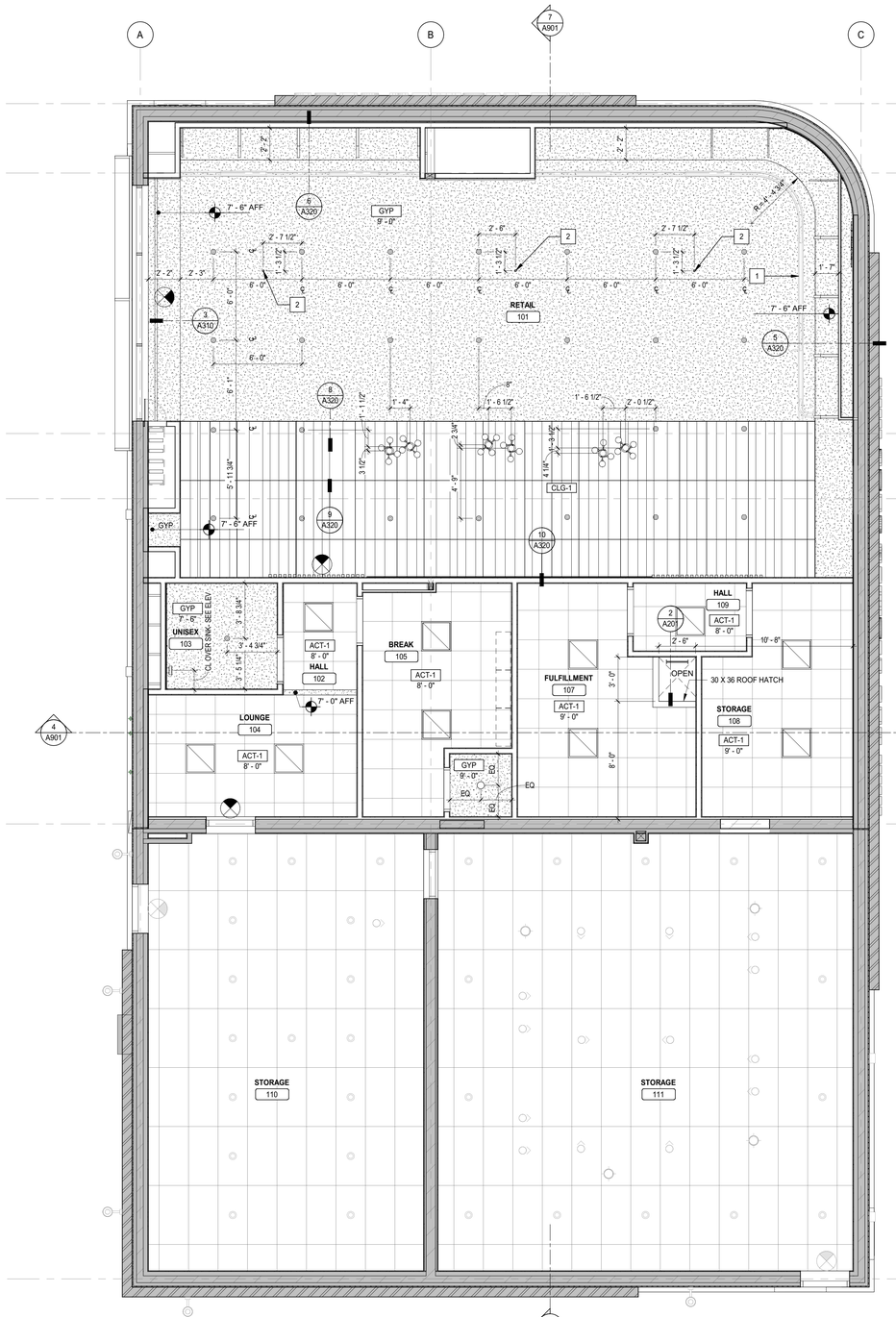
NOTE: ALL KEYNOTES MAY NOT BE USED ON THIS SHEET

NO.	DESCRIPTION
1	PROVIDE POWER FOR LIT SIGNAGE AT SOFFIT - SEE ENGINEERING DRAWINGS
2	POWER DROP DOWN FOR FLOOR DISPLAYS; VERIFY WITH OWNER

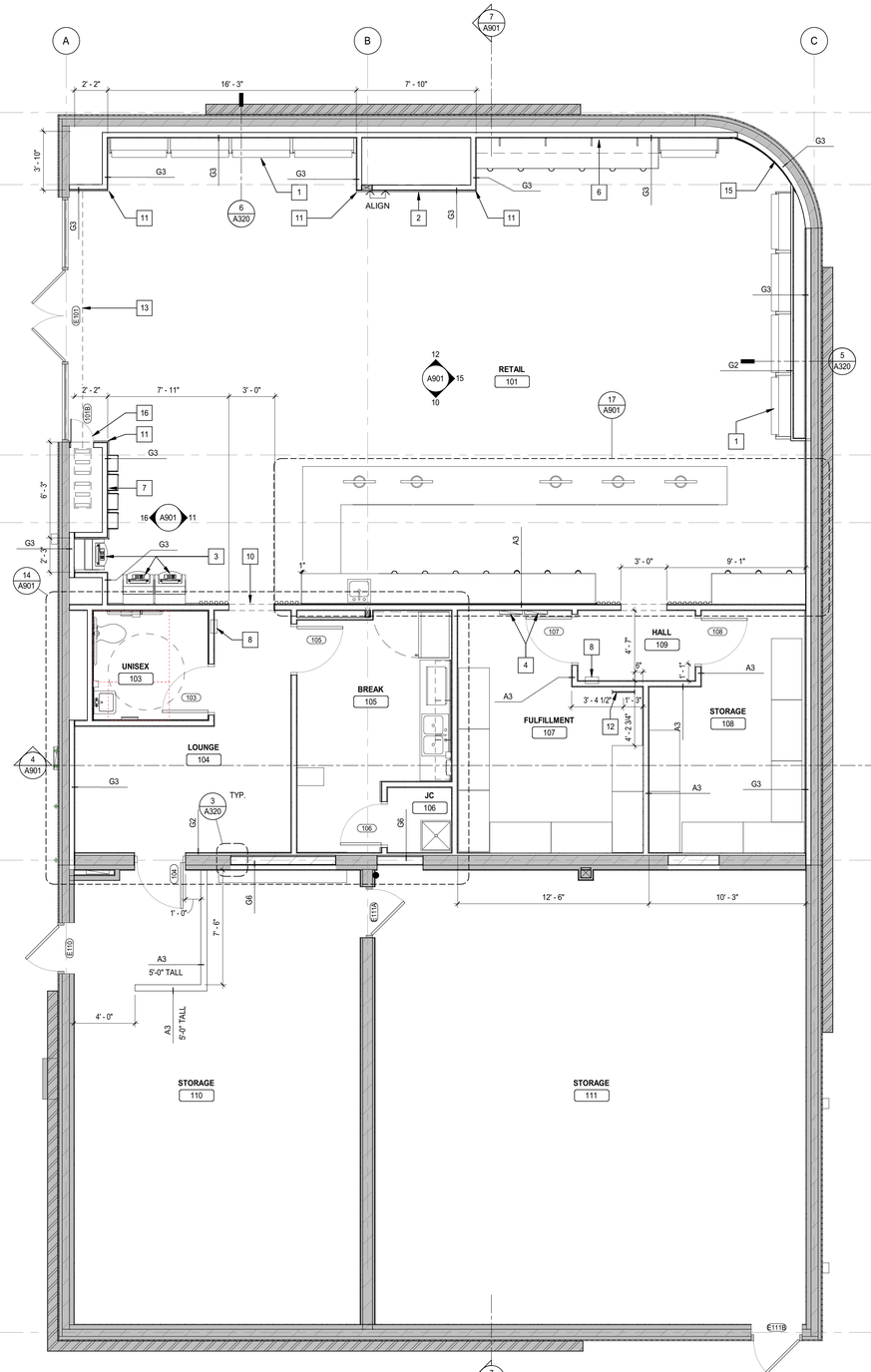
KEYNOTES - FLOOR PLAN

NOTE: ALL KEYNOTES MAY NOT BE USED ON THIS SHEET

NO.	DESCRIPTION
1	NEW DISPLAY SHELVING MILLWORK - SEE ELEVATIONS
2	PROVIDE IN-WALL BACKING FOR OWNER PROVIDED WALL MOUNTED MONITOR INSTALLATION. PROVIDE OUTLET AT 60" AFF.
3	ATM- COORDIANTE WITH OWNER WHETHER RE-INSTALLED FROM DEMOLITION OR NEW.
4	RELOCATED ELECTRICAL PANELS
5	RE-INSTALL SALVAGED SAFE FROM DEMO
6	PROVIDE PAINTABLE OUTLET COVERS AT THIS LOCATION. PAINT TO MATCH ADJACENT WALL.
7	NEON SIGNAGE LOCATION AT 72" AFF TO BOTTOM OF FIXTURE. PROVIDE PAINTABLE OUTLET COVER TO MATCH ADJACENT PARTITION- SEE ELECTRICAL. SIGNAGE TO BE PROVIDED BY OWNER.
8	SEMI-RECESSED FIRE EXTINGUISHER CABINET
10	3'-0" x 7'-0" AFF GYP FINISHED OPENING. SEE ELEVATION FOR MORE INFORMATION.
11	BLACK METAL CORNER GUARD, AS INDICATED.
12	ROOF LADDER
13	SIDE-FOLDING OPEN AIR SECURITY GRILLE - REF. DOOR SCHEDULE
15	PROVIDE IN-WALL BLOCKING FOR GREEN WALL- PROVIDED BY OTHERS.
16	PROVIDE 18" W x 7'-6" ACCESS DOOR FOR SECURITY GRILLE



2 01 - REFLECTED CEILING PLAN
1/4" = 1'-0"



1 01 - FIRST FLOOR PLAN
1/4" = 1'-0"

GENERAL NOTES - FLOOR PLAN

A. PLAN DIMENSIONS LOCATING INTERIOR PARTITIONS ARE TO FACE OF FINISH, UNO. DO NOT SCALE DRAWINGS, WRITTEN DIMENSIONS GOVERN. ALL PARTITION LOCATIONS SHALL BE AS SHOWN ON FLOOR PLAN. IN CASE OF CONFLICT NOTIFY THE ARCHITECT. FLOOR PLAN BY ARCHITECT TAKES PRECEDENCE OVER ALL OTHER PLANS.

B. PLAN DIMENSIONS LOCATING DOOR AND WINDOW OPENINGS ARE ROUGH OPENING DIMENSIONS.

C. SEE SHEET A310 FOR DOOR TYPES AND A320 FOR PARTITION TYPES.

D. ALL DIMENSIONS TO THE EXTERIOR WINDOW WALL ARE TO THE INSIDE OF FACE OF SILL, U.N.O.

E. CLEAR DIMENSIONS ARE TO BE EXACT WITHIN 1/8". ALONG FULL HEIGHT AND FULL WIDTH OF WALLS. CONTRACTOR SHALL NOT ADJUST ANY DIMENSIONS MARKED "CLEAR" OR "CLR" WITHOUT WRITTEN INSTRUCTION FROM THE ARCHITECT AND/OR OWNER.

F. ALL WORK SHALL BE ERECTED AND INSTALLED PLUMB, LEVEL, SQUARE AND TRUE, AND IN PROPER ALIGNMENT.

G. ALL MILLWORK TO BE FASTENED TO THE PARTITION. PROVIDE BLOCKING FOR ALL MILLWORK NOT SUPPORTED ON FLOOR SLAB. ALL CONCEALED LUMBER & BLOCKING TO BE FIRE TREATED.

H. ALL DOORS SHALL HAVE 1'-6" CLEARANCE ON STRIKE/PULL SIDE OF DOOR. VERIFY AND ADVISE BUILDING OWNER OF EXCEPTIONS PRIOR TO CLOSING OUT PARTITIONS.

I. ALL SURFACES SHALL BE PREPARED TO RECEIVE THE SPECIFIED FINISH. ALL GYPSUM BOARD PARTITIONS SHALL BE TAPED AND SANDED SMOOTH AND PREPARED TO RECEIVE THE SPECIFIED FINISH. PAINT GRADE WOODWORK SHALL BE HAND SANDED BETWEEN COATS AND DUSTED CLEAN. ALL HOLES, PITCH POCKETS OR SAPPY PORTIONS SHALL BE SCRAPED AND SHELLACKED, OR SEALED WITH KNOT SEALER.

J. INTERIOR GYPSUM BOARD SURFACES SHALL BE WIPED WITH A DAMP CLOTH JUST PRIOR TO APPLICATION OF THE FIRST COAT, IN ORDER TO LAY FLAT ANY NAP WHICH MAY HAVE FORMED IN SANDING PROCESS.

K. ALL EXPOSED GYPSUM BOARD EDGES TO HAVE METAL EDGE TRIM, EXCEPT WHERE A CORNER GUARD IS SPECIFIED.

L. ALL INTERSECTIONS OF FLOOR FINISH MATERIALS SHALL BE LOCATED DIRECTLY UNDER CENTER OF DOOR, WHERE OCCURS. PROVIDE TRANSITION STRIP AT ALL INTERSECTIONS OF MATCH EXISTING BUILDING STANDARD.

M. TRIM THE BOTTOMS OF DOOR TO CLEAR THE TOP OF FINISHED FLOOR, AS APPLICABLE. BY 1/4" INCH MAXIMUM, U.N.O. VERIFY SLAB CONDITIONS, TRIM EACH DOOR TO FIT CONDITION. WHERE RADICAL VARIATION IN FLOOR ELEVATION EXIST, DOOR SHALL BE ORDERED WITH BOTTOM STILE SIZED TO ACCOMODATE THESE UNDERCUT CONDITIONS.

N. FLUSH MOUNTED OR SEMI-RECESSED FIRE EXTINGUISHER CABINETS AS REQUIRED PER CODE. CABINETS MUST MEET ADA GUIDELINES REGARDING PROJECTIONS AND MOUNTING HEIGHT. IF SEMI-RECESSED CABINETS ARE USED MAINTAIN REQUIRED FIRE-RATING FOR PARTITION OR STRUCTURAL ELEMENT. CABINETS ARE TO BE PAINTED BLACK. FINAL LOCATIONS ARE TO BE VERIFIED WITH THE CITY FIRE INSPECTOR.

O. SEE G010 FOR TYPICAL DEVICE MOUNTING HEIGHTS AT DOORS.

P. UPON COMPLETION, REMOVE ALL PAINT FROM WHERE IT HAS SPILLED, SPLASHED OR SPLATTERED ON EXPOSED ADJACENT SURFACES. EXAMINE ALL FINISH SURFACES AFTER COMPLETION OF WORK INCLUDING MILLWORK INSTALLATION AND PROCEED WITH "TOUCHUP" AS REQUIRED.

Q. CONTRACTOR SHALL BE RESPONSIBLE FOR ALLOWING FOR DELIVERY LEAD TIMES FOR ALL FINISHES WITHIN THE CONSTRUCTION SCHEDULE. ALL DELIVERY TIMES MUST BE CONFIRMED, AND ANY EXCESSIVE LEAD TIME MUST BE BROUGHT TO THE BUILDING OWNER'S ATTENTION IMMEDIATELY TO ALLOW FOR RE-SPECIFICATION IF NECESSARY.

R. PAINT CEILING ACCESS PANELS WHERE THEY OCCUR TO MATCH ADJACENT CEILING FINISH.

S. FINISH INTERIOR OF ALL EXTERIOR WALLS FROM SLAB TO DECK. PROVIDE METAL STUD FURRING AND GYPSUM BOARD, TYP. VERIFY & REPLACE INSULATION TO MEET IECC REQUIREMENTS, AS NEEDED.

T. PROVIDE ADD ALTERNATE FOR NEW STOREFRONT THROUGHOUT WITH NEW STOREFRONT DOORS.

U. CLEAN ALL EXTERIOR WINDOWS. PROVIDE NEW INTERIOR PAINT FINISH THROUGHOUT.

V. PROVIDE NEW INSULATION ABOVE CEILING IN ENTIRE BUILDING.

GENERAL NOTES - RCP

A. NOTIFY ARCHITECT OF ANY CONFLICTS OF LIGHT FIXTURE LOCATIONS WITH MAIN RUNNERS, DUCTS, STRUCTURES, HVAC AND/ OR (E) CONDUIT, PRIOR TO FRAMING FOR LIGHTS. ANY DISCREPANCIES BETWEEN ARCHITECT'S CEILING GRID LOCATIONS AND ACTUAL FIELD CONDITIONS ARE TO BE CLARIFIED WITH THE BUILDING OWNER PRIOR TO FRAMING.

B. ALL SPECIFIC INFORMATION CONCERNING INSTALLATION OF VARIOUS ABOVE-CEILING ELEMENTS ARE TO BE FOUND IN THE HVAC, PLUMBING, FIRE PROTECTION, ELECTRICAL AND LIGHTING DIFFUSERS, RETURNS, SPRINKLERS, LIGHT SWITCHES, THERMOSTATS, FIRE ALARMS, STROBES, ETC.

C. REFER TO FIRE MARSHAL FOR ALL LIFE SAFETY DEVICES REQUIRED BY CODE AND ALL EMERGENCY LIGHT FIXTURES.

D. FURNISH AND INSTALL UNDERWRITERS LABORATORIES INC. (UL) LABELED DEVICES THROUGHOUT.

E. INSTALL LIGHT FIXTURES WITH PROTECTIVE FILM OR SIMILAR TO COVER OVER LOUVER, LENS, BAFFLE AND THE LIKE, TO AVOID FIXTURE SOILING OR DAMAGE, FIXTURES SHALL BE MAINTAINED CLEAN AND AS NEW. LAMPS SHALL BE NEW AT PROJECT COMPLETION. RELAMP AS REQUIRED SO ALL LAMP COLORS ARE CONSISTENT.

F. LIGHT FIXTURES, EXIT SIGNS, SPRINKLERS AND OTHER CEILING ELEMENTS SHALL BE LOCATED IN CENTER OF INDIVIDUAL CEILING TILE, U.N.O.

G. PROVIDE CEILING ACCESS AS REQUIRED FOR EQUIPMENT AND SYSTEM MAINTENANCE AND MATCH ADJACENT CEILING FINISH, U.N.O.

H. ALL SOFFITS AND CEILING HEIGHTS ARE DIMENSIONED FROM TOP OF FINISHED FLOOR TO BOTTOM OF GYP. BD. OR CEILING TILE AND SHALL ALLOW FOR THICKNESS OF ALL FLOOR FINISHES. VERIFY HEIGHTS IN FIELD.

I. SWITCH LOCATION: UNLESS NOTED OTHERWISE, SWITCHES SHALL BE SET 3'-6" FROM CENTER LINE OF PLATE TO FLOOR, AND THOSE AT DOOR LOCATIONS TO BE WITHIN 7" FROM LATCH SIDE OF OPEN DOOR. VARIANCES FROM THESE STANDARDS TO BE REVIEWED WITH BUILDING OWNER PRIOR TO INSTALLATION.

J. LIGHTING FIXTURE LOCATIONS TO BE PER LOCATION SHOWN ON ARCHITECTURAL DRAWINGS. DEVIATION FROM PLAN DUE TO MECHANICAL INTERFERENCE MUST BE APPROVED BY ARCHITECT.

K. CARRYING CHANNELS SHALL BE SECURELY SUPPORTED BY HANGERS FASTENED TO SLAB SUPPORTS AND PROPERLY SPACED TO SUPPORT LIGHT FIXTURES AND ACOUSTIC CEILING ASSEMBLY.

L. CEILING LIGHT SLACK WIRES TO BE RUN TO TRUSS TOP CORDS. NO SLACK WIRES ARE TO BE ATTACHED TO 2X SUBPURLINS.

M. RECEPTACLE AMD SWITCH PLATE COVERS TO MATCH WALL COLOR- WHITE ON WHITE WALLS, MATTE BLACK ON FINISHED WALLS. NOTIFY ARCHITECT OF ANY INCONSISTENCIES.

N. METAL SUSPENSION SYSTEMS FOR ACOUSTICAL TILE AND LAY-IN PANEL CEILINGS SHALL BE INSTALLED IN ACCORDANCE WITH IBC SECTION 808 AND ASTM C635 AND C636. CEILING LOADS NOT TO EXCEED 5 POUNDS PER SQUARE FOOT.

O. ALLOWANCE SHALL BE MADE FOR LATERAL MOVEMENT OF THE SYSTEM. MAIN RUNNERS AND CROSS RUNNERS MAY BE ATTACHED AT TWO ADJACENT WALLS WITH CLEARANCE BETWEEN THE WALL AND THE RUNNERS MAINTAINED AT THE OTHER TWO WALLS. VERTICAL SUPPORT SHALL BE PROVIDED AS REQUIRED IN UBC STD. WITH THE ADDED REQUIREMENTS THAT DISCONTINUOUS ENDS OF CROSS RUNNERS AND MAIN RUNNERS BE VERTICALLY SUPPORTED WITHIN 8" OF SUCH DISCONTINUITIES AS MAY OCCUR WHERE THE CEILING IS DISRUPTED BY A WALL. SEE DETAIL 3/A201.

P. LIGHTING FIXTURES AND AIR DIFFUSERS SHALL BE SUPPORTED DIRECTLY BY WIRES TO THE STRUCTURE ABOVE.

Q. HANGER SHALL BE ATTACHED TO BOTTOM OF STEEL JOISTS WITH APPROVED TYPE ATTACHMENT

R. ALL GYP. BOARD SOFFIT TO BE PAINTED (PT-1) AT FACE AND BOTTOM, U.N.O.

RSP Architects
502 South College Avenue Suite 203 Tempe, AZ 85281-3714
480.889.2000 480.889.2099 fax www.rsparch.com

Consultants

REGISTERED ARCHITECT CERTIFICATE NO. 35561 MICHAEL L. KRAGER 13OCT2022 Signed... ARIZONA, U.S.A.

Project For

THE FLOWER SHOP PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd
Ste. 1 and Ste. 2,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	JJZ/DM
Checked By	SL
Date	08/23/2022

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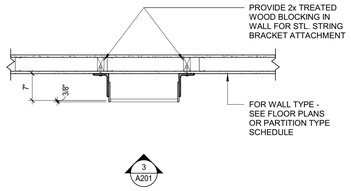
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Sheet Issues / Revisions

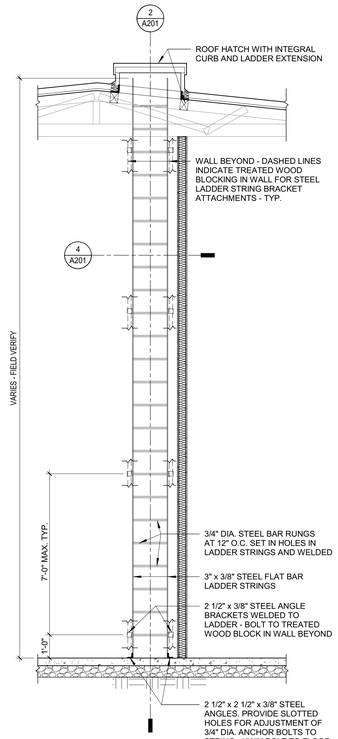
No.	Date	Description

FLOOR PLAN & REFLECTED CEILING PLAN

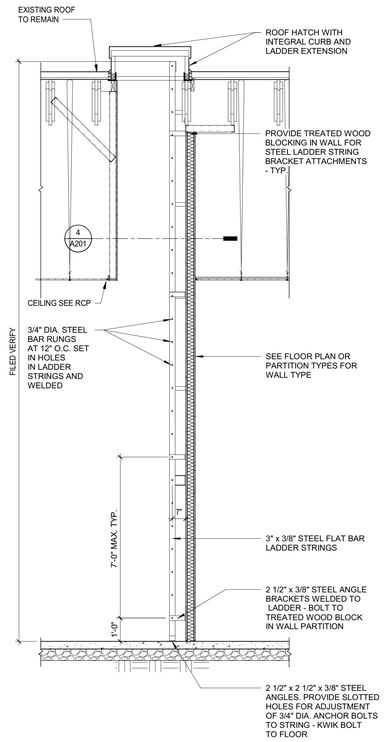
A101



4 PLAN AT VERTICAL LADDER
3/4" = 1'-0"



3 ELEVATION AT VERTICAL LADDER
3/8" = 1'-0"

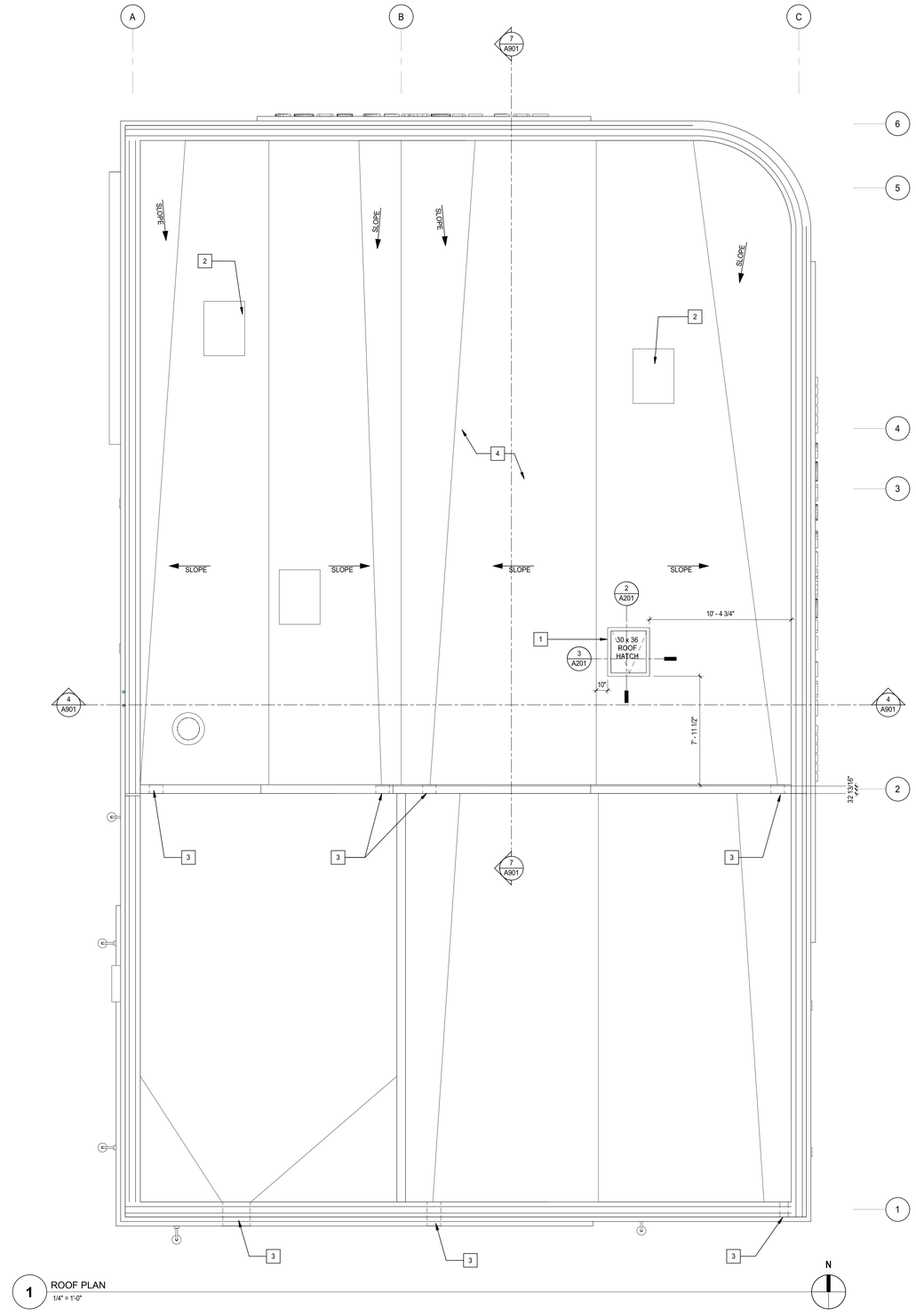


2 SECTION AT VERTICAL LADDER
3/8" = 1'-0"

KEYNOTES - ROOF PLAN

NOTE: ALL KEYNOTES MAY NOT BE USED ON THIS SHEET

NO.	DESCRIPTION
1	NEW ROOF HATCH: REMOVE, REPLACE, PATCH, AND REPAIR MATERIALS AND SURFACES CUT OR DAMAGED DURING REROOFING, BY METHODS AND WITH MATERIALS SO AS NOT TO VOID EXISTING ROOFING SYSTEM WARRANTY
2	EXISTING RTU
3	EXISTING SCUPPER THROUGH PARAPET
4	EXISTING ROOF TO REMAIN. PATCH AND REPAIR AS NECESSARY.



1 ROOF PLAN
1/4" = 1'-0"

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO.
35861
MICHAEL L. KRAGER
08/23/2022
Signed
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	Author
Checked By	Checker
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Sheet Issues / Revisions

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	7/29/22	PERMIT PACKAGE

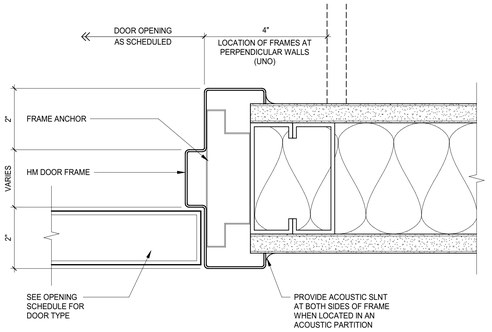
ROOF PLAN

A201

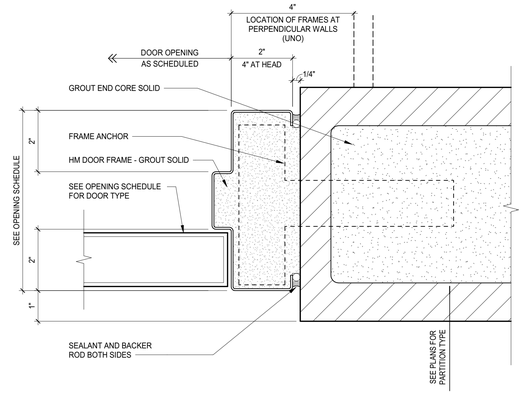
MATERIAL SCHEDULE- RETAIL PHASE

ABC

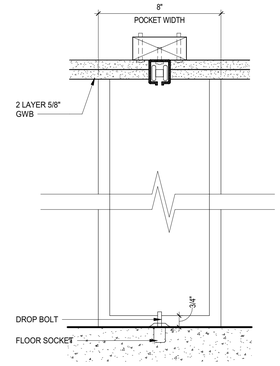
Number	Manufacturer	Model	Color	Size	Installation	Comments
ACCENT CEILING						
CLG-1	TURF	SOFTSPAN 48	D07 DARK WALNUT	4'x8' PANEL	SUSPEND STAINLESS STEEL CABLE AND MODULE JOINING BRACKTS, PER MFR. PROVIDE END CAPS AT EXPOSED EDGES	SHANNON CURATOLO, shannon@saleswestgroup.net
BASE						
VB-1	JOHNSONITE	TRADITIONAL WALL BASE	48 GREY	4" H, CONTINUOUS ROLL	WALL, PER MFR.	MICHAEL RAEL, 602.540.6663
CARPET						
CPT-1	JJ FLOORING	TBD	TBD	24" x 24"	TBD	JACOB SEGALL, 410.733.2006, jacob.segall@jjflorinq.com
FIBER REINFORCED PANEL						
FRP-1	MARLITE	STANDARD FRP	WHITE P 100	4'X8'	WALL	PROVIDE PVC TRIM
GLAZING						
GL-1			Green			New Storefront and glass (green)
GROUND CONCRETE						
CONC-1	-	GROUND & SEALED CONCRETE	-	-	FLOOR	-
HARDWARE						
HW-1	HICKORY HARDWARE	COTTAGE CABINET PULL, P3077-VB	VINTAGE BRONZE	3 INCH	SEE DETAIL, PER MFR	TBD
HW-1(R)	HICKORY HARDWARE	COTTAGE CABINET PULL, P3077-VB	VINTAGE BRONZE	3 INCH	SEE DETAIL, PER MFR	TBD
HW-2	DOUG MOCKETT	DROP EDGE DRAWER PULL-DP269B	BRUSHED BRASS (4B)	7 7/8" X 1 29/32"	SEE DETAIL, PER MFR	TBD
METAL LAMINATE						
ML-1	FORMICA	DECOMETAL	BLACK PATINA M9422	TBD	SEE ELEVATIONS	-
METAL TRIM						
MTR-1	SCHLUTER	RENO-U	BURSHED ANTIQUE BRONZE ANODIZED	TBD	SEE DETAIL, PER MFR	TBD
MTR-2	SCHLUTER	JOLLY	BRUSHED BRASS	3/8"	FLOOR, AT TRANSITION BETWEEN TL-1(R) & TL-2(R)	-
MTR-2(R)	SCHLUTER	DECO	TBD	9mm	FLOOR, AT TRANSITION BETWEEN TL-1(R) & TL-2(R)	-
MTR-3	SCHLUTER	RENO-TK	BRUSHED ANTIQUE BRONZE ANODIZED	1"	SEE ELEVATIONS	-
MTR-4	SCHLUTER	JOLLY	DARK ANTHRACITE	TBD	SEE DETAIL, PER MFR	TBD
MTR-5	SCHLUTER	DILEX AHK	TSDA: TRENDLINE: DARK ANTHRACITE	TBD	FLOOR	-
PAINT						
PT-1	SHERWIN WILLIAMS	SW7646	FIRST STAR	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-2	SHERWIN WILLIAMS	-	TFS MIX (REF. AHWATUKEE)	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-3	SHERWIN WILLIAMS	SW0016	BILLIARD GREEN	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-4	SHERWIN WILLIAMS	SW6060	MOROCCAN BROWN	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-5	SHERWIN WILLIAMS	SW6057	MALTED MILK	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-6	SHERWIN WILLIAMS	SW9137	NIEBLA AZUL	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-7	SHERWIN WILLIAMS	SW7747	RECYCLED GLASS	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PT-8	SHERWIN WILLIAMS	SW6271	EXPRESSIVE PLUM	-	(2) COATS EGGSHELL AT TYPICAL WALLS, (2) COATS FLAT AT CEILINGS, (2) COATS SEMI-GLOSS AT RESTROOMS	-
PLASTIC LAMINATE						
PL-1	LAMINART (WILSONART COMPANY)	DECORATIVE, STANDARD GRADE	ITALIAN WALNUT 3134-E	4'X8'	SEE ELEVATIONS	-
PL-2	FORMICA	COLOR CORE 2	BLACK, MATTE	4'X8'	VERTICAL SURFACES- SEE ELEVATIONS	-
QUARTZ						
QZ-1	CAMBRIA	SIGNATURE SERIES	ROSE BAY	-	EASED EDGE	MELISSA FEDOCK, 480.428.9143, melissa.fedock@cambriausa.com
QZ-2	CORIAN	WILSONART	VESUVIUS Q1017	-	EASED EDGE	
TILE						
TL-1	CONCEPT SURFACES	ENDURING	STATURIO BIANCO	12"X24"X 3/8"	FLOOR, FLOAT, AS NEEDED, TO ALIGN WITH ACCENT TILE	GROUT COLOR: CUSTOM SNOW WHITE 11, MATERIAL TRANSITION- SEE DETAIL
TL-2	BF CERAMICS	TIMELESS: MADRID	B88TDM	8"X8"	FLOOR & WALL- SEE FINISH PLANS & ELEVATIONS	GROUT COLOR: CUSTOM DELORIAN GRAY 165, MATERIAL TRANSITION- SEE DETAIL
VINYL COMPOSITE TILE						
VCT-1	JOHNSONITE	AZTERRA	AT111	12"X12"	FLOOR	MICHAEL RAEL, 60.2540.6663
WALLCOVERING						
WC-1	MAHARAM	BURST	300053-002 KASHA	-	WALL-BEHIND DISPLAYS ONLY	-
WC-2	3M	DI-NOC	WG-1196	-	SLAT WALL- SEE ELEVATION & DETAIL	-
WOOD VENEER						
WD-1	SPECIES: WALNUT	TBD	TBD	-	POS FRONT	-



2 HOLLOW METAL DOOR JAMB (HEAD SIM)
6" = 1'-0"



1 HOLLOW METAL DOOR JAMB AT MASONRY WALL (HEAD SIM)
6" = 1'-0"



3 SECURITY GRILLE DETAIL
3" = 1'-0"

Set: 1.0
Doors: 107, 108

3	Hinge, Std. Wt.	TA2714 4.5 x 4.5	D4 MK
1	Storeroom Lock	CLX3357 NZD CTB	BSP RU
1	Final Keyed Core	CR8000	BLK RU
1	Electric Strike	1600-CS-LMS	BSP HS
1	ElectroLynx Adaptor	2004M(for electric strike)	HS
1	SMART Pac Bridge Rectifier	2005MD (for electric strike)	HS
1	Surface Closer	DC8210 693	RU
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	BSP RO
1	Floor Stop	440	BSP RO
3	Silencers	608 GRY	RO
1	Frame Wiring Harness	QC-C1500P	MK
1	Door Wiring Harness	QC-Cxxx	MK
1	Card Reader	By Security Integrator	OT
1	Position Switch	DPS-M/W-BK	SU
1	Power Supply	By Security Integrator	SU

Notes: Field verify and modify exsiting opening to receive new hardware.

Operation Description:
Door normally closed, latched, and locked.
Presenting valid credential or key override to unlock.
Door remains locked when power fails.
Free to egress at all times.

Set: 2.0
Doors: 105, 106

3	Hinge, Hvy Wt	T4A3786 NRP	US10BE MK
1	Storeroom Lock	CLX3357 NZD CTB	613 RU
1	Final Keyed Core	CR8000	613 RU
1	Surface Closer/Stop	DC8210 A4	690 RU
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	US10BE RO
3	Silencer	608-RKW	RO
	S88D	PE	

Set: 3.0
Doors: 103

3	Hinge, Hvy Wt.	T4A3786	US10BE MK
1	Privacy Lock w/ IND	ML2068 NSA V21 CTB	613 RU
1	Final Keyed Core	CR8000	613 RU
1	Surface Closer	DC8210	690 RU
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	US10BE RO
1	Wall Stop	406	US10BE RO
1	Gasketing	S88D	PE

Set: 4.0
Doors: 104

1	Elec Continuous Hinge	DFM_SLF-HD1/3 EL-EPT	PE
1	Elec CVR Exit	8600 MLR M2 MEC ELX	313 AD
1	Cylinder Escutcheon	8650	313 AD
1	Final Keyed Core	CR8000	613 RU
1	Cylinder	CR1080 A03 CTB	613E RU
1	Door Pull	BF158	US10BE RO
1	Surface Closer/Stop/Hold	DC8210 A5	613E RU
1	Threshold	271D	PE
1	Frame Wiring Harness	QC-C1500P	MK
1	Door Wiring Harness	QC-Cxxx	MK
1	Card Reader	By Security Integrator	OT
1	Position Switch	DPS-M/W-BK	SU
1	Power Supply	AQL (R8E1 as req.)	SU

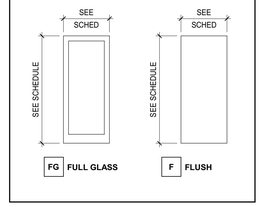
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Door remains locked when power fails.
Free to egress at all times
Field verify and modify exsiting opening to receive new hardware.

DOOR SCHEDULE

NO.	DOOR ROOM NAME	DOOR W	H	T	TYPE	MATL.	FIN.	GLAZIING	PAIR	FRAME TYPE	MATL.	FIN.	SPECS. HDWR.	COMMENTS
101B	RETAIL	16' - 4 3/4"	7' - 6"				CLR. ANOD.							SECURITY GRILLE. MFR.: OVERHEAD DOOR. MODEL: 683 STANDARD BODY.
103	UNISEX	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM	PT			F1	HM	PT	SET 3.0	
104	STORAGE	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM							SET 1.0	REINSTALL SALVAGED DOOR
105	BREAK	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM	PT			F1	HM	PT	SET 2.0	
106	BREAK	2' - 8"	7' - 0"	0' - 1 3/4"	F	HM	PT			F1	HM	PT	SET 4.0	
107	FULFILLMENT	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM	PT			F1	HM	PT	SET 2.0	
108	STORAGE	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM	PT			F1	HM	PT	SET 1.0	
E101	RETAIL	6' - 0"	7' - 0"	0' - 1 1/2"	FG	AL			•	EXIST	EXIST			PROVIDE "DOOR TO REMAIN UNLOCKED DURING BUISINESS HOURS" SIGNAGE
E110	STORAGE	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM				EXIST	EXIST			NO HARDWARE CHANGE
E111A	STORAGE	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM				EXIST	EXIST			NO HARDWARE CHANGE
E111B	STORAGE	3' - 0"	7' - 0"	0' - 1 3/4"	F	HM				EXIST	EXIST			NO HARDWARE CHANGE

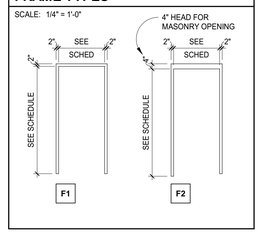
DOOR TYPES

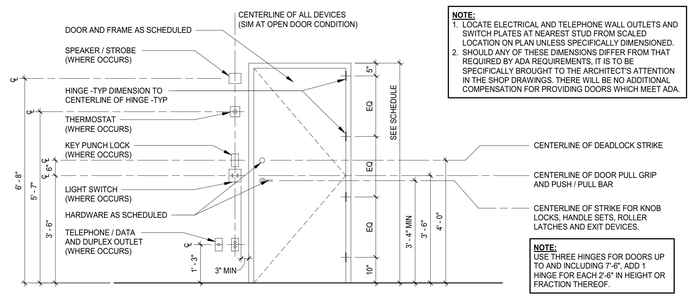
SCALE: 1/4" = 1'-0"



FRAME TYPES

SCALE: 1/4" = 1'-0"





NOTE:
1. LOCATE ELECTRICAL AND TELEPHONE WALL OUTLETS AND SWITCH PLATES AT NEAREST STUD FROM SCALED LOCATION ON PLAN UNLESS SPECIFICALLY DIMENSIONED.
2. SHOULD ANY OF THESE DIMENSIONS DIFFER FROM THAT REQUIRED BY ADA REQUIREMENTS, IT IS TO BE SPECIFICALLY BROUGHT TO THE ARCHITECT'S ATTENTION IN THE SHOP DRAWINGS. THERE WILL BE NO ADDITIONAL COMPENSATION FOR PROVIDING DOORS WHICH MEET ADA.

4 TYPICAL HARDWARE LOCATIONS
1/2" = 1'-0"

RSP Architects
502 South College Avenue Suite 203 Tempe, AZ 85281-3714
480.889.2000 480.889.2099 fax www.rsparch.com

Consultants

REGISTERED ARCHITECT CERTIFICATE NO. 35561 MICHAEL L. KRAGER Date Signed 13OCT2022 ARIZONA, U.S.A.

Project For

THE FLOWER SHOP PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd Ste. 1 and Ste. 2, Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	-
Checked By	-
Date	08/23/2022

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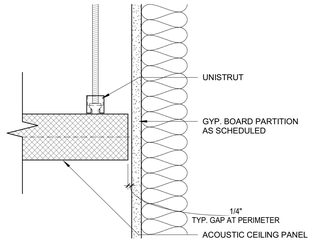
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Sheet Issues / Revisions

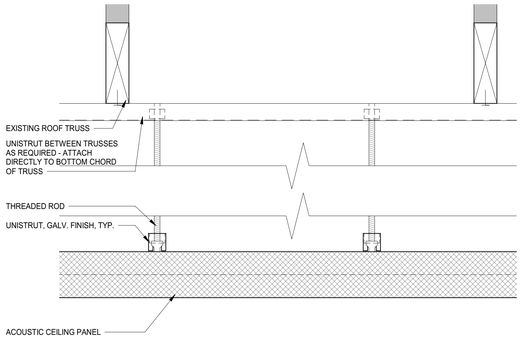
No.	Date	Description

SCHEDULES AND DOOR DETAILS

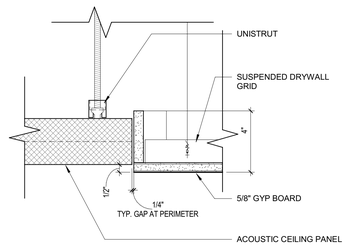
A310



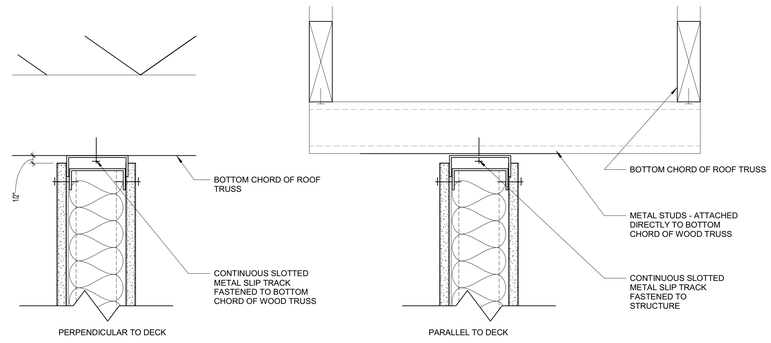
10 ACOUSTIC PANEL DETAIL AT WALL
3" = 1'-0"



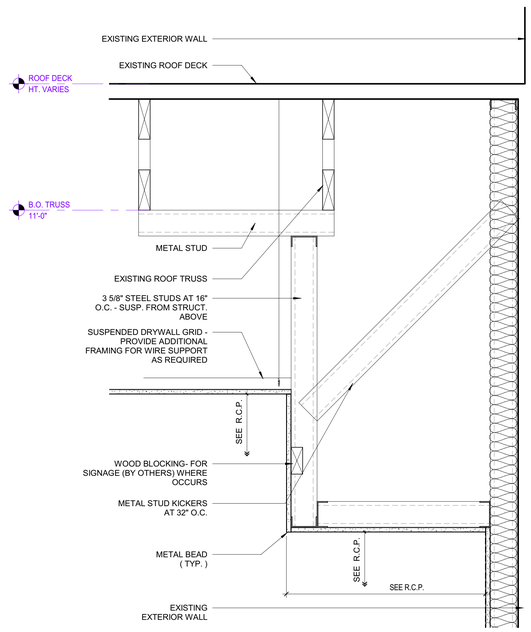
9 ACOUSTIC PANEL DETAIL
3" = 1'-0"



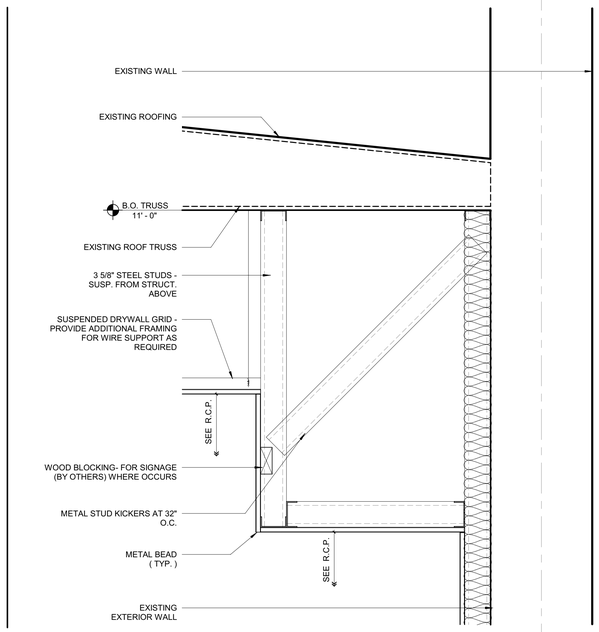
8 ACOUSTIC PANEL DETAIL AT GYP. BD. CEILING
3" = 1'-0"



7 SLIP TRACK DETAIL
3" = 1'-0"



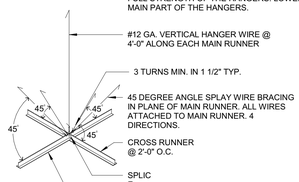
6 GYP. BD. SOFFIT - PARALLEL TO ROOF TRUSS
1 1/2" = 1'-0"

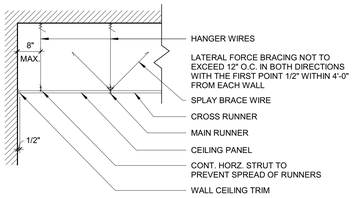


5 GYP. BD. SOFFIT - PERP. TO ROOF TRUSS
1 1/2" = 1'-0"

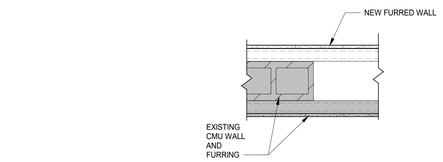
NOTES:
METAL SUSPENSION SYSTEMS FOR ACOUSTICAL TILE AND FOR LAY-IN PANEL CEILINGS SHALL BE INSTALLED IN ACCORDANCE WITH 1994 INIFORM BUILDING CODE CH. 16

HANGERS FOR SUSPENDED CEILINGS SHALL BE NOT LESS THAN THAT SET FORTH IN U.B.C. FASTENED TO OR EMBODIED IN THE STRUCTURAL FRAMES, MASONRY OR CONCRETE. HANGERS SHALL BE SADDLE-TIED AROUND MAIN RUNNERS TO DEVELOP THE FULL STRENGTH OF THE HANGERS. LOWER ENDS OF FLAT CHANNELS OR BENT TIGHTLY AROUND RUNNERS AND BOLTED TO THE MAIN PART OF THE HANGERS.

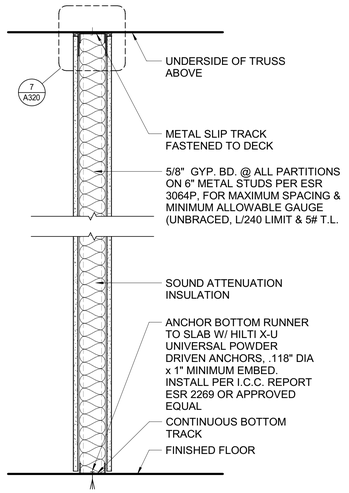




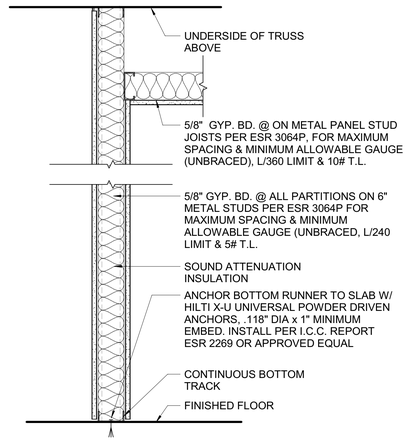
4 SUSPENDED CEILING DETAIL, TYP.
1 1/2" = 1'-0"



3 EXISTING CMU OPENING INFILL
1" = 1'-0"

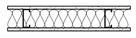


2 PARTITION TYPE A3
1 1/2" = 1'-0"



1 PARTITION TYPE A W/CEILING
1 1/2" = 1'-0"

TYPE A - FULL HEIGHT (NOT RATED) TO UNDERSIDE OF TRUSS



DESCRIPTION:
5/8" GYPSUM BOARD
METAL STUD AT 16" O.C.
5/8" GYPSUM BOARD
SEE 1 / A320

PTN TYPE	STUD WIDTH	PTN WIDTH	UL DESIGN	FIRE RATING	STC RATING W/O INSUL	STC RATING W/ INSUL	NOTES
A2	2 1/2"	3 3/4"	NA	NA	40	49	
A3	3 5/8"	4 7/8"	NA	NA	40	49	SEE 2 / A320
A4	4"	5 1/4"	NA	NA	40	49	
A6	6"	7 1/4"	NA	NA	40	49	

TYPE G - PARTIAL HEIGHT (FULL HEIGHT STUD WITH PARTIAL HEIGHT GYP BD)



DESCRIPTION:
5/8" GYPSUM BOARD
METAL STUD AT 16" O.C.

PTN TYPE	STUD WIDTH	PTN WIDTH	UL DESIGN	FIRE RATING	STC RATING W/O INSUL	STC RATING W/ INSUL	NOTES
G1	1 5/8"	2 1/4"	NA	NA			
G2	2 1/2"	3 1/8"	NA	NA			
G3	3 5/8"	4 1/4"	NA	NA			PROVIDE INSULATION AT ALL EXTERIOR WALLS AND SOUTH WALL ON GRID 2
G4	4"	4 5/8"	NA	NA			
G6	6"	6 5/8"	NA	NA			

RSP Architects
502 South College Avenue Suite 203 Tempe, AZ 85281-3714
480.889.2000 480.889.2099 fax www.rsparch.com

Consultants

REGISTERED ARCHITECT CERTIFICATE NO. 35861 MICHAEL L. KRAGER 22NOV2022 ARIZONA, U.S.A.

Project For

THE FLOWER SHOP PHOENIX RETAIL INTERIOR PHASE

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Project No.	3378.002.00
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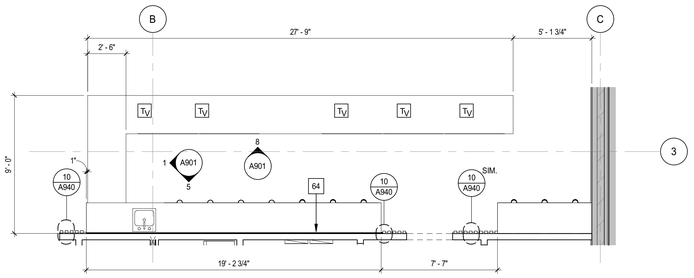
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Sheet Issues / Revisions

No.	Date	Description
1	11/22/22	CITY COMMENTS
	7/29/22	PERMIT PACKAGE

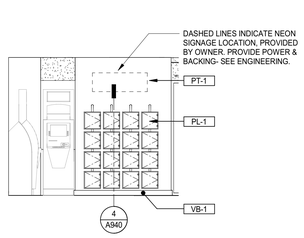
PARTITION TYPES AND DETAILS

A320

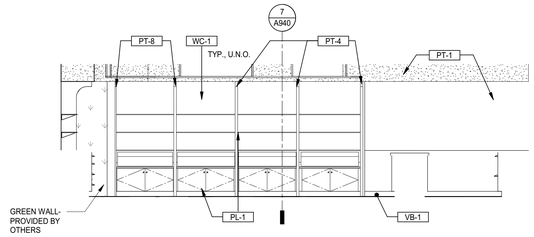


17 01 - POS ENLARGED PLAN
1/4" = 1'-0"

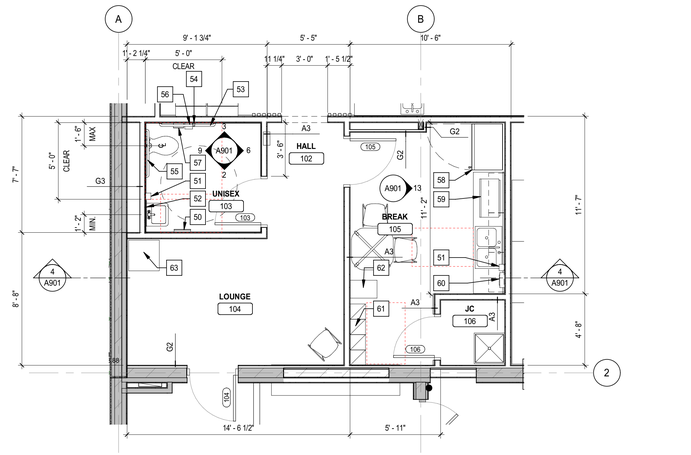
#	KEYNOTES - ENLARGED PLANS
NOTE: ALL KEYNOTES MAY NOT BE USED ON THIS SHEET	
NO.	DESCRIPTION
50	SEMI-RECESSED PAPER TOWEL DISPENSER WITH INTEGRAL WASTE DISPOSAL: BOBRICK 38032
51	WALL-MOUNT MANUAL SOAP DISPENSER: BOBRICK B-2111
52	HIGH QUALITY MIRROR WITH POLISHED EDGES (1'-8" WIDE X 3'-6" TALL), MOUNTED AT 3'-4" AFF
53	42" GRAB BAR: BOBRICK B-6806
54	18" VERTICAL GRAB BAR: BOBRICK B-6806
55	36" GRAB BAR: BOBRICK B-6806
56	SURFACE-MOUNT TOILET PAPER DISPENSER: GEORGIA PACIFIC PART #56796, BRUSH STEEL FINISH
57	SURFACE-MOUNT NAPKIN DISPOSAL: BOBRICK B-254
58	RE-INSTALL SALVAGED REFRIGERATOR FROM DEMO
59	MICROWAVE TO BE PROVIDED BY OWNER
60	BOBRICK B-9262 WALL MOUNTED PAPER TOWEL DISPENSER
61	TWO TIER LOCKERS. PROVIDE PER 2010 ADA STANDARDS. SEE FURNITURE PLAN FOR MORE INFO.
62	WATER DISPENSER- SEE ENGINEERING DRAWINGS
63	RELOCATE SAFE FROM DEMO- SEE FURNITURE PLAN FOR MORE INFO
64	PROVIDE IN-WALL BACKING & POWER FOR GREEN WALL & BACKLIT LOGO- SEE ENGINEERING



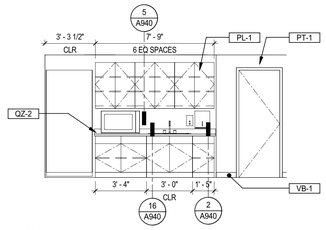
16 BOX DISPLAY ELEVATION
1/4" = 1'-0"



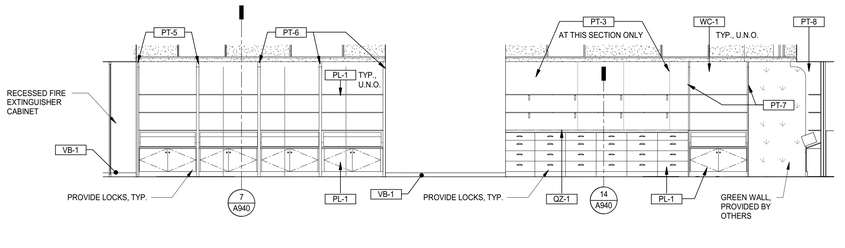
15 RETAIL ELEVATION- EAST
1/4" = 1'-0"



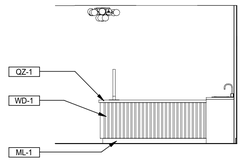
14 01 - ENLARGED RESTROOM PLANS
1/4" = 1'-0"



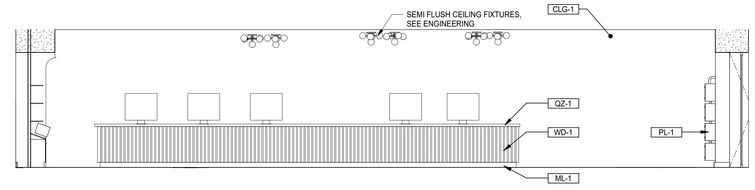
13 BREAKROOM- EAST
1/4" = 1'-0"



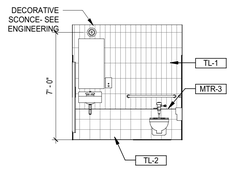
12 RETAIL DISPLAY- SOUTH ELEVATION
1/4" = 1'-0"



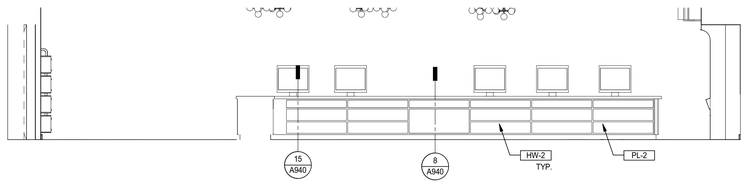
11 POS EAST ELEVATION
1/4" = 1'-0"



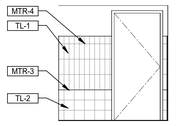
10 POS ELEVATION
1/4" = 1'-0"



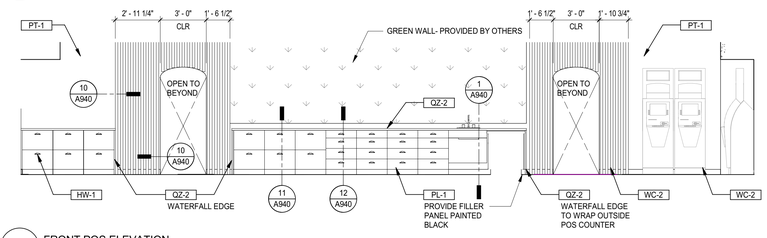
9 UNISEX- WEST
1/4" = 1'-0"



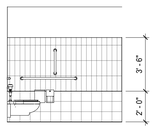
8 BACK COUNTER POS ELEVATION
1/4" = 1'-0"



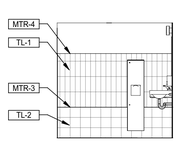
6 UNISEX- EAST
1/4" = 1'-0"



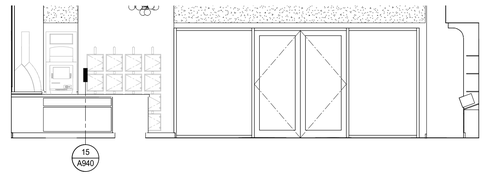
5 FRONT POS ELEVATION
1/4" = 1'-0"



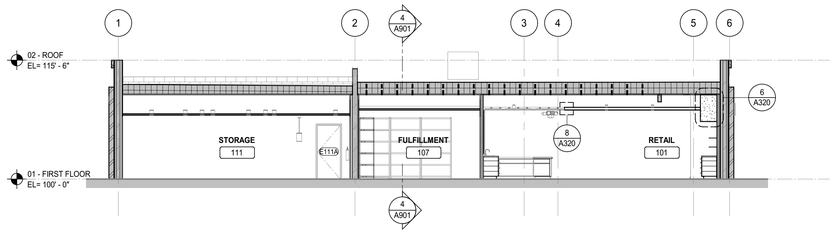
3 UNISEX- NORTH
1/4" = 1'-0"



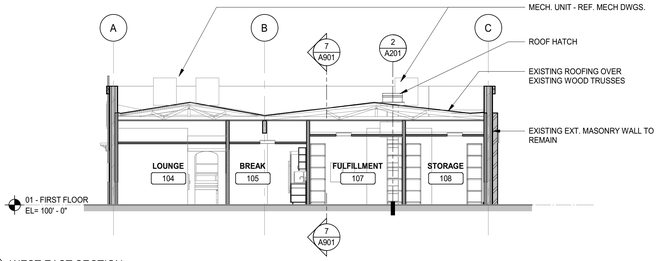
2 UNISEX-SOUTH
1/4" = 1'-0"



1 RETAIL DISPLAY- WEST ELEVATION
1/4" = 1'-0"



7 SOUTH NORTH SECTION
1/8" = 1'-0"



4 WEST EAST SECTION
1/8" = 1'-0"

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO. 25561
MICHAEL L. KRAGER
Date Signed 10OCT2022
ARIZONA, U.S.A.

Project For

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PHOENIX RETAIL INTERIOR PHASE

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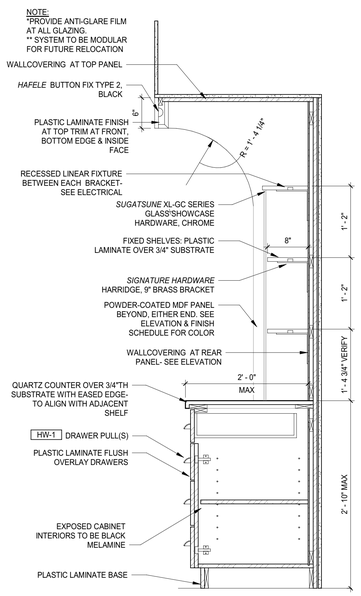
Sheet Issues / Revisions

No.	Date	Description

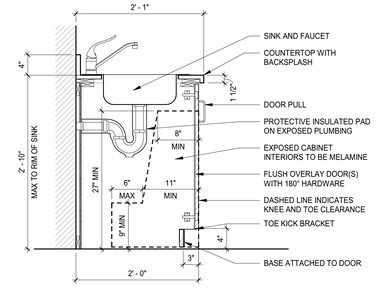
BUILDING SECTIONS, INTERIOR ELEVATIONS & ENLARGED PLANS

A901

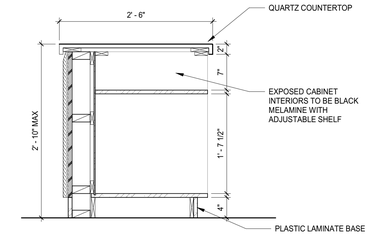
City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22



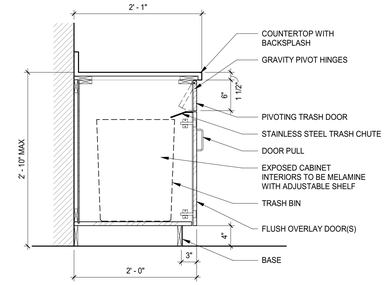
14 BDD - RETAIL SHELVING WITH DRAWERS
1" = 1'-0"



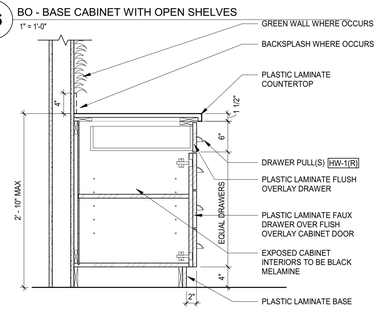
16 BSK - SINK CABINET WITH INTEGRATED KICK
1" = 1'-0"



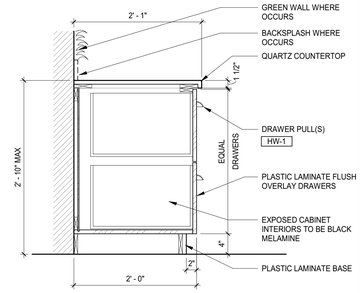
15 BO - BASE CABINET WITH OPEN SHELVES
1" = 1'-0"



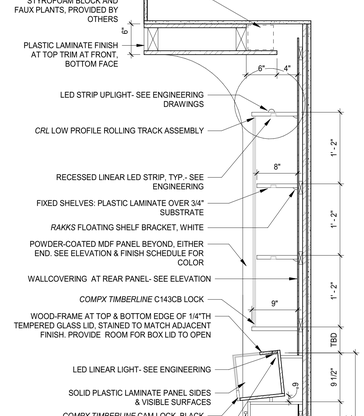
13 BT1 - TRASH CABINET WITH PIVOT DOOR
1" = 1'-0"



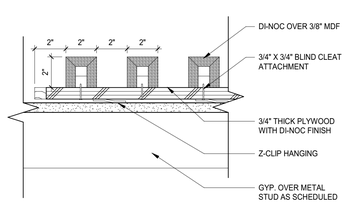
12 4D - FOUR DRAWER CABINET
1" = 1'-0"



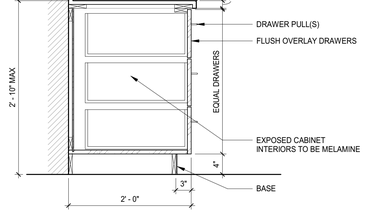
11 3DE - TWO DRAWER POS CABINET
1" = 1'-0"



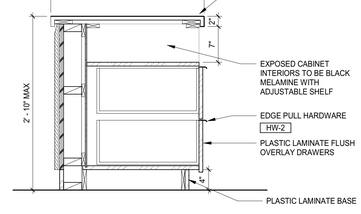
7 BDD - RETAIL DISPLAY W BOXES
1" = 1'-0"



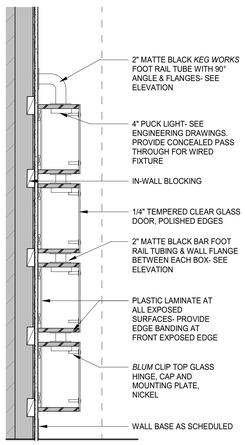
10 SW-SLAT WALL DETAIL
3" = 1'-0"



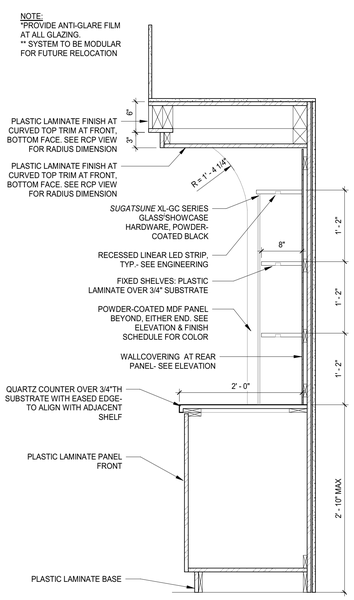
9 3DE - THREE DRAWER BREAKROOM CABINET
1" = 1'-0"



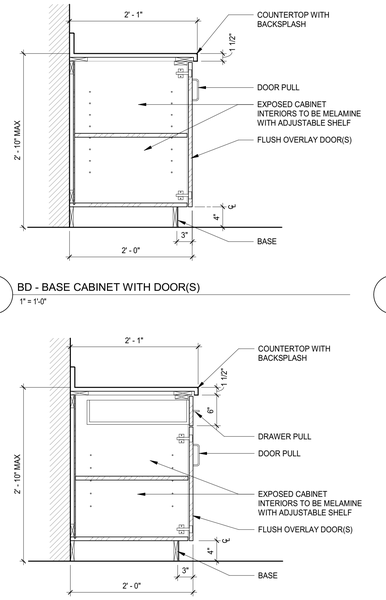
8 BP - BASE CABINET WITH PULL OUT DRAWER
1" = 1'-0"



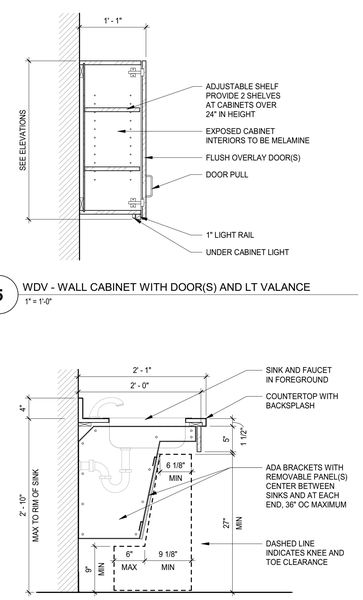
4 SPECIAL WALL DISPLAY
1" = 1'-0"



3 BDD - RETAIL CORNER DISPLAY
1" = 1'-0"



6 BD - BASE CABINET WITH DOOR(S)
1" = 1'-0"



5 WDV - WALL CABINET WITH DOOR(S) AND LT VALANCE
1" = 1'-0"

2 BC - BASE CABINET WITH DOOR(S) AND A DRAWER
1" = 1'-0"

1 OS3 - SINK WITH ADA BRACKET
1" = 1'-0"

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO.
35561
MICHAEL L. KRAGER
08 August 2022
Signed
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	Author
Checked By	Checker
Date	08/23/2022

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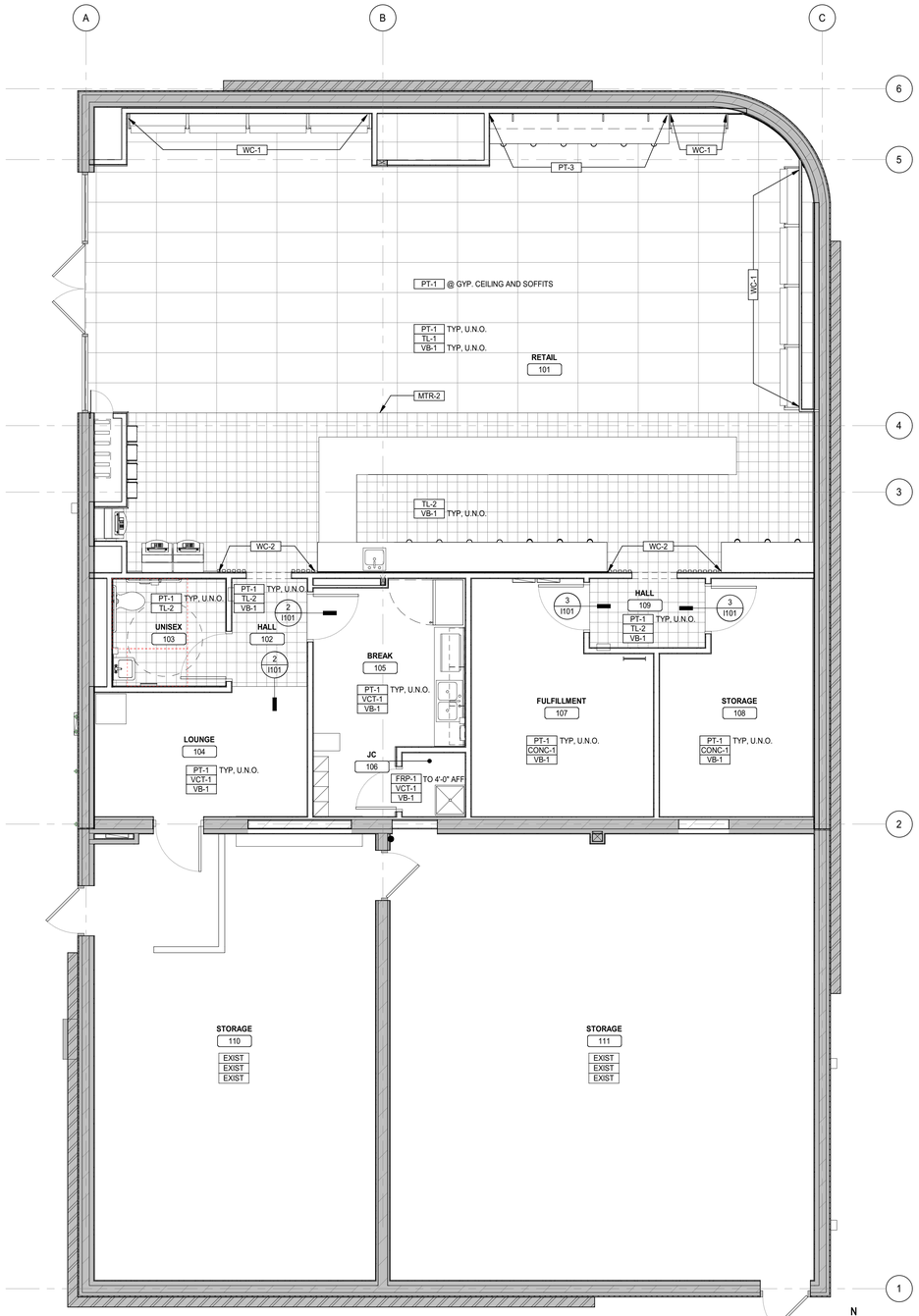
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Sheet Issues / Revisions

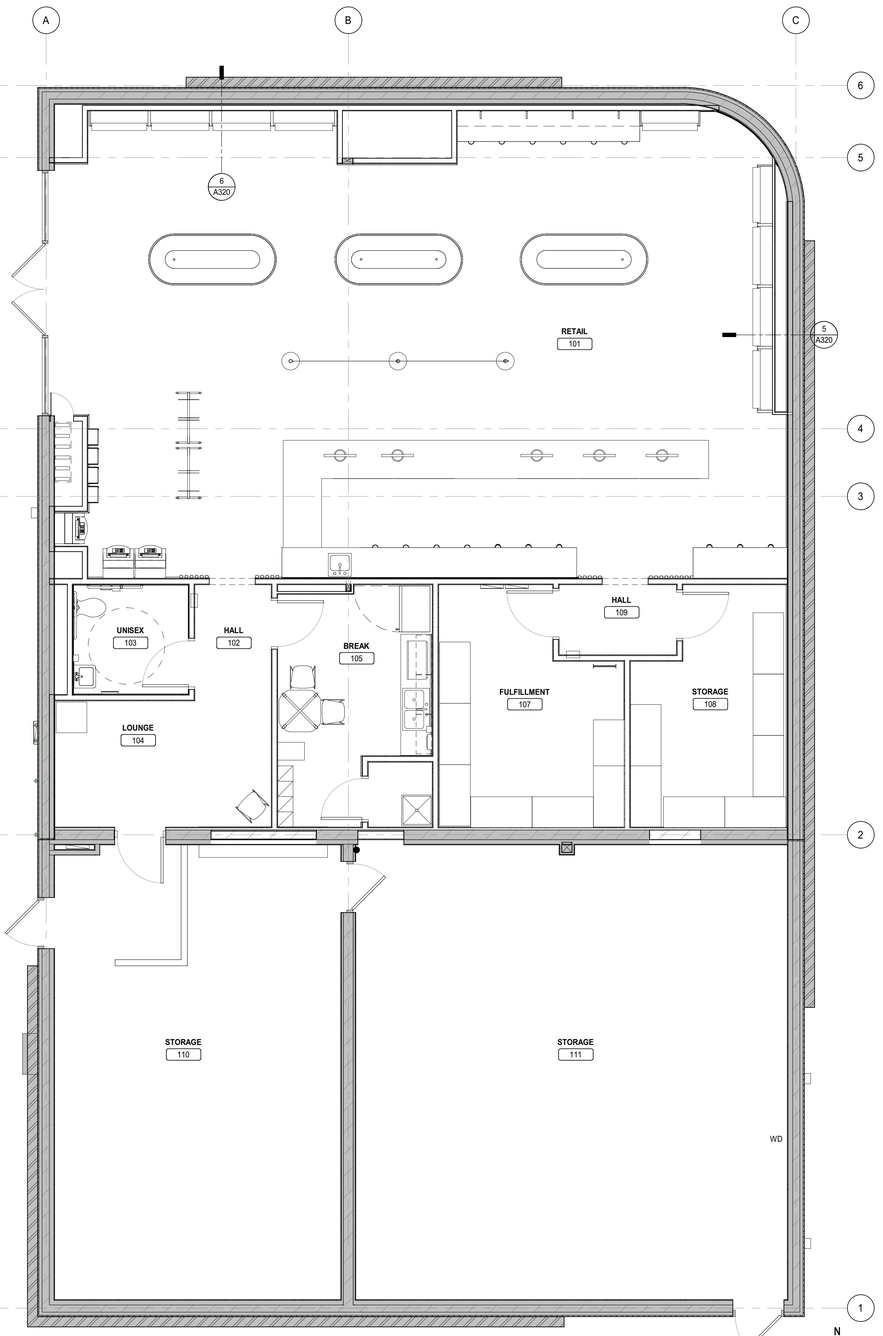
No.	Date	Description
	7/29/22	PERMIT PACKAGE

ARCHITECTURAL MILLWORK

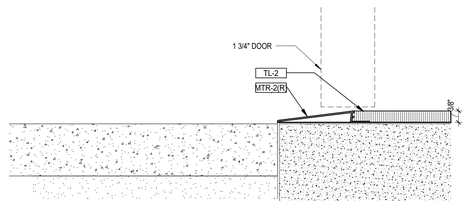
A940



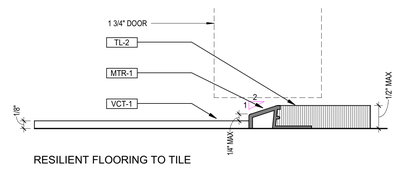
5 01 - FIRST FLOOR PLAN FINISHES PLAN
1/4" = 1'-0"



4 FIRST FLOOR FURNITURE PLAN "FOR REFERENCE ONLY"
1/4" = 1'-0"

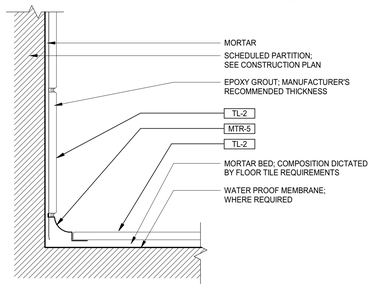


3 TILE TO CONCRETE TRANSITION
6" = 1'-0"



RESILIENT FLOORING TO TILE

2 RESILIENT TO TILE TRANSITION
12" = 1'-0"



1 TILE BASE TRANSITION DETAIL
6" = 1'-0"

GENERAL NOTES - FINISH PLAN

A. REFER TO SHEET I000 FOR FINISH AND MATERIAL SPECIFICATIONS.
B. INSTALL FINISHES PER MANUFACTURERS RECOMMENDATIONS, UNO.
C. FLOORING TRANSITIONS TO OCCUR AT CENTERLINE OF DOOR OPENINGS, UNO.
D. COORDINATE AVAILABILITY OF ALL MATERIALS SPECIFIED AND NOTIFY ARCHITECT OF ANY PRODUCTS WITH EXCESSIVE LEAD TIMES.
E. FIELD VERIFY ALL EXISTING CONDITIONS AND IDENTIFY DISCREPANCIES WITH PLANS AS SHOWN.
F. REFER TO ARCHITECTURAL SHEETS AND ELEVATIONS FOR ADDITIONAL INFORMATION ON FINISHES AND LOCATIONS.
G. CONTRACTOR TO ENSURE ALL SUBSTRATES ARE FLUSH AND SMOOTH FOR FINAL WALL, CEILING AND FLOOR FINISH MATERIAL INSTALLATION.

FINISH KEY - FLOOR PLAN

INTERIOR FINISH NUMBERING SYSTEM:

PT-1	- WALL FINISH
CPT-1	- FLOOR FINISH
VB-1	- BASE FINISH

RSP Architects
502 South College Avenue Suite 203 Tempe, AZ 85281-3714
480.889.2000
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www.rsparch.com

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REGISTERED ARCHITECT
CERTIFICATE NO. 35561
MICHAEL L. KRAGER
13OCT2022
Date Signed
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd
Ste. 1 and Ste. 2,
Phoenix, AZ 85008

Project No.	3378.002.00
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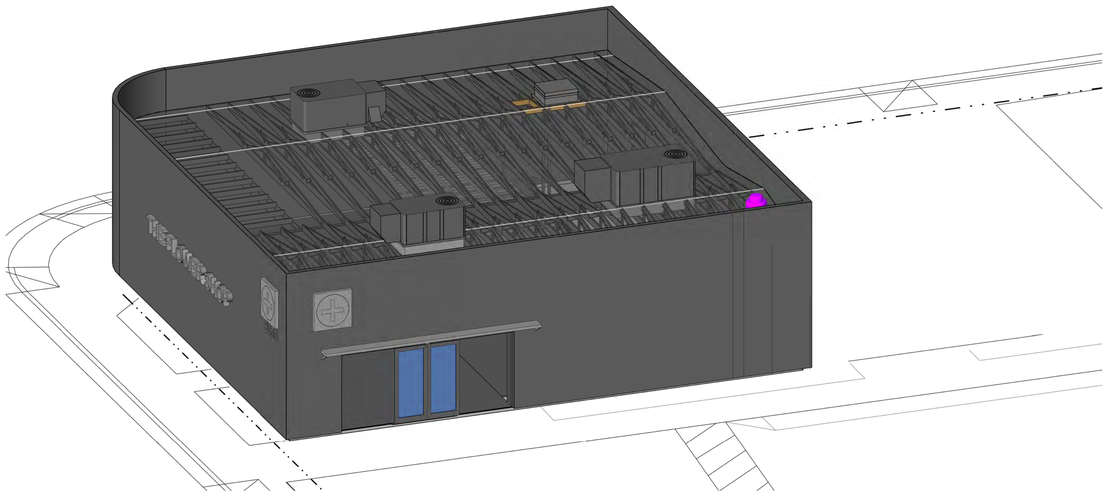
FIRST FLOOR FINISH PLAN

I101

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

SHEET LIST	
SHEET #	SHEET NAME
S001	LEGEND SHEET
S002	GENERAL STRUCTURAL NOTES
S201	FOUNDATION PLAN
S202	ROOF FRAMING PLAN
S501	FOUNDATION DETAILS
S701	FRAMING DETAILS

1 INTERIOR PHASE
THE STRUCTURAL SCOPE FOR THIS PHASE INCLUDES:
1. INFILL OF EXISTING OPENINGS
2. INFILL OF EXISTING OPENINGS
3. ADDITION OF ROOF HATCH
4. ADDITION OF NEW ROOFTOP UNIT



1 3D MODEL - FOR REFERENCE ONLY

STRUCTURAL ABBREVIATIONS:

A	
ADDL	ADDITIONAL
ADJ	ADJACENT
ALT	ALTERNATE
ALUM	ALUMINUM
AR	ANCHOR ROD
ARCH	ARCHITECT
B	
BDE	BOTTOM OF DECK ELEVATION
BFE	BOTTOM OF FOOTING ELEVATION
BM	BEAM
BOT	BOTTOM
BP	BEARING PLATE / BASE PLATE
BR	BOTTOM REINFORCING
BTWN	BETWEEN
C	
CANTL	CANTILEVER
CFS	COLD-FORMED STEEL
CGS	CENTER OF GRAVITY STRAND
CIP	CAST IN PLACE
CJ	CONTROL JOINT
CJP	COMPLETE JOINT PENETRATION
CL	CENTER LINE
CLR	CLEAR
CMU	CONCRETE MASONRY UNIT
COL	COLUMN
CONC	CONCRETE
CONN(S)	CONNECTION(S)
CONST	CONSTRUCTION
CONT	CONTINUOUS
D	
d	NAIL DIAMETER
db	BAR DIAMETER
DBA	DEFORMED BAR ANCHOR
DBL	DOUBLE
DEG	DEGREE
DEMO	DEMOLITION
DF	DOUGLAS FIR-LARCH
DIA	DIAMETER
DIAG	DIAGONAL
DIM	DIMENSION
DL	DEAD LOAD
E	
EA	EACH
EF	EACH FACE
EL	ELEVATION
ELEC	ELECTRICAL
ELEV	ELEVATOR
EJ	EXPANSION JOINT
EMBED	EMBEDMENT
EQ	EQUAL
EQUIP	EQUIPMENT
ES	EACH SIDE
EW	EACH WAY
E-W	EAST - WEST DIRECTION
(E)	EXISTING
EXP	EXPANSION
F	
FDN	FOUNDATION
FD	FLOOR DRAIN
FFE	FINISHED FLOOR ELEVATION
FLR	FLOOR
FS	FOOTING STEP
FT	FEET
FTG	FOOTING
FV	FIELD VERIFY
G	
GA	GAGE/GAUGE
GALV	GALVANIZED
GB	GRADE BEAM
GC	GENERAL CONTRACTOR
GLB	GLUE LAMINATED BEAM
GR	GRADE
GSN	GENERAL STRUCTURAL NOTES
GWB	GYPSUM WALL BOARD

H	
HK	HOOK
HORIZ	HORIZONTAL
HSA	HEADED STUD ANCHOR
HSS	HOLLOW STRUCTURAL SHAPE
HT	HEIGHT
I	
ID	INSIDE DIAMETER
INCL	INCLUDE
ISF	INSIDE FACE
J	
JT	JOINT
JBE	JOIST BEARING ELEVATION
K	
K	KIPS
KLF	KIPS PER LINEAL FOOT
KSF	KIPS PER SQUARE FOOT
KSI	KIPS PER SQUARE INCH
KO	KNOCK OUT
L	
LB(S)	POUND(S)
LL	LIVE LOAD
LLH	LONG LEG HORIZONTAL
LLV	LONG LEG VERTICAL
LONG	LONGITUDINAL
LSL	LAMINATED STRAND LUMBER
LSH	LONG SIDE HORIZONTAL
LSV	LONG SIDE VERTICAL
LWT	LIGHT WEIGHT
LVL	LAMINATED VENEER LUMBER
M	
MAX	MAXIMUM
MECH	MECHANICAL
MEP	MECHANICAL, ELECTRICAL & PLUMBING
MEZZ	MEZZANINE
MFR	MANUFACTURER
MIN	MINIMUM
MISC	MISCELLANEOUS
MSR	MACHINE STRESS RATED
MTL	METAL
N	
NIC	NOT IN CONTRACT
N-S	NORTH - SOUTH DIRECTION
NTS	NOT TO SCALE
NWT	NORMAL WEIGHT
O	
OC	ON CENTER
OD	OUTSIDE DIAMETER
OSF	OUTSIDE FACE
OPNG	OPENING
OPP	OPPOSITE
O/O	OUT TO OUT
P	
PAF	POWER ACTUATED FASTENER
PC	PRECAST CONCRETE
PL	PLATE
PLF	POUNDS PER LINEAL FOOT
PLYWD	PLYWOOD
PRE FAB	PREFABRICATED
PROJ	PROJECTION
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PSL	PARALLEL STRAND LUMBER
PT	POST TENSIONED

Q	
QTY	QUANTITY
R	
R	RADIUS
RD	ROOF DRAIN
REF	REFERENCE
REINF	REINFORCEMENT/REINFORCING
REQD	REQUIRED
REV	REVISION
RO	ROUGH OPENING
RSS	RUGGED STRUCTURAL SCREW
RTU	ROOF TOP UNIT
S	
SB	SOIL BORING
SC	SLIP CRITICAL
SCHED	SCHEDULE
SER	STRUCTURAL ENGINEER OF RECORD
SF	SQUARE FOOT
SIM	SIMILAR
SL	SNOW LOAD
SOG	SLAB ON GRADE
SPA	SPACES
SPEC	SPECIFICATION
SPF	SPRUCE PINE FIR
SS	STAINLESS STEEL
SSLT	SHORT-SLOT LOAD TRANSVERSE
STD	STANDARD
STIFF	STIFFENER
STL	STEEL
STRUCT	STRUCTURE / STRUCTURAL
SYM	SYMMETRICAL
SYP	SOUTHERN YELLOW PINE
T	
T/G	TONGUE AND GROOVED
TBE	TOP OF BEAM ELEVATION
TDE	TOP OF DECK ELEVATION
TEMP	TEMPORARY
TFE	TOP OF FOOTING ELEVATION
TGBE	TOP OF GRADE BEAM ELEVATION
TPCE	TOP OF PILE CAP ELEVATION
TPCPE	TOP OF PRECAST PLANK ELEVATION
TPE	TOP OF PIER ELEVATION
TR	TOP REINFORCING
TRANS	TRANSVERSE
TSE	TOP OF SHEATHING ELEVATION
TSE	TOP OF SLAB ELEVATION
TSE	TOP OF SUBFLOOR ELEVATION
TWE	TOP OF WALL ELEVATION
TYP	TYPICAL
U	
UNO	UNLESS NOTED OTHERWISE
URM	UNREINFORCED MASONRY
V	
VERT	VERTICAL
W	
W/	WITH
W/O	WITHOUT
WD	WOOD
WF	WIDE FLANGE
WL	WIND LOAD
WP	WORK POINT
WT	WEIGHT
WWF	WELDED WIRE FABRIC

MARKS AND SYMBOLS LEGEND:

MARKS:

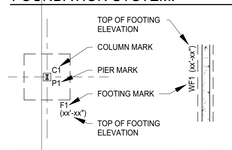
B1001	CONCRETE BEAM MARK NUMBER
B1001-PT	POST TENSIONED CONCRETE BEAM MARK NUMBER
BP1	BEARING / BASE PLATE MARK NUMBER
BRF1	BRACE FRAME MARK NUMBER
BR1	MILD STEEL BOTTOM REINFORCING MARK NUMBER
C1	COLUMN MARK NUMBER
CC1	CONCRETE COLUMN MARK NUMBER
CW1	CONCRETE WALL NUMBER
D1	STEEL DECK MARK NUMBER
DC1	DROP CAPITAL MARK NUMBER
DP1	DRILLED PIER MARK NUMBER
DPC1	DRILLED PIER CAP NUMBER
EP1	EMBEDDED PLATE MARK NUMBER
F1	SPREAD FOOTING MARK NUMBER
GB1	GRADE BEAM MARK NUMBER
H1	HEADER MARK NUMBER
HCP	HOLLOW CORE PLANK
HD1	HOLD DOWN MARK NUMBER
J10	JOIST MARK NUMBER
L1	LINTEL MARK NUMBER
LC1	LIGHT GAGE COLUMN MARK NUMBER
MC1	MASONRY COLUMN MARK NUMBER
MF1	MOMENT FRAME MARK NUMBER
MW1	MASONRY WALL NUMBER
P1	PIER MARK NUMBER
PC1	PILE CAP MARK NUMBER
RD1	ROOF DECK MARK NUMBER
S1	SLAB MARK NUMBER
SC1	STEEL COLUMN MARK NUMBER
SR1	STUD RAIL REINFORCING MARK NUMBER
SW1	SHEAR WALL MARK NUMBER
T1	TRUSS MARK NUMBER
TR1	MILD STEEL TOP REINFORCING MARK NUMBER
W1	WALL MARK NUMBER
WC1	WOOD COLUMN MARK NUMBER
WF1	WALL FOOTING MARK NUMBER
WO1	WEB OPENING

GENERAL SYMBOLS:

- APPROXIMATE LOCATION OF DRAIN TILE
- MATCH LINE
- LINE OF DEMOLITION
- (?'-?") SLAB STEP LOCATION WITH ELEVATIONS
- SLAB STEP LOCATION
- CHANGE IN SLAB SLOPE
- CHANGE IN SLAB THICKNESS
- 1 KEYNOTE MARK NUMBER
- ? NEW BUILDING GRID LINE
- ? EXISTING BUILDING GRID LINE
- ELEVATION MARKER
- SHADED AREA INDICATES CUT THROUGH EXISTING CONSTRUCTION
- SHADED AREA INDICATES PROJECTION OF EXISTING CONSTRUCTION
- W1 WALL MARK NUMBER OR WALL TYPE
- APPROXIMATE LOCATION OF UTILITY PIPE PENETRATION THROUGH FOUNDATION WALL
- FS FOOTING STEP LOCATION
- SB1 APPROXIMATE LOCATION OF SOIL BORING
- COMPRESSION PILE
- TENSION / COMPRESSION PILE
- TEST PILE
- SPAN DIRECTION OF ELEMENT
- EXTENT OF ELEMENT
- CONTINUOUS EXTENT OF ELEMENT
- # Sxxx DETAIL CALLOUT
- # Sxxx ELEVATION CALLOUT

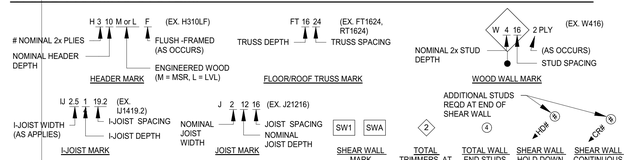
PLAN SYMBOLS LEGEND:

FOUNDATION SYSTEM:



SPREAD FOOTING WALL FOOTING

WOOD FRAMING SYSTEM:



RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099 fax
Tempe, AZ 85281-3714 www.rsparch.com

Consultants
MEYER | BORGMAN | JOHNSON
STRUCTURAL DESIGN + ENGINEERING
TEL: 480.747.6727 www.mbjeng.com 116 EAST UNIVERSITY DRIVE SUITE 300 TEMPE, AZ 85281
SEE STRUCTURE.

Certification

Registered Professional Engineer Certificate No. 24045 KURT F. KINDERMANN Date Signed 11/23/22 ARIZONA, U.S.A.

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	P21.215
Drawn By	JRT
Checked By	CC
Date	

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Sheet Issues / Revisions

No.	Date	Description
1	11/23/22	City Comments
	7/29/22	PERMIT PACKAGE

LEGEND SHEET

S001

TYPICAL NOTES:
These notes specify the requirements for the design represented in these documents. The construction and materials shall comply with all the pertinent codes and references, plans, and details, including (but not limited to) those shown in architectural, civil, mechanical and electrical drawings.

The contractor shall verify all dimensions and existing conditions in the field that affect construction prior to commencing work on the affected element or shop drawing submittals. Resolve any discrepancies with the architect prior to construction.

The contract structural drawings and specifications represent the completed structure. The contractor is responsible for bracing and shoring (without overstressing) all structural elements as necessary at any stage of construction until completion of the project. The Structural Engineer is not responsible for the contractor's means, methods, sequences or procedures of construction. Contractor shall recognize and consider effects of thermal movements of structural elements during construction period.

The contractor is solely responsible for site safety including all temporary precautionary measures and safety programs. Site observation visits by the Structural Engineer do not include review of the contractor's safety precautions.

Refer to architectural, mechanical and electrical drawings for locations, elevations, dimensions, and details of sleeves, inserts, openings, recesses, curbs, housekeeping pads, etc. that are not shown on the structural drawings and do not damage structural members.

Information shown in the structural drawings regarding existing conditions represents the current and general field conditions related to the new work, to the best of our knowledge. Report all discrepancies (unforeseen conditions) to the Architect for resolution prior to performing related new work.

Requests for information shall be submitted in writing and shall reference the part of the construction documents that is in question.

SHOP DRAWINGS:

Submit shop drawing schedule with construction schedule that includes consideration for review period. See specification for additional information.

General contractor shall submit shop drawings in digital format for structural review. Digital drawing shall meet the following requirements.
1. Adobe extension (.pdf)
2. All pages are rotated, printed to scale.
3. All transmittals shall be located as the first page of the submittal or as a separate file within one digital package.
4. Contractor digital review comments and their digital stamp shall be attached. Our review will not occur until the contractor has reviewed, coordinated with other trades and provided shop stamp.
5. MBJ will mark-up the digital set in red and return a digital file via email, ftp site or other means.

DESIGN CODES AND STANDARDS:

2018 International Building Code, as amended and adopted by the City of Phoenix, Arizona.

MATERIAL PROPERTIES:

Structural Composite Lumber:
Doug Fir (DF) No. 2 or better: (Joists, Ledgers and Headers): Fb 900 psi, Fv 180 psi, E 1,600,000 psi

Cold-formed Steel Framing (Fy):
Studs, Joists, Braces-54 mils and thicker: 50,000 psi ASTM A653
Studs, Joists, Braces-43 mils and thinner: 33,000 psi ASTM A653
Track, Channels and Accessories 54 mils and thicker: 50,000 psi ASTM A653
Track, Channels and Accessories 43 mills and thinner: 33,000 psi ASTM A653

DESIGN LOADS:
LATERAL LOADS:
Risk Category: II
Wind Loads:
Primary Frame Wind Data:
Basic Wind Speed: V ult = 101 mph

Exposure Category: C
Internal Pressure Coefficient (Gcpi): +0.18 or –0.18

Components and Cladding Wind Loads:
Exterior Component/Cladding: Supplier to develop based on code criteria and indicate on shop drawings.

Seismic Loads:
Primary Seismic Data:
Mapped Spectral Response Accelerations: Ss: 0.183 S1: 0.065

Site Class: C
Site Coefficients: Fa = 1.6, Fv = 2.4

Design Spectral Acceleration Parameters: Sds: 0.195 Sd1: 0.104

Importance Factor: 1.0
Seismic Design Category: B

Basic Seismic-Force- Resisting System: Ordinary reinforced masonry shear walls

Response Modification Factor: R = 2.0
Overstrength Factor: ΩO = 2.0
Seismic Response Coefficient: Cs = 0.098
Ultimate Design Base Shear: V = 0.098(W)

Analysis Procedure: Equivalent Lateral Force Procedure

GRAVITY LOADS:
Roof Dead Load (Assumed): 20 psf

Roof Live Load:
Live Load, (reducible): 20 psf
Mechanical and Electrical Equipment Units: Refer to drawings, for the units' locations, sizes, and weights.
Future Mechanical and Electrical Units: This project is not designed for future units.

COLD-FORMED METAL FRAMING:
The design and connection detailing of all cold-formed material including, but not limited to exterior studs, headers, jambs and are as indicated on the drawings. The minimum design criteria for all systems is as follows:

Studs in exterior walls shall be spaced at 16" o.c. unless noted otherwise.

Studs shall be cold rolled steel, galvanized, C shape, with minimum 1 5/8" flange and minimum 1/2" return. They are to be punched for utility access and galvanized to G60 coating per ASTM 525.

At all openings in exterior walls provide a minimum two studs full wall height each side of opening and a minimum one additional stud each side for lintel bearing.

Anchor bottom track to concrete or masonry with minimum 5/32" x 1 1/4" power driven fasteners at 16" o.c.

Top and bottom tracks shall be cold rolled or break formed steel, galvanized U shaped and minimum 18 gauge and as noted on the drawings.

Cold-formed metal framing fasteners shall be minimum #10 self-drilling sheet metal screws, 16 threads per inch, with low profile head. Provide a minimum of two screws per connection unless noted otherwise.

Fasten cold-formed framing to wood with minimum #10 x 1 7/8" bugle head wood screws. Pre-drill holes in metal studs. Provide a minimum of two screws per connection unless noted otherwise.

All framing components shall be squarely cut for attachment to perpendicular members. Stud ends must seat tightly into tracks for all bearing applications.

At all wall elements, provide 1 1/2"-16 gauge horizontal channel bridging to prevent stud rotation. For all axial loaded walls, space bridging at 4'-0" o.c. For all non-load bearing exterior walls, space bridging at 5'-0" o.c.

All designations are in accordance with the Steel Stud Manufacturers Association (SSMA).

Refer to architectural drawings and specification for size, extent, and location of interior non-bearing cold-formed framing not shown on the structural drawings. Interior cold-formed framing shall be a minimum of 33 mil (20 GA) and spaced at 16" o.c. (maximum). Provide 2-1/2" deep leg tracks at interior framing deflection tracks.

Temporary bracing shall be furnished by the supplier and framing installer and maintained until permanent systems providing lateral stability are in place.

Welding shall conform to the American Welding Society (AWS) "Structural Welding Code – Sheet Steel, D1.3 – Current Edition." Welders shall be qualified in accordance with AWS D1.3 and shall be experienced in cold-formed welding.

All cold-formed material to be welded must be nominal 16 gauge or thicker.

Touch up all cold-formed material at welds with zinc-rich paint.

Align load bearing wall studs with floor or roof joists.

Splices in studs, joists, and headers, are not permitted, unless approved in writing by the structural engineer.

Framing components may be pre-assembled into panels prior to erecting. Prefabricated panels shall be square, with components attached in a manner that prevents racking.

WOOD FRAMING:
DIMENSION LUMBER:
With Type 3A or 3B construction, all lumber in exterior walls shall be fire retardant treated wood.

All member sizes given in the drawings are nominal dimensions.

All lumber shall be kiln-dried, maximum moisture content 15% and grade marked according to the National Forest Products Association Regulations.

All joists (greater than 2x8) shall be supported laterally at the ends and at each support by solid blocking except where ends of joists are nailed to a header, band or rim joist or to an adjoining stud. Solid blocking shall be not less than 2" in thickness and the full depth of the joist.

Wood joists shall bear on the full width of supporting members, stud walls, beams, etc., unless otherwise noted.

Do not notch or cut joist unless approved by the Structural Engineer of Record.

All beams and joists not bearing on supporting members shall be framed with "Simpson Strong- Tie" joist hangers or equal. Use type "LUS" (or equal) for single 2x's and double 2x's and type "UTF" for framing to trusses where required. The joist hangers shall be nailed using special nails supplied by the hanger manufacturer. Proposed nail type substitutions shall conform to the ICC report for equal or greater load capacity and shall be submitted with the ICC report to the Structural Engineer of Record for written approval.

Wood headers or posts made up of 2 or more 2x's shall be spiked together per the nailing schedule.

Provide minimum 2 - 2x trim studs at bearing ends of all headers. Where posts are shown on drawings, headers shall bear fully on the posts.

Wood columns shall have solid vertical blocking through the floors to the support below.

All holes drilled through studs or posts in walls shall strictly conform to the detail in the drawings. Wood columns are not to be notched or drilled for utilities.

For walls 10'-0" and greater provide blocking at mid-height for construction stability.

All walls shall have single bottom plate and double top plate.

Double top plate splices shall lap 4'-0" and be nailed with 16 - 0.131" x 3" nails equally spaced with 1 1/2" end distance, unless noted otherwise on plan.

Unless otherwise noted, bottom plates of all exterior stud walls and interior bearing walls shall be anchored to new concrete with 1/2" diameter anchor bolts, at 4'-0" oc, or with equivalent anchors, as approved by the structural engineer.

All exterior lumber and all lumber in contact with concrete or masonry, or exposed to the exterior shall be treated Southern Yellow Pine. Each sill plate shall have a minimum of 2 anchor rods with an anchor rod located within 12" of each end.

All connectors in contact with treated lumber shall have corrosion protection.

For nailing/fastener schedule refer to the drawings.

WOOD FASTENERS – NAILING:
Framing nail sizes specified on the drawings are based on the following specification U.N.O.:

Size	Length	Diameter
6d common	2"	0.113"
8d common	2 1/2"	0.131"
10d common	3"	0.148"
12d common	3 1/4"	0.148"
16d common	3 1/2"	0.162"

Size	Length	Diameter
6d box	2"	0.099"
8d box	2 1/2"	0.113"
10d box	3"	0.128"
16d box	3 1/4"	0.135"

Size	Length	Diameter
6d cooler	1 7/8"	0.092"
8d cooler	2 3/8"	0.113"

All framing nails shall conform to ASTM F667, "Standard Specification for Power Driven Fasteners: Nails, Spikes and Staples" and NER-272 "Power Driven Staples and Nails for Use in All Types of Building Construction". Cooler nails shall comply with ASTM C514.

Refer to nailing schedule in the drawings for nail size and spacing at a specified condition.

Nails shall be identified by labels attached to their containers that show the Manufacturer's name and NES report number, nail shank diameter, and length. Submit this information prior to framing.

If the Contractor proposes the use of alternate nails, they shall submit prior to construction nail specifications with certified calculations showing structural equivalence to the Structural Engineer of Record for review and approval.

Nails fastening APA rated plywood sheathing shall be driven flush to the face of sheathing with no counter sinking permitted. Renail sheathing as necessary to comply.

WOOD FASTENERS – STRUCTURAL WOOD SCREWS:
Structural wood screws as specified in the drawings refer to threaded steel screws that are self-drilling, dowel-type fasteners used primarily for wood-to-wood connections. These carbon steel screws are manufactured by a cold-formed process and are heat-treated with rolled threads. No pre-drilling is required.

Screws are specified in the drawings per nominal diameter and length. The diameter refers to a nominal measure of the threads, which is larger than the unthreaded shaft of the fastener. Length specified does not include fastener head. Actual dimensions and available lengths vary with Manufacturer.

Acceptable products are listed below. Contractor may submit alternate products for approval by Structural Engineer of Record.

The following minimum dimensions and material properties shall apply:

Size specified	Minimum Shank:	Root Diameters (in)	Acceptable Products
1/4" Diam	0.169"	0.150"	GRK RSS
5/16" Diam	0.189"	0.172"	GRK RSS Simpson SDWH FastenMasterTimberlok
3/8" Diam	0.219"	0.191"	GRK RSS Simpson SDWS FastenmasterLedgerlok

Minimum Allowable Tension strength of fastener (lbs):
1/4" Diameter 1112 lbs
5/16" Diameter 1210 lbs
3/8" Diameter 1505 lbs

Minimum Allowable Shear strength of fastener (lbs):
1/4" Diameter 754 lbs
5/16" Diameter 770 lbs
3/8" Diameter 910 lbs

Minimum Bending Yield Strength: 165,000 psi

RSP Architects
502 South College Avenue, Suite 203, Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants
MEYER | BORGMAN | JOHNSON
STRUCTURAL DESIGN + ENGINEERING
TEL: 480.747.6727 www.mbjeng.com
116 EAST UNIVERSITY DRIVE SUITE 300 TEMPE, AZ 85281
SEE STRUCTURE.

Certification
Registered Professional Engineer Certificate No. 24045 KURT F. KINDERMANN Date Signed 11/23/22 ARIZONA, U.S.A.

Project For
THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE
3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	P21.215
Drawn By	JRT
Checked By	CC
Date	

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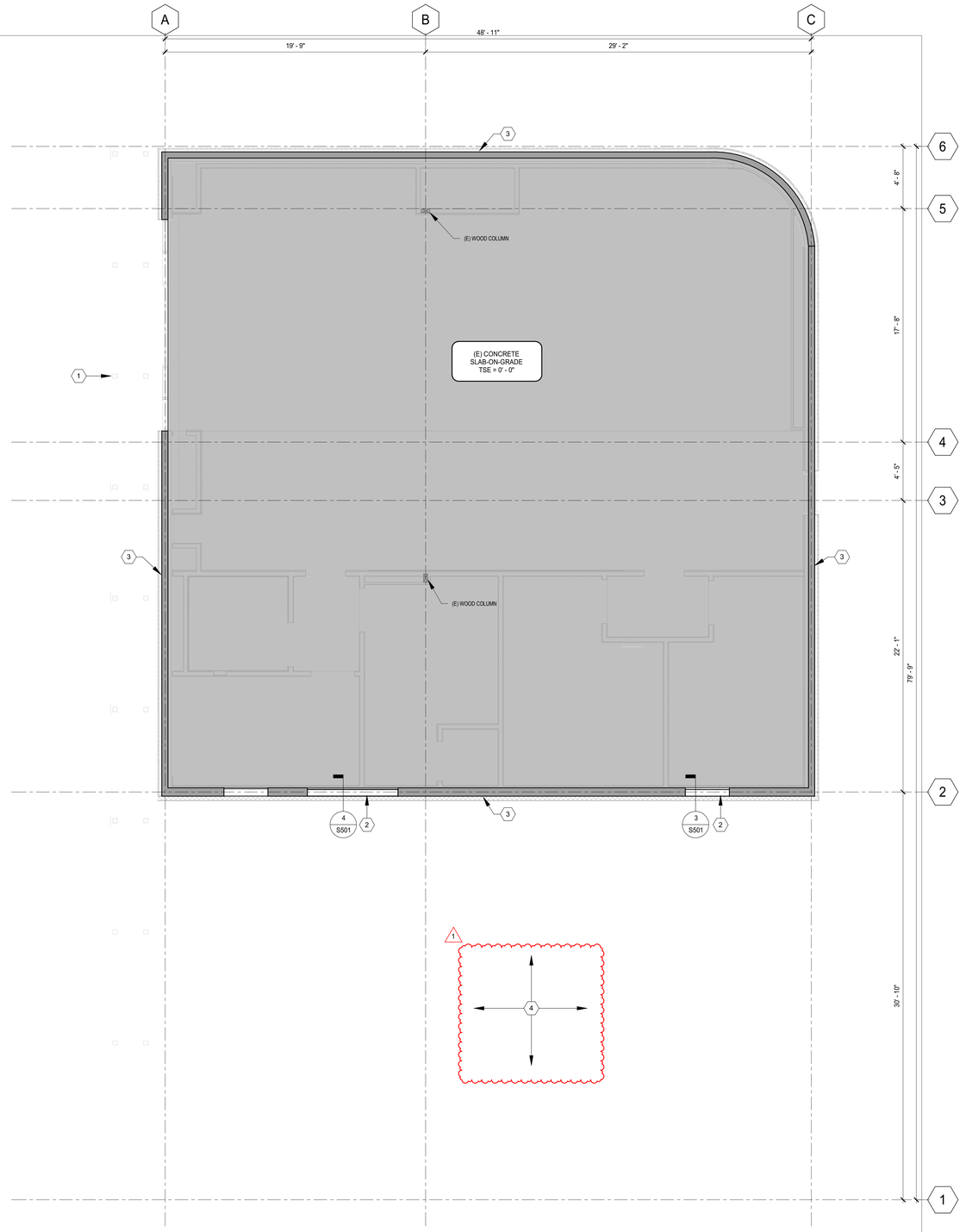
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Sheet Issues / Revisions

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1	11/23/22	City Comments
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GENERAL STRUCTURAL NOTES

S002



1 FOUNDATION PLAN
1/4" = 1'-0"

NORTH

REFERENCE NOTES:

1. SEE SHEET S001 FOR SHEET INDEX, TYPICAL ABBREVIATIONS AND LEGENDS.
2. SEE SHEET S002 FOR GENERAL STRUCTURAL NOTES.
3. SEE SHEET S003 FOR STRUCTURAL SPECIAL INSPECTIONS.
4. SEE SHEET S500 SERIES FOR FOUNDATION AND CONCRETE FRAMING DETAILS.
5. SEE SHEET S700 SERIES FOR FRAMING DETAILS.

PLAN NOTES (UNLESS NOTED OTHERWISE):

1. PROJECT DATUM ELEVATION 0' - 0" = EXISTING INTERIOR SLAB-ON-GRADE. SEE ARCHITECTURAL DRAWINGS.
2. (E) TOP OF FOOTING ELEVATION (TFE) TO BE FIELD VERIFY.
3. TOP OF SLAB-ON-GRADE ELEVATION (TSE) = 0' - 0"
4. 5" THICK SLAB-ON-GRADE. REINFORCE WITH #4 AT 12" OC EW.
5. SEE ARCHITECTURAL DRAWINGS FOR ADDITIONAL INFORMATION, INCLUDING LOCATIONS AND DIMENSIONS OF RAMPS, SLAB SLOPES, SLAB STEPS AND SLAB DEPRESSIONS.
6. SEE CIVIL / LANDSCAPE DRAWINGS FOR PAVING AND SITE DETAILS AT THE BUILDING EXTERIOR.

KEYNOTES:

1 NEW TRELLIS FOOTINGS, POSTS, AND FRAMING IN FUTURE EXTERIOR PHASE.

2 (E) DOOR AND WINDOW TO BE STUD INFILL.

1 3 (E) 8" MASONRY WALL. GENERAL CONTRACTOR TO FIELD VERIFY.

4 EXISTING BUILDING SOUTH OF GRID LINE 2 NOT WITHIN STRUCTURAL SCOPE OF INTERIOR PHASE. SEE ARCHITECTURAL DRAWINGS FOR LOCATION AND EXTENTS

RSP

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants
MEYER | BORGMAN | JOHNSON
STRUCTURAL DESIGN + ENGINEERING
TEL: 480.747.6727
www.mbjeng.com
116 EAST UNIVERSITY DRIVE
SUITE 300
TEMPE, AZ 85281
SEE STRUCTURE.

Certification

Registered Professional Engineer
Certificate No.
24045
KURT F. KINDERMANN
Date Signed 11/23/22
ARIZONA, U.S.A.

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
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Project No. P21.215
Drawn By JRT
Checked By CC
Date

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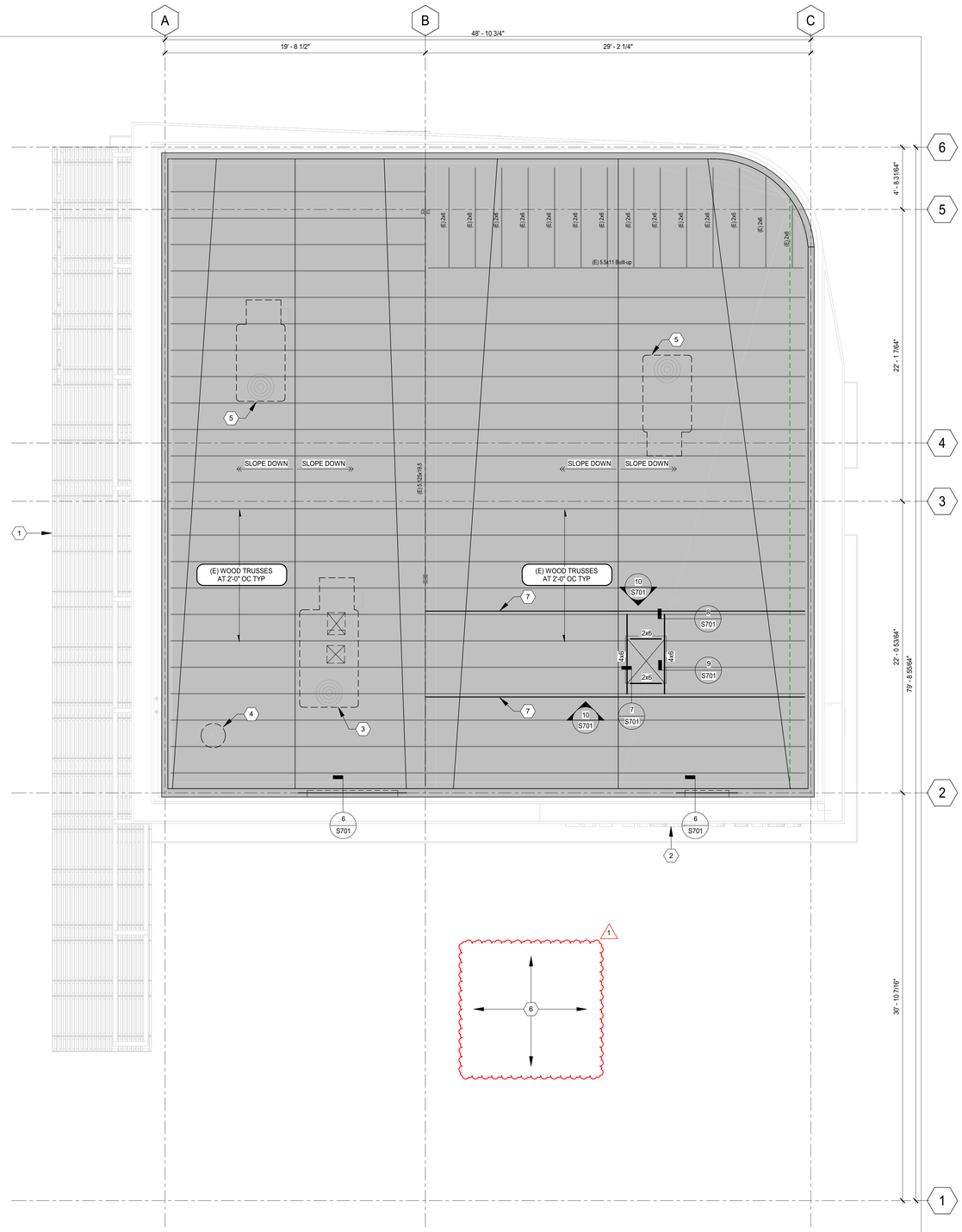
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FOUNDATION PLAN

S201

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22



1 ROOF FRAMING PLAN
1/4" = 1'-0"

NORTH

REFERENCE NOTES:

1. SEE SHEET S001 FOR SHEET INDEX, TYPICAL ABBREVIATIONS AND LEGENDS.
2. SEE SHEET S002 FOR GENERAL STRUCTURAL NOTES.
3. SEE SHEET S003 FOR STRUCTURAL SPECIAL INSPECTIONS.
4. SEE SHEET S500 SERIES FOR FOUNDATION AND CONCRETE FRAMING DETAILS.
5. SEE SHEET S700 SERIES FOR FRAMING DETAILS.

PLAN NOTES (UNLESS NOTED OTHERWISE):

1. TOP OF SHEATHING ELEVATION (TSE) = VARIES. REFER TO ARCHITECTURAL DRAWINGS FOR FINISHED ROOF ELEVATION.
2. ROOF SHEATHING CONSIST OF 5/8" APA RATED PLYWOOD/OSB, GENERAL CONTRACTOR TO VERIFY.
3. VERIFY SIZE, LOCATION AND NUMBER OF ALL OPENINGS THROUGH ROOF WITH ARCHITECTURAL, MECHANICAL AND ELECTRICAL DRAWINGS. SEE DETAILS FOR REQUIRED FRAMING AT OPENINGS.
4. VERIFY LOCATION, QUANTITY, SIZE AND OPERATING WEIGHT OF ROOFTOP UNITS WITH ARCHITECTURAL AND MECHANICAL DRAWINGS.

KEYNOTES:

1. NEW STEEL TRELLIS IN FUTURE EXTERIOR PHASE.
2. NEW WALL SIGNAGE SUPPORT AND CONNECTIONS IN FUTURE EXTERIOR PHASE.
3. ROOFTOP MECHANICAL UNIT. FOR EXACT SIZE AND LOCATION - SEE MECHANICAL AND ARCHITECTURAL DRAWINGS. STRUCTURAL DESIGN WEIGHT = 400 LBS
4. EXHAUST FAN. FOR EXACT SIZE AND LOCATION - SEE MECHANICAL AND ARCHITECTURAL DRAWINGS. STRUCTURAL DESIGN WEIGHT = 50 LBS
5. EXISTING ROOFTOP MECHANICAL UNIT TO REMAIN.
6. EXISTING BUILDING SOUTH OF GRID LINE 2 NOT WITHIN STRUCTURAL SCOPE OF INTERIOR PHASE. SEE ARCHITECTURAL DRAWINGS FOR LOCATION AND EXTENTS.
7. REINFORCE EXISTING WOOD TRUSS - SEE DETAIL 10/ S701 .

RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099 fax
Tempe, AZ 85281-3714 www.rsparch.com

Consultants
MEYER | BORGMAN | JOHNSON
STRUCTURAL DESIGN + ENGINEERING
TEL: 480.747.6727 www.mbjeng.com
116 EAST UNIVERSITY DRIVE SUITE 300 TEMPE, AZ 85281
SEE STRUCTURE.

Certification

Registered Professional Engineer Certificate No. 24045 KURT F. KINDERMANN Date Signed 11/23/22 ARIZONA, U.S.A.

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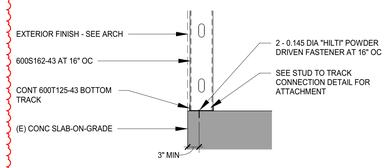
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ROOF FRAMING PLAN

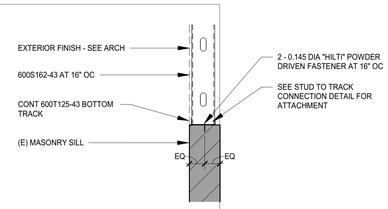
S202

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22



NOTES:
1. 3/4" MIN EMBED REQUIRED FOR ALL FASTENERS.
2. MAINTAIN 4" MIN SPACING FOR ALL FASTENERS.

3 / S501 BASE DETAIL AT CONCRETE CONSTRUCTION
NO SCALE



NOTES:
1. 1" MIN EMBED REQUIRED FOR ALL FASTENERS.
2. MAINTAIN 4" MIN SPACING FOR ALL FASTENERS.

4 / S501 BASE DETAIL AT WINDOW SILL
3/4" = 1'-0"

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants
MEYER | BORGMAN | JOHNSON
STRUCTURAL DESIGN + ENGINEERING
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116 EAST UNIVERSITY DRIVE
SUITE 300
TEMPE, AZ 85281
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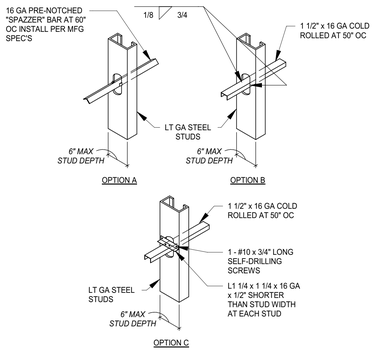
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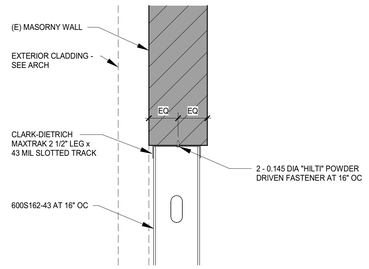
FOUNDATION DETAILS

S501

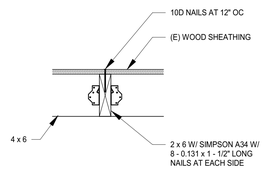
City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22



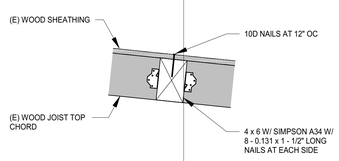
5 S701 BRIDGING ATTACHMENT
NO SCALE



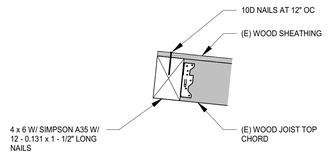
6 S701 STUD WALL TO MASONRY WALL CONNECTION
1 1/2" = 1'-0"



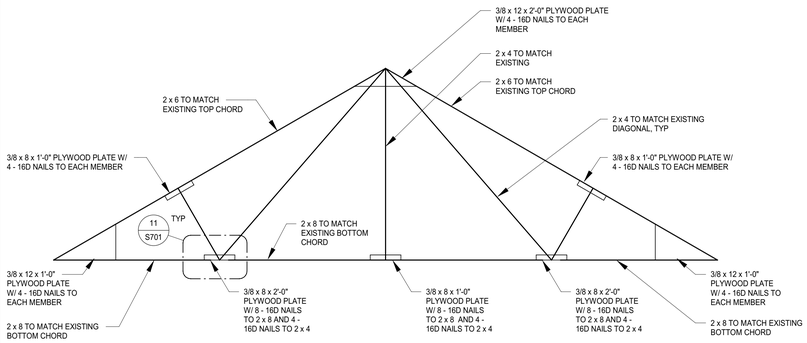
7 S701 WOOD BEAM TO WOOD BEAM CONNECTION
1 1/2" = 1'-0"



8 S701 WOOD BEAM TO (E) WOOD TRUSS CONNECTION
1 1/2" = 1'-0"

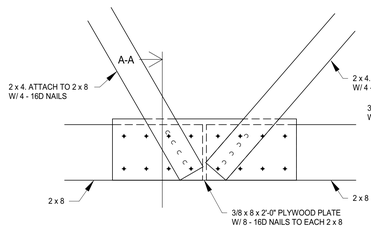


9 S701 (E) WOOD TRUSS TO WOOD BEAM CONNECTION
1 1/2" = 1'-0"

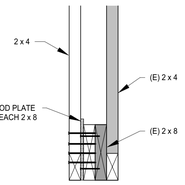


NOTES:
1. NAIL ALL TOP AND BOTTOM CHORD REINFORCING MEMBERS TO EXISTING CHORD WITH 16D NAILS AT 6" OC - STAGGERED.
2. NAIL ALL DIAGONAL WEB REINFORCING MEMBERS TO EXISTING MEMBER WITH 16D NAILS AT 6" OC - STAGGERED.

10 S701 (E) WOOD TRUSS STRENGTHENING
NO SCALE



11 S701 (E) PLYWOOD PLATE AT BOTTOM CHORD SPLICE
1 1/2" = 1'-0"



SECTION A-A

RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099 fax
Tempe, AZ 85281-3714 www.rsparch.com

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STRUCTURAL DESIGN + ENGINEERING
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FRAMING DETAILS

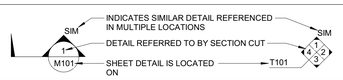
S701

VIEW KEY

NAME — LEVEL NAME
10'-0" — HEIGHT ABOVE PROJECT 0'-0"

1 — INDICATES NOTE USED TO DESCRIBE ADDITIONAL INFORMATION ABOUT WORK REQUIRED, SPECIFIC TO THE SHEET AND/OR DETAIL





LINE TYPE AND TAG KEY:

NEW WORK BY THIS CONTRACTOR (WIDE LINE)
——— NEW
- - - - - - - - - EXISTING TO BE REMOVED (SHORT DASHED PATTERN)
— — — — NEW UNDERFLOOR OR UNDERGROUND (LONG DASHED PATTERN)

EXISTING TO REMAIN OR WORK BY OTHERS (NARROW LINE)
——— EXISTING
- - - - - - - - - EXISTING TO BE REMOVED BY OTHERS (SHORT DASHED PATTERN)
— — — — EXISTING UNDERFLOOR OR UNDERGROUND (LONG DASHED PATTERN)

HALFTONING DOES NOT MODIFY SCOPE.

'TAG'-E — TAGS WITH DASH 'E' INDICATES THE REFERENCED OBJECT IS EXISTING

TAG-1 — UNDERLINED TAG INDICATES OBJECT IS IN-SCOPE. IF NEW, ADDITIONAL INFORMATION IS AVAILABLE IN A SCHEDULE, MATERIAL LIST, OR SYMBOL LIST

INDICATES AN EXISTING SYSTEM'S POINT OF CONNECTION/REMOVAL

CONTRACTOR ABBREVIATION KEY

ABBR:	DESCRIPTION:
E.C.	ELECTRICAL CONTRACTOR
G.C.	GENERAL CONTRACTOR
M.C.	MECHANICAL CONTRACTOR
P.C.	PLUMBING CONTRACTOR
T.C.	TECHNOLOGY CONTRACTOR

MECHANICAL SYMBOL LIST

NOT ALL SYMBOLS MAY APPLY.

SYMBOL:	DESCRIPTION:
BD	BOILER BLOW DOWN
BF	BOILER FEED WATER
CA	COMPRESSED AIR
CBR	CHILLED BEAM RETURN
CBS	CHILLED BEAM SUPPLY
CR	CONDENSER WATER RETURN
CS	CONDENSER WATER SUPPLY
CS15	CLEAN STEAM - NUMBER INDICATES PRESSURE IN PSIG.
CWR	CHILLED WATER RETURN
CWS	CHILLED WATER SUPPLY
DPP	DRAIN
FOR	FUEL OIL RETURN
FOS	FUEL OIL SUPPLY
G	NATURAL GAS
GV	GAS REGULATOR VENT
GWR	GLYCOL WATER RETURN
GWS	GLYCOL WATER SUPPLY
HCR	HEATING/CHILLED WATER RETURN
HCS	HEATING/CHILLED WATER SUPPLY
HG	REFRIGERANT HOT GAS
HPC	HIGH PRESSURE CONDENSATE
HPS	HIGH PRESSURE STEAM
HWR	HEATING WATER RETURN
HWS	HEATING WATER SUPPLY
LCS	LOW PRESSURE CLEAN STEAM
LIQ	REFRIGERANT LIQUID
LPC	LOW PRESSURE CONDENSATE
LPS	LOW PRESSURE STEAM
LWR	LOOP WATER RETURN
LWS	LOOP WATER SUPPLY
MV	MEDICAL VACUUM
PC	PUMPED CONDENSATE
PD	PUMPED DISCHARGE
RCR	RADIANT COOLING RETURN
RCS	RADIANT COOLING SUPPLY
RWR	REHEAT WATER RETURN
RWS	REHEAT WATER SUPPLY
SUC	REFRIGERANT SUCTION
SV	SAFETY RELIEF VENT
VAC	LAB VACUUM
	PIPE CAP
	PIPE DOWN
	PIPE UP OR UP/DOWN
	PITCH PIPE IN DIRECTION
	DIRECTION OF FLOW IN PIPE
	DIELECTRIC CONNECTION
	UNION/FLANGE
	SHUTOFF VALVE NORMALLY OPEN
	SHUTOFF VALVE NORMALLY CLOSED
	THROTTLING VALVE
	BALANCING VALVE (NUMBER INDICATES GPM)
	AUTOMATIC BALANCING VALVE
	MIXING VALVE
	CONTROL VALVE (THREE-WAY)
	CONTROL VALVE (TWO-WAY)
	SOLENOID VALVE
	CHECK VALVE
	BACKFLOW PREVENTER
	SAFETY/RELIEF VALVE
	PRESSURE REDUCING VALVE (LIQUID/GAS)
	PRESSURE REDUCING VALVE (STEAM)
	TRIPLE DUTY VALVE (ANGLE TYPE)
	TRIPLE DUTY VALVE (IN-LINE TYPE)
	PUMP
	VACUUM BREAKER
	"WYE" - STRAINER
	"WYE" - STRAINER W/SHUTOFF VALVE AND HOSE CONNECTION WITH CAP
	BASKET STRAINER
	FLEXIBLE CONNECTION
	PRESSURE/TEMPERATURE TEST PLUG
	REDUCER - REFERENCE SPECIFICATION FOR CONCENTRIC/ECCENTRIC AND FOT/FOB
	SUCTION DIFFUSER WITH SUPPORT FOOT
	AUTOMATIC AIR VENT
	MANUAL AIR VENT
	DRAIN VALVE WITH HOSE CONNECTION AND CAP
P	PRESSURE SENSOR (FURNISHED WITH BALL VALVE)
P	PRESSURE GAUGE (FURNISHED WITH BALL VALVE)
	DIFFERENTIAL PRESSURE SENSOR
SP	STATIC SWITCH
FM	FLOW METER
F	FLOW SWITCH
FS	FLOW SENSOR
T T-*	STEAM TRAP (REFER TO SCHEDULE)
T-*	F&T STEAM TRAP (REFER TO SCHEDULE)
T-*	INVERTED BUCKET STEAM TRAP (REFER TO SCHEDULE)
	ALIGNMENT GUIDE
	PIPE ANCHOR
EJ-# (#.#")	EXPANSION JOINT #.#" IS THE EXPANSION TRAVEL INCHES
M	METER

MECHANICAL SYMBOL LIST

NOT ALL SYMBOLS MAY APPLY.

SYMBOL:	DESCRIPTION:
	DIRECTION OF AIR FLOW
	FLEXIBLE DUCT
	MANUAL VOLUME DAMPER
R	RISE IN DIRECTION OF AIR FLOW
D	DROP IN DIRECTION OF AIR FLOW
	DUCT CAP
	DUCT DOWN
	DUCT UP
	SUPPLY/OUTSIDE AIR DUCT SECTION
	RETURN AIR DUCT SECTION
	EXHAUST/RELIEF AIR DUCT SECTION
	4-WAY DIFFUSER WITH BLANKOFF IN ONE DIRECTION
SD-1 6/115	AIR TERMINAL PROPERTIES SYMBOL NECK SIZE/CFM
###	TERMINAL AIR BOX (REFER TO SCHEDULE)
###	TERMINAL AIR BOX w/REHEAT COIL (REFER TO SCHEDULE)
	FAN POWERED TERMINAL AIR BOX w/REHEAT COIL (REFER TO SCHEDULE)
H	HUMIDIFIER
	OPPOSED BLADE DAMPER (REFER TO SCHEDULE)
	PARALLEL BLADE DAMPER (REFER TO SCHEDULE)
	DIFFERENTIAL PRESSURE SENSOR
H	HUMIDISTAT SENSOR
	HUMIDISTAT / SENSOR
C	CARBON MONOXIDE SENSOR
C_2	CARBON DIOXIDE SENSOR
O	OCCUPANCY SENSOR
P	PRESSURE SENSOR/MONITOR
P	PRESSURE SENSOR (DUCT MOUNTED)
T	THERMOSTAT/SENSOR
T	TEMPERATURE SENSOR
T	THERMOSTAT/SENSOR WITH HEAVY DUTY ENCLOSURE
T	TEMPERATURE SENSOR WITH WELL
T	THERMOMETER WITH WELL (DIAL TYPE)
	THERMOMETER WITH WELL (FILLED TYPE)
XX-Y	AIRFLOW MEASUREMENT SYMBOL XX - AHU SYMBOL Y - SEQUENTIAL NUMBER

MECHANICAL RENOVATION NOTES:

THESE NOTES APPLY TO ALL MECHANICAL SHEETS AND TRADES, INCLUDING BUT NOT LIMITED TO, FIRE PROTECTION, PLUMBING, MEDICAL GAS, VENTILATION, PIPING AND TEMPERATURE CONTROL.

1. EXISTING CONDITIONS ARE SHOWN BASED ON INFORMATION OBTAINED FROM FIELD SURVEYS, EXISTING BUILDING DOCUMENTS, AND STAFF. VERIFY EXISTING CONDITIONS AND REPORT ANY CONFLICTS BEFORE PROCEEDING.
2. NOT ALL EXISTING DUCTWORK AND PIPING IS SHOWN. VERIFY EXISTING CONDITIONS BEFORE STARTING WORK. NOTIFY ENGINEER OF ANY CONFLICTS WITH NEW WORK.
3. FIELD VERIFY THE AVAILABLE CLEARANCES FOR DUCTWORK AND PIPING BEFORE FABRICATION. RISES AND DROPS MAY BE NECESSARY BECAUSE OF EXISTING FIELD CONDITIONS.
4. EACH CONTRACTOR SHALL FIELD VERIFY ACCESSIBILITY TO THE AREA OF THEIR WORK AND SHALL NOTIFY THE ARCHITECT/ENGINEER PRIOR TO BIDDING IF OTHER UTILITIES ARE REQUIRED TO BE REMOVED OR RELOCATED TO ALLOW ACCESS TO THEIR AREA OF WORK.
5. THE GENERAL CONTRACTOR IS RESPONSIBLE FOR CUTTING, REMOVAL AND PATCHING OF ROOFS, WALLS, AND FLOORS ASSOCIATED WITH WORK BY ALL CONTRACTORS. CONTRACTORS SHALL NOTIFY THE GC OF AFFECTED AREAS PRIOR TO BIDDING] [EACH CONTRACTOR SHALL CUT AND PATCH ROOFS, WALLS, AND FLOORS ASSOCIATED WITH THEIR WORK.]
6. THE GENERAL CONTRACTOR IS RESPONSIBLE FOR REMOVAL AND REPLACEMENT OF CEILINGS, CEILING TILES, AND CEILING GRIDS ASSOCIATED WITH AREAS OF WORK BY ALL CONTRACTORS. NOTIFY THE GENERAL CONTRACTOR OF AFFECTED AREAS PRIOR TO BIDDING.
7. WHERE EXISTING MECHANICAL SYSTEMS ARE LOCATED IN AREAS THAT CONFLICT WITH NEW EQUIPMENT, PIPING, OR DUCTWORK TO BE INSTALLED, EACH CONTRACTOR SHALL EITHER ARRANGE NEW EQUIPMENT, PIPING, OR DUCTWORK IN SUCH A FASHION THAT IT DOES NOT CONFLICT WITH EXISTING SYSTEMS, OR REWORK EXISTING MECHANICAL SYSTEMS TO ALLOW FOR INSTALLATION OF NEW EQUIPMENT, PIPING, OR DUCTWORK.
8. PROVIDE TEMPORARY CONNECTIONS TO MAINTAIN EXISTING SYSTEMS IN SERVICE DURING CONSTRUCTION. MAINTAIN ACCESS TO EXISTING MECHANICAL INSTALLATIONS THAT REMAIN ACTIVE.
9. OBTAIN PERMISSION FROM OWNER BEFORE SHUTTING DOWN ANY SYSTEM FOR ANY REASON. MAINTAIN SERVICE TO ALL COMPONENTS THAT ARE TO REMAIN UNTIL NEW SYSTEMS ARE INSTALLED.
10. MAINTAIN EXISTING SYSTEM IN SERVICE UNTIL NEW SYSTEM IS COMPLETE AND READY FOR TIE IN AND SWITCHOVER. DRAIN SYSTEM ONLY TO MAKE SWITCHOVERS AND CONNECTIONS. OBTAIN PERMISSION FROM OWNER BEFORE PARTIALLY OR COMPLETELY DRAINING SYSTEM. MAKE CHANGEOVER TO NEW SYSTEMS WITH MINIMUM OUTAGE.

MECHANICAL ABBREVIATION KEY

ABBR:	DESCRIPTION:
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
C	COMMON
CO	CLEANOUT
CFSD	CONTROL/FIRE/SMOKE DAMPER
DPG (0-2")	DIFFERENTIAL PRESSURE GAUGE (RANGE)
DPS	DIFFERENTIAL PRESSURE SWITCH
EA	EXHAUST/RELIEF AIR
ECFSD	EXISTING CONTROL FIRE SMOKE DAMPER
EFD	EXISTING FIRE DAMPER
EFSD	EXISTING FIRE SMOKE DAMPER
EP	ELECTRICAL TO PNEUMATIC VALVE
ESD	EXISTING SMOKE DAMPER
FD	FIRE DAMPER
FOB	FLAT ON BOTTOM
FOT	FLAT ON TOP
FSD	FIRE/SMOKE DAMPER
MA	MIXED AIR
MV	MIXING VALVE
N.C.	NORMALLY CLOSED
NIC	NOT IN CONTRACT
N.O.	NORMALLY OPEN
OA	OUTSIDE AIR
PS	PRESSURE SWITCH
RA	RETURN AIR
SA	SUPPLY AIR
SCCR	SHORT CIRCUIT CURRENT RATING
SD	SMOKE DAMPER
TAB	TERMINAL AIR BOX
TD	TRANSFER DUCT
TYP	TYPICAL
UC-1	DOOR UNDERCUT BY OTHERS (1" TYPICAL)
UON	UNLESS OTHERWISE NOTES

TAB POST-CONSTRUCTION NOTES:

1. AFTER CONSTRUCTION ACTIVITIES ARE COMPLETE, TESTING, ADJUSTING (TAB) AND BALANCING CONTRACTOR SHALL REBALANCE AIR HANDLING UNITS AND EXHAUST FANS AS REQUIRED TO ACHIEVE THE NEW AIRFLOW VALUES SHOWN ON THE CONSTRUCTION DRAWINGS.
2. AREAS SERVED BY THIS EQUIPMENT WHICH WERE NOT RENOVATED SHALL BE RE-BALANCED TO THE AIRFLOW RATES MEASURED BEFORE THE RENOVATION OCCURRED (REFER TO THE FINAL PRE- DEMOLITION REPORT).
3. TAB CONTRACTOR SHALL COMPILE AND SUBMIT COPIES OF THE FINAL POST-CONSTRUCTION TAB REPORT AS REQUIRED BY SECTION 23 05 93.
4. THE FINAL POST CONSTRUCTION REPORT SHALL INCLUDE ALL ITEMS REQUIRED IN THE SPECIFICATIONS.

PIPING GENERAL NOTES:

1. THE SIZE OF BRANCH PIPING TO TERMINAL HEATING DEVICES AND COILS SHALL BE 3/4" UNLESS NOTED OTHERWISE.
2. PIPE DRAIN LINES FROM EQUIPMENT TO NEAREST FLOOR DRAIN.
3. INSTALL ALL REFRIGERANT LIQUID AND SUCTION PIPING SIZED PER EQUIPMENT MANUFACTURER RECOMMENDATIONS.

VENTILATION GENERAL NOTES:

1. UNLESS NOTED OTHERWISE, THE SIZE OF EACH BRANCH DUCT TO A TERMINAL AIR BOX (TAB) SHALL MATCH THE INLET SIZE UNLESS THE BRANCH IS GREATER THAN 6FEET IN LENGTH, IN WHICH CASE THE BRANCH DUCT SHALL BE SIZED AT A PRESSURE DROP OF 0.07"W.C. PER 100' OF DUCTWORK
2. UNLESS NOTED OTHERWISE, THE SIZE OF EACH BRANCH DUCT TO AN AIR TERMINAL SHALL MATCH THE INLET SIZE.
3. ALIGN TEMPERATURE SENSORS WITH LIGHT SWITCHES AND WHEN IN CLOSE PROXIMITY TO EACH OTHER.
4. PROVIDE ACCESS DOORS AT ALL DUCT MOUNTED EQUIPMENT.
5. EXISTING AIR INLET AND OUTLET CFM SHOWN ON DRAWINGS ARE FROM EXISTING DRAWINGS, AND ARE FOR REFERENCE ONLY. CONTRACTOR SHALL USE PRE-BALANCE VALUES, AND NOT EXISTING CFM SHOWN ON DRAWINGS.
6. CONTRACTOR MAY REUSE PORTIONS OF EXISTING DUCT PROVIDED SIZES AND PRESSURE CLASSES ARE CORRECT, DUCT IS THOROUGHLY CLEANED AND FREE OF DEFECTS, AND ALL TRANSVERSE JOINTS, LONGITUDINAL SEAMS, AND DUCT WALL PENETRATIONS ARE SEALED AS SPECIFIED FOR NEW DUCTWORK.

MECHANICAL GENERAL NOTES:

THESE NOTES APPLY TO ALL MECHANICAL SHEETS AND TRADES, INCLUDING BUT NOT LIMITED TO, FIRE PROTECTION, PLUMBING, MEDICAL GAS, VENTILATION, PIPING AND TEMPERATURE CONTROL.

1. DRAWINGS SHOWING LOCATIONS OF EQUIPMENT, DUCTWORK, PIPING, ETC. ARE DIAGRAMMATIC AND MAY NOT ALWAYS REFLECT EXACT INSTALLATION CONDITIONS. DRAWINGS SHOW THE GENERAL ARRANGEMENT OF DUCTWORK, PIPING, EQUIPMENT, ETC., AND MAY NOT INCLUDE ALL OFFSETS AND FITTINGS REQUIRED FOR COMPLETE INSTALLATION. THE DRAWINGS SHALL BE FOLLOWED AS CLOSELY AS ACTUAL BUILDING CONSTRUCTION AND THE WORK OF OTHERS WILL PERMIT.
2. DO NOT SCALE DRAWINGS. VERIFY ALL DIMENSIONS AND CLEARANCES FROM ARCHITECTURAL, STRUCTURAL, SUBMITTALS, AND OTHER APPROPRIATE DRAWINGS OR PHYSICALLY AT SITE. REVIEW ALL DRAWINGS, INCLUDING THOSE OF OTHER TRADES.
3. COORDINATE ALL WORK WITH ALL OTHER TRADES PRIOR TO INSTALLATION TO PROVIDE CLEARANCES REQUIRED FOR OPERATION, MAINTENANCE, CODE COMPLIANCE, AND TO VERIFY NON-INTERFERENCE WITH OTHER WORK. DO NOT FABRICATE PRIOR TO VERIFICATION OF NECESSARY CLEARANCES FOR ALL TRADES. BRING ANY INTERFERENCES OR CONFLICTS TO THE ATTENTION OF THE ARCHITECT/ENGINEER BEFORE PROCEEDING WITH FABRICATION OR EQUIPMENT ORDERS.
4. REVIEW SPACE REQUIREMENTS OF EQUIPMENT SPECIFIED OR SUBSTITUTED AND MAKE REASONABLE ACCOMMODATIONS IN LAYOUT AND POSITIONING TO PROVIDE PROPER ACCESS.
5. ANY CHANGES REQUIRED TO ELIMINATE CONFLICTS OR THAT RESULT FROM A FAILURE TO COORDINATE SHALL BE MADE BY THE CONTRACTOR WITHOUT ADDITIONAL COST OR EXPENSE TO OTHERS.
6. EACH CONTRACTOR IS RESPONSIBLE FOR ALL COSTS ASSOCIATED WITH ELECTRICAL CHANGES REQUIRED FOR EQUIPMENT PROPOSED THAT DIFFERS FROM THE BASIS OF DESIGN.
7. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN, ELECTRICAL, TECHNOLOGY AUDIO/VISUAL, AND OTHER MECHANICAL PLANS FOR EXACT LOCATIONS OF ALL CEILING MOUNTED DEVICES, OTHER THAN SPRINKLERS.
8. EACH CONTRACTOR IS RESPONSIBLE FOR DAMAGE CAUSED BY THEIR ACTIONS TO WALLS, FLOORS, CEILINGS, AND ROOFS. THE CONTRACTOR WHOSE WORK CAUSES DAMAGE IS RESPONSIBLE FOR PATCHING TO MATCH ORIGINAL CONSTRUCTION, FIRE RATING, AND FINISH.
9. IN AREAS WITH DRYWALL CEILINGS COORDINATE LOCATIONS OF ACCESS PANELS WITH THE GC FOR ACCESS TO VALVES, DUCTWORK ACCESSORIES, DAMPERS, ETC. COORDINATE PANEL TYPE AND COLOR WITH ARCHITECT. NOTIFY THE GC OF THE REQUIRED ACCESS PANELS PRIOR TO BIDDING.
10. SEAL ALL FLOOR, WALL, AND ROOF PENETRATIONS AIRTIGHT WHERE CONDUITS, PIPING, AND DUCTS PENETRATE. PENETRATIONS THROUGH EXTERIOR WALLS AND ROOF SHALL BE SEALED AIRTIGHT WITH WATERPROOFING MATERIALS RECOMMENDED BY MANUFACTURER FOR OUTDOOR USE.
11. CAULK ALL PIPE AND DUCT PENETRATIONS OF FULL HEIGHT NON-FIRE RATED WALL, PARTITION, FLOOR, AND ROOF ASSEMBLIES. THIS IS ESSENTIAL TO PREVENT NOISE TRANSMISSION FROM ONE ROOM TO ANOTHER AND TO PROVIDE THE DESIRED NC LEVELS WITHIN ROOMS.
12. WHERE PIPES AND DUCTS ARE SHOWN TO PENETRATE FLOORS, PROVIDE SLEEVED OPENINGS WITH THE TOP EDGE RAISED ABOVE FLOOR SURFACE IN ACCORDANCE WITH ALL RELEVANT SPEC SECTIONS. SEAL SLEEVE PERIMETER TO BE WATERTIGHT.
13. EQUIPMENT SIZES AND SERVICE CLEARANCE REQUIREMENTS VARY AMONG DIFFERENT MANUFACTURERS. CONSULT APPROVED SHOP DRAWINGS FOR EQUIPMENT SIZES AND REQUIRED SERVICE CLEARANCES. COORDINATE WITH LAYOUT OF EQUIPMENT PADS, PIPING, DUCTWORK, ETC.
14. DO NOT BLOCK TUBE PULL OR EQUIPMENT SERVICE CLEARANCES.
15. MAINTAIN A MINIMUM WORKING CLEARANCE OF 3'-6" IN FRONT OF ALL ELECTRICAL EQUIPMENT REQUIRING MAINTENANCE, INSPECTION, AND TESTING INCLUDING BUT NOT LIMITED TO PANELS, DISTRIBUTION PANELS, SWITCHBOARDS, MOTOR CONTROL CENTERS, TRANSFORMERS, EQUIPMENT DISCONNECTS AND STARTERS.
16. MAINTAIN THE DEDICATED ELECTRICAL EQUIPMENT SPACE DEFINED BY THE WIDTH / DEPTH OF ELECTRICAL EQUIPMENT MEASURED FROM THE FLOOR TO A HEIGHT 6'-0" ABOVE THE EQUIPMENT OR THE STRUCTURAL CEILING, WHICHEVER IS LOWER. SYSTEMS FOREIGN TO THE ELECTRICAL DISTRIBUTION SYSTEM ARE NOT ALLOWED IN THE DEDICATED ELECTRICAL SPACE INCLUDING; DUCTWORK, PIPING, ETC.
17. PROVIDE CONCRETE EQUIPMENT PAD FOR ALL FLOOR MOUNTED EQUIPMENT. PAD SHALL EXTEND MINIMUM 6" BEYOND ALL SIDES OF EQUIPMENT.
18. DO NOT SUPPORT EQUIPMENT, PIPING, OR DUCTWORK FROM METAL DECKING OR OTHER NON-STRUCTURAL BUILDING ELEMENTS. ANCHORS EMBEDDED IN CONCRETE SHALL BE CRACKED CONCRETE APPROVED IN ACCORDANCE WITH SPECIFICATIONS.

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M500	MECHANICAL SCHEDULES
M001	MECHANICAL SPECIFICATIONS
GRAND TOTAL: 6	

IMEG
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

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REF. SCALE IN INCHES

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.co

Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008



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Sheet Issues / Revisions

No.	Date	Description

MECHANICAL COVERSHEET

M000

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

MECHANICAL SPECIFICATIONS

PART ONE - GENERAL

1. CODE USED IN DESIGN: IBC 2018, IMC 2018, IPC-2018, UMC-2018, UPC-2018, IECC-2018, IFGC 2018
2. THE OWNER HAS CONTRACT LANGUAGE THAT NEEDS TO BE READ PRIOR TO BID SUBMISSION AS THERE ARE ITEMS THAT MAY SUPPLEMENT OR SUPERSEDE ITEMS NOTED HEREIN. THE OWNER'S CONTRACT DOCUMENTS HAS INFORMATION ON HOW WORK IS TO BE PERFORMED, HOW DOCUMENT SUBMITTALS ARE PROVIDED, RECORD DOCUMENTS ARE SUBMITTED, ETC. SEE THE ARCHITECTURAL DOCUMENTS FOR ADDITIONAL DIVISION 1 INFORMATION.
3. ALL WORK SHALL BE DONE IN ACCORDANCE WITH THE LATEST ADOPTED EDITIONS OF THE APPLICABLE INTERNATIONAL BUILDING CODE (IBC), LOCAL MECHANICAL CODE (UMC, IMC, ETC.), LOCAL PLUMBING CODE (UPC, IPC, ETC.), NATIONAL ELECTRIC CODES (NEC) AND ALL OTHER APPLICABLE FEDERAL, STATE, AND LOCAL REGULATIONS.
4. THE CONTRACTOR MUST ARRANGE A VISIT TO THE WORK/SITE PRIOR TO BID SUBMISSION TO FULLY UNDERSTAND THE EXISTING CONDITIONS. THE DRAWINGS ARE DIAGRAMMATIC AND SHOW THE WORK INTENT BUT NOT NECESSARILY ALL EXISTING OBSTRUCTIONS, PIPE OR DUCT BENDS. DETERMINING SITE CONDITIONS AND ADJUSTING THE INSTALLATION IS THE RESPONSIBILITY OF THE CONTRACTOR.
5. THE CONTRACTOR SHALL PROVIDE THE WORK SHOWN ON THE DRAWINGS AND SPECIFIED FOR THEIR INDIVIDUAL SECTIONS OF WORK. THE WORD "WORK" SHALL MEAN ALL LABOR, TRANSPORTATION, MATERIAL, EQUIPMENT, TOOLS, INSTALLATION, SUPERVISION AND ANY OTHER INCIDENTAL ITEMS OR SERVICES NECESSARY FOR THE PROPER INSTALLATION AND OPERATION OF THE COMPLETE SYSTEMS, WHICH SHALL BE PROVIDED WHETHER OR NOT SPECIFICALLY INDICATED OR NOTED.
6. ALL GENERAL CONDITIONS, SPECIAL REQUIREMENTS OR GENERAL REQUIREMENTS OF THE CONSTRUCTION SPECIFICATIONS ARE MADE PART OF THIS SPECIFICATION AND HAVE THE SAME FORCE AND EFFECT AS IF COMPLETELY REPRODUCED.
7. THE WORD "PROVIDE" SHALL MEAN FURNISH AND INSTALL, MAKE ALL FINAL CONNECTIONS AND LEAVE IN AN APPROVED COMPLETE OPERATING CONDITION.
8. THE CONTRACTOR SHALL BE RESPONSIBLE FOR PAYING ALL FEES AND OBTAINING ALL PERMITS AND INSPECTIONS REQUIRED FOR THE WORK.
9. THE CONTRACTOR SHALL CAREFULLY EXAMINE ALL CONTRACT DOCUMENTS. THE CONTRACTOR SHALL COORDINATE THE WORK WITH ALL OTHER TRADES INCLUDING, BUT NOT LIMITED TO, THE CONTRACT DOCUMENTS, SHOP DRAWINGS, ETC. FOR ALL GENERAL CONSTRUCTION, STRUCTURAL, MECHANICAL, ELECTRICAL AND SPECIALTY CONTRACTOR WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER FITTING OF MATERIAL INTO THE BUILDING AS PLANNED, WITHOUT INTERFERENCE WITH OTHER WORK, AND SHALL MAKE REASONABLE MODIFICATIONS IN THE LAYOUTS NEEDED TO PREVENT CONFLICT WITH OTHER TRADES, TO PROVIDE ACCESS AND FOR THE PROPER EXECUTION OF THE WORK.
10. DRAWINGS ARE DIAGRAMMATIC AND SCHEMATIC IN NATURE, AND INDICATE THE TYPE, SIZE, ARRANGEMENT AND LOCATION OF MATERIALS AND EQUIPMENT. WORK INCLUDES CERTAIN COMPONENTS, APPURTENANCES AND RELATED SPECIALTIES THAT MAY NOT BE SHOWN. CONTRACTOR SHALL PROVIDE ALL NECESSARY ITEMS TO COMPLETE THE WORK ACCORDING TO INDUSTRY STANDARDS. IT IS THE INTENT OF THE DRAWINGS AND SPECIFICATIONS TO CALL OUT FOR FINISHED WORK, TESTED AND READY FOR OPERATION. DO NOT SCALE DRAWINGS. ARRANGEMENT OF EQUIPMENT AND ROUTING OF PIPES AND DUCTWORK, ETC. INDICATED ON DRAWINGS SHALL BE ROUTED PLUMB AND AT RIGHT ANGLES TO BUILDING CONSTRUCTION AND MAY REQUIRE MODIFICATION DUE TO UNFORESEEN CONDITIONS AND REQUIRE ON SITE REVISIONS DURING CONSTRUCTION. (SEE ALSO "BIDDING"). WITH THE DRAWINGS BEING DIAGRAMMATIC THEY DO NOT ILLUSTRATE POTENTIAL WORKING PLATFORMS FOR EQUIPMENT SERVICE THAT MAY BE REQUIRED TO SATISFY OSHA COMPLIANCE - THESE PLATFORMS ARE TO BE PROVIDED BY THE CONTRACTOR.
11. ALL WORK REQUIRED FOR IDENTICAL/SIMILAR ITEMS SHOWN ON THE DRAWINGS SHALL BE PROVIDED, ALTHOUGH EACH SPECIFIC IDENTICAL/SIMILAR ITEM MAY NOT BE SHOWN IN DETAIL.
12. THE CONTRACTOR SHALL SUBMIT ELECTRONIC PDF SHOP DRAWINGS AND TECHNICAL DATA SHEETS FOR ALL EQUIPMENT AND MATERIALS SPECIFIED HEREIN TO THE ENGINEER. THE ENGINEER SHALL REVIEW SHOP DRAWINGS AND TECHNICAL DATA SHEETS FOR CONFORMANCE WITH THE CONTRACT DOCUMENTS AND ISSUE A WRITTEN ASSESSMENT TO THE OWNER PRIOR TO COMMENCEMENT OF WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL ENGINEERING FEES NECESSARY TO CHANGE PERMIT DOCUMENTS BASED ON ALTERNATE SUBMITTAL PACKAGES/EQUIPMENT SUBSTITUTIONS.
13. SPECIFIED EQUIPMENT SHALL BE CONSIDERED BASE BID. ANY APPROVED ALTERNATE MANUFACTURERS PRODUCT SHALL BE LISTED AS A FEE ADDITION/REDUCTION AS A SEPARATE LINE ITEM AT BID. A WRITTEN DESCRIPTION OF PRODUCT DIFFERENCES MUST BE PROVIDED FOR EVALUATION OR THE ALTERNATE PRODUCT WILL BE REJECTED. THE REQUIREMENTS OF PARA. 14.1 BELOW APPLIES TO PRE-APPROVED ALTERNATE MANUFACTURER EQUIPMENT. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL ENGINEERING FEES NECESSARY TO CHANGE PERMIT DOCUMENTS BASED ON ALTERNATE SUBMITTAL PACKAGES/EQUIPMENT SUBSTITUTIONS.
14. ALL SUBSTITUTIONS SHALL BE SUBMITTED TO THE ENGINEER FOR CONSIDERATION PRIOR TO BIDDING. THE PROPOSED SUBSTITUTE PRODUCT SHALL BE LISTED AS A FEE ADDITION/REDUCTION. A WRITTEN DESCRIPTION OF PRODUCT AND INSTALLATION DIFFERENCES MUST BE PROVIDED FOR EVALUATION OR THE SUBSTITUTE PRODUCT WILL BE REJECTED. THE OWNER'S REPRESENTATIVE SHALL PRE-APPROVE ANY PROPOSED SUBSTITUTION IN WRITING. IF APPROVED, THE PRODUCT FALLS UNDER THE RULES OF PARA. 13 ABOVE.
    14.1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING ALL REQUIREMENTS ASSOCIATED WITH SUBSTITUTED EQUIPMENT OR MATERIALS WITH OTHER BUILDING TRADES, INCLUDING, BUT NOT LIMITED TO, ELECTRICAL, STRUCTURAL, OR ARCHITECTURAL ELEMENTS. SUBSTITUTED EQUIPMENT, ANYTHING DIFFERENT FROM SPECIFIED ON THE DOCUMENTS, MUST BE IDENTIFIED AS SUCH DURING THE SUBMITTAL PROCESS. THE CONTRACTOR SHALL IDENTIFY AND ANNOTATE ALL REVISIONS REQUIREMENTS PER BUILDING TRADE ON THE SHOP DRAWINGS. THE CONTRACTOR SHALL ALSO IDENTIFY ALL COSTS, DEBITS OR CREDITS, IN WRITING FOR THE PROPOSED CHANGES PER BUILDING TRADE.
15. SHOP DRAWING REVIEW DOES NOT RELIEVE THE CONTRACTOR FROM BASE BID, ALTERNATE OR SUBSTITUTE EQUIPMENT COORDINATION REQUIREMENTS.
16. UPON COMPLETION OF CONSTRUCTION,
    16.1. THE CONTRACTOR SHALL SUPPLY THE ENGINEER WITH AN ELECTRONIC CAD AND PDF SET OF AS-BUILT DOCUMENTS ACCURATELY SHOWING THE MATERIALS AND EQUIPMENT AS INSTALLED.
    16.2. THE CONTRACTOR SHALL PROVIDE THE BUILDING OWNER OR REPRESENTATIVE WITH AN ELECTRONIC (PDF) MANUAL WITH DETAILED OPERATING AND MAINTENANCE INSTRUCTIONS FOR EACH PIECE OF EQUIPMENT PROVIDED, WITH CONTENT MEETING THE REQUIREMENTS NOTED BELOW:
        16.2.1. SUBMITTAL DATA STATING EQUIPMENT SIZE AND SELECTED OPTIONS.
        16.2.2. MANUFACTURER'S OPERATION MANUALS AND MAINTENANCE MANUALS. REQUIRED ROUTINE MAINTENANCE ACTIONS SHALL BE CLEARLY IDENTIFIED.
        16.2.3. NAME, ADDRESS AND CONTACT NUMBER FOR AT LEAST ONE SERVICE AGENCY.
        16.2.4. HVAC AND SERVICE HOT WATER CONTROLS SYSTEM MAINTENANCE AND CALIBRATION INFORMATION, INCLUDING WIRING DIAGRAMS, SCHEMATICS AND CONTROL SEQUENCE DESCRIPTIONS. DESIRED OR FIELD DETERMINED SET-POINTS SHALL BE PERMANENTLY RECORDED ON A CONTROLS DRAWING AT CONTROL DEVICES OR IN SYSTEM PROGRAMMING INSTRUCTIONS.
        16.2.5. A NARRATIVE OF HOW EACH SYSTEM IS INTENDED TO OPERATE, INCLUDING RECOMMENDED SET-POINTS.
        16.2.6. COPIES OF GUARANTIES AND/OR WARRANTIES.
17. ALL MATERIALS AND WORKMANSHIP SHALL BE GUARANTEED FOR A MINIMUM OF ONE (1) YEAR FROM DATE OF ACCEPTANCE BY OWNER. REFRIGERATION COMPRESSORS SHALL BE GUARANTEED FOR A MINIMUM OF FIVE (5) YEARS FROM DATE OF OWNER'S ACCEPTANCE. IN ADDITION, THE CONTRACTOR SHALL GUARANTEE THAT THE INSTALLATION WHEN OPERATED IN ACCORDANCE WITH THE CONTRACTOR'S INSTRUCTIONS WILL DEVELOP CAPACITY AND CHARACTERISTICS AS SPECIFIED AND WILL FULFILL EACH AND EVERY REQUIREMENT OF THE DRAWINGS AND SPECIFICATIONS. SHOULD THE INSTALLATION IN ANY WAY FAIL TO DO SO, THE CONTRACTOR WILL WITHOUT DELAY AND WITHOUT COST TO THE OWNER, PROVIDE WHATEVER ADDITIONAL EQUIPMENT, MATERIAL, AND LABOR REQUIRED TO CORRECT THE DEFICIENCY AND COMPLY WITH THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATIONS.
18. CONTRACTOR SHALL CHECK AND VERIFY ALL SIZES, DIMENSIONS, AND CONDITIONS BEFORE STARTING ANY WORK. ANY DEVIATIONS OR PROBLEMS SHALL BE TRANSMITTED TO THE ENGINEER FOR REVIEW.
19. PROVIDE BASE AND COUNTER FLASHING FOR ITEMS PENETRATING THE ROOF OR EXTERIOR WALLS.
20. STARTERS, VFDs, DISCONNECT SWITCHES AND CONTROLS FOR MOTORS IF NOT UNIT MOUNTED AND/OR SUPPLIED BY THE EQUIPMENT MANUFACTURER, UNLESS NOTED SPECIFICALLY OTHERWISE SHALL FOLLOW:
    20.1. VFDs TO BE SUPPLIED BY THE MECHANICAL CONTRACTOR AND INSTALLED BY THE ELECTRICAL CONTRACTOR. FINAL LOCATIONS COORDINATED WITH THE ENGINEER. WIRING BETWEEN THE VFD AND THE MOTOR SHALL BE SHIELDED POWER CABLE DESIGNED FOR VFD APPLICATIONS, GROUNDED AT BOTH ENDS.
    20.2. UNLESS NOTED OTHERWISE, LOOSE MOTOR STARTERS, COMBINATION STARTERS, DISCONNECT SWITCHES, MOTOR RATED SWITCHES, TOGGLE SWITCHES, ETC. TO BE SUPPLIED AND INSTALLED BY THE ELECTRICAL CONTRACTOR.
    20.3. CONTROL AND INTERLOCKING WIRING SHALL BE FURNISHED AND INSTALLED BY CONTRACTOR PERFORMING CONTROLS WORK. (SEE AUTOMATIC TEMPERATURE CONTROLS SECTION FOR ADDITIONAL INFORMATION WITH REGARD TO THIS WIRING RULE.)
21. ALL WORK SHOWN IS NEW UNLESS NOTED OTHERWISE.
22. MAINTAIN OCCUPANCY AND FIRE WALL SEPARATION INTEGRITY AS REQUIRED. REFER TO ARCHITECTURAL PLANS FOR LOCATIONS OF ALL OCCUPANCY/FIREWALL SEPARATIONS AND SPECIFIC DETAILS FOR CONSTRUCTION. PROVIDE ALL NECESSARY FIRE AND SMOKE FIRE DAMPERS, ACCESS DOORS, CAULKING, ETC. FOR APPROVED INSTALLATION.
23. IECC COMPLIANCE: THE CONTRACTOR IS RESPONSIBLE FOR COMPLYING WITH AND PERFORMING ALL REQUIREMENTS AND WORK SET FORTH IN THE IECC COMPLIANCE CERTIFICATE THAT IS INCLUDED IN THESE DOCUMENTS.

COORDINATION DRAWINGS

1. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR CREATING "ALL TRADES" COORDINATION DRAWINGS. THIS WORK APPLIES TO ABOVE CEILINGS, IN SHAFTS AND ON THE ROOF. THESE DRAWINGS ARE TO BE PREPARED IN 3D OR REVIT. 2D AUTOCAD DRAWINGS WILL BE AVAILABLE FROM THE DESIGN TEAM AND CAN BE UTILIZED TO ASSIST IN THE PREPARATION OF THE COORDINATION DRAWINGS. THESE DRAWINGS SHALL BE COMPLETED PRIOR TO WORK BEING INSTALLED IN THE FIELD FOR THE LOCATIONS NOTED ABOVE. AS-BUILT DOCUMENTS ARE NOT AVAILABLE FOR THE WORK LOCATION. THE CONTRACTOR WILL NEED TO PERFORM FIELD WORK TO LOCATE STRUCTURAL CONSTRAINTS AND EXISTING SERVICES THAT ARE REQUIRED TO REMAIN SO THAT ACCURATE DOCUMENTS CAN BE CREATED. THE ENGINEER WILL ASSIST WHERE NEEDED TO HELP IDENTIFY EQUIPMENT THAT NEEDS TO STAY WITHIN THE CEILING SPACE TO SERVICE OTHER SPACES. EACH TRADE INVOLVED IS TO PROVIDE THEIR DOCUMENTS, IN AUTOCAD FORMAT, TO THE MECHANICAL CONTRACTOR FOR INSERTION TO THE COMMON FILE. (DURING THE BID PROCESS THE MECHANICAL CONTRACTOR IS TO ENSURE THE GC IS AWARE OF THIS WORK WHERE THEY HAVE RESPONSIBLE CHARGE.) THE OWNER MAY PRE-PURCHASE THE SERVICES OF A FIRE SPRINKLER AGENCY, BUT THE INSTALLATION OF THE SPRINKLER PIPING WITH REGARD TO SCHEDULE, LOCATION AND INSTALLATION WILL FALL UNDER THE DIRECTION OF THE GC AND SPRINKLER PIPING MUST BE INCLUDED IN THE COORDINATION DOCUMENTS. AT A MINIMUM THE FOLLOWING TRADES ARE TO BE INVOLVED IN THE COORDINATION DRAWINGS: MECHANICAL, PLUMBING, ELECTRICAL, LOW VOLTAGE, BAS, SPRINKLER, CARPENTRY, ANY OTHER TRADE THAT WILL WORK IN THE AFFECTED AREAS.

BIDDING

1. THE CONTRACTOR SHALL VISIT THE SITE PRIOR TO SUBMITTING A BID TO BECOME FAMILIAR WITH THE EXISTING CONDITIONS. THE CONTRACTOR SHALL COMPARE THE WORK SPECIFIED IN THE CONTRACT DOCUMENTS WITH THE EXISTING CONDITIONS. THE CONTRACTOR SHALL IDENTIFY AND NOTATE ALL WORK OR CONDITIONS THAT ARE DIFFERENT FROM THE CONTRACT DOCUMENTS OR THEIR INTENT. THE CONTRACTOR SHALL, UPON DISCOVERY, IMMEDIATELY NOTIFY AND REPORT, IN WRITING, ANY DISCREPANCIES TO THE ENGINEER. NO EXTRAS OR CHANGE ORDERS WILL BE ALLOWED FOR FAILURE TO PERFORM THE PRE-BID SITE VISIT.
2. BASE PROPOSAL ON MANUFACTURER NAMES LISTED UNLESS "OR EQUAL" IS INDICATED. PROVIDE SUBSTITUTION REQUESTS A MINIMUM OF FIVE (5) BUSINESS DAYS PRIOR TO BID DATE CLOSING TO ALLOW TIME FOR DUE CONSIDERATION OF PROPOSED ALTERNATE. DETERMINATION OF SUBSTITUTION OF EQUIPMENT RESTS SOLELY WITH THE ENGINEER.

COMMISSIONING (NEEDED IF GREATER THAN 480 MBH COOLING OF 600 MBH HEATING)

1. IBC PROJECTS: MECHANICAL AND ELECTRICAL SYSTEMS DEFINED WITHIN THE IECC AND LOCAL AMENDMENTS. COMMISSIONING MUST BE PERFORMED BY AN APPROVED THIRD PARTY CONTRACTOR.
2. THE COMMISSIONING AGENT IS RESPONSIBLE FOR THE DEVELOPMENT OF THE COMMISSIONING PLAN AND THE COMMISSIONING PLAN IS TO BE SUBMITTED TO THE ENGINEER AND THE LOCAL AHJ AS A DEFERRED SUBMITTAL.
3. FOR THIS PROJECT THE OWNER HAS RETAINED AN APPROVED THIRD PARTY COMMISSIONING AGENT. THE COMMISSIONING AGENT (CXA) RETAINED BY THE OWNER WILL BE RESPONSIBLE FOR THE DEVELOPMENT OF THE COMMISSIONING PLAN AND REQUIREMENTS ASSOCIATED WITH THE IECC. EVERY CONTRACTOR WORKING ON THE PROJECT WILL BE RESPONSIBLE FOR WORKING WITH THE CXA AND DEMONSTRATING THEIR SYSTEMS PERFORMANCE AS DIRECTED BY THE CXA.

PART TWO - PRODUCTS

HVAC EQUIPMENT

1. PROVIDE HVAC EQUIPMENT AS SPECIFIED AND/OR SCHEDULED HEREIN AND IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. EQUIPMENT SHALL OPERATE ACCORDING TO THE MANUFACTURER'S "OWNER'S OPERATING AND MAINTENANCE MANUAL" TROUBLE-FREE PRIOR TO STARTING TEST AND BALANCE (TAB) WORK.

DUCTWORK

1. DUCTWORK, UNLESS LISTED OTHERWISE IN THIS SECTION, SHALL BE ASTM A653/A653M GALVANIZED SHEET METAL, LOCK-FORMING QUALITY HAVING A ZINC COATING OF 0.90 OZ PER SQ. FT. (G90) EQUALLY APPLIED TO EACH SURFACE, TESTED PER ASTM A90. DUCTWORK IS TO BE INSTALLED ACCORDING TO ASHRAE RECOMMENDATIONS AND SMACNA DUCT CONSTRUCTION STANDARDS. NO SHEETMETAL DUCTWORK TO BE LESS THAN 26 GA.
2. ROUND DUCTWORK: 8"Ø AND UNDER CAN BE SPIRAL OR SNAP-LOCK, >8"Ø TO BE SPIRAL CONSTRUCTION.
3. PROVIDE MANUAL VOLUME DAMPERS WITH LOCKING QUADRANTS AND IDENTIFYING RIBBONS AT DAMPER HANDLES FOR AIR BALANCING EACH BRANCH DUCT TAKE-OFF OR PIECE OF AIR DISTRIBUTION EQUIPMENT. NOT ALL DAMPERS MAY BE INDICATED ON THE DOCUMENTS.
4. SEAL ALL DUCT PENETRATIONS THROUGH WALLS, FLOOR AND ROOF. SEAL ALL TRANSVERSE DUCT SEAMS WITH APPROVED MASTIC. DUCT TAPES SHALL NOT BE ALLOWED FOR RIGID DUCTWORK.
5. SUPPLY, OUTSIDE AIR AND RETURN DUCTWORK SHALL BE INSULATED WITH FLEXIBLE GLASS FIBER INSULATION MEETING ANSI/ASTM C612, MAXIMUM 'K' VALUE OF 0.29 AT 75° F, WITH FOIL-KRAFT FLAME RESISTANT VAPOR BARRIER, MINIMUM 3/4 #/CUFT. DENSITY. BELOW ARE MINIMUM R VALUES FOR DUCTWORK INSULATION WHERE NOT OTHERWISE SPECIFICALLY SPECIFIED, PER ASHRAE 90.1-2016.
    5.1. SUPPLY/RETURN - HEATING & COOLING

CLIMATE ZONE	EXTERIOR, ATTICS, PARKING GARAGE, CRAWL SPACE.	UNCONDITIONED & BURIED DUCTS	INDIRECTLY CONDITIONED, RETURN AIR PLENUM. (C) (NO INSUL. ON RETURN)
0 TO 4	R-8	R-6	R-1.9
5 TO 8	R-12	R-6	R-1.9

    5.2. SUPPLY/RETURN - HEATING ONLY

CLIMATE ZONE	EXTERIOR, ATTICS, PARKING GARAGE, CRAWL SPACE.	UNCONDITIONED & BURIED DUCTS	INDIRECTLY CONDITIONED, RETURN AIR PLENUM. (C) (NO INSUL. ON RETURN)
0 TO 1	NONE	NONE	NONE
2 TO 4	R-6	R-6	R-1.9
5 TO 8	R-12	R-6	R-1.9

    5.3. SUPPLY/RETURN - COOLING ONLY

CLIMATE ZONE	EXTERIOR, ATTICS, PARKING GARAGE, CRAWL SPACE.	UNCONDITIONED & BURIED DUCTS	INDIRECTLY CONDITIONED, RETURN AIR PLENUM. (C) (NO INSUL. ON RETURN)
0 TO 6	R-8	R-6	R-1.9
7 TO 8	R-1.9	R-1.9	R-1.9

    5.4. GENERAL NOTES
    A. INTERIOR RETURN DUCTWORK, IN INDIRECTLY CONDITIONED RETURN AIR PLENUM SPACE - NO INSULATION REQUIRED.
    B. EXTERIOR DUCTWORK TO BE INSULATED WITH URETHANE OR POLYSTYRENE FOIL FACED RIGID BOARD, TOP OF DUCT TO HAVE A WATERSHED DESIGN, A WEATHER PROOF COVER IS TO BE APPLIED USING MFM FLEXCLAD-400 (ALUMINUM).
    C. SUPPLY DUCTWORK TO HAVE A MINIMUM OF 1.5" OF INSTALLED THICKNESS INSULATION REGARDLESS OF LOCATION.
    D. IN HUMID LOCATIONS, SUCH AS POOLS, DUCT INSULATION SHALL BE AP ARMAFLEX (ASTM E 84 25/50 RATED), TWO LAYERS, OVERLAPPING SEAMS, 1.5 TO 2" TOTAL THICKNESS).
6. ALL DUCTWORK SIZES SHOWN ARE FREE AREA DIMENSIONS. EXHAUST DUCTWORK SHALL BE UNINSULATED.
7. THE INTERIOR OF SUPPLY AND RETURN DUCTWORK VISIBLE BEHIND DEVICES (GRDs), SHALL BE PAINTED FLAT BLACK.
8. LINE DUCTWORK FIFTEEN FEET UPSTREAM AND DOWNSTREAM OF ALL FANS (EXCEPT FOR TYPE I HOOD, TYPE II EXHAUST OR EVAPORATIVE COOLING) AND WHERE INDICATED WITH 1" THICK, 1.5# DENSITY DUCT LINER. LINING SHALL BE APPLIED TO DUCTWORK WITH FIRE RESISTANT ADHESIVES, (FOSTER 85-10 OR EQUAL) AND COPPER OR CADMIUM PLATED MECHANICAL FASTENERS, (GRAHAM, OMARK OR EQUAL). ALL DUCT SIZES INDICATED ARE CLEAR INSIDE.
9. FLEXIBLE DUCTWORK WHERE INDICATED ON THE DRAWINGS SHALL BE INSULATED, WITH PLASTIC VAPOR BARRIER AT INTERIOR AND EXTERIOR, STEEL WIRE COIL REINFORCED. JOINTS SHALL BE BAND-CLAMPED, MASTIC-DUCT SEALER AND TAPE SEALED TO MAINTAIN INTEGRITY OF VAPOR BARRIER. FLEXIBLE INSTALLATION SHALL BE SUPPORTED TO ELIMINATE SAGS. FLEXIBLE GLASS FIBER INSULATION SHALL HAVE A MAXIMUM 0.23 K VALUE AT 75°F. INSTALLATION SHALL MEET ASHRAE & SMACNA STANDARDS. UNLESS NOTED OTHERWISE MAXIMUM LENGTH IS 5'-0" AND TWO 45° BENDS.
10. FIRE DAMPERS SHALL BE DYNAMIC (RATED FOR SYSTEM VELOCITY) AND MEET UL 555 AND SHALL HAVE BLADES OUT OF AIR STREAM IN COILED POSITION. FUSIBLE LINK SHALL BE RATED AT 165°F.
11. COMBINATION FIRE/SMOKE DAMPERS SHALL BE DYNAMIC (RATED TO SYSTEM VELOCITY) MEET UL 555S.
12. FLAMMABLE EXHAUST DUCT SHALL BE PER LOCAL CODES AND NFPA 30. HIGHLIGHTS ARE LISTED BELOW:
    12.0.1. 18 GA STAINLESS STEEL, WITH CONTINUOUS EXTERNAL WELDED JOINTS. ELBOWS TO BE RADIUS BACK AND THROAT, R/D≥1.5. MITERED ELBOWS NOT ALLOWED. FABRICATE AND INSTALL IN ACCORDANCE WITH SMACNA DUCT CONSTRUCTION STANDARDS, NFPA 30 AND LOCAL MECHANICAL CODE.
    12.1. WHERE FLAMMABLE EXHAUST DUCTWORK PASSES THROUGH A VERTICAL FIRE RESISTANCE RATED ASSEMBLY, THE DUCT SHALL BE ENCLOSED IN A CONTINUOUS ENCLOSURE EXTENDING FROM THE FIRST PENETRATED FIRE BARRIER THROUGH TO THE EXTERIOR, TO MAINTAIN THE FIRE RESISTANCE RATING OF THE HIGHEST FIRE BARRIER PENETRATED.
18. DUCTWORK TO BE CONSTRUCTED TO SMACNA AND ASHRAE DUCT CONSTRUCTION STANDARDS.

DUCTWORK SYSTEMS: (4)	PRESSURE (3)	SMACNA CLASS SEAL	LEAKAGE CLASS (5) ROUND	RECT.
CONSTANT VOLUME	2	A	2	4
RETURN	2	B	4	8
EXHAUST	2	B	4	8
EXHAUST (POSITIVE)	3	A	2	4
EXHAUST - LAB HOODS (6)	4	WELDED	2	3
TYPE 1 (GREASE)	3	WELDED	2	3
TYPE II (7)	2	A	2	3

PRESSURE CLASS OPTIONS: 1/2", 1", 2", 3", 4", 6" 10"
NOTE (1): DOWNSTREAM of VAV BOX
NOTE (2): UPSTREAM of VAV BOX
NOTE (3): THESE ARE MINIMUMS, REFER TO EQUIPMENT SCHEDULES & SUBMITTAL DOCUMENTS. IF ESP MEETS OR EXCEEDS THESE FIGURES, INCREASE PRESSURE CLASS TO NEAREST CLASS THAT IS 0.5" W.C. OVER THE LISTED ESP. WHERE INFORMATION IS NOT PROVIDED THE CONTRACTOR MUST SUBMIT A RFI.
NOTE (4): WHEN USED AS PART OF A SMOKE CONTROL OR REMOVAL SYSTEM SHALL, AT A MINIMUM, BE SMACNA PRESSURE CLASS 3, SEAL CLASS A
NOTE (5): LEAKAGE CLASS IS CFM LEAKAGE/100 SQ.FT. @ 1" H/2O
NOTE (6): UNLESS NOTED OTHERWISE FUME EXHAUST DUCTS AND SUPPORTS TO BE FABRICATED FROM 316 STAINLESS STEEL, MINIMUM 18-GA. SEAMS AND JOINTS TO BE WELDED LIQUID/AIR TIGHT.
NOTE (7): WHERE MOISTURE MAY BE CONVEYED MUST BE LIQUID TIGHT AND SLOPED BACK TO THE HOOD. OTHER LOW POINTS TO BE DRAINED.

DUCTWORK ELBOWS: ROUND:	FPM	MINIMUM RADIUS/DIA RATIO
	TO 1000	0.75
	1,001 TO 1,500	1
	1,500+	1.5(2)

ASPECT RATIO, W/D

RECTANGULAR: R/D	0.25	0.5	1	2	3	4
0.0 (3)	(1)	(1)	(1)	(1)	(1)	(1)
0.5	(1)	(1)	(1)	(1)	(1)	(1)
1	(1)	(1)	(1)	(1)	(1)	(1)
1.5 (2), (5)	(1)	(1)	(4)	(4)	(4)	(4)
2	(1)	(4)	(4)	(4)	(4)	(4)
3	(1)	(4)	(4)	(4)	(4)	(4)

NOTE (1): MUST HAVE AIRFOIL TURNING VALVES
NOTE (2): STANDARD/DEFAULT CENTERLINE RADIUS
NOTE (3): MITRED ELBOW
NOTE (4): TURNING VANES NOT REQUIRED
NOTE (5): THIS R/D MUST BE USED FOR TYPE I GREASE DUCTS, TURNING VANES NOT ALLOWED

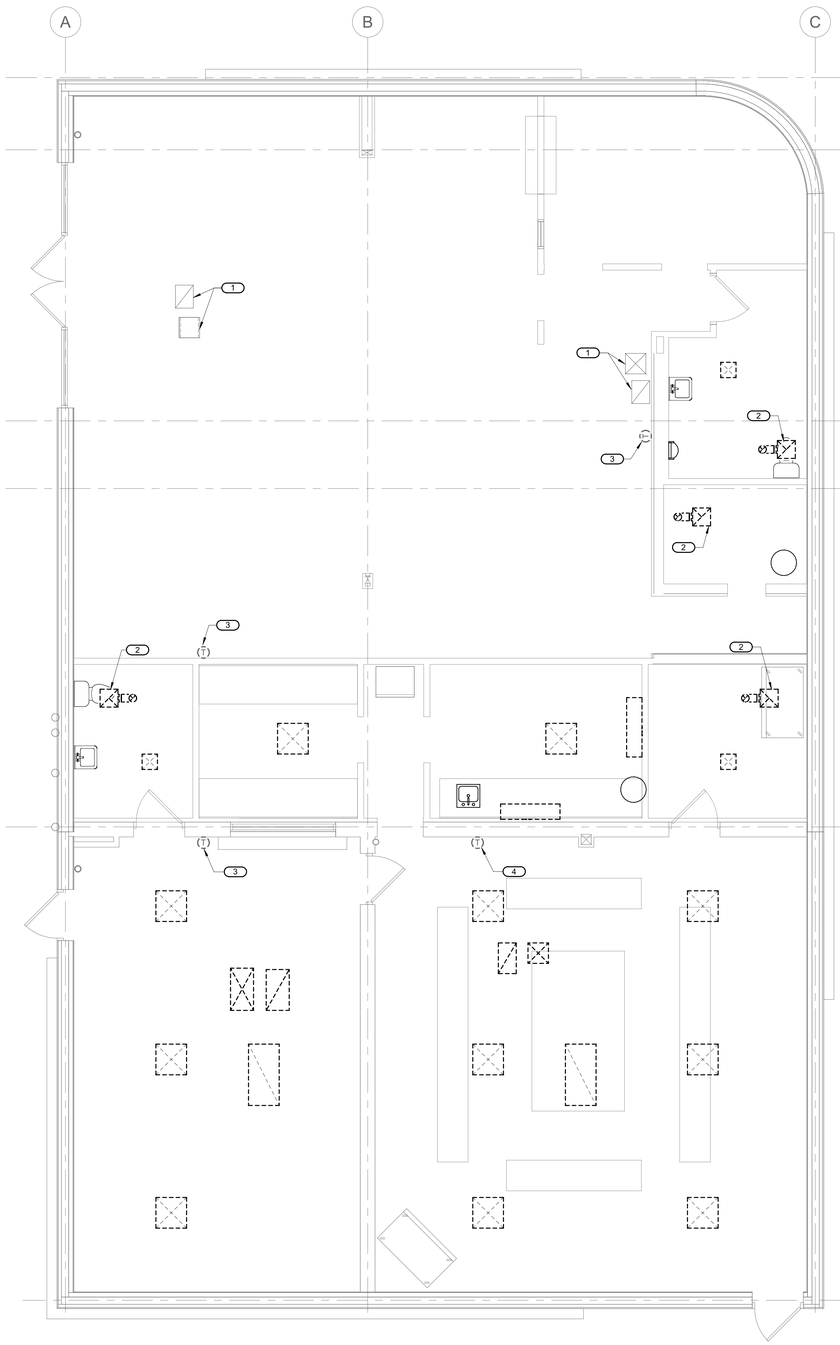
MECHANICAL PRODUCTS

1. DIFFUSERS, REGISTERS AND GRILLES: MAXIMUM SOUND PRESSURE LEVELS SHALL NOT EXCEED NC 30. COORDINATE FINISH AND MOUNTING TYPE WITH ARCHITECT. ACCEPTABLE MANUFACTURERS: TITUS, NAILOR, KRUEGER, TUTTLE AND BAILEY, PRICE, METALAIRE.
2. CONTROL DAMPERS: LEAKAGE CLASS 1A/1. EQUAL TO RUSKIN CD-60 (CD-50 IN WET LOCATIONS)
3. PIPE HANGERS: PIPE SIZES 1/2" TO 1 1/2": MALLEABLE IRON, CARBON STEEL, ADJUSTABLE SWIVEL, SPLIT RING. PIPE SIZES OVER 2" (UNLESS NOTED OTHERWISE): CARBON STEEL, ADJUSTABLE, CLEVIS. PIPE SIZES CHILLED WATER 8" AND OVER, HEATING WATER 6" AND OVER, STEAM (SUPPLY & CONDENSATE) 4" AND OVER: ADJUSTABLE STEEL YOKE, CAST IRON ROLL, DOUBLE HANGER.
4. PIPING:
    4.2. EQUIPMENT DRAIN OVERFLOWS AND CONDENSATE DRAIN PIPING: TYPE "M" COPPER (ASTM B-88), WROUGHT FITTINGS (ASME B16.22), JOINTS: ANSI/ASTM B32, SOLDER: 95/5 TIN/ANTIMONY, 0.2% MAX. LEAD.
6. PIPE INSULATION: GLASS FIBER INSULATION WITH A MAXIMUM K VALUE NOTED BELOW AT 75 DEGREES F. OUTDOOR INSULATION THICKNESS SHALL BE DOUBLE INDOOR THICKNESS WITH A MAXIMUM THICKNESS OF 5". INTERIOR APPLICATIONS SHALL HAVE KRAFT REINFORCED FOIL VAPOR BARRIER WITH ONE PIECE PREMOLDED PVC JACKETS FOR FITTINGS. EXTERIOR APPLICATIONS SHALL HAVE STUCCO EMBOSSED ALUMINUM JACKETS. ACCEPTABLE MANUFACTURERS: OWENS CORNING, CERTAINTEED, JOHNS MANVILLE, KNAUF.

FLUID TEMP RANGE °F	INSUL. CONDUCTIVITY BTU-IN/(HR-SQ.FT.-°F)	INSULATION THICKNESS NOMINAL PIPE SIZE (IN) <1.5	1.5 & >
> 350	0.32	5	5
251-350	0.29	4	4.5
201-250	0.27	2.5	3
141-200	0.25	1.5	2
< 140	0.22	1	1.5

OUTER INSUL. DIAMETER	MINIMUM ALUMINUM JACKET THICKNESS (INCHES) RIGID INSULATION	NON-RIGID INSULATION
≤ 8"	0.016	0.016
> 8"-11"	0.016	0.020
> 11"-24"	0.016	0.024
> 24"-36"	0.020	0.032
> 36"	0.024	0.040

    6.1. WHERE THE CLIMATE DICTATES, CONDENSATE DRAIN PIPING TO BE INSULATED.
    6.2. PROVIDE METAL SADDLES AND RIGID INSULATION AT HANGERS WHERE SYSTEM WEIGHT COMPRESSES INSULATION.
7. DUCT MOUNTED SMOKE DETECTORS:
    7.1. WHEN THE DUCT TYPE SMOKE DETECTOR IS REQUIRED TO BE PART OF THE DESIGN BUILD FIRE ALARM SYSTEM: SHALL BE FURNISHED BY THE FIRE ALARM CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR AND WIRED BY THE FIRE ALARM CONTRACTOR.
    7.2. WHEN THE DUCT TYE DETECTOR IS NOT PART OF THE FIRE ALARM SYSTEM: THE DUCT MOUNTED SMOKE DETECTORS SHALL BE FURNISHED, INSTALLED AND WIRED BY THE MECHANICAL CONTRACTOR. (NORMALLY PROVIDED AT 24 VAC.)
    7.3. DUCT MOUNTED SMOKE DETECTORS SHALL BE INSTALLED ON AIR MOVING EQUIPMENT THAT EXCEEDS 2,000 CFM AND ON AIR MOVING EQUIPMENT UNDER 2,000 CFM THAT SUPPLIES A COMMON SPACE AND THE TOTAL CFM CAPACITY OF THE EQUIPMENT SERVING THE SPACE EXCEEDS 2,000 CFM.
    7.4. DUCT MOUNTED DETECTORS SHALL BE WIRED TO SHUT DOWN THE ASSOCIATED AIR MOVING EQUIPMENT ON ALARM.
    7.5. DETECTORS TO BE MOUNTED IN THE SUPPLY AIR DUCTWORK.
8. FANS, ACCEPTABLE MANUFACTURERS: GREENHECK, COOK, TWIN CITY, ACME, PENN-BERRY. FAN DESIGN RP



GENERAL NOTES:

1. REMOTE VOLUME DAMPERS (RVD) ARE REQUIRED FOR ALL DAMPERS INSTALLED ABOVE INACCESSIBLE CEILINGS. COORDINATE EXACT LOCATION OF ALL RVDS WITH ARCHITECT PRIOR TO INSTALLATION.
2. VERIFY LOCATION OF ALL THERMOSTATS/SENSORS WITH ARCHITECT PRIOR TO INSTALLATION. MOUNT ALL THERMOSTATS/SENSORS @48" A.F.F. IN ACCORDANCE WITH ADA STANDARDS (COORDINATE WITH ARCHITECTURAL DRAWINGS).
3. VERIFY AND COORDINATE FRAME AND BORDER TYPE REQUIREMENTS FOR AIR DEVICES WITH ARCHITECTURAL CEILING PLANS PRIOR TO ORDERING. COORDINATE AIR DEVICE FINISHES WITH ARCHITECT.
4. DUCT SIZES SHOWN ARE THE CLEAR INSIDE DIMENSIONS.
5. THE MECHANICAL CONTRACTOR SHALL VERIFY THE LOCATION OF ALL ROOF MOUNTED EQUIPMENT AND ROOF PENETRATIONS WITH ARCHITECT AND STRUCTURAL ENGINEER PRIOR TO COMMENCING WORK.
6. THE MECHANICAL CONTRACTOR SHALL COORDINATE LOCATION AND ROUTING OF HVAC EQUIPMENT AND DUCTWORK WITH OTHER TRADES PRIOR TO COMMENCING WORK.
7. ALL EXHAUST OUTLETS SHALL BE LOCATED MIN. OF 10'-0" FROM ANY OUTSIDE AIR INTAKES.
8. THE CUTTING, NOTCHING AND BORING OF HOLES IN FLOOR JOIST AND WALL STUDS SHALL BE IN ACCORDANCE WITH THE LATEST APPROVED EDITION OF THE INTERNATIONAL BUILDING CODE.
9. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL CUTTING AND PATCHING AS REQUIRED TO ACCOMMODATE HIS WORK.
10. REFER TO THE MECHANICAL DIAGRAMS THAT APPLY TO THE WORK ON THIS DRAWING. THESE DIAGRAMS PROVIDE GUIDANCE AS TO INSTALLATION INTENT AND DO NOT NECESSARILY SHOW ALL COMPONENTS REQUIRED.
11. CONTRACTOR TO COORDINATE WHAT TIME OF DAY THE WORK IS TO OCCUR, HOW MANY SHIFTS PER DAY, IF WEEKENDS, NIGHTS, HOLIDAYS ARE REQUIRED PRIOR TO SUBMITTING BID.
12. CONTRACTOR TO GIVE WRITTEN NOTIFICATION OF ANY INTERRUPTION OF SERVICES A MINIMUM OF THREE WORKING DAYS PRIOR TO COMMENCING WORK.

SHEET NOTES:

1. EXISTING DUCTWORK AND DIFFUSERS/GRILLES TO BE REMOVED DOWNSTREAM OF THIS LOCATION. REFER TO NEW WORK PLAN FOR RECONNECTION.
2. EXISTING CEILING EXHAUST FAN, ASSOCIATED DUCTWORK, AND VENT CAP TO BE REMOVED.
3. EXISTING MANUFACTURERS THERMOSTAT TO BE RELOCATED. REFER TO NEW WORK PLAN.
4. EXISTING MANUFACTURERS THERMOSTAT TO BE REMOVED.
5. RELOCATED MANUFACTURERS THERMOSTAT. PROVIDE WIRING (IN CONDUIT) TO UNIT INDICATED.
6. CONNECT NEW DUCTWORK TO EXISTING SUPPLY AND RETURN DUCT MAINS DOWN FROM ROOF.
7. NEW SUPPLY AND RETURN DUCT MAINS DOWN FROM ROOF.
8. NEW 12"X12" EXHAUST DUCT UP THROUGH ROOF TO ROOF MOUNT EXHAUST FAN.
9. 3/4" CONDENSATE DRAIN LINE UP THROUGH ROOF.
10. 3/4" CONDENSATE DRAIN LINE DOWN IN WALL. TERMINATE ABOVE MOP SINK WITH AIR GAP REQUIRED.
11. INSTALL GLASFLOSS CARBOTRON Z-LINE CARBON FILTER, 12X12X2, WITH FILTER BOX ACCESS.

NORTH 2 **LEVEL 01 VENTILATION DEMO PLAN**
1/4" = 1'-0"

NORTH 1 **LEVEL 01 VENTILATION PLAN**
1/4" = 1'-0"

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.co

Consultants

MECHANICAL ENGINEERING CERTIFICATE
34046
MARIO P. TORREGROSSA
11/21/2022
REGISTERED PROFESSIONAL ENGINEER
ARIZONA, U.S.A.
EXPIRES 09/30/2023

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE
3155 E. McDowell Rd.
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description
1	11/22/22	City Comments

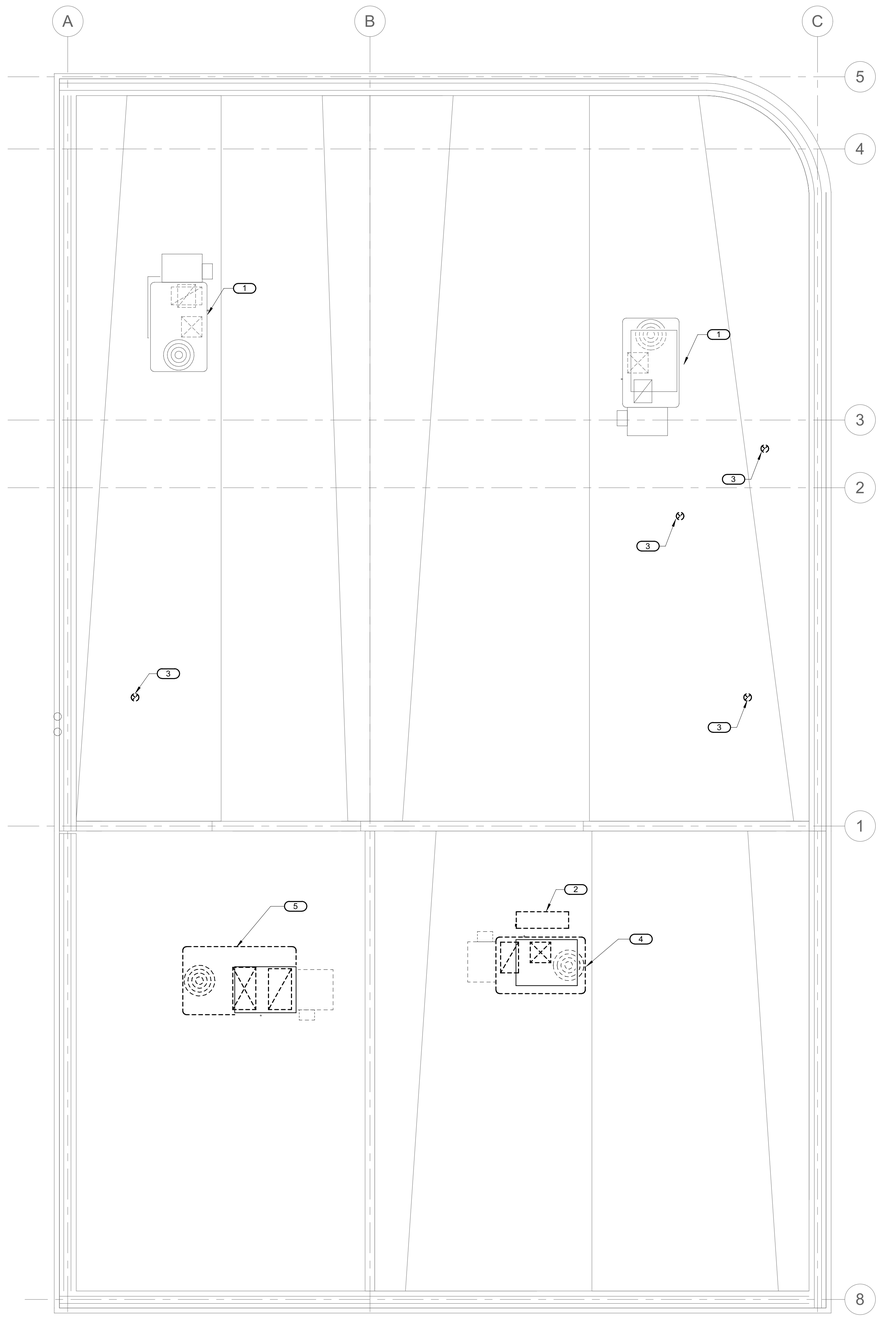
IMEG
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

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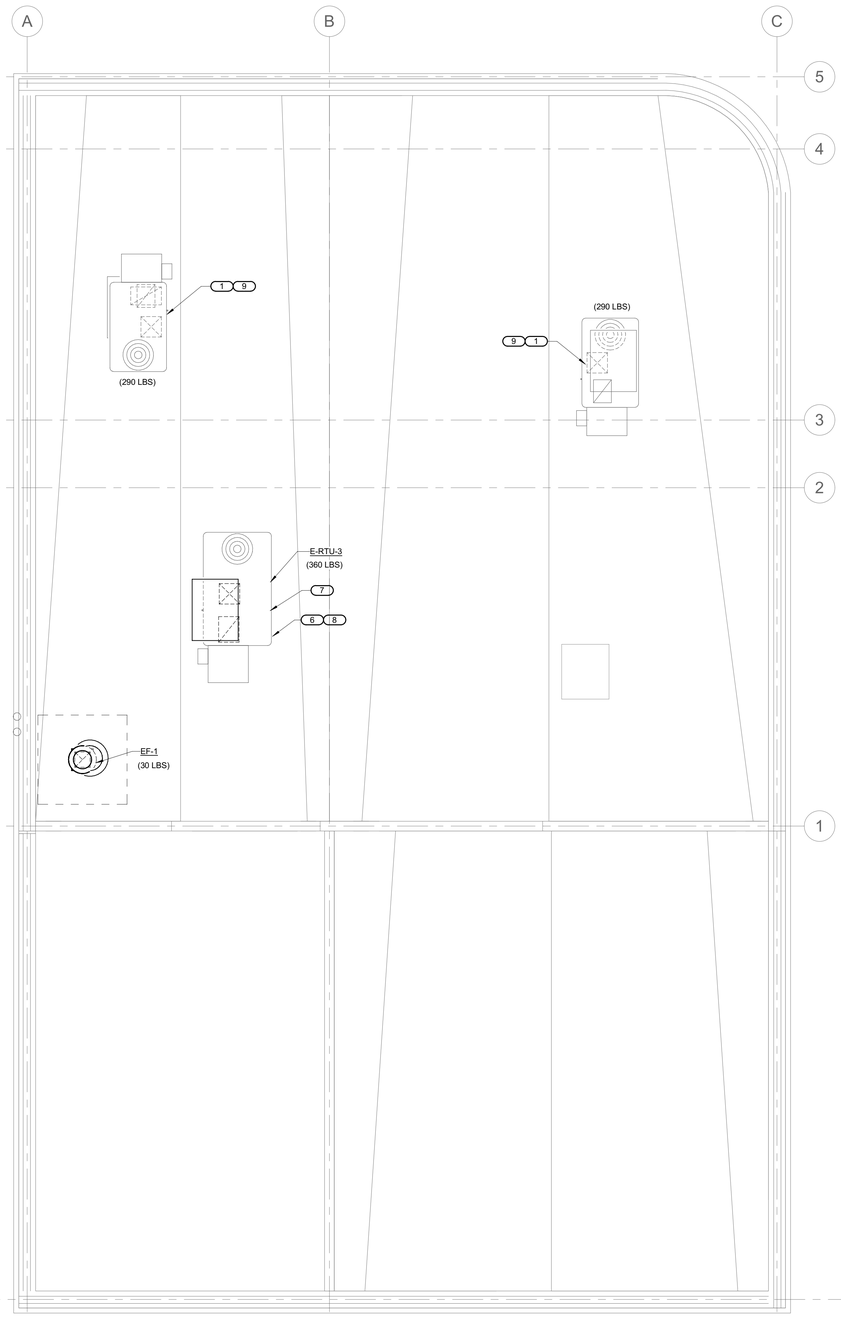
MECHANICAL PLANS

M100



ROOF VENTILATION DEMO PLAN

1/4" = 1'-0"



ROOF VENTILATION PLAN

1/4" = 1'-0"

GENERAL NOTES:

1. REMOTE VOLUME DAMPERS (RVD) ARE REQUIRED FOR ALL DAMPERS INSTALLED ABOVE INACCESSIBLE CEILINGS. COORDINATE EXACT LOCATION OF ALL RVDS WITH ARCHITECT PRIOR TO INSTALLATION.
2. VERIFY LOCATION OF ALL THERMOSTATS/SENSORS WITH ARCHITECT PRIOR TO INSTALLATION. MOUNT ALL THERMOSTATS/SENSORS @48" A.F.F. IN ACCORDANCE WITH ADA STANDARDS (COORDINATE WITH ARCHITECTURAL DRAWINGS).
3. VERIFY AND COORDINATE FRAME AND BORDER TYPE REQUIREMENTS FOR AIR DEVICES WITH ARCHITECTURAL CEILING PLANS PRIOR TO ORDERING. COORDINATE AIR DEVICE FINISHES WITH ARCHITECT.
4. DUCT SIZES SHOWN ARE THE CLEAR INSIDE DIMENSIONS.
5. THE MECHANICAL CONTRACTOR SHALL VERIFY THE LOCATION OF ALL ROOF MOUNTED EQUIPMENT AND ROOF PENETRATIONS WITH ARCHITECT AND STRUCTURAL ENGINEER PRIOR TO COMMENCING WORK.
6. THE MECHANICAL CONTRACTOR SHALL COORDINATE LOCATION AND ROUTING OF HVAC EQUIPMENT AND DUCTWORK WITH OTHER TRADES PRIOR TO COMMENCING WORK.
7. ALL EXHAUST OUTLETS SHALL BE LOCATED MIN. OF 10'-0" FROM ANY OUTSIDE AIR INTAKES.
8. THE CUTTING, NOTCHING AND BORING OF HOLES IN FLOOR JOIST AND WALL STUDS SHALL BE IN ACCORDANCE WITH THE LATEST APPROVED EDITION OF THE INTERNATIONAL BUILDING CODE.
9. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL CUTTING AND PATCHING AS REQUIRED TO ACCOMMODATE HIS WORK.
10. REFER TO THE MECHANICAL DIAGRAMS THAT APPLY TO THE WORK ON THIS DRAWING. THESE DIAGRAMS PROVIDE GUIDANCE AS TO INSTALLATION INTENT AND DO NOT NECESSARILY SHOW ALL COMPONENTS REQUIRED.
11. CONTRACTOR TO COORDINATE WHAT TIME OF DAY THE WORK IS TO OCCUR, HOW MANY SHIFTS PER DAY, IF WEEKENDS, NIGHTS, HOLIDAYS ARE REQUIRED PRIOR TO SUBMITTING BID.
12. CONTRACTOR TO GIVE WRITTEN NOTIFICATION OF ANY INTERRUPTION OF SERVICES A MINIMUM OF THREE WORKING DAYS PRIOR TO COMMENCING WORK.

SHEET NOTES:

1. EXISTING ROOFTOP UNIT TO REMAIN.
2. EXISTING CONDENSING UNIT TO BE REMOVED. REMOVE ALL ASSOCIATED REFRIGERANT PIPING.
3. EXISTING VENT CAP TO BE REMOVED.
4. EXISTING ROOFTOP UNIT TO BE REMOVED (3 TON). REMOVE ALL ASSOCIATED DUCTWORK.
5. EXISTING ROOFTOP UNIT (4 TON) TO BE RELOCATED. SEE NEW WORK PLAN FOR NEW LOCATION. REMOVE ALL ASSOCIATED DUCTWORK.
6. EXISTING ROOFTOP UNIT (4 TON) RELOCATED TO SERVE NEW SPACE.
7. 3/4" TRAPPED CONDENSATE DOWN THROUGH ROOF TO LEVEL BELOW.
8. RE-BALANCE EXISTING ROOFTOP UNIT TO OUTSIDE AIR VALUE OF 130 CFM.
9. RE-BALANCE EXISTING ROOFTOP UNIT TO OUTSIDE AIR VALUE OF 220 CFM.

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.co
Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008



EXP. 03-31-2023

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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IMEG
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
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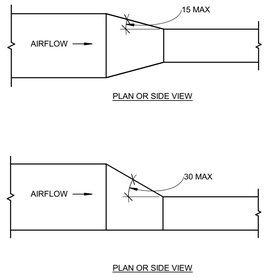
0 1 2 3
REF. SCALE IN INCHES

MECHANICAL
ROOF PLANS

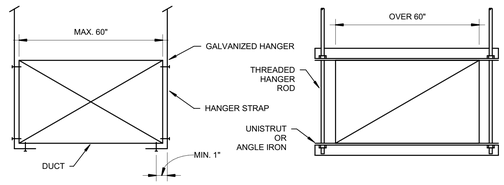
M101

BIM 360://Flower Shop - 10827 S 51st St./MEP20_21006821-00_Flower Shop McDowell_C.rvt
8/22/2022 11:38:17 AM

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

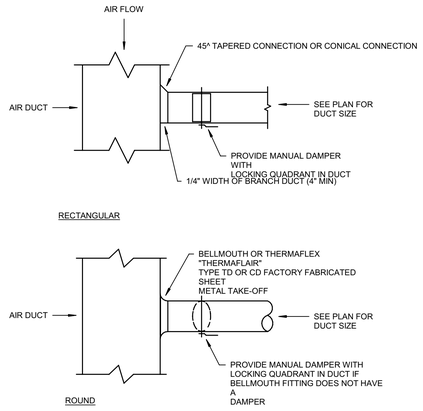


1 DUCT TRANSITION
NO SCALE

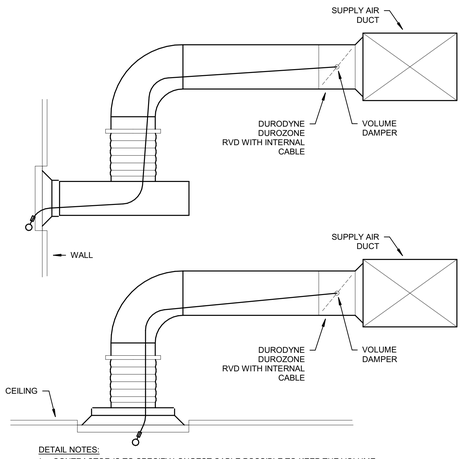


DETAIL NOTES:
1. ON DUCTS OVER 48" WIDE, BOTTOM SHALL BE BRACE BY ANGLE. FOR CROSS SECTION AREA MORE THAN 8 SQUARE FEET, DUCT SHALL BE BRACED BY ANGLES ON ALL FOUR SIDES.
2. SUPPORTS SHALL BE SPACED AND SIZED AS PER SMACNA STANDARDS.

2 DUCT HANGER SUPPORT
NO SCALE

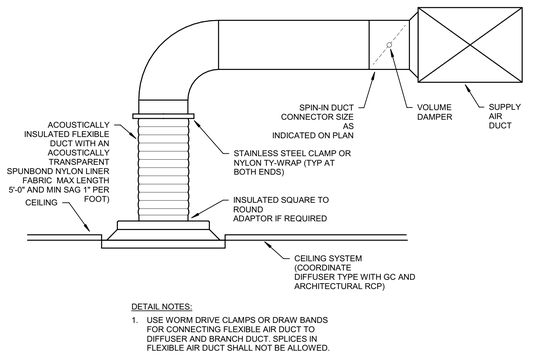


3 TYPICAL BRANCH DUCT TAKE-OFF
NO SCALE



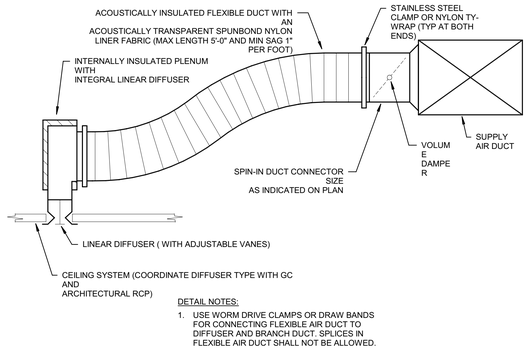
DETAIL NOTES:
1. CONTRACTOR IS TO SPECIFY LONGEST CABLE POSSIBLE TO KEEP THE VOLUME DAMPER AS FAR AWAY AS POSSIBLE FROM AIR DEVICE. THE END OF THE VABLE IS TO BE LOCKED AND HIDDEN BEHIND THE AIR DEVICE AFTER THE AIR DEVICE HAS BEEN BALANCED.
2. INSTALL AS PER MANUFACTURER'S INSTRUCTIONS.

4 REMOTE VOLUME DAMPER (RVD)
NO SCALE



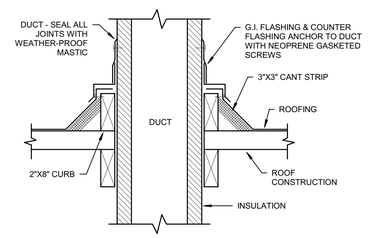
DETAIL NOTES:
1. USE WORM DRIVE CLAMPS OR DRAW BANDS FOR CONNECTING FLEXIBLE AIR DUCT TO DIFFUSER AND BRANCH DUCT. SPLICES IN FLEXIBLE AIR DUCT SHALL NOT BE ALLOWED.

5 CEILING DIFFUSER WITH FLEX DUCT
NO SCALE

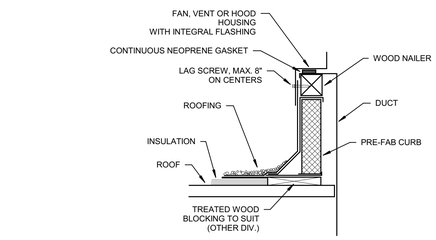


DETAIL NOTES:
1. USE WORM DRIVE CLAMPS OR DRAW BANDS FOR CONNECTING FLEXIBLE AIR DUCT TO DIFFUSER AND BRANCH DUCT. SPLICES IN FLEXIBLE AIR DUCT SHALL NOT BE ALLOWED.

6 LINEAR DIFFUSER WITH FLEX DUCT
NO SCALE

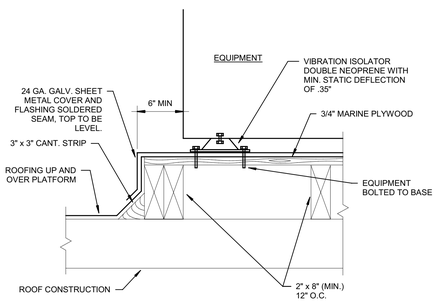


8 DUCT THROUGH ROOF
NO SCALE

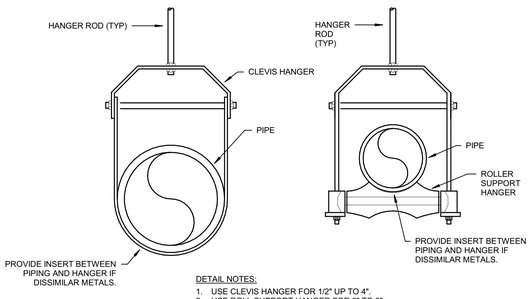


DETAIL NOTE:
1. INSTALL CURB PER MANUFACTURER RECOMMENDATIONS

9 ROOF MOUNTED EQUIPMENT CURB
NO SCALE

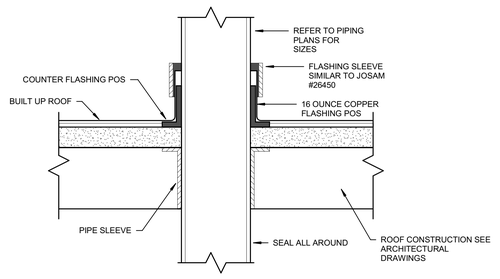


10 EQUIPMENT BASE
NO SCALE

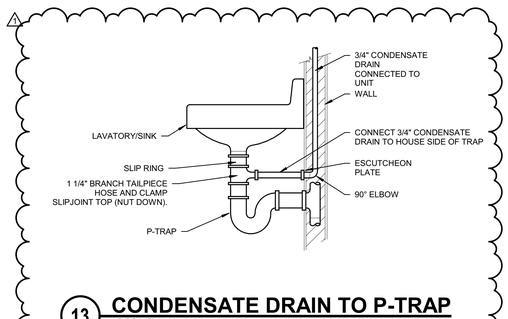


DETAIL NOTES:
1. USE CLEVIS HANGER FOR 1/2" UP TO 4".
2. USE ROLL SUPPORT HANGER FOR 6" TO 8".
3. PIPE 10" AND LARGER SHALL HAVE ROLLERS. SUPPORT HANGER WITH DUAL RODS. FASTEN TO TOP CHORD OF STRUCTURAL ELEMENTS.

11 PIPE HANGER FOR UNINSULATED PIPING
NO SCALE



12 PIPE THROUGH ROOF
NO SCALE



13 CONDENSATE DRAIN TO P-TRAP
NO SCALE

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RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099
Tempe, AZ 85281-3714 fax
www.rsparch.co
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MECHANICAL CERTIFICATE No. 34046 MARIO P. TORREGROSSA 11/21/2022 REGISTERED PROFESSIONAL ENGINEER ARIZONA, U.S.A.
EXPIRES 09/30/2023

Project For
THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE
3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description
1	11/22/22	City Comments

MECHANICAL DIAGRAMS

M400

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

FAN SCHEDULE1

NOTES:
1. PROVIDE MOTOR WITH THERMAL OVERLOADS.
2. DISCONNECT SWITCH PROVIDED BY ELECTRICAL CONTRACTOR.
3. PROVIDE BACKDRAFT DAMPER.
4. PROVIDE UNIT MOUNTED SPEED CONTROLLER.
5. EXHAUST FAN SHALL OPERATE CONTINUOUSLY DURING OCCUPIED HOURS.
6. CONTROLLED VIA TIMECLOCK (COORDINATE WITH ELECTRICAL).

			FAN			ELECTRICAL							
TAG NAME	AREA SERVED	DRIVE TYPE	CFM	ESP	RPM	BHP	MHP	VOLTAGE	PHASES	WEIGHT	MANUFACTURER	MODEL	NOTES
EF-1	TOILET/JANITOR/RETAIL	DIRECT	510	0.50	1619	0.08	0.2	120	1	30	GREENHECK	G-090-VG	1, 2, 3, 4, 5, 6

AIR TERMINAL SCHEDULE

NOTES:
1.CONTRACTOR SHALL DETERMINE PROPER BORDER TYPE TO MATCH CEILING CONSTRUCTION.
2.REFER TO DRAWINGS FOR NECK SIZE. ALL BRANCH DUCTWORK TO AIR TERMINALS SHALL BE NECK SIZE UNLESS NOTED OTHERWISE.

TAG NAME	FACE SIZE (IN.) (NOTE 2)	TYPE	BORDER (NOTE 1)	MATERIAL	FINISH	MANUFACTURER	MODEL	NOTES
EG-1	12x12	PERFORATED FACE	SURFACE	STEEL	WHITE	TITUS	50F	1, 2
EG-2	INLET +2	PERFORATED FACE	SIDEWALL	STEEL	WHITE	TITUS	350RL	1, 2
RG-1	24x24	PERFORATED FACE	LAY-IN	STEEL	WHITE	TITUS	350RL	1, 2
RG-2	24x24	PERFORATED FACE	LAY-IN	STEEL	WHITE	TITUS	350RL	1, 2
SD-1	12x12	LOUVER FACE	SURFACE	STEEL	WHITE	TITUS	TDC	1, 2
SD-2	24x24	PANEL FACE	LAY-IN	STEEL	WHITE	TITUS	TMS	1, 2
SD-3	24x24	PANEL FACE	LAY-IN	STEEL	WHITE	TITUS	TMS	1, 2

LINEAR DIFFUSER SCHEDULE

NOTES:
1.CONTRACTOR SHALL DETERMINE PROPER MARGIN STYLE TO MATCH CEILING CONSTRUCTION.
2.PROVIDE WITH CONCEALED FASTENERS.
3.DIFFUSER/GRILLE MOUNTED IN VERTICAL FACE OF ARCHITECTURAL SOFFIT.

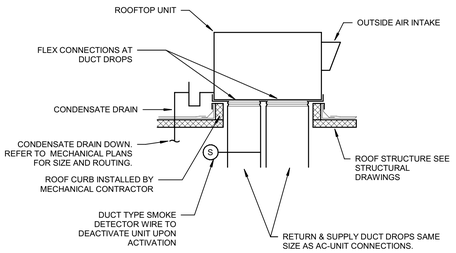
TAG NAME	MATERIAL	SLOT WIDTH	NO. OF SLOTS	LENGTH	PLENUM REQUIRED	PLENUM INSULATION TYPE	PLENUM INLET SIZE	PATTERN CONTROL REQUIRED	BALANCING DAMPER REQUIRED	FINISH	MANUFACTURER	MODEL	NOTES
LD-1	ALUMINUM	1"	4	4'-0"	Yes	WRAPPED	12"	Yes	Yes	WHITE	TITUS	ML-39	1, 2, 3
LR-1	ALUMINUM	1"	4	4'-0"	Yes	WRAPPED	12"	Yes	Yes	WHITE	TITUS	MLR-39	1, 2, 3

VENTILATION CALCULATION (IMC_2018)

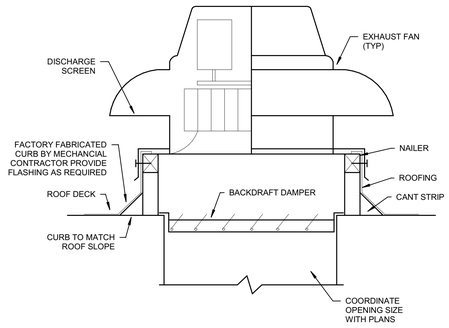
SPACE NAME	SPACE AREA	OCCUPANT DENSITY	CALCULATED OCCUPANTS	FIXED SEATS	USE OCCUPANCY	OUTDOOR AIR (Rp)	OUTDOOR AIR (Ra)	EXHAUST AIR	DISTRIBUTION EFFECTIVENESS	PLUMBING FIXTURE COUNT	FIXTURE EXHAUST AIR	REQUIRED OUTDOOR AIR	USE OUTDOOR AIR	USE EXHAUST AIR
[N/A]	*[SQFT]*	*[#/1000 SQFT]*				*[CFM/Person]*	*[CFM/SQFT]*	*[CFM/SQFT]*	*[N/A]*	*[N/A]*	*[CFM/Fixture]*	*[CFM]*	*[CFM]*	*[CFM]*
BREAK	135	50.0	6.8		6.8	5.0	0.06	0.0	0.8			52	60	0
STOCK/FULFILLMENT	168	5.0	0.8		0.8	5.0	0.06	0.0	0.8			18	20	0
OFFICE	117	5.0	0.6		0.6	5.0	0.06	0.0	0.8			12	20	0
PUBLIC RR	55	0.0	0.0		0.0	0.0	0.00	0.0	0.8	1	100	0	0	100
RETAIL	1460	15.0	21.9		21.9	7.5	0.12	0.0	0.8			424	430	0
JANITOR	18	0.0	0.0		0.0	0.0	0.00	0.0	0.8	1	100	0	0	100
STORAGE	145	0.0	0.0		0.0	0.0	0.12	0.0	0.8			22	30	0
HALL	70	0.0	0.0		0.0	0.0	0.06	0.0	0.8			5	10	0
												TOTAL:	570	200

AIR BALANCE SCHEDULE

OUTSIDE AIR	+ 570 CFM
EXHAUST AIR	- 200 CFM
RELIEF AIR	- 310 CFM
BALANCE	+ 60 CFM
(MAINTAIN 10% POSTITIVE)	



1 **ROOFTOP AIR CONDITIONING UNIT**
NO SCALE



2 **ROOF EXHAUST FAN**
NO SCALE

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
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SUITE 320
SCOTTSDALE, AZ 85258
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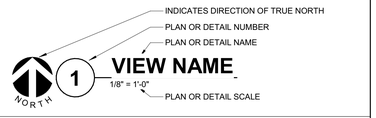
MECHANICAL SCHEDULES

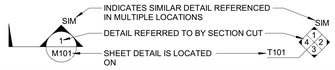
M500

VIEW KEY

NAME — LEVEL NAME
16'-0" — HEIGHT ABOVE PROJECT 0'-0"

1 — INDICATES NOTE USED TO DESCRIBE ADDITIONAL INFORMATION ABOUT WORK REQUIRED, SPECIFIC TO THE SHEET AND/OR DETAIL





LINE TYPE AND TAG KEY:

NEW WORK BY THIS CONTRACTOR (WIDE LINE)
——— NEW
- - - - - - - - EXISTING TO BE REMOVED (SHORT DASHED PATTERN)
— — — — NEW UNDERFLOOR OR UNDERGROUND (LONG DASHED PATTERN)

EXISTING TO REMAIN OR WORK BY OTHERS (NARROW LINE)
——— EXISTING
- - - - - - - - EXISTING TO BE REMOVED BY OTHERS (SHORT DASHED PATTERN)
— — — — EXISTING UNDERFLOOR OR UNDERGROUND (LONG DASHED PATTERN)

HALFTONING DOES NOT MODIFY SCOPE.

'TAG'-E — TAGS WITH DASH 'E' INDICATES THE REFERENCED OBJECT IS EXISTING

TAG-1 — UNDERLINED TAG INDICATES OBJECT IS IN-SCOPE. IF NEW, ADDITIONAL INFORMATION IS AVAILABLE IN A SCHEDULE, MATERIAL LIST, OR SYMBOL LIST

INDICATES AN EXISTING SYSTEM'S POINT OF CONNECTION/REMOVAL

FIRE / SMOKE BARRIER DESIGNATIONS

THE LINE TYPES SHOWN ARE FOR THE CONVENIENCE OF THE CONTRACTOR. THE CONTRACTOR SHALL VERIFY RATINGS WITH THE LATEST SET OF ARCHITECTURAL PLANS AND FURNISH ALL MATERIALS REQUIRED TO COMPLY WITH THOSE RATINGS WHETHER SHOWN OR NOT.

ALL [FLOOR, FLOOR CEILING, AND ROOF CEILING] ASSEMBLIES SHALL BE DESIGNATED AS [1], [2], [3], [4] HOUR FIRE [/SMOKE], BARRIER(S), UNLESS NOTED OTHERWISE ON THE PLANS. RATINGS WERE ACQUIRED FROM THE ARCHITECTURAL PLANS DATED [**/**/**].
NTD: COORDINATE WITH ARCHITECT

FIRE PARTITION	- - - - - - -

City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

PLUMBING SPECIFICATIONS

PART ONE - GENERAL

1. THE CONTRACTOR SHALL PROVIDE THE WORK SHOWN ON THE DRAWINGS AND SPECIFIED FOR THEIR INDIVIDUAL SECTIONS OF WORK. THE WORD "WORK" SHALL MEAN ALL LABOR, TRANSPORTATION, MATERIAL, EQUIPMENT, TOOLS, INSTALLATION, SUPERVISION AND ANY OTHER INCIDENTAL ITEMS OR SERVICES NECESSARY FOR THE PROPER INSTALLATION AND OPERATION OF THE COMPLETE SYSTEMS, WHICH SHALL BE PROVIDED WHETHER OR NOT SPECIFICALLY INDICATED OR NOTED.
2. ALL GENERAL CONDITIONS, SPECIAL REQUIREMENTS OR GENERAL REQUIREMENTS OF THE CONSTRUCTION SPECIFICATIONS ARE MADE PART OF THIS SPECIFICATION AND HAVE THE SAME FORCE AND AFFECT AS IF COMPLETELY REPRODUCED.
3. THE WORD "PROVIDE" SHALL MEAN FURNISH AND INSTALL, MAKE ALL FINAL CONNECTIONS AND LEAVE IN AN APPROVED COMPLETE OPERATING CONDITION.
4. ALL WORK SHALL BE DONE IN ACCORDANCE WITH THE LATEST ADOPTED EDITIONS OF THE APPLICABLE INTERNATIONAL BUILDING CODE (IBC), INTERNATIONAL MECHANICAL CODE (IMC), INTERNATIONAL PLUMBING CODE (IPC), NATIONAL ELECTRIC CODES (NEC) AND ALL OTHER APPLICABLE FEDERAL, STATE, AND LOCAL REGULATIONS.
5. THE CONTRACTOR SHALL BE RESPONSIBLE FOR PAYING ALL FEES AND OBTAINING ALL PERMITS AND INSPECTIONS REQUIRED FOR THE WORK.
6. THE CONTRACTOR SHALL CAREFULLY EXAMINE ALL CONTRACT DOCUMENTS. THE CONTRACTOR SHALL COORDINATE THE WORK WITH ALL OTHER TRADES INCLUDING, BUT NOT LIMITED TO, THE CONTRACT DOCUMENTS, SHOP DRAWINGS, ETC., FOR ALL GENERAL CONSTRUCTION, STRUCTURAL, MECHANICAL, ELECTRICAL AND SPECIALTY CONTRACTOR WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER FITTING OF MATERIAL INTO THE BUILDING AS PLANNED, WITHOUT INTERFERENCE WITH OTHER WORK, AND SHALL MAKE REASONABLE MODIFICATIONS IN THE LAYOUTS NEEDED TO PREVENT CONFLICT WITH OTHER TRADES, TO PROVIDE ACCESS AND FOR THE PROPER EXECUTION OF THE WORK.
7. DRAWINGS ARE DIAGRAMMATIC AND SCHEMATIC IN NATURE, AND INDICATE THE TYPE, SIZE, ARRANGEMENT AND LOCATION OF MATERIALS AND EQUIPMENT. WORK INCLUDES CERTAIN COMPONENTS, APPURTENANCES AND RELATED SPECIALTIES THAT MAY NOT BE SHOWN. CONTRACTOR SHALL PROVIDE ALL NECESSARY ITEMS TO COMPLETE THE WORK ACCORDING TO INDUSTRY STANDARDS. IT IS THE INTENT OF THE DRAWINGS AND SPECIFICATIONS TO CALL OUT FOR FINISHED WORK, TESTED AND READY FOR OPERATION. DO NOT SCALE DRAWINGS. ARRANGEMENT OF EQUIPMENT AND ROUTING OF PIPES AND DUCTWORK, ETC. INDICATED ON DRAWINGS SHALL BE ROUTED PLUMB AND AT RIGHT ANGLES TO BUILDING CONSTRUCTION AND MAY REQUIRE MODIFICATION DUE TO UNFORESEEN CONDITIONS AND REQUIRE ON SITE REVISIONS DURING CONSTRUCTION. (SEE ALSO "BIDDING").
8. ALL WORK REQUIRED FOR IDENTICAL ITEMS SHOWN ON THE DRAWINGS SHALL BE PROVIDED, ALTHOUGH EACH SPECIFIC IDENTICAL ITEM MAY NOT BE SHOWN IN DETAIL.
9. THE CONTRACTOR SHALL SUBMIT TECHNICAL DATA SHEETS FOR ALL EQUIPMENT AND MATERIALS SPECIFIED HEREIN PER THE REQUIREMENTS OF THE ARCHITECT'S GENERAL PROVISIONS.
10. ALL SUBSTITUTIONS SHALL BE SUBMITTED FOR CONSIDERATION PER THE REQUIREMENTS OF THE ARCHITECT'S GENERAL PROVISIONS.
11. UPON COMPLETION OF CONSTRUCTION, THE CONTRACTOR SHALL SUPPLY COMPLETE AS-BUILT DOCUMENTS ACCURATELY SHOWING THE MATERIALS AND EQUIPMENT AS INSTALLED.
12. ALL MATERIALS AND WORKMANSHIP SHALL BE GUARANTEED PER ARCHITECTS GENERAL CONDITIONS. REFRIGERATION COMPRESSORS SHALL BE GUARANTEED FOR A MINIMUM OF FIVE (5) YEARS FROM DATE OF OWNER'S ACCEPTANCE. IN ADDITION, THE CONTRACTOR SHALL GUARANTEE THAT THE INSTALLATION WHEN OPERATED IN ACCORDANCE WITH THE CONTRACTOR'S INSTRUCTIONS WILL DEVELOP CAPACITY AND CHARACTERISTICS AS SPECIFIED AND WILL FULFILL EACH AND EVERY REQUIREMENT OF THE DRAWINGS AND SPECIFICATIONS. SHOULD THE INSTALLATION IN ANY WAY FAIL TO DO SO, THE CONTRACTOR WILL, WITHOUT DELAY OR WITHOUT COST TO THE OWNER, PROVIDE WHATEVER ADDITIONAL EQUIPMENT, MATERIAL, AND LABOR REQUIRED TO CORRECT THE DEFICIENCY AND COMPLY WITH THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATIONS.
13. CONTRACTOR SHALL CHECK AND VERIFY ALL SIZES, DIMENSIONS, AND CONDITIONS BEFORE STARTING ANY WORK. ANY DEVIATIONS OR PROBLEMS SHALL BE TRANSMITTED TO THE ENGINEER FOR REVIEW.
14. PROVIDE BASE AND COUNTER FLASHING FOR ITEMS PENETRATING THE ROOF OR EXTERIOR WALLS.
15. STARTERS AND CONTROLS FOR MOTORS, ETC. TO BE FURNISHED BY MECHANICAL CONTRACTOR. ELECTRICAL CONTRACTOR TO INSTALL THE AFOREMENTIONED ITEMS, AND FURNISH ALL POWER WIRING. ALL CONTROL AND INTERLOCKING WIRING SHALL BE FURNISHED AND INSTALLED BY MECHANICAL CONTRACTOR.
16. ALL WORK SHOWN IS NEW UNLESS NOTED OTHERWISE.
17. MAINTAIN OCCUPANCY AND FIRE WALL SEPARATION INTEGRITY AS REQUIRED. REFER TO ARCHITECTURAL PLANS FOR LOCATIONS OF ALL OCCUPANCY/FIREWALL SEPARATIONS AND SPECIFIC DETAILS FOR CONSTRUCTION. PROVIDE ALL NECESSARY FIRE AND SMOKE FIRE DAMPERS, ACCESS DOORS, CAULKING, ETC. FOR APPROVED INSTALLATION.

BIDDING

1. THE CONTRACTOR SHALL VISIT THE SITE PRIOR TO SUBMITTING A BID TO BECOME FAMILIAR WITH THE EXISTING CONDITIONS. THE CONTRACTOR SHALL COMPARE THE WORK SPECIFIED IN THE CONTRACT DOCUMENTS WITH THE EXISTING CONDITIONS. THE CONTRACTOR SHALL IDENTIFY AND NOTATE ALL WORK OR CONDITIONS THAT ARE DIFFERENT FROM THE CONTRACT DOCUMENTS OR THEIR INTENT. THE CONTRACTOR SHALL, UPON DISCOVERY, IMMEDIATELY NOTIFY AND REPORT, IN WRITING, ANY DISCREPANCIES TO THE ENGINEER. NO EXTRAS OR CHANGE ORDERS WILL BE ALLOWED FOR FAILURE TO PERFORM THE PRE-BID SITE VISIT.
2. BASE PROPOSAL ON MANUFACTURER NAMES LISTED UNLESS "OR EQUAL" IS INDICATED. PROVIDE SUBSTITUTION REQUESTS A MINIMUM OF FIVE (5) BUSINESS DAYS PRIOR TO BID DATE CLOSING TO ALLOW TIME FOR DUE CONSIDERATION OF PROPOSED ALTERNATE. DETERMINATION OF SUBSTITUTION OF EQUALITY RESTS SOLELY WITH THE ENGINEER.

PART TWO - PRODUCTS

PLUMBING EQUIPMENT

1. PROVIDE PLUMBING EQUIPMENT AS SPECIFIED AND/OR SCHEDULED HEREIN AND IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. EQUIPMENT SHALL OPERATE ACCORDING TO THE MANUFACTURER'S "OWNER'S OPERATING AND MAINTENANCE MANUAL" TROUBLE FREE AND CONFORMING TO THE ONE-YEAR WARRANTEE.

PLUMBING PRODUCTS

1. DOMESTIC WATER PIPING: (ABOVE GROUND) SHALL BE OF CPVC MEETING ASTM D1784 CELL CLASS 23447-B WITH SCHEDULE 80 PIPE MEETING ASTM D 1785 WITH CPVC SCHEDULE 80 SOCKET WELD FITTINGS MEETING ASTM F 439 (UNDER GROUND: PROTECTED FROM SOIL) TYPE "K" COPPER BELOW GRADE (ASTM B-88), HARD DRAWN, WROUGHT FITTINGS (ASME B16.22) JOINTS: AWS A5.8, BCuP SILVER BRAZE.
2. DOMESTIC WASTE & VENT PIPING MATERIALS: ABOVE AND BELOW GROUND SHALL BE PVC SCHEDULE 40 DWV ASTM D2665 PIPE WITH PVC MEETING ASTM B1784, "SOLID WALL" CELL CLASS 12454-B WITH ASTM 2665 SOCKET FITTINGS WITH SOLVENT WELD JOINTS IS ALSO PERMITTED WHERE APPROVED BY CODE.
3. PIPE INSULATION: ALL DOMESTIC HOT WATER PIPING ABOVE GROUND SHALL BE INSULATED WITH 1" THICK FIBERGLASS PIPE INSULATION WITH ALL-SERVICE JACKET AND MAXIMUM K VALUE OF 0.27 AT 75°F.
4. PIPE HANGERS: PIPE SIZES 1/2" TO 1 1/2": MALLEABLE IRON, CARBON STEEL, ADJUSTABLE SWIVEL, SPLIT RING. PIPE SIZES 2" TO 4": CARBON STEEL, ADJUSTABLE, CLEVIS. PIPE SIZES 6" AND OVER: ADJUSTABLE STEEL YOKE, CAST IRON ROLL, DOUBLE HANGER.

NOTES

1. DIELECTRIC FITTINGS SHALL BE USED WHEREVER DISSIMILAR METALS ARE JOINED.
2. ALL UNDERGROUND WATER PIPING SHALL BE WRAPPED WITH TAPE, ENCASED WITH A FOAM, COATED WITH A PROTECTIVE COATING, OR ROUTED THROUGH CONDUIT FOR PASSIVE CATHODIC PROTECTION. NO PIN HOLE LEAKS IN PROTECTIVE COATING OR TAPE ARE ALLOWED.
3. PROVIDE ACCESS PANELS IN CEILING TO ACCESS VALVES WHERE REQUIRED.
4. PLUMBING FIXTURES: PROVIDE CHROME PLATED ANGLE STOPS WITH ESCUTCHEON PLATES AT PLUMBING FIXTURES. ALL PLUMBING FIXTURES SHALL COMPLY WITH LOCAL REGULATIONS AND ADOPTED WATER CONSERVATION CODES.
5. DISINFECT ALL POTABLE WATER SYSTEMS IN ACCORDANCE WITH PLUMBING CODE AND/OR, AWWA STANDARD. PROVIDE WRITTEN CONFIRMATION TO OWNERS REPRESENTATIVE THAT THIS WORK HAS BEEN COMPLETED.

PART THREE - EXECUTION

1. THE CONTRACTOR SHALL PROVIDE ALL SLEEVES, OPENINGS, CUTTING AND PATCHING NECESSARY FOR THE INSTALLATION OF THE WORK. CUTTING AND PATCHING SHALL BE DONE BY WORKMEN SKILLED IN THE TRADES REQUIRED AND PAID BY THE CONTRACTOR REQUIRING THE WORK COMPLETED.
2. THE CONTRACTOR SHALL PROVIDE ALL RIGGING, HANDLING OF MATERIALS AND EQUIPMENT, AND THE NECESSARY PROTECTION FOR MATERIALS AND EQUIPMENT.
3. THE CONTRACTOR WILL PROTECT THE WORK AND MATERIAL AGAINST DIRT, THEFT, INJURY OR DAMAGE UNTIL ACCEPTED BY OWNER. ALL WORK SHALL BE TURNED OVER TO OWNER CLEAN AND IN NEW CONDITION.
4. PIPES AND/OR CONDUITS PASSING THROUGH WALL, FLOORS AND PARTITIONS SHALL BE PROVIDED WITH SLEEVES. SLEEVES PASSING THROUGH WATER PROOFING OR DAMP PROOFING SHALL BE WATER TIGHT. SLEEVES PASSING THROUGH FIRE RATED CONSTRUCTION SHALL BE FIRE PROOFED WITH MATERIAL APPROVED FOR THE FIRE RATING OF THE SEPARATION AREA AND U.L. LISTED.
5. EACH CONTRACTOR SHALL PROVIDE ALL FOUNDATIONS, HANGERS, AND SUPPORTS FOR ALL EQUIPMENT SUPPLIED AND/OR INSTALLED UNDER THEIR WORK. ANY EQUIPMENT WITH MOVING PARTS SHALL BE PROVIDED WITH VIBRATION ISOLATION AND FLEXIBLE CONNECTIONS TO PIPING IF APPLICABLE.
6. WHERE PIPES OR CONDUITS PASS THROUGH WALLS, FLOORS, OR CEILINGS IN FINISHED AREAS, THEY SHALL BE FURNISHED WITH ESCUTCHEON PLATES (COLOR PER ARCHITECT AND/OR INTERIOR DESIGNER).
7. AT THE CONCLUSION OF THE JOB, EACH PIECE OF EQUIPMENT, VALVE, SWITCH, STARTER, PANEL, PIPE LINE, CONDUIT, ETC., SHALL BE CLEARLY IDENTIFIED WHETHER EXPOSED OR CONCEALED, COVERED OR UNCOVERED, IN ACCORDANCE WITH OSHA AND ANSI REGULATIONS. IDENTIFY PIPES NEAR EACH VALVE WITH "BRANDY-PERMA' CODE PIPE TAPE" OR T. & B. WESTLINE "TEL-A-PIPE" INDICATING DIRECTION OF FLOW, SERVICE, ZONE, AND SIZE. TAPE SHALL BE APPLIED TO PIPE, CONDUIT, OR COVERING. VALVES AND CONTROLS SHALL BE IDENTIFIED BY 2-INCH LACQUERED BRASS TAGS WITH STAMPED LETTERS FASTENED WITH "S" HOOKS OR CHAINS. EQUIPMENT IS TO BE IDENTIFIED AS TO FUNCTION AND PURPOSE BY MEANS OF PERMANENTLY ATTACHED LAMINATED ENGRAVED PHENOLIC NAMEPLATES WITH BEVELED EDGES, AND WHITE LETTERS ON BLACK BACKGROUND. (NO ADHESIVE LABELS ALLOWED).
8. AT THE CONCLUSION OF THE WORK, ALL EQUIPMENT AND SYSTEMS SHALL BE BALANCED, ADJUSTED, AND TESTED TO PROVIDE A QUIET-OPERATING, STABLE, AND SAFELY OPERATING SYSTEM(S). DEMONSTRATE OPERATION OF ALL SYSTEMS TO THE OWNER'S DESIGNATED REPRESENTATIVE. THE TEST AND BALANCE WORK SHALL BE PERFORMED IN ACCORDANCE WITH NEBB OR AABC STANDARDS, BY INDEPENDENT, APPROVED, AND CERTIFIED TEST AND BALANCE PERSONNEL.
9. PIPE HANGERS: PIPE SIZES 1/2" TO 1 1/2" - 6'-0" MAX SPACING, 3/8" ROD DIAMETER; PIPE SIZES 2" TO 3" - 10'-0" MAX SPACING, 1/2" ROD DIAMETER; PIPE SIZES 4 TO 6"-10'-0" MAX SPACING, 5/8" ROD DIAMETER. DAMPERS.
10. WATER PROOFING AND FLASHING OF PIPE PENETRATIONS THROUGH EXTERIOR WALL AND ROOF SHALL BE BY THIS CONTRACTOR. PLUMBING CONTRACTOR SHALL COORDINATE LOCATIONS AND METHODS WITH GENERAL CONTRACTOR PRIOR TO CONSTRUCTION OF ROOF DECK.
11. CONTRACTOR SHALL OBTAIN FROM THE ARCHITECT THE EXACT LOCATION OF EQUIPMENT, PLUMBING FIXTURES, FLOOR DRAINS AND ANY OTHER APPARATUS SPECIFIED IN THESE DRAWINGS.
12. PROVIDE CLEAN OUTS IN SANITARY, WASTE AND DRAIN LINES AS SHOWN AND AS REQUIRED BY LOCAL CODE. ALL CLEANOUTS SHALL BE READILY ACCESSIBLE.

IMEG
www.imegcorp.com
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771

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RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Professional Engineer
Robert J. Kiesl
63742
08-22-22
Registered
ARIZONA U.S.A.

EXP. 03-31-2023

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description

PLUMBING SPECIFICATIONS

P001

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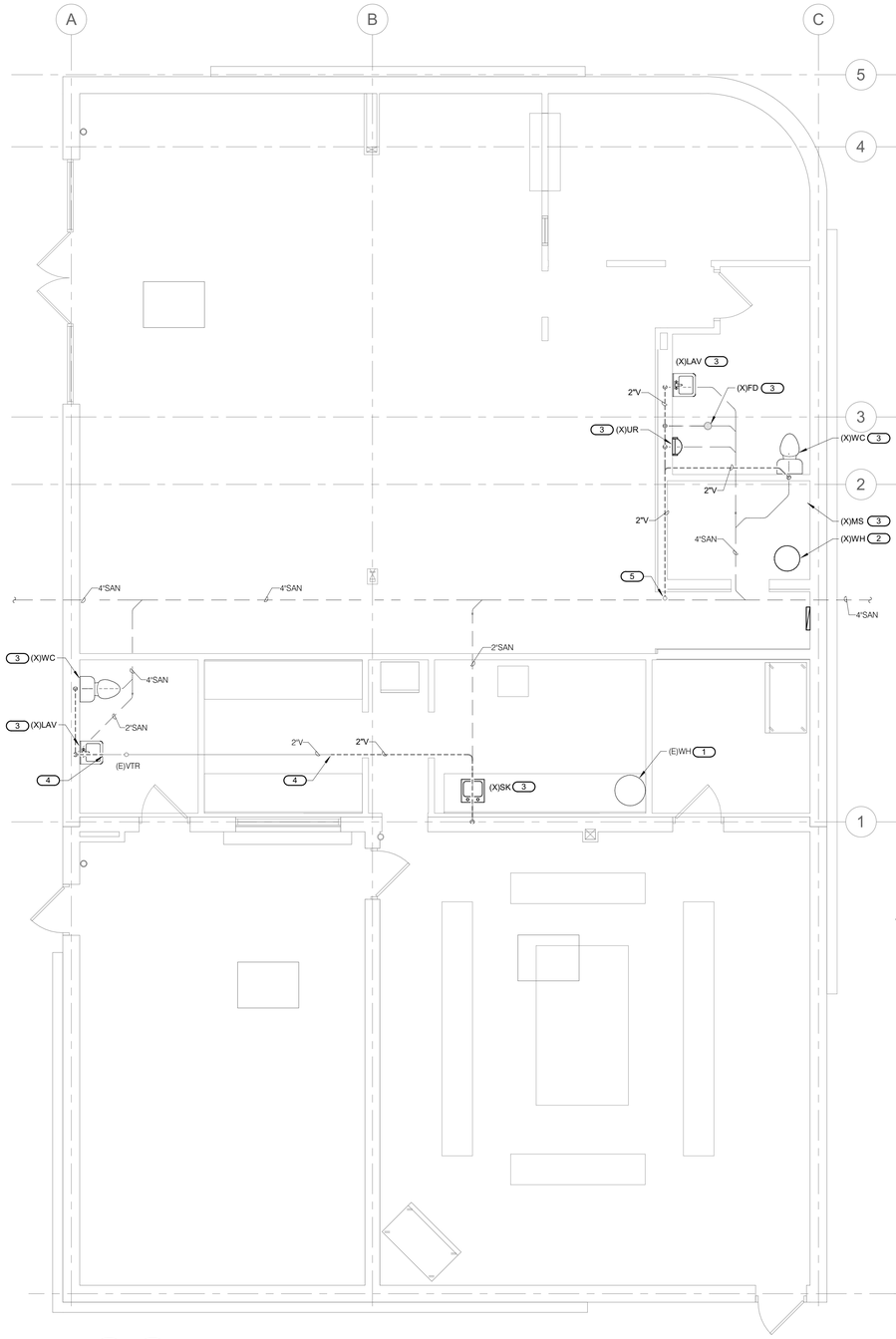
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GENERAL NOTES: DEMOLITION

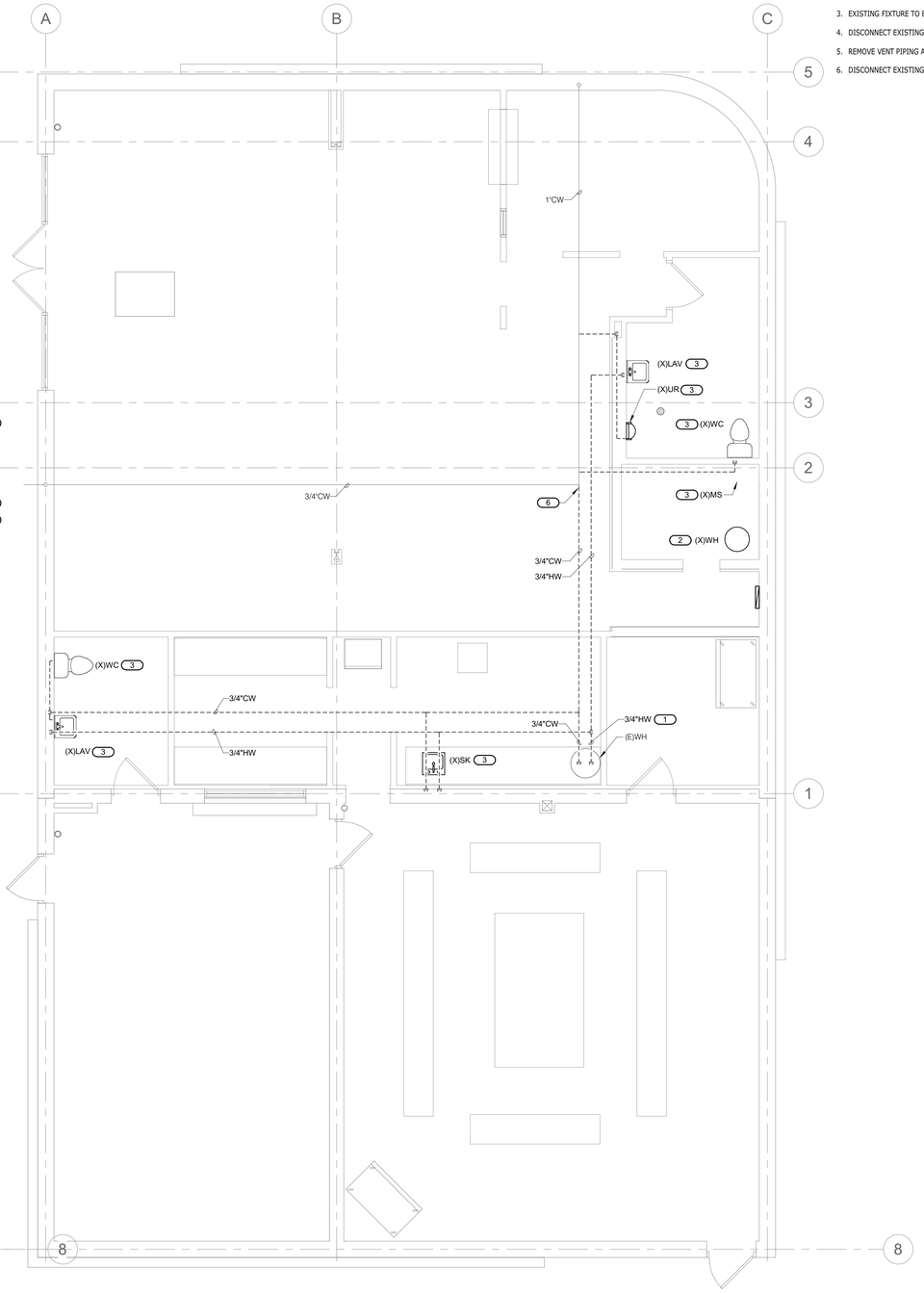
1. EQUIPMENT AND PIPING LOCATIONS SHOWN FROM BEST AVAILABLE INFORMATION. CONTRACTOR SHALL FIELD VERIFY SIZES AND LOCATIONS.
2. EQUIPMENT THAT IS BEING REMOVED SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND SHALL BE REMOVED FROM THE JOB SITE, EXCEPT EQUIPMENT SELECTED BY OWNER. OWNER SELECTED EQUIPMENT WILL BE TAGGED AND SHALL BE MOVED BY CONTRACTOR TO OWNER'S STORAGE ON SITE.
3. WHERE PIPING OR DUCTWORK IS TO BE CUT OFF AT A POINT, IT SHALL BE CAPPED OR BLANKED OFF AT THAT POINT. INSULATION ON REMAINING PIPE OR DUCT TO BE REPAIRED TO NEW CONDITION.
4. PIPING CONNECTED TO EQUIPMENT THAT IS BEING REMOVED SHALL BE CUT AND CAPPED IN WALLS, FLOORS OR CEILING SO AS NOT TO INTERFERE WITH NEW CONSTRUCTION OR EQUIPMENT.

SHEET NOTES: DEMOLITION

1. EXISTING WATER HEATER TO BE RELOCATED. DISCONNECT EXISTING PIPING AND ACCESSORIES FROM EXISTING WATER HEATER.
2. EXISTING WATER HEATER TO BE REMOVED.
3. EXISTING FIXTURE TO BE REMOVED. DEMOLISH EXISTING PIPING ABOVE GRADE AND CAP AT FLOOR.
4. DISCONNECT EXISTING 2" VENT IN CEILING AND PREPARE REMAINING PIPING FOR NEW CONNECTION.
5. REMOVE VENT PIPING AS SHOWN AND CAP VENT ABOVE ROOF.
6. DISCONNECT EXISTING CW LINE IN CEILING AND PREPARE REMAINING CW LINE FOR NEW CONNECTION.



2 NORTH **LEVEL 01 WASTE AND VENT DEMO PLUMBING PLAN**
1/4" = 1'-0"



1 NORTH **LEVEL 01 WATER DEMO PLUMBING PLAN**
1/4" = 1'-0"

RSP Architects
502 South College Avenu 480.889.2000
Suite 203 480.889.2099
Tempe, AZ 85281-3714 fax
www.rsparch.co
Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

REGISTERED PROFESSIONAL ENGINEER (61742) robert j. kiesl 07-29-22 ARIZONA U.S.A.
EXP. 03-31-2023

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	07/29/2022

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SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

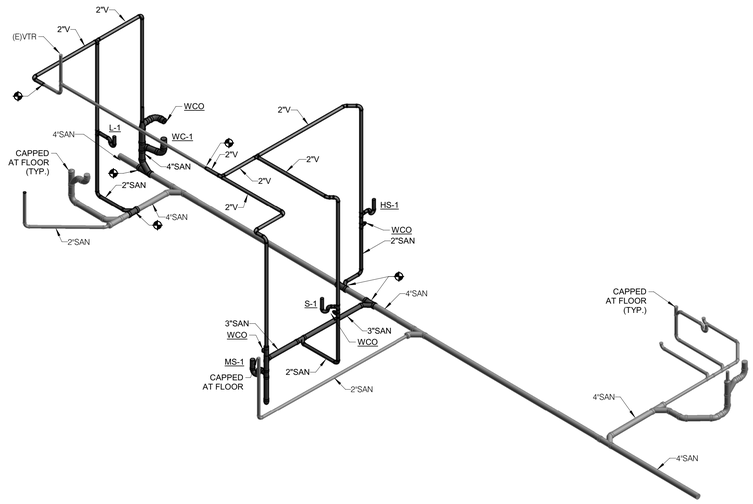
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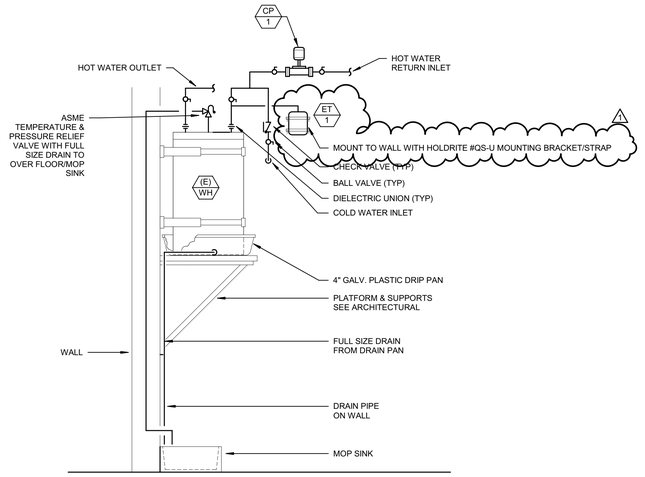
PLUMBING DEMO PLANS

P100

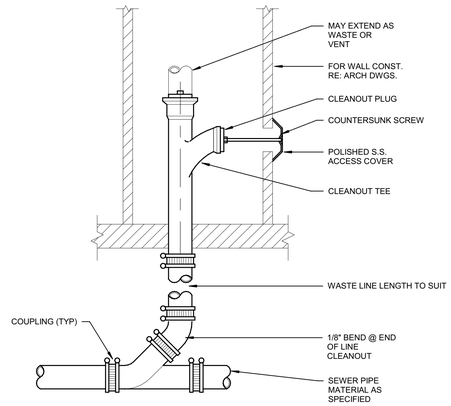
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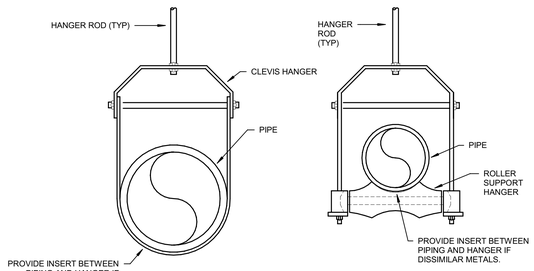
1 PLUMBING WASTE AND VENT DIAGRAM
NO SCALE



2 ELECTRIC WATER HEATER W/ RECIRC (ELEVATED LOW BOY)
NO SCALE



3 WALL CLEANOUT
NO SCALE



DETAIL NOTES:
1. USE CLEVIS HANGER FOR 1/2" UP TO 4".
2. USE ROLL SUPPORT HANGER FOR 6" TO 8".
3. PIPE 10" AND LARGER SHALL HAVE ROLLERS. SUPPORT HANGER WITH DUAL RODS. FASTEN TO TOP CHORD OF STRUCTURAL ELEMENTS.

4 PIPE HANGER FOR UNINSULATED PIPING
NO SCALE

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.co

Consultants

REGISTERED ARCHITECT
CERTIFICATE NO.
34046
MARIO P. TORREGROSSA
11/21/2022
STATE OF ARIZONA, U.S.A.
EXPIRES 09/30/2023

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	Author
Checked By	Checker
Date	08/23/2022

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IMEG
9000 E PIMA CENTER PARKWAY
SUITE 335
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

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PLUMBING DIAGRAMS

P400

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EXISTING ELECTRIC WATER HEATER SCHEDULE

MARK	GENERAL DATA					ELECTRICAL				OPERATING WEIGHT (LBS)	REMARKS
	MANUFACTURER MODEL	LOCATION	STORAGE CAPACITY (GAL)	RECOVERY RATE (GPH)	TEMP RISE (°F)	INPUT (KW)	QTY OF ELEMENTS	AMPS	V/PH/HZ		
(E)WH	RHEEM #XE38S06ST45U1	JANITOR	38	21	90	4.5	2	18.75	240/1/60	425	1, 2, 3

1. TP&R VALVE.
2. NON-SIMULTANEOUS OPERATION.
3. MOUNT ON WALL ABOVE MOP SINK. PROVIDE WITH HOLDRITE MODEL #50-SWHP-W MOUNTING PLATFORM. MOUNT TO WALL PER MANUFACTURER'S RECOMMENDATIONS.

PLUMBING FIXTURE SCHEDULE

MARK	FIXTURE	ROUGH-IN (INCHES)						REMARKS
		C.W.	TW	HW	V	S/W	TRAP	
WC-1	WATER CLOSET	3/4	-	-	2	4	-	FLOOR MOUNTED
L-1	LAVATORY	1/2	-	1/2	1 1/2	2	1 1/4 X 1 1/2	WALL HUNG - ACCESSIBLE
HS-1	SINK	3/4	-	3/4	2	2	1 1/4 X 1 1/2	DROP-IN
S-1	SINK	3/4	-	3/4	2	2	1 1/4 X 1 1/2	DROP-IN DOUBLE
MS-1	MOP SINK	3/4	-	3/4	2	3	-	FLOOR MOUNT

PROVIDE STOPS ON ALL FIXTURES NOT FURNISHED WITH INTEGRAL STOPS.
REFER TO PLANS FOR TEMPERED/HOT WATER LOCATIONS.

PLUMBING FIXTURE SPECIFICATIONS

MARK	DESCRIPTION
WC-1	WATER CLOSET (ACCESSIBLE) - AMERICAN STANDARD YORKVILLE 209AA137,FLOOR-MOUNTED ELONGATED BOWL. 1.28 GPF. SEAT: AMERICAN STANDARD #5901100 1/4 TURN ANGLE STOPS, BRAIDED STAINLESS STEEL SUPPLY.
L-1	LAVATORY (ACCESSIBLE) - LUCERNE 0355.012 WALL-HUNG LAVATORY "20-1/2 X 18-1/4", VITREOUS CHINA, FRONT OVERFLOW, FAUCET: CHICAGO FAUCETS #EVR-A12A-11ABCP, DECK MOUNTED TOUCHLESS FAUCET. 1/4 TURN ANGLE STOPS, BRAIDED STAINLESS STEEL SUPPLIES AND SEMI-CAST P-TRAP. PROVIDE ADA PIPE PROTECTION. PROVIDE WITH ASSE 1070 BELOW DECK THERMOSTATIC MIXING VALVE SET AT 110°F. REFER TO ARCHITECT'S ACCESSIBLY DETAILS PRIOR TO ROUGH-IN.
HS-1	SINK (ADA) - ELKAY #LRADQ1716S5, 17" x 16" x 15 1/2"D, 18 GAUGE, SINGLE FAUCET HOLE. FAUCET: CHICAGO FAUCETS #116.201.AB.1, BATTERY POWERED ELECTRONIC SENSOR OPERATED, 0.5 GPM NON. AERATING SPRAY 1/4 TURN ANGLE STOPS, BRAIDED STAINLESS STEEL SUPPLIES AND SEMI-CAST P-TRAP. PROVIDE ADA PIPE PROTECTION. PROVIDE WITH ASSE 1070 BELOW DECK THERMOSTATIC MIXING VALVE SET AT 110°F.
S-1	SINK (ADA) - ELKAY #LRAD291860, 29" x 18" x 6"D, DOUBLE BOWL, 18 GAUGE S.S., DROP-IN, SINGLE FAUCET HOLE. FAUCET: CHICAGO FAUCETS #434-ABCP, 8 1/4" REACH, SWING SPOUT, 1.5 GPM PULL-DOWN TYPE. PROVIDE WITH ASSE 1070 APPROVED BELOW DECK THERMOSTIC MIXING VALVE SET AT 110°F. 1/4 TURN ANGLE STOPS, BRAIDED STAINLESS STEEL SUPPLIES AND SEMI-CAST P-TRAP.
MS-1	MOP SINK - FIAT MODEL MSB2424, 24" X 24" X 10" DEEP, FLOOR MOUNTED, POLYMER CONSTRUCTION, CHICAGO 897-CP SERVICE FAUCET WITH WALL BRACE, HOSE END SPOUT, VACUUM BREAKER & INTEGRAL STOPS, 832-AA HOSE & BRACKET, 889-CC MOP HANGER, E-77-AA VINYL RIM GUARDS & MS2424 STAINLESS STEEL WALL PROTECTION PANELS. COMPLETE WITH P-TRAP.

RECIRCULATING PUMP SCHEDULE

MARK	MANUFACTURER MODEL	SERVICE	FLOW GPM	HEAD FT	MOTOR		REMARKS
					HP	V/PH/HZ	
CP 1	B & G NBF-8S/LW	DHW	1	3	39 WATTS	120/1/60	1

1. PROVIDE 24 HOUR TIME CLOCK TO CONTROL HOURS OF OPERATION.

EXPANSION TANK SCHEDULE

MARK	MANUFACTURER MODEL	LOCATION	TANK VOLUME (GAL)	ACCEPTANCE VOLUME (GAL)	AIR PRE-CHARGE (PSI)	OPERATING WEIGHT (LBS)	NOTES
ET 1	THERM-X-TROL ST-5	JANITOR	2	.9	70	20	1, 2, 3, 4

1. BLADDER TYPE.
2. 150 PSI WORKING PRESSURE FOR POTABLE HOT WATER USE.
3. APPLICATION PER FDA REQUIREMENTS.
4. MOUNT HIGH ON WALL WITH HOLDRITE #QS-U BRACKET AND STRAPS.

PLUMBING FIXTURE CALCULATIONS

FIXTURE TYPE	QTY	D.F.U. (EA)	TOTAL D.F.U.	HOT S.F.U. (EA)	COLD S.F.U. (EA)	COMBINED S.F.U. (EA)	TOTAL S.F.U. (HOT)	TOTAL S.F.U. (COLD)	TOTAL SERVICE S.F.U.
PRIVATE SINK (BAR, KITCHEN OR BREAKROOM)	2	2.0	4.0	1.00	1.00	1.40	2	2	2.8
PRIVATE LAVATORY	1	1.0	1.0	0.50	0.50	0.70	0.5	0.5	0.7
SERVICE SINK (MOP BASIN)	1	2.0	2.0	2.25	2.25	3.00	2.25	2.25	3.0
WALL HY DRANT / HOSE BIBB	1	0.0	0.0	0.00	2.50	2.50	0	2.5	2.5
WALL HY DRANT / HOSE BIBB (ADDITIONAL)	1	0.0	0.0	0.00	1.00	1.00	0	1	1.0
PRIVATE WATER CLOSET (FLUSH TANK)	1	3.0	3.0	0.00	2.20	2.20	0	2.2	2.2
TOTAL UNITS:	7		10.0				4.8	10.5	12.2

DOMESTIC WATER BRANCH PIPE SIZING CALC

PROJECT: THE FLOWER SHOP - MCDOWELL DATE: 04-09-2021
PROJECT NO.: 22004950.00 BY: IMEG CHECKED BY: PE
FLOW TEST COMPLETED ON 04/09/2022
DOMESTIC WATER PIPING SIZING BASED OFF OF IPC APPENDIX E
ESTIMATED CONNECTED WATER SUPPLY FIXTURE UNITS =12.2 WSFU (10 GPM)
EXISTING 1" METER
*PRESSURE READING TAKEN DOWNSTREAM OF SITE WATER METER / BACKFLOW PREVENTER

STATIC HEAD = 15 FT. x 0.43 = 6.45 PSI
PRESSURE AVAIL= 69 (SERVICE PRESS.) - (6.45 + 25 (REQ. PRESS.) + 0 METER/BFP= 37.55
LONGEST RUN = 100 FT.
25 % FITTING LENGTH = 25 FT.
TOTAL LENGTH = 125 FT.
ALLOWABLE FRICTION LOSS/100 FT. = $\frac{(37.55) * 100 \text{ PSI}}{125}$ = 30.04 "(6 PSI / 100' USED)

COLD WATER	MAX. FLOW (GPM)	FLUSH TANK F.U.	FLUSH VALVE F.U.	VELOCITY (FPS)
1/2"	2.7	2	-	3.5
3/4"	7	8	-	4.7
1"	15	21	-	5.7

HOT WATER	MAX. FLOW (GPM)	FLUSH TANK F.U.	FLUSH VALVE F.U.	VELOCITY (FPS)
1/2"	2.7	2	-	3.5
3/4"	7	8	-	4.7
1"	13	18	-	5.0

RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099
Tempe, AZ 85281-3714 fax
www.rsparch.co

Consultants

REGISTERED PROFESSIONAL ENGINEER
34046
MARIO P. TORREGROSSA
11/21/2022
ARIZONA, U.S.A.
EXPIRES 09/30/2023

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description
1	11/22/22	City Comments

IMEG
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

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0 1 2 3
REF. SCALE IN INCHES

PLUMBING SCHEDULES

P500

ELECTRICAL SYMBOL LIST

SYMBOL:	TAG:	DESCRIPTION:
GB	GB	GROUND BUS
IBT	IBT	INTERSYSTEM BONDING TERMINATION
E / E	ECONN	ELECTRICAL CONNECTION
J / J	JB	JUNCTION BOX
	FB-# or PT-#	FLOOR BOX or POKE THROUGH
RI	RI-TECH	TECHNOLOGY OUTLET ROUGH-IN
RI	RI-TECH-C	TECHNOLOGY ROUGH-IN, CEILING
W/RI	RI-TECH-W	TECHNOLOGY ROUGH-IN, WALL PHONE
TV	RI-TV	TV ANTENNA OUTLET ROUGH-IN
	WM-#	MULTI OUTLET SYSTEM
	WW-#	ELECTRICAL WIREWAY w/ DEVICES SHOWN
	PANEL '###'	PANELBOARD - RECESS MOUNT
	PANEL '###'	PANELBOARD - SURFACE MOUNT
	MX-#/MS-# /CB-#/CS-#	MANUAL SWITCH / STARTER / COMBINATION STARTER/ CIRCUIT BREAKER. REFER TO DISC/STA SCHEDULE
	DS-#/FDS-#/DSS-#	DISCONNECT. REFER TO DISC/STA SCHEDULE
S		FIRE ALARM DUCT SMOKE DETECTOR

ELECTRICAL SYMBOL LIST

SYMBOL:	TAG:	DESCRIPTION:
	REC-DUP	DUPLEX RECEPTACLE, 125V
	REC-DUP-GFI	DUPLEX GFI RECEPTACLE, 125V
W	REC-DUP-WP	DUPLEX GFI WEATHERPROOF RECEPTACLE 125V
U	REC-USB	DUPLEX RECEPTACLE, USB CHARGING
	REC-QUAD	QUAD RECEPTACLE, 125V
	REC-QUAD-GFI	QUAD GFI RECEPTACLE, 125V
U	REC-QUAD-USB	QUAD RECEPTACLE, USB 125V
W	REC-QUAD-WP	QUAD GFI WEATHERPROOF RECEPTACLE, 125V

ELECTRICAL SYMBOL LIST

SYMBOL:	TAG:	DESCRIPTION:
$	SW-1P	SWITCH - SINGLE POLE
$_W	SW-1P-WP	SWITCH - WEATHERPROOF
$_3	SW-3W	SWITCH - THREE WAY
$_4	SW-4W	SWITCH - FOUR WAY
$_D	SW-D-LED	DIMMER - LED
$_D3	SW-D3-LED	DIMMER - LED - 3-WAY
$_O	SW-OD	DIMMER - WALL DIMMER OCCUPANCY SENSOR
LS	SW-LS	DAYLIGHT LEVEL SENSOR
SW	SW	WALL CONTROL STATION

ELECTRICAL SYMBOL LIST

SYMBOL:	TAG:	SPEC SECTION:	DESCRIPTION:
	REFER TO LUMINAIRE SCHEDULE		LINEAR LUMINAIRES
			TROFFER
			WALL SCONCE LUMINAIRE
			DOWNLIGHT LUMINAIRE
			AIMABLE OR WALL WASH LUMINAIRE
			INDUSTRIAL LUMINAIRE
			WALL BRACKET LUMINAIRE
			POLE MOUNTED LUMINAIRE
			SINGLE FACE EXIT SIGN
			DOUBLE FACE EXIT SIGN
			WALL/CEILING EMERGENCY EXIT SIGN
			EMERGENCY UNIT

ELECTRICAL EQUIPMENT TAGS

TAG:	DESCRIPTION:
MX-#	MANUAL SWITCH, REFER TO DISCONNECT AND STARTER SCHEDULE
R-#	RELAY
SB-#	SWITCHBOARD
VFD-#	VARIABLE FREQUENCY DRIVE - REFER TO VFD SCHEDULE

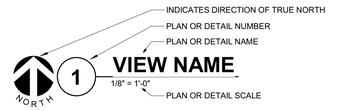
ELECTRICAL ABBREVIATION KEY

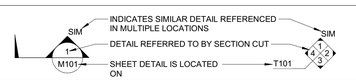
ABBR:	DESCRIPTION:
AFF	ABOVE FINISHED FLOOR
C	CONDUIT
GFI	GROUND FAULT INTERRUPTER
N.C.	NORMALLY CLOSED
NIC	NOT IN CONTRACT
N.O.	NORMALLY OPEN
SV	SOLENOID VALVE
TYP	TYPICAL
UON	UNLESS OTHERWISE NOTED

VIEW KEY

NAME — LEVEL NAME
'10'-0"' — HEIGHT ABOVE PROJECT 0'-0"

1 — INDICATES NOTE USED TO DESCRIBE ADDITIONAL INFORMATION ABOUT WORK REQUIRED. SPECIFIC TO THE SHEET AND/OR DETAIL





LINE TYPE AND TAG KEY:

NEW WORK BY THIS CONTRACTOR (WIDE LINE)
- NEW
- EXISTING TO BE REMOVED (SHORT DASHED PATTERN)
- NEW UNDERFLOOR OR UNDERGROUND (LONG DASHED PATTERN)

EXISTING TO REMAIN OR WORK BY OTHERS (NARROW LINE)
- EXISTING
- EXISTING TO BE REMOVED BY OTHERS (SHORT DASHED PATTERN)
- EXISTING UNDERFLOOR OR UNDERGROUND (LONG DASHED PATTERN)

HALFTONING DOES NOT MODIFY SCOPE.

'TAG'-E — TAGS WITH DASH 'E' INDICATES THE REFERENCED OBJECT IS EXISTING

TAG-1 — UNDERLINED TAG INDICATES OBJECT IS IN-SCOPE. IF NEW, ADDITIONAL INFORMATION IS AVAILABLE IN A SCHEDULE, MATERIAL LIST, OR SYMBOL LIST

INDICATES AN EXISTING SYSTEM'S POINT OF CONNECTION/REMOVAL

LUMINAIRE SYMBOL KEY

SYMBOL:	DESCRIPTION:
	NORMAL BRANCH LUMINAIRE
	BATTERY PACK EQUIPPED BRANCH LUMINAIRE UNSWITCHED FOR NIGHT LIGHT, UNLESS NOTED 'SE'

CONTRACTOR ABBREVIATION KEY

ABBR:	DESCRIPTION:
E.C.	ELECTRICAL CONTRACTOR
G.C.	GENERAL CONTRACTOR
M.C.	MECHANICAL CONTRACTOR
P.C.	PLUMBING CONTRACTOR
T.C.	TECHNOLOGY CONTRACTOR

COMcheck Software Version 4.1.5.4

Interior Lighting Compliance Certificate

Project Information

Energy Code: 2018 IECC
Project Title: THE FLOWER SHOP PHOENIX EXTERIOR IMPROVEMENTS
Project Type: New Construction

Construction Site: 3155 E. McDOWELL
Owner/Agent:
Designer/Contractor:

Additional Efficiency Package(s)
Credits: 1.0 Required 1.0 Proposed
Reduced Lighting Power, 1.0 credit

Allowed Interior Lighting Power

A Area Category	B Floor Area (ft2)	C Allowed Watts / ft2	D Allowed Watts (B X C)
1-Retail	2261	0.93	2157
		Total Allowed Watts =	2157

Proposed Interior Lighting Power

A Fixture ID : Description / Lamp / Wattage Per Lamp / Ballast	B Lamps / Fixture	C # of Fixtures	D Fixture Watt.	E (C X D)
1-Retail				
L1: L1: TROFFER 2X2: Other:	1	6	32	190
L2-4: L2-4: STRIP LIGHT: Other:	1	4	32	127
L3: L3: STRIP LIGHT: Other:	1	12	23	276
L4A: L4A: DOWNLIGHT: Other:	1	23	35	805
L4B: L4B: DOWNLIGHT: Other:	1	1	16	16
L5: L5: DOWNLIGHT: Other:	1	6	4	27
L2-2: L2-2: STRIP LIGHT: Other:	1	1	18	18
L6: L6: VANITY LIGHT: Other:	1	1	10	10
L7-L7: DOWNLIGHT: Other:	1	4	4	16
L8: L8: UNDERCABINET LIGHTS: Other:	1	3	11	32
		Total Proposed Watts =		1518

Interior Lighting PASSES: Design 30% better than code

Interior Lighting Compliance Statement

Compliance Statement: The proposed interior lighting design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed interior lighting systems have been designed to meet the 2018 IECC requirements in COMcheck Version 4.1.5.4 and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Hugo A. Chairez — Name - Title
Hugo A. Chairez — Signature
07/27/2022 — Date

Project Title: THE FLOWER SHOP PHOENIX EXTERIOR IMPROVEMENTS
Data filename: \\files\Active\Projects\2021\21006821.00\Design\Electrical\Flower Shop_Aman.cck
Report date: 07/27/22
Page 1 of 6

ELECTRICAL RENOVATION NOTES:

THESE NOTES APPLY TO ALL ELECTRICAL SHEETS AND TRADES, INCLUDING BUT NOT LIMITED TO, LIGHTING, POWER, AND SYSTEMS.

1. EXISTING CONDITIONS ARE SHOWN BASED ON INFORMATION OBTAINED FROM FIELD SURVEYS, EXISTING BUILDING DOCUMENTS, AND STAFF. VERIFY EXISTING CONDITIONS AND REPORT ANY CONFLICTS BEFORE PROCEEDING.
2. NOT ALL EXISTING EQUIPMENT, LUMINAIRES, AND CONDUIT ARE SHOWN. VERIFY EXISTING CONDITIONS AND REPORT ANY CONFLICTS WITH NEW WORK BEFORE STARTING WORK.
3. EACH CONTRACTOR SHALL FIELD VERIFY ACCESSIBILITY TO THE AREA OF THEIR WORK AND SHALL NOTIFY THE GENERAL CONTRACTOR PRIOR TO BIDDING IF OTHER UTILITIES ARE REQUIRED TO BE REMOVED OR RELOCATED TO ALLOW ACCESS TO THEIR AREA OF WORK.
4. THE GENERAL CONTRACTOR IS RESPONSIBLE FOR CUTTING, REMOVAL AND PATCHING OF ROOFS, WALLS, AND FLOORS ASSOCIATED WITH WORK BY ALL CONTRACTORS SHALL NOTIFY THE GC OF AFFECTED AREAS PRIOR TO BIDDING] [EACH CONTRACTOR SHALL CUT AND PATCH ROOFS, WALLS, AND FLOORS ASSOCIATED WITH THEIR WORK.
5. THE GENERAL CONTRACTORIS RESPONSIBLE FOR REMOVAL AND REPLACEMENT OF CEILINGS, CEILING TILES, AND CEILING GRIDS ASSOCIATED WITH AREAS OF WORK BY ALL CONTRACTORS. NOTIFY THE GENERAL CONTRACTOR OF AFFECTED AREAS PRIOR TO BIDDING.
6. WHERE EXISTING ELECTRICAL SYSTEMS ARE LOCATED IN AREAS THAT CONFLICT WITH NEW EQUIPMENT, PIPING, OR DUCTWORK TO BE INSTALLED, EACH CONTRACTOR SHALL EITHER ARRANGE NEW EQUIPMENT, CONDUIT, OR DUCTWORK IN SUCH A FASHION THAT IT DOES NOT CONFLICT WITH EXISTING SYSTEMS, OR REWORK EXISTING ELECTRICAL SYSTEMS TO ALLOW FOR INSTALLATION OF NEW EQUIPMENT, PIPING, OR DUCTWORK.

ELECTRICAL LIGHTING DEMOLITION NOTES:

1. THE ELECTRICAL LIGHTING DRAWINGS INDICATE EXISTING ELECTRICAL ITEMS TO BE REMOVED. THE DRAWINGS ARE INTENDED TO INDICATE THE SCOPE OF WORK REQUIRED AND DO NOT INDICATE EVERY BOX, CONDUIT, OR WIRE THAT MUST BE REMOVED. THE CONTRACTOR SHALL VISIT THE SITE PRIOR TO SUBMITTING A BID AND VERIFY EXISTING CONDITIONS.
2. EQUIPMENT REMOVAL IN CERTAIN LOCATIONS MAY REQUIRE THE INSTALLATION OF A JUNCTION BOX TO RECONNECT CIRCUITS THAT REMAIN IN OPERATION. EXTEND CONDUIT AND WIRING AS REQUIRED TO MAINTAIN POWER TO REMAINING EQUIPMENT.
3. BALLASTS MANUFACTURED PRIOR TO 1980 CONTAIN PCBs AND SHALL BE DISPOSED OF BY A FEDERAL OR STATE E.P.A. APPROVED METHOD AND IN ACCORDANCE WITH SPECIFICATIONS.
4. HID AND FLUORESCENT LAMPS CONTAIN MERCURY AND SHALL BE DISPOSED OF BY A FEDERAL OR STATE E.P.A. APPROVED METHOD AND IN ACCORDANCE WITH SPECIFICATIONS.
5. REUSE EXISTING CONDUIT, CIRCUITS AND LIGHTING CONTROL WHERE POSSIBLE. PROVIDE NEW CONDUIT AND WIRE WHERE SHOWN, MISSING OR REQUIRED TO INSTALL THE NEW LIGHT FIXTURES.
6. WHERE REMOVED EXTERIOR LIGHT FIXTURE IS NOT BEING REPLACED. PROVIDE WATERPROOF GROMMETS, SEALS OR PLUGS TO COVER EXISTING HOLES IN POLES.
7. VERIFY MANUFACTURERS INSTALLATION GUIDELINES WITH EXISTING FIELD CONDITIONS PRIOR TO BIDDING AND ORDERING NEW LIGHT FIXTURES AND INSTALLATION MATERIAL.
8. MATCH EXISTING PAINTED SURFACES. WHERE REPLACED LUMINAIRE DOES NOT FULLY COVER EXISTING JUNCTION BOX OR PAINTED SURFACE. PROVIDE CUSTOM BACK PLATE WHERE NECESSARY TO COVER ANY FIELD CONDITIONS THAT WOULD ALLOW INTRUSION OF WATER AND CAULK WHERE NECESSARY.

TYPICAL NEW CONSTRUCTION:

1. COORDINATE LUMINAIRES IN MECHANICAL ROOMS WITH DUCTWORK, PIPING AND ANY MECHANICAL EQUIPMENT. PROVIDE LUMINAIRE WITH CHAINS OR HANGER KIT WHERE REQUIRED.
2. COORDINATE ALL LUMINAIRE LOCATIONS WITH STRUCTURAL TRUSS AND BEAMS PRIOR TO ROUGH-IN.
3. LUMINAIRES SHALL BE TARGETED AND ADJUSTED BY THE CONTRACTOR UNDER THE OBSERVATION AND IN COMPLIANCE WITH RECOMMENDATIONS OF THE ENGINEER. ALL LABOR AND MATERIAL COSTS MADE NECESSARY BY THIS REQUIREMENT SHALL BE INCLUDED.

TYPICAL REMODEL:

1. ALL LUMINAIRES SHOWN TO BE DEMOLISHED SHALL BE DISPOSED OF IF NOT REQUIRED BY OWNER FOR STOCK. CONFIRM WITH OWNER PRIOR TO DISPOSAL IF THE LAMPS, LENS OR SUBSET OF LUMINAIRES SHOULD BE TURNED OVER FOR ATTIC STOCK.
2. REMOVE EXISTING LUMINAIRES AND WALL SWITCHES WHERE SHOWN. LOCATE AND IDENTIFY ELECTRICAL CIRCUIT SERVING REMOVED LUMINAIRES FOR REUSE WITH NEW DEVICES.
3. COORDINATE HOURS OF ACCESS WITH OWNER.
4. EXISTING EMERGENCY LIGHTING AND EXIT SIGNAGE SHALL REMAIN.
5. MATCH EXISTING FACEPLATE FINISH AND TYPE FOR ALL LOCATIONS WHERE NEW WALL CONTROL DEVICE IS BEING INSTALLED.
6. WHERE WALL SWITCH DEVICE IS REMOVED AND NOT REPLACED. PROVIDE WITH BLANK SWITCH PLATE.
7. EXPOSED 3/4" CONDUIT TO NEW OR EXISTING FIXTURES OR DEVICES IS ACCEPTABLE AS LONG AS IT IS INSTALLED IN A NEAT AND ORDERLY METHOD AND MEETS ADOPTED CODES. COORDINATE NEW RUNS WITH OWNER PRIOR TO INSTALLATIONS.
8. PROVIDE (1) UNSWITCHED LEG FROM PANEL SERVING THE EMERGENCY FIXTURES TO THE SENSOR LEG SERVING THE NEW BATTERY BACK UP IN NEW LUMINAIRES.
9. NEW OCCUPANCY SENSORS TO BE INSTALLED IN A MANUAL ON/AUTO OFF' CONFIGURATION.
10. COORDINATE LOCATIONS OF NEW LUMINAIRES WITH EXISTING DUCT, PIPING, STRUCTURAL AND CEILING MOUNTED DEVICES.

ELECTRICAL GENERAL NOTES:

1. "NL" INDICATES LUMINAIRE IS UNSWITCHED FOR NIGHT LIGHT.
2. SHADED LUMINAIRE OR DEVICE INDICATES LUMINAIRE OR DEVICE IS CONNECTED TO AN EMERGENCY CIRCUIT.
3. REFER TO SHEET E-101 FOR LIGHTING CONTROL SCHEDULE.
4. REFER TO SHEET E-101 FOR LUMINAIRE SCHEDULE.
5. (Z###) INDICATES THE LIGHTING ZONES FOR THE SPACE. PROVIDE SEPARATE CONTROL OF EACH CONTROLLED ZONE. LUMINAIRES ASSOCIATED WITH THE SAME ZONE SHALL OPERATE TOGETHER WITHIN THE SAME PROGRAMMED SCENE.
6. VACANCY/OCCUPANCY SENSOR LAYOUT: SENSORS ARE SHOWN ON THE PLANS FOR DESIGN INTENT AND MAY NOT REPRESENT EVERY DEVICE. PROVIDE MANUFACTURER SPECIFIC FLOOR PLAN LAYOUTS SHOWING LOCATION, ORIENTATION, AND COVERAGE AREA OF EACH CONTROL DEVICE, SENSOR, AND CONTROLLER/INTERFACE. AREAS REQUIRING MULTIPLE SENSOR DEVICES FOR APPROPRIATE COVERAGE, SUBMIT SPECIFIC MANUFACTURER-APPROVED SENSOR LAYOUT AS AN OVERLAY DIRECTLY ON THE PROJECT DRAWINGS, EITHER IN PRINT OR APPROVED ELECTRONIC FORM.

LUMINAIRE KEY:

LUMINAIRE — F1 = FIXTURE TAG
1 = CIRCUIT NUMBER
a = SWITCH DESIGNATION
NL = SUBSCRIPT (IF APPLICABLE)
Z = ZONE DESIGNATION

*IF LABEL IS ORIENTED HORIZONTALLY A SLASH WILL SEPARATE THIS INFORMATION. EX: F1 / 1 / a / NL

DEVICE KEY:

DEVICE — A = MOUNTING (IF APPLICABLE)
1 = CIRCUIT NUMBER

*IF LABEL IS ORIENTED HORIZONTALLY A SLASH WILL SEPARATE THIS INFORMATION. EX: A / 1

ELECTRICAL MOUNTING SUBSCRIPT KEY:

A	MOUNT AT +6" TO CENTERLINE ABOVE COUNTER OR BACKSPLASH
C	MOUNT AT CEILING
H	MOUNT ORIENTED HORIZONTALLY
L	MOUNT IN CASEWORK
M	MOUNT IN MODULAR FURNITURE
R	MOUNT IN SURFACE RACEWAY
EWC	ELECTRIC WATER COOLER

ELECTRICAL INSTALLATION NOTES:

1. THE COMPLETE INSTALLATION SHALL BE IN ACCORDANCE WITH THE ADA STANDARDS FOR ACCESSIBLE DESIGN. REFER TO THE ADA GUIDELINES FOR ALL CONFIGURATION DETAILS ON THIS PAGE FOR ADDITIONAL INFORMATION.
2. CIRCUIT NUMBERS ARE SHOWN FOR CIRCUIT IDENTIFICATION. CIRCUITING SHALL AGREE WITH NUMBERING ON THE PANEL PROVIDED. COMMON NEUTRALS MAY NOT BE USED FOR BRANCH CIRCUITS. BALANCE THE LOAD ON PANEL AS EVENLY AS POSSIBLE BETWEEN EACH PHASE.
3. FLUSH MOUNT ALL LIGHTING CONTROL DEVICES AT +42" FROM FLOOR (CENTERLINE DIMENSION), EXCEPT WHERE OTHERWISE NOTED. DEVICES MAY BE SURFACE MOUNTED WHEN CONDUIT IS SPECIFIED EXPOSED.
4. FLUSH MOUNT ALL DUPLEX RECEPTACLES AND TECHNOLOGY OUTLETS AT +18" FROM FLOOR (CENTERLINE DIMENSION), EXCEPT WHERE OTHERWISE NOTED. RECEPTACLES AND OUTLETS MAY BE SURFACE MOUNTED WHEN CONDUIT IS SPECIFIED EXPOSED. MOUNT EXTERIOR LOCATED RECEPTACLES WITH WHILE-IN-USE COVERS AT +20" FROM FINISHED GRADE (CENTER DIMENSIONS) TO MAINTAIN INSTALLATION ADA COMPLIANCE.
5. ALL MATERIALS USED TO SEAL PENETRATIONS OF FIRE RATED WALLS AND FLOORS SHALL BE TESTED AND CERTIFIED AS A SYSTEM PER ASTM E814 STANDARDS FOR FIRE TESTS OF THROUGH-PENETRATION FIRESTOPS.
6. CONNECTION FOR ELECTRIC WATER COOLERS (EWC) SHALL BE A JUNCTION BOX CONCEALED BEHIND WATER COOLER ACCESS PLATE OR BE A GFI RECEPTACLE LOCATED DIRECTLY BELOW AND CENTERED ON EWC. CONTRACTOR SHALL VERIFY TYPE OF EWC TO BE INSTALLED.
7. MOUNT ALL FIRE ALARM PULL STATIONS AT +42" FROM FLOOR (CENTERLINE DIMENSION) EXCEPT WHERE OTHERWISE NOTED.
8. INSTALL ALL WALL MOUNTED FIRE ALARM NOTIFICATION DEVICES AT 90" ABOVE FINISHED FLOOR OR 6" BELOW THE CEILING, WHICHEVER IS LOWER, EXCEPT WHERE OTHERWISE NOTED. HEIGHT SHALL BE MEASURED TO THE TOP OF THE DEVICE.
9. CONTRACTOR SHALL COORDINATE THE LOCATION OF ALL CEILING MOUNTED DEVICES AND EQUIPMENT WITH LUMINAIRES, SPRINKLER, AND CEILING DIFFUSERS. CENTER ALL DEVICES IN CEILING TILE PATTERN. SMOKE DETECTORS AND OCCUPANCY/VACANCY SENSORS SHALL BE LOCATED NO CLOSER THAN 3 FEET TO AN AIR SUPPLY DIFFUSER OR RETURN GRILLE.
10. ELECTRICAL EQUIPMENT SHALL BE MOUNTED TO AVOID IMPEDANCE OF, OPERATION OF, AND/OR ACCESS TO ELECTRICAL AND MECHANICAL EQUIPMENT. ALL MOUNTING OF ELECTRICAL EQUIPMENT SHALL BE APPROVED IN ADVANCE BY THE OTHER CONTRACTOR.
11. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL OPENINGS REQUIRED IN WALLS. ALL OPENINGS SHALL BE REPAIRED TO MATCH EXISTING BY A QUALIFIED CONTRACTOR AT THE EXPENSE OF THIS CONTRACTOR. ALL CONDUITS THROUGH WALLS SHALL BE GROUTED OR SEALED INTO OPENINGS.
12. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN AND OTHER ELECTRICAL PLANS FOR EXACT LOCATIONS OF ALL CEILING MOUNTED DEVICES, OTHER THAN SPRINKLERS.

LED LUMINAIRE SCHEDULE

CATALOG NUMBER SHALL NOT BE CONSIDERED COMPLETE AND MATERIAL SHALL NOT BE ORDERED BY MANUFACTURER AND CATALOG NUMBER ONLY. THE COMPLETE DESCRIPTION AND THE SPECIFICATION SHALL BE COORDINATED WITH THE CATALOG NUMBER TO DETERMINE THE EXACT MATERIAL AND ACCESSORIES TO BE ORDERED. THE FIRST MANUFACTURER LISTED IS THE BASIS OF DESIGN.

VERIFY AND COORDINATE ALL CEILING TYPES WITH LUMINAIRE MOUNTING AND TRIM REQUIREMENTS PRIOR TO THE RELEASE OF THE LUMINAIRE ORDER.
CONFIRM ALL COLORS AND FINISHES OF ALL LUMINAIRE COMPONENTS WITH ARCHITECT AND INTERIOR DESIGNER PRIOR TO THE RELEASE OF THE LUMINAIRE ORDER.
UNLESS INDICATED ON LIGHTING PLANS OR BELOW, REFER TO ARCHITECTURAL AND INTERIOR DESIGN ELEVATIONS, SECTIONS AND DETAILS FOR ALL SUSPENDED AND WALL MOUNTED LUMINAIRE MOUNTING HEIGHTS.

ITEM	DESCRIPTION	WATT ANSI WATTS	WATT PER	LED DELIVERED LUMENS (MIN)	DRIVER VOLTS	DRIVER TYPE	MANUFACTURER AND MODEL
EX	EXIT SIGN.	5 W	FIX	-	120 V	-	LITHONIA #EDG/EDGR SERIES
L1	2'X2' TROFFER.	32 W	FIX	4025	120 V	0-10V	LITHONIA #EPANL-2X2-4000LM-80CRI-35K-ZT-MVOLT SERIES
L1E	2'X2' TROFFER WITH EMERGENCY BATTERY BACKUP.	32 W	FIX	4025	120 V	0-10V	LITHONIA #EPANL-2X2-4000LM-80CRI-35K-ZT-MVOLT-E10WCP SERIES
L2-2	2' UTILITY STRIP LIGHT.	19 W	FIX	2372	120 V	0-10V	LITHONIA #CLX-L24-2500LM-SEF-FDL-MVOLT-GZ10-35K-80CRI SERIES
L2-4	4' UTILITY STRIP LIGHT.	32 W	FIX	4801	120 V	0-10V	LITHONIA #CLX-L48-5000LM-SEF-FDL-MVOLT-35K-80CRI SERIES
L2E-4	4' UTILITY STRIP LIGHT WITH EMERGENCY BATTERY BACKUP.	32 W	FIX	4801	120 V	0-10V	LITHONIA #CLX-L48-5000LM-SEF-FDL-MVOLT-35K-80CRI-PS1050 SERIES
L3	LINEAR LED WITH REMOTE DIMMABLE DRIVER.	23 W	FIX	1257	120 V	0-10V	LUMINII KENDO #KRS-LENGTH-72HO-35K-F-HS-XX SERIES LUMINII KENDO #K45M-LENGTH-72HO-35K-RF-CB-XX SERIES
L4A	4.5" DIAMETER DOWNLIGHT WITH SHALLOW HOUSING.	36 W	FIX	2678	120 V	0-10V	USAI #B4RDF-36E1-35KS-90-S-FINISH-FINISH-FT-UNV-D6E SERIES
L4AE	4.5" DIAMETER DOWNLIGHT WITH SHALLOW HOUSING AND EMERGENCY BATTERY BACKUP.	36 W	FIX	2678	120 V	0-10V	USAI #B4RDF-36E1-35KS-90-S-FINISH-FINISH-FT-UNV-D6E-EM SERIES
L4B	4.5" DIAMETER DOWNLIGHT.	16 W	FIX	1072	120 V	0-10V	USAI #B4RDF-16C3-35KS-90-S-FINISH-FINISH-NC-UNV-D6E SERIES
L5	LED FLUSHMOUNT.	5 W	FIX	1917	120 V	0-10V	TECH LIGHTING #700FMVGO-FINISH-LED930 SERIES
L6	BATHROOM SCONCE.	11 W	FIX	138	120 V	-	TOM DIXON #STW02WUL SERIES
L7	TASK LIGHTING WITH REMOTE DRIVER.	5 W	FIX	272	120 V	ELV	WAC #HR-AC70-CS-FINISH-3500K SERIES
L8	18" UNDERCABINET.	8 W	FIX	546	120 V	-	JUNO #UCES-18IN-SWW4-90CRI-WH-M6 SERIES

LIGHTING CONTROL SCHEDULE

AREA	PANEL	CIRCUIT	ZONE	CONTROL PROTOCOL	REMARKS
RETAIL: FRONT-OF-HOUSE - FLOOR	A	1	z1	0-10V	1,2,3,4
RETAIL: FRONT-OF-HOUSE - FLOOR	A	1	z1E	0-10V	1,2,3,4,5
RETAIL: FRONT-OF-HOUSE - COUNTER	A	1	z2	0-10V	1,2,3,4
RETAIL: FRONT-OF-HOUSE - COUNTER	A	1	z2E	0-10V	1,2,3,4,5
RETAIL: FRONT-OF-HOUSE - COUNTER	A	1	z3	ELV	1,2,3,4
RETAIL: FRONT-OF-HOUSE - CASEWORK LINEAR	A	3	z4	0-10V	1,2,3,4
RETAIL: FRONT-OF-HOUSE - CASEWORK DOWNLIGHT	A	1	z5	ELV	1,2,3,4
BUILDING EXTERIOR	A	7	z6	0-10V	2,3,4

LIGHTING CONTROL NOTES:

1. PROVIDE RAISE/LOWER CONTROLS AT LOW VOLTAGE LIGHTING CONTROL STATION FOR ZONE.
2. CONNECT TIME CLOCK. OPERATE PER OWNER DEFINED HOURS OF OPERATION.
3. PROVIDE TIME CLOCK OVERRIDE AT LOCAL CONTROLLER.
4. DIM TO 30% OUTPUT AFTER HOURS.
5. PROVIDE UL924 RELAYS FOR DIMMED EMERGENCY ZONES.

LIGHTING CONTROL GENERAL NOTES:

1. LIGHTING CONTROL SYSTEM SHALL BE NLIGHT BY ACUITY OR APPROVED EQUAL.
2. PROVIDE ALL NECESSARY COMPONENT FOR A FULLY FUNCTIONAL LIGHTING CONTROL SYSTEM.
3. PROVIDED DEDICATED POWER PACK PER ZONE.
4. PROVIDE LIGHTING CONTROL PLAN AS PART OF LIGHTING CONTROL SUBMITTAL.

ELECTRICAL SHEET INDEX

E000	ELECTRICAL COVERSHEET
E001	ELECTRICAL SPECIFICATIONS
E101	ELECTRICAL PLANS
E102	ELECTRICAL ROOF PLAN
E100	ELECTRICAL DEMO PLANS
E301	SINGLE LINE DIAGRAM AND CALCULATIONS
GRAND TOTAL: 6	

IMEG
9000 E PIMA CENTER PARKWAY SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

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REF. SCALE IN INCHES

RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099
Tempe, AZ 85281-3714 fax
www.rsparch.co

Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

JUSTIN C. DAGERMAN 32644 — 08-22-22 — ARIZONA, U.S.A.
EXP. 09-30-2023

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description

ELECTRICAL COVERSHEET

E000

PART ONE - GENERAL

1.1. THE WORK: ALL WORK SHALL BE NEW UNLESS OTHERWISE NOTED. THE CONTRACTOR SHALL PROVIDE THE WORK SHOWN ON THE DRAWINGS AND SPECIFIED FOR ITS INDIVIDUAL SECTIONS OF WORK. THE WORD "WORK" IS DEFINED AS ALL LABOR, TRANSPORTATION, MATERIAL, EQUIPMENT, TOOLS, INSTALLATION, SUPERVISION AND ANY OTHER INCIDENTAL ITEMS OR SERVICES NECESSARY FOR THE PROPER INSTALLATION AND OPERATION OF THE COMPLETE SYSTEMS, WHICH SHALL BE PROVIDED BY THIS CONTRACTOR WHETHER OR NOT SPECIFICALLY INDICATED OR NOTED.

1.2. RESPONSIBILITY: THIS CONTRACTOR IS SOLELY RESPONSIBLE FOR THE ACTIONS OF ITS PERSONNEL, SUPPLIERS, AND SUB-CONTRACTORS. THIS CONTRACTOR SHALL BE RESPONSIBLE FOR THE PERFORMANCE OF ALL WORK AS MAY BE REQUIRED TO ACCOMMODATE OR SUPPORT THE ELECTRICAL WORK. EXAMPLES: PAINTING, STRUCTURAL SUPPORTS, CUTTING AND PATCHING, EXCAVATION AND BACKFILL, CONCRETE PADS, ROOF JACKS, ETC. REQUIRING THIS CONTRACTOR'S ENGAGEMENT OF APPROPRIATE TRADES TO PERFORM SUCH WORK FOR THE PROPER INSTALLATION AND OPERATION OF COMPLETE ELECTRICAL SYSTEMS.

1.3. MINIMUM REQUIREMENTS: THESE SPECIFICATIONS ESTABLISH THE MINIMUM REQUIREMENTS FOR THE WORK AND MATERIALS, EQUIPMENT AND METHODS TO BE PROVIDED. THE DRAWINGS MAY INDICATE REQUIREMENTS WHICH EXCEED THESE MINIMUMS.

1.4. GENERAL CONDITIONS: ALL GENERAL CONDITIONS, SPECIAL REQUIREMENTS OR GENERAL REQUIREMENTS OF THE CONSTRUCTION SPECIFICATIONS ARE MADE PART OF THIS SPECIFICATION AND HAVE THE SAME FORCE AND EFFECT AS IF COMPLETELY REPRODUCED.

1.5. DEFINITIONS:
AHJ: AUTHORITY HAVING JURISDICTION.
ASSEMBLY: AN INSTALLATION OR SYSTEM OF MULTIPLE COMPONENTS REQUIRING MULTIPLE CONNECTIONS. (EXAMPLES: TRASH COMPACTOR, MOTORIZED DOOR, HVAC SPLIT SYSTEM, ETC.).
EQUAL: ACCEPTED BY THE ENGINEER AS EQUAL.
FF&E: FURNISHINGS, FIXTURES AND EQUIPMENT - PROVIDED BY OTHERS AT JOBSITE. RECEIVE, PROTECT, STORE, ASSEMBLE, INSTALL AND CONNECT. PROVIDE MINIMUM 5x STRUCTURAL BACKING. (EXAMPLES: CHANDELIERS, PROJECTORS, ETC.).
PROVIDE: FURNISH, INSTALL, ACTIVATE, AND COMMISSION.

1.6. CODES: ALL WORK SHALL BE DONE IN ACCORDANCE WITH THE LATEST ADOPTED EDITIONS OF THE NATIONAL ELECTRICAL CODE (NEC) AND ALL OTHER APPLICABLE FEDERAL, STATE, AND LOCAL REGULATIONS.

1.7. PERMITS: PAY ALL FEES AND OBTAIN ALL PERMITS AND INSPECTIONS REQUIRED FOR THE WORK.

1.8. DRAWINGS: DRAWINGS ARE DIAGRAMMATIC AND SCHEMATIC IN NATURE, AND INDICATE THE TYPE, SIZE, ARRANGEMENT AND LOCATIONS OF MATERIALS AND EQUIPMENT. WORK INCLUDES CERTAIN COMPONENTS, APPURTENANCES, AND RELATED SPECIALTIES THAT MAY NOT BE SHOWN. PROVIDE ALL NECESSARY ITEMS TO COMPLETE THE WORK ACCORDING TO INDUSTRY STANDARDS. IT IS THE INTENT OF THE DRAWINGS AND SPECIFICATIONS TO REQUIRE FINISHED WORK, TESTED AND READY FOR OPERATION. DO NOT SCALE DRAWINGS. ARRANGEMENT OF EQUIPMENT AND ROUTING OF FEEDERS AND BRANCH CIRCUITING SHALL BE PLUMB AND AT RIGHT ANGLES TO BUILDING CONSTRUCTION, AND MAY REQUIRE MODIFICATION DUE TO UNFORESEEN CONDITIONS REQUIRING ONSITE REVISIONS DURING CONSTRUCTION.

1.9. COORDINATION: THIS PROJECT REQUIRES A HIGH LEVEL OF COORDINATION AND COOPERATION WITH OWNER, ARCHITECT, OTHER TRADES, VENDORS, AND SPECIALTY CONTRACTORS. CAREFULLY EXAMINE ALL CONTRACT DOCUMENTS INCLUDING, BUT NOT LIMITED TO, SHOP DRAWINGS, ETC. FOR ALL GENERAL CONSTRUCTION, STRUCTURAL, MECHANICAL, PLUMBING, ELECTRICAL, AND SPECIALTY CONTRACTOR WORK. PRIOR TO ROUGH-IN, COORDINATE THE WORK WITH ALL OTHER TRADES, TAKING RESPONSIBILITY FOR THE PROPER FITTING OF MATERIAL INTO THE BUILDING AS PLANNED WITHOUT INTERFERENCE WITH OTHER WORK. ESTABLISH AND VERIFY LOCATIONS, HEIGHTS, CONNECTION METHODS, ETC. WITH EQUIPMENT INSTALLER (AND OWNER, ARCHITECT, AND/OR INTERIOR DESIGNER FOR FF&E ITEMS), AND MAKE REASONABLE MODIFICATIONS IN THE LAYOUTS NEEDED TO PREVENT CONFLICTS WITH OTHER TRADES IN ORDER TO PROVIDE ACCESS FOR THE PROPER EXECUTION OF THE WORK.

1.10. IDENTICAL: ALL WORK REQUIRED FOR IDENTICAL ITEMS AND ASSEMBLIES OF THE PROJECT SHALL BE PROVIDED, ALTHOUGH EACH SPECIFIC IDENTICAL ITEM MAY NOT BE SHOWN IN DETAIL.

1.11. VERIFICATION: CHECK AND VERIFY ALL SIZES, DIMENSIONS, AND CONDITIONS BEFORE STARTING ANY WORK. ANY DEVIATION(S) OR PROBLEM(S) SHALL BE TRANSMITTED TO THE ENGINEER FOR REVIEW.

1.12. CONNECTIONS: CONNECT ALL EQUIPMENT, SYSTEMS, AND ASSEMBLIES PROVIDED BY OTHERS INCLUDING CONTROLS, SAFETY DEVICES AND INTERCONNECTIONS. EXCEPTION: DO NOT INTERCONNECT THE CONTROL SYSTEMS OF THOSE MECHANICAL AND PLUMBING SYSTEMS WHICH ARE SPECIFICALLY NOTED TO BE THE RESPONSIBILITY OF THOSE TRADES. PROVIDE FUSIBLE DISCONNECT SWITCHES AND MOTOR STARTERS FOR ALL EQUIPMENT EXCEPT THOSE ITEMS WHICH ARE SPECIFICALLY LISTED WITH INTEGRAL STARTERS/DISCONNECT SWITCHES. WHERE STARTERS AND/OR DISCONNECT SWITCHES ARE FURNISHED TOGETHER WITH EQUIPMENT, RECEIVE, INSTALL, AND CONNECT THOSE ITEMS.

1.13. SUBMITTAL: SUBMIT TO THE ENGINEER COMPLETE ELECTRONIC SETS OF SHOP DRAWINGS AND TECHNICAL DATA SHEETS FOR ALL EQUIPMENT AND MATERIALS SPECIFIED HEREIN. THE ENGINEER SHALL REVIEW SHOP DRAWINGS AND TECHNICAL DATA SHEETS FOR CONFORMANCE WITH THE CONTRACT DOCUMENTS AND ISSUE A WRITTEN ASSESSMENT TO THE OWNER PRIOR TO COMMENCEMENT OF WORK. THE ENGINEER'S FAILURE TO CORRECT ERRORS IN THE SUBMITTAL SHALL NOT RELIEVE THE CONTRACTOR OF THE OBLIGATION TO PERFORM THE WORK AS SHOWN AND/OR SPECIFIED. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL ENGINEERING FEES NECESSARY TO CHANGE PROJECT DOCUMENTS BASED ON ALTERNATE SUBMITTAL PACKAGES/EQUIPMENT SUBSTITUTIONS.

1.14. OR-EQUAL SUBSTITUTIONS: ALL PROPOSED "OR EQUAL" SUBSTITUTIONS SHALL BE SUBMITTED TO THE ENGINEER FOR CONSIDERATION PRIOR TO BIDDING AND AFTER ALL REQUIREMENTS ASSOCIATED WITH SUBSTITUTED EQUIPMENT AND/OR MATERIALS HAVE BEEN COORDINATED WITH OTHER BUILDING TRADES, INCLUDING ALL MECHANICAL, STRUCTURAL, AND/OR ARCHITECTURAL ELEMENTS. THE OWNER'S REPRESENTATIVE SHALL PRE-APPROVE ANY PROPOSED SUBSTITUTION IN WRITING. IDENTIFY AND ANNOTATE ALL REVISED REQUIREMENTS PER BUILDING TRADE ON THE SHOP DRAWINGS. ALSO IDENTIFY ALL COST DEBITS OR CREDITS IN WRITING FOR THE PROPOSED CHANGES PER BUILDING TRADE AND SUMMARIZE THESE AS A TOTAL NET-TO-OWNER CHARGE OR CREDIT FOR CONSIDERATION.

1.15. AS-BUILT: UPON COMPLETION OF CONSTRUCTION, SUPPLY THE ENGINEER WITH AS-BUILT DOCUMENTS ACCURATELY SHOWING THE MATERIALS AND EQUIPMENT AS INSTALLED. PROVIDE OPERATION AND MAINTENANCE MANUAL(S) CONTAINING APPROVED SHOP DRAWINGS, OPERATING AND MAINTENANCE INSTRUCTION FOR SWITCHGEAR, LIGHTING FIXTURES, CONTROLS, AND SPECIALTY EQUIPMENT.

1.16. SPECIAL INSPECTIONS: OBTAIN A SPECIAL INSPECTION BY A QUALIFIED INDEPENDENT THIRD-PARTY INDIVIDUAL(S), FIRM(S), OR TESTING AGENCY PER PBCC SECTION 2703. THE SPECIAL INSPECTOR SHALL FURNISH A REPORT TO THE ENGINEER FOR REVIEW AND APPROVAL OF THE FOLLOWING ITEM(S):
ITEM:
SWITCHBOARDS, PANELBOARDS AND MCCS RATED OVER 1000A OR OVER 600V

1.17. GUARANTEE: ALL MATERIALS AND WORKMANSHIP SHALL BE GUARANTEED FOR A MINIMUM OF ONE (1) YEAR FROM DATE OF ACCEPTANCE BY OWNER (LONGER IF REQUIRED BY GENERAL AND/OR SPECIAL CONDITIONS). IN ADDITION, THE INSTALLATION SHALL BE GUARANTEED TO PERFORM AS SPECIFIED AND FULFILL EACH AND EVERY REQUIREMENT OF THE DRAWINGS AND SPECIFICATIONS WHEN OPERATED IN ACCORDANCE WITH THE CONTRACTOR'S INSTRUCTIONS. SHOULD THE INSTALLATION IN ANY WAY FAIL TO DO SO, THE CONTRACTOR WILL, WITHOUT DELAY AND WITHOUT COST TO THE OWNER, PROVIDE WHATEVER ADDITIONAL EQUIPMENT, MATERIAL, AND LABOR REQUIRED TO CORRECT THE DEFICIENCY AND COMPLY WITH THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATIONS. WHERE SPECIFIED EQUIPMENT HAS A LONGER GUARANTEE PERIOD, THE TERMS OF THAT GUARANTEE SHALL GOVERN (EXAMPLE: LED SYSTEM WITH 5 YEAR GUARANTEE). INCANDESCENT LAMPS ARE EXEMPT BUT SHALL BE NEW AND UNUSED AT THE TIME OF FINAL ACCEPTANCE.

1.18. IECC COMPLIANCE: COMPLY WITH ALL REQUIREMENTS SET FORTH IN THE IECC COMPLIANCE CERTIFICATE INCLUDED IN THESE DOCUMENTS. HIRE A COMMISSIONING AGENT TO COMPLY WITH AND PERFORM ALL ASPECTS OF SECTION C408 OF THE 2018 IECC.

BIDDING

1.19. SITE VISIT: CONTRACT DOCUMENTS INDICATE NEW WORK TO BE PERFORMED AND DO NOT PURPORT TO SHOW ALL EXISTING CONDITIONS. VISIT THE SITE PRIOR TO SUBMITTING A BID TO BECOME FAMILIAR WITH EXISTING CONDITIONS. COMPARE THE WORK SPECIFIED IN THE CONTRACT DOCUMENTS AGAINST EXISTING CONDITIONS, AND IDENTIFY AND ANNOTATE ALL WORK OR CONDITIONS THAT ARE DIFFERENT FROM THE CONTRACT DOCUMENTS OR THEIR INTENT. UPON DISCOVERY, IMMEDIATELY NOTIFY AND REPORT IN WRITING ANY DISCREPANCIES TO THE ENGINEER. NO EXTRAS OR CHANGE ORDERS WILL BE ALLOWED FOR FAILURE TO PERFORM THE PRE-BID SITE VISIT.

1.20. BASIS OF PROPOSAL: PROPOSAL SHALL BE BASED ON MANUFACTURERS AND MODELS AS LISTED UNLESS "OR EQUAL" IS INDICATED. PROVIDE SUBSTITUTION REQUESTS A MINIMUM OF FIVE (5) BUSINESS DAYS PRIOR TO BID DATE CLOSING TO ALLOW TIME FOR DUE CONSIDERATION OF PROPOSED ALTERNATE AND SUBSEQUENT NOTIFICATION TO ALL OTHER BIDDERS IN THE EVENT SUBSTITUTION IS DEEMED ACCEPTABLE. DETERMINATION OF SUBSTITUTION EQUALITY RESTS SOLELY WITH THE ENGINEER.

1.21. VALUE ENGINEERING (V.E.) INITIATIVES: IN ADDITION TO THE "AS SPECIFIED/OR EQUAL" BASE BID, A COST REDUCTION INITIATIVE(S) MAY BE PROPOSED BASED ON SUBSTITUTIONS OF EQUIPMENT, MATERIALS, AND/OR METHODS. EACH SUCH PROPOSAL SHALL INCLUDE A DATA SHEET(S) ON THE SPECIFIED ITEM(S), THE PROPOSED SUBSTITUTE(S), AND THE NET CREDIT TO THE OWNER, INCLUDING ALL CREDITS AND CHARGES FROM ALL MEMBERS OF THE CONSTRUCTION TEAM. THE ENGINEER WILL REVIEW AND RENDER AN OPINION TO THE OWNER. IF THE V.E. INITIATIVE IS DECLINED, PROVIDE THE SPECIFIED EQUIPMENT/MATERIAL/METHOD. IF THE V.E. INITIATIVE IS ACCEPTED, AND IF SUCH ACCEPTANCE RESULTS IN A REQUIREMENT TO REVISE ANY DESIGN DOCUMENTS, THE CHARGES FOR THESE REVISIONS SHALL BE BILLED TO THE CONTRACTOR AND THE INVOICING SHALL BE SETTLED BEFORE THE PROJECT IS SIGNED OFF FOR FINAL ACCEPTANCE.

1.22. BIDDING: THE CIVIL, ARCHITECTURAL, MECHANICAL AND/OR INTERIOR DRAWINGS CONTAIN DETAILED DESCRIPTIONS, CIRCUITING, AND CONNECTION REQUIREMENTS WHICH ARE PART OF THIS CONTRACTOR'S RESPONSIBILITIES. DO NOT SUBMIT BIDS ON THIS PROJECT PRIOR TO REVIEWING ALL PROJECT DRAWINGS, SPECIFICATIONS AND ADDENDA.

PART TWO - PRODUCTS

2.1. EQUIPMENT STANDARDS: ALL MATERIALS AND EQUIPMENT SHALL BE NEW AND OF THE HIGHEST QUALITY AVAILABLE ("SPECIFICATION GRADE"). EQUIPMENT SHALL BE CONSTRUCTED TO NEMA STANDARDS AND SHALL BE LABELED FOR THEIR INTENDED PURPOSE BY A RECOGNIZED TESTING AGENCY ACCEPTABLE TO THE AHJ (U.L., CSA, ETL, ETC.).

2.2. ACCEPTABLE MANUFACTURERS AND SUPPLIERS: WHERE EQUIPMENT AND MATERIALS ARE NOT SPECIFIED BY NAME THEY ARE DEEMED TO GENERIC, SUBJECT TO THE REQUIREMENTS LISTED HEREIN. THESE MANUFACTURERS ARE CONSIDERED CAPABLE OF OFFERING EQUIVALENT PRODUCTS. MINIMUM STANDARD IN ALL INSTANCES IS COMMERCIAL GRADE:
LIGHT FIXTURES: ACUITY, COOPER, HUBBELL, THOMAS
WIRING DEVICES: HUBBELL, LEVITON, LEGRAND, WIREMOLD

2.3. CIRCUITING: ALL WIRING SHALL BE IN CONDUIT, CONCEALED EXCEPT WHERE NOTED. EMT WITH STEEL INSULATED THROAT SET SCREW FITTINGS MAY BE USED IN DRY, PROTECTED INTERIOR LOCATIONS. PVC SCHEDULE 40 SHALL BE USED BELOW GRADE AT MINIMUM -24". WRAPPED RIGID ELBOWS AND RISERS SHALL BE USED FOR ALL THROUGH-GRADE TRANSITIONS AND STUB-UPS. RGS OR IMC CONDUIT WITH THREADED FITTINGS SHALL BE USED IN ALL LOCATIONS WHERE EXPOSED TO THE ELEMENTS OR SUBJECT TO PHYSICAL DAMAGE. IMC OR RIGID CONDUIT BELOW GRADE SHALL BE HALF-LAP WRAPPED WITH 20 MIL PVC TAPE. TYPE ENT RACEWAY IS NOT ALLOWED. CONNECT RECESSED AND SUSPENDED LIGHTING FIXTURES, MOTORIZED AND/OR VIBRATING EQUIPMENT WITH STEEL FLEX OR SEALTITE CONDUIT. ALL CONDUIT SHALL HAVE PULL CORD IF OTHERWISE EMPTY.

2.4. MC CABLE: MC CABLE MAY BE USED IN LOCAL 1- AND 2-CIRCUIT APPLICATIONS ACCEPTABLE TO THE AHJ. HOMERUNS AND FEEDERS SHALL BE CONDUIT AND WIRE.

2.5. WIRING: ALL WIRE SHALL BE COPPER UNLESS OTHERWISE NOTED, STRANDED IN SIZES #8 AWG AND LARGER. WHERE ALUMINUM IS INDICATED, WIRE SHALL BE COMPACTED-STRAND TYPE WITH JOINT COMPOUND AT TERMINATIONS. INSULATION SHALL BE TYPE THWN OR THHN (XHHW FOR ALUMINUM). ALUMINUM CONDUCTORS SHALL NOT BE USED IN SIZES SMALLER THAN #1/0 (100A EQUIPMENT FEEDER), AND WHEN USED SHALL BE TERMINATED IN INSULATED COMPRESSION-TYPE CU/AL FITTING. SINGLE PHASE BRANCH CIRCUITS SHALL INCLUDE A SEPARATE NEUTRAL WIRE WITH EACH PHASE WIRE. NEUTRAL SHALL BE WHITE WITH COLOR STRIPE MATCHING COLOR OF PHASE WIRE.

2.6. FUSES AND CIRCUIT BREAKERS: FUSES AND CIRCUIT BREAKERS SHALL BE SIZED PER ACTUAL RESPECTIVE APPLICATION (i.e., MOTOR CIRCUIT PROTECTOR, GROUND FAULT CIRCUIT INTERRUPTER, ARC FAULT CIRCUIT INTERRUPTER, ETC.). FUSES SHALL BE DUAL ELEMENT, CURRENT-LIMITING, AND SHALL BE INTERCHANGEABLE BETWEEN FRAME SIZES WITH STANDARD FACTORY FUSE REDUCERS. PROVIDE LOCKABLE SPARE FUSE CABINET WITH (3) SPARE FUSES OF EACH SIZE USED.

2.7. PANELBOARDS: PANELS SHALL HAVE COPPER BUS AND HARDWARE, BOLT-ON CIRCUIT BREAKERS, FLUSH MONO-FLAT TRIM, PIANO HINGED DOORS AND COVER (DOOR-IN-DOOR) WITH LOCKABLE MASTER-KEYED FLUSH LATCHES. FLUSH-MOUNTED PANELS SHALL HAVE EMPTY CONDUITS STUBBED TO ACCESSIBLE ATTIC SPACE: (1) 3/4" CONDUIT FOR EACH THREE (3) SPARE/SPACE CIRCUITS.

2.8. SAFETY SWITCHES: SWITCHES SHALL BE GENERAL DUTY UP TO 250 VOLTS, HEAVY DUTY ABOVE 250 VOLTS. FUSIBLE SWITCHES SHALL BE FUSED PER THE NAMEPLATE REQUIREMENTS OF THE EQUIPMENT BEING CONNECTED.

2.9. MOTOR STARTERS: STARTERS SHALL BE MINIMUM NEMA SIZE 1 WITH INTEGRAL CONTROL TRANSFORMER, RED NEON "RUN" PILOT LIGHT AND "ON-OFF-AUTO" SELECTOR SWITCH ON COVER. OVERLOAD DEVICES SHALL BE SIZED PER THE NAMEPLATE AMPERAGE OF THE EQUIPMENT BEING CONTROLLED.

2.10. RATINGS: ALL ELECTRICAL EQUIPMENT SHALL BE FULLY RATED FOR BRACING IN EXCESS OF THE MAXIMUM AVAILABLE FAULT CURRENT CALCULATED AND SHOWN AT THE EQUIPMENT CONNECTION POINT WITHIN THE DISTRIBUTION SYSTEM. MINIMUM RATING SHALL BE 10K AIC.

2.11. WIRING DEVICES: WIRING DEVICES (SWITCHES, RECEPTACLES, ETC.) SHALL BE SPECIFICATION GRADE "DECORA" STYLE, MINIMUM 20-AMP RATED. COVER PLATES SHALL BE NYLON. DEVICE AND PLATE COLOR(S) SHALL BE AS SPECIFIED BY ARCHITECT OR INTERIOR DESIGNER - VERIFY PRIOR TO COMMENCEMENT OF WORK. WIRING DEVICES EXPOSED TO THE ELEMENTS SHALL HAVE WEATHERPROOF-IN-USE LOCKABLE COVERS. RAISED STEEL BOX COVERS MAY BE USED IN UTILITY AREAS.

2.12. TAMPERPROOF: ALL EQUIPMENT AND CIRCUITING ACCESSIBLE BY THE PUBLIC SHALL BE DEMONSTRATED TO BE TAMPERPROOF AND VANDAL RESISTANT. OPENABLE DEVICES AND EQUIPMENT SHALL BE PAD LOCKABLE.

PART THREE - EXECUTION

3.1. GROUNDING: GROUND ALL EQUIPMENT AND SYSTEM NEUTRAL IN ACCORDANCE WITH THE REQUIREMENTS OF NEC ARTICLE 250. PROVIDE CODE-SIZED EQUIPMENT GROUNDING CONDUCTOR IN ALL FEEDERS AND BRANCH CIRCUIT RACEWAYS. WHERE ISOLATED GROUNDS ARE INDICATED, PROVIDE INSULATED CONDUCTOR (GREEN WITH YELLOW STRIPE).

3.2. WORKMANSHIP: THE WORK SHALL BE INSTALLED PARALLEL AND AT RIGHT ANGLES TO THE BUILDING LINES, LEVEL AND PLUMB. THE WORK SHALL BE WELL SUPPORTED AND SOLIDLY MOUNTED. DRESS AND TIE WIRING IN PANELBOARDS AND SWITCHGEAR. THE WORK SHALL BE LEFT CLEAN WITH NO DIRT, DENTS, ABRASIONS, PAINT SPLATTERS OR OTHER IRREGULARITIES.

3.3. FIRE STOPPING: ALL PENETRATED FIRE RATED SURFACES SHALL BE FIRE SEALED WITH APPROVED U.L. LISTED SEALANTS AS LISTED WITHIN ARCHITECTURAL SPECIFICATIONS. DO NOT EXCEED MAXIMUM ALLOWABLE SURFACE PENETRATIONS DEPENDENT ON RATING OF SURFACES. REFER TO ARCHITECTURAL DRAWINGS FOR DETERMINATION OF PENETRATION LOCATIONS THROUGH FIRE RATED ASSEMBLIES.

3.4. SUPPORTS AND HANGERS: PROVIDE 3" HIGH HOUSEKEEPING CONCRETE PAD BENEATH FLOOR MOUNTED EQUIPMENT, EXTENDING 3" BEYOND EQUIPMENT FOOTPRINT. SUPPORT AND ALIGN ALL RACEWAYS, CABINETS, BOXES, BACK BOXES, FIXTURES, AND EQUIPMENT FROM STRUCTURE. SECURE ALL SUPPORTING METHODS BY MEANS OF TOGGLE BOLTS IN HOLLOW MASONRY, EXPANSION BOLTS IN SOLID MASONRY, CONCRETE PRESET INSERTS OR EXPANSION BOLTS IN CONCRETE, MACHINE SCREWS OR BOLTS IN METAL, AND WOOD SCREWS IN WOOD CONSTRUCTION. ALL SUPPORTING SYSTEMS AND COMPONENTS SHALL BE RATED FOR A MINIMUM OF FIVE (5) TIMES THE ACTUAL LOAD.

3.5. SLEEVES AND PENETRATIONS: PENETRATIONS OF ALL SURFACES SHALL BE PROVIDED WITH SLEEVES THAT SHALL BE SEALED WITH LIKE MATERIALS AND SHALL BE FINISHED WITH ESCUTCHEON PLATES. PENETRATIONS BELOW GRADE LEVEL SHALL BE WATERTIGHT. PENETRATIONS AT EXTERIOR WALLS SHALL BE WEATHERPROOF. ROOF PENETRATIONS SHALL BE FLASHED AND COUNTER FLASHED.

3.6. EXPANSION AND CONTRACTION: RACEWAYS PASSING THROUGH BUILDING EXPANSION JOINTS, ON ROOF, AND IN AREAS OF TEMPERATURE VARIATIONS GREATER THAN 30°F SHALL BE INSTALLED WITH EXPANSION FITTINGS.

3.7. IDENTIFICATION: IDENTIFY ALL EQUIPMENT, SWITCHBOARD CIRCUITS AND ELECTRICALLY-CONNECTED EQUIPMENT WITH ENGRAVED NAMEPLATES. BOXES SHALL BE MARKED WITH PANEL AND CIRCUIT NUMBERS (PERMANENT PEN ACCEPTABLE ABOVE CEILING). NAMEPLATES SHALL BE FASTENED WITH A MINIMUM OF TWO (2) SCREWS. PANEL DIRECTORIES SHALL BE TYPED. CONDUCTORS SHALL BE TAGGED WITH CIRCUIT NUMBERS AT SOURCE, JUNCTION BOXES, AND ALL OUTLET BOXES WITH PERMANENT ADHESIVE MARKER STRIP. PANEL DIRECTORIES SHALL BE TYPED. IDENTIFY WIRING DEVICES WITH SELF ADHESIVE CLEAR SATIN FINISH LABELS WITH SOURCE AND CIRCUIT NUMBER.

3.8. ELECTRICALLY-OPERATED EQUIPMENT: VERIFICATION AND SUBSTITUTION: FEEDERS AND OVER-CURRENT DEVICES (INCLUDING STARTERS, DISCONNECTS, ETC.) HAVE BEEN DESIGNED BASED ON INFORMATION PROVIDED BY THE RESPONSIBLE CONSULTANT AND/OR DESIGNATED SUPPLIER. PRIOR TO ROUGH-IN, COORDINATE WITH THE APPROPRIATE TRADE AND/OR INSTALLER TO DETERMINE THAT THE ACTUAL NAMEPLATE ELECTRICAL REQUIREMENTS MATCH THIS DESIGN. ALL ADDITIONAL ELECTRICAL COSTS RELATED TO THE CONNECTION OF EQUIPMENT WHICH VARIES FROM THE ORIGINAL SPECIFICATIONS SHALL BE RESOLVED WITHIN THE CONSTRUCTION TEAM AT NO ADDITIONAL COST TO THE OWNER.

3.9. COMMUNICATIONS SYSTEMS: THE ELECTRICAL CONTRACTOR SHALL PROVIDE OUTLETS AND RACEWAYS FOR COMMUNICATION SYSTEMS AS INDICATED HEREIN, INCLUDING TELEPHONE, DATA, POINT-OF-SALE, SOUND, SECURITY, AUDIO/VISUAL, CCTV, MATV, ETC. CABLING AND DEVICES SHALL BE INSTALLED AND TERMINATED BY OTHERS.

PART FOUR - SPECIAL SYSTEMS

4.1. DESIGN/BUILD FIRE ALARM SYSTEM: THESE DOCUMENTS DO NOT INDICATE DEVICES, OUTLETS, CONNECTIONS, AND CIRCUITRY NECESSARY FOR A COMPLETE FIRE ALARM SYSTEM. PROVIDE A COMPLETE, NEW FIRE ALARM DETECTION AND ALARM SYSTEM WITH CLASS 1 CIRCUITING INCLUDING, BUT NOT LIMITED TO, INITIATING DEVICES, DUCT DETECTORS, ADA HORN/STROBES, ETC. WHICH SHALL BE IN FULL COMPLIANCE WITH ALL LOCAL, STATE, AND ADA REQUIREMENTS. CONTROL PANEL SHALL INCLUDE INTEGRAL STANDBY BATTERIES, CHARGER, AND MUNICIPAL TIE MODULE OR AGENCY APPROVED AUTO-DIALER CONNECTED TO THE TELEPHONE SYSTEM (CONNECTION AND MONITORING CHARGES BY OWNER). SUBMIT PROPOSED DESIGN AND OBTAIN FIRE MARSHAL APPROVED SHOP DRAWINGS PRIOR TO COMMENCEMENT OF WORK. AFTER RECEIPT OF PLAN APPROVAL BY THE FIRE MARSHAL, PROVIDE ONE (1) SET OF STAMPED DRAWINGS (PRINT OR ELECTRONIC COPY) ALONG WITH AN APPROVED EQUIPMENT SUBMITTAL TO THE ELECTRICAL ENGINEER. ALL CONNECTIONS TO SYSTEM SHALL BE PERFORMED BY FACTORY-CERTIFIED TECHNICIAN AND SHALL BE ACCEPTED BY OWNER'S SYSTEM-MONITORING AGENCY.

4.2. THIRD PARTY TESTING: PROVIDE ALL ASSOCIATED COSTS FOR THIRD PARTY TESTING OF ALL EQUIPMENT, CONDUCTORS, GROUND FAULT, GROUND FAULT COORDINATION STUDY WITH REPORT PREPARATION, ETC. AS REQUIRED BY THE NEC, AHJ AND ALL OTHER GOVERNING AUTHORITIES.

IMEG
www.imegcorp.com
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771

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REF. SCALE IN INCHES

RSP Architects
502 South College Avenue
Suite 203
Tempe, AZ 85281-3714
480.889.2000
480.889.2099 fax
www.rsparch.com

Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

JUSTIN C. DAGERMAN
No. 52467
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REGISTERED PROFESSIONAL ENGINEER
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EXP. 09-30-2023

Project No.	3378.002.00
Drawn By	IMEG
Checked By	IMEG
Date	08/23/2022

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Sheet Issues / Revisions

No.	Date	Description

ELECTRICAL SPECIFICATIONS

E001

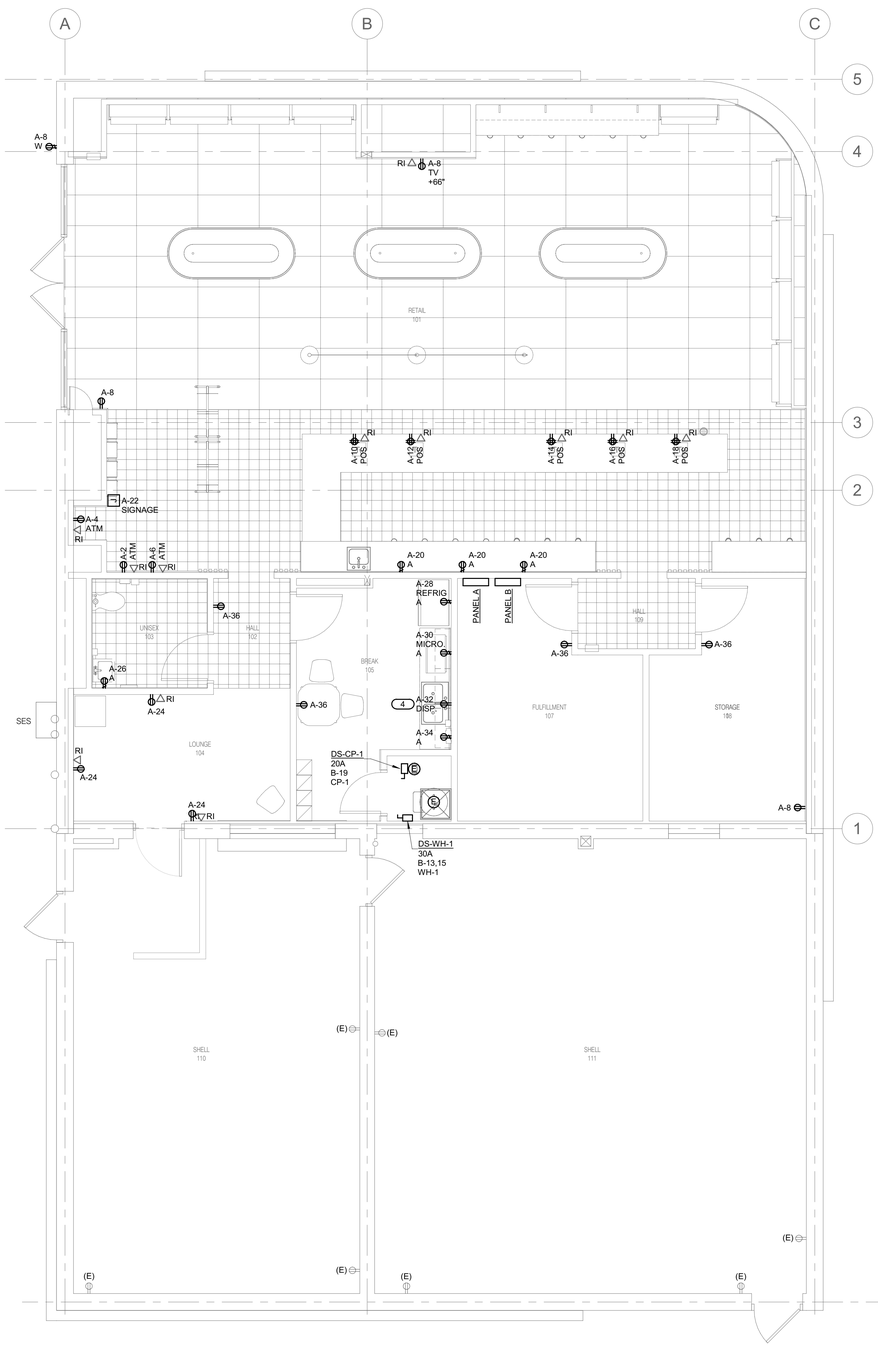
City of Phoenix Plan #: 2207827-LPRR Date: 12/12/22

SHEET NOTES:

1. PROVIDE WEATHERPROOF (NEMA-3R) ENCLOSURES FOR ALL ELECTRICAL EQUIPMENT LOCATED OUTDOORS. PROVIDE WEATHERPROOF WHILE-IN-USE COVER FOR ALL RECEPTACLES LOCATED OUTDOORS.

KEYNOTES: (#)

1. LOW VOLTAGE LIGHTING CONTROL STATION. INTERCONNECT WITH LIGHTING CONTROL SYSTEM.
2. PROVIDE CONNECTION AT CEILING FOR ILLUMINATED DISPLAY CASES. COORDINATE EXACT REQUIREMENTS PRIOR TO INSTALLATION.
3. PROVIDE CONNECTION AT CEILING FOR ILLUMINATED SIGNAGE. COORDINATE EXACT REQUIREMENTS PRIOR TO INSTALLATION.
4. PROVIDE HALF-SWITCHED RECEPT FOR GARBAGE DISPOSAL.



1 LEVEL 01 ELECTRICAL PLAN

1/4" = 1'-0"



2 LEVEL 01 LIGHTING PLAN

1/4" = 1'-0"

IMEG
9000 E PIMA CENTER PARKWAY
SUITE 320
SCOTTSDALE, AZ 85258
P: 480.222.3771
www.imegcorp.com

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RSP Architects
502 South College Avenue 480.889.2000
Suite 203 480.889.2099
Tempe, AZ 85281-3714 fax
www.rsparch.com

Consultants

Project For

THE FLOWER SHOP
PHOENIX RETAIL INTERIOR PHASE

3155 E. McDowell Rd,
Phoenix, AZ 85008

JUSTIN C. DAGERMAN 52647 REGISTERED ARCHITECT ARIZONA U.S.A.
08-22-22
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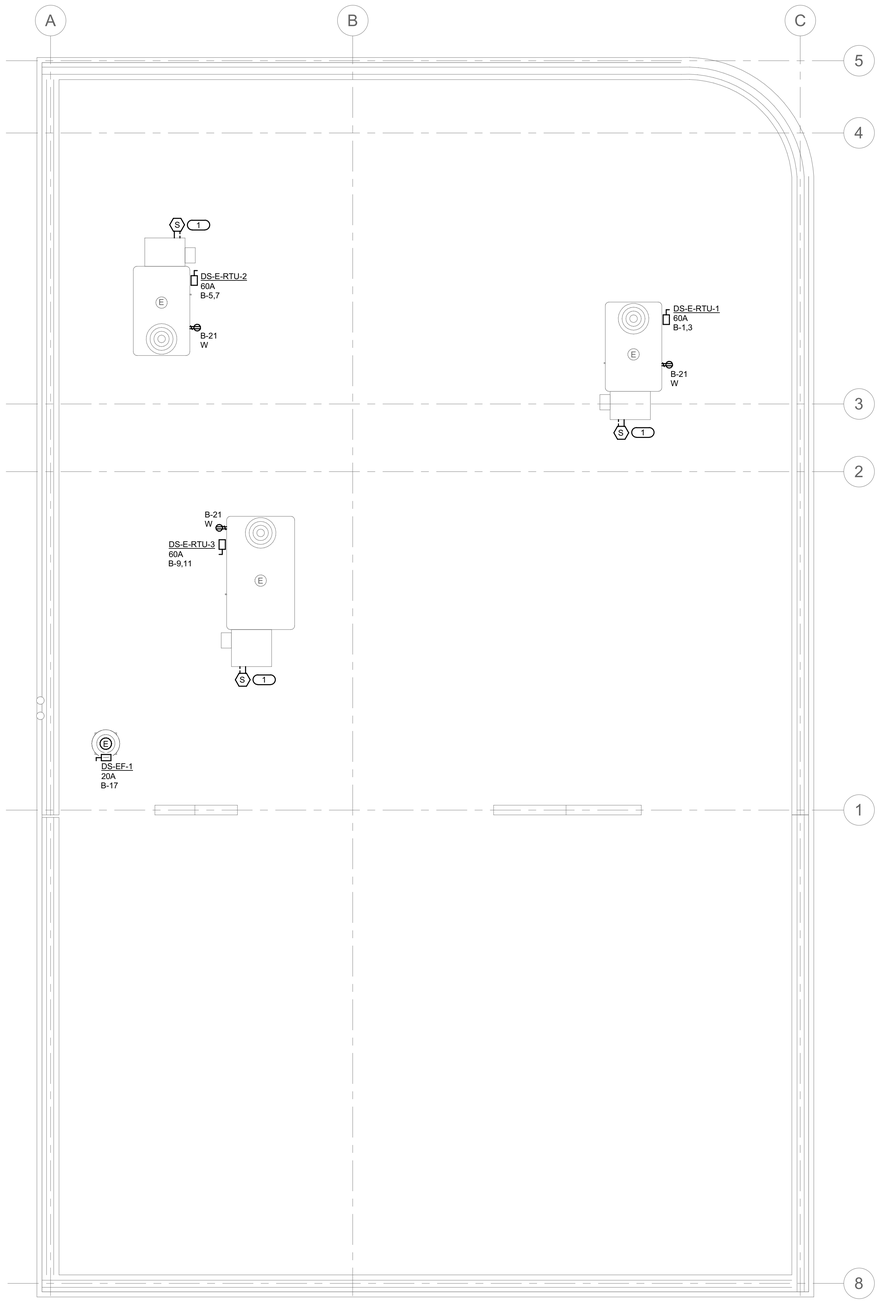
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Sheet Issues / Revisions

No.	Date	Description

ELECTRICAL PLANS

E101



ELECTRICAL CONNECTION SCHEDULE

ITEM	VOLTAGE	LOAD CLASS.	APPARENT LOAD	FLA	MOCP	OCPD	CIRCUIT NUMBER	WIRE AND RACEWAY
B								
CP-1	120 V, 1Ø	Motor	0.04 kVA	0.3 A	15 A	20 A	19	2#12 & 1#12 EGC IN 3/4" C.
E-RTU-1	240 V, 1Ø	HVAC	4.80 kVA	20 A	35 A	35 A	1,3	2#8 & 1#10 EGC IN 3/4" C.
E-RTU-2	240 V, 1Ø	HVAC	4.80 kVA	20 A	35 A	35 A	5,7	2#8 & 1#10 EGC IN 3/4" C.
E-RTU-3	240 V, 1Ø	HVAC	6.00 kVA	25 A	45 A	45 A	9,11	2#6 & 1#8 EGC IN 1" C.
EF-1	120 V, 1Ø	Motor	0.53 kVA	4.4 A	15 A	20 A	17	2#12 & 1#12 EGC IN 3/4" C.
WH-1	240 V, 1Ø	Other	4.50 kVA	18.8 A	25 A	25 A	13,15	2#10 & 1#10 EGC IN 3/4" C.

DISCONNECT AND STARTER SCHEDULE

NOTE: ALL DISCONNECTS (EXCEPT MANUAL STARTERS) SHALL BE HEAVY DUTY TYPE.

DISCONNECT TYPE:	ACCESSORIES & OPTIONS
FU - FUSED	SA - STANDARD ACCESSORIES (INCLUDES *...
NF - NON-FUSED	
CB - CIRCUIT BREAKER	
STARTER TYPE:	
MX - MANUAL SWITCH	

ITEM	DISCONNECT TYPE & RATING: TYPE	RATING	CIRCUIT VOLTAGE	POLES	ENCLOSURE	REQUIRED ACCESSORIES & OPTIONS	COMMENTS
DS-E-RTU-2	FU	60 A	240 V	2	NEMA 3R	SA	
DS-E-RTU-1	FU	60 A	240 V	2	NEMA 3R	SA	
DS-E-RTU-3	FU	60 A	240 V	2	NEMA 3R	SA	
DS-CP-1	MX	20 A	120 V	1	NEMA 1	SA	
DS-EF-1	MX	20 A	120 V	1	NEMA 1	SA	
DS-WH-1	NF	30 A	240 V	2	NEMA 1	SA	

SHEET NOTES:

1. PROVIDE WEATHERPROOF (NEMA-3R) ENCLOSURES FOR ALL ELECTRICAL EQUIPMENT LOCATED OUTDOORS. PROVIDE WEATHERPROOF WHILE-IN-USE COVER FOR ALL RECEPTACLES LOCATED OUTDOORS.

KEYNOTES: (#)

1. CONNECT SYSTEM POWERED DUCT DETECTOR.

NORTH 1 **ROOF ELECTRICAL PLAN**
1/4" = 1'-0"

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Suite 203
Tempe, AZ 85281-3714
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SCOTTSDALE, AZ 85258
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ELECTRICAL ROOF PLAN

E102

PANEL A

MOUNTING: SURFACE
ENCLOSURE: NEMA PB 1
FED FROM: 200 A/2P @ SES
LOCATION: Space 2

SOLID NEUTRAL GROUND BUS

MAIN: 200 A MLO
VOLTS: 120/240 Single
PHASE: 1
WIRE: 3
SCCR: 35 kA
ISC: 28.45 kA

NOTES:

KEY	CKT NO.	LOAD DESCRIPTION	OCPD AMPS	P	WIRE SIZE H	N	G	A	B	WIRE SIZE G	N	H	P	OCPD AMPS	LOAD DESCRIPTION	CKT NO.	KEY
	1	FRONT-OF-HOUSE - LIGHTING	20 A	1	10	10	10	1.09	1.2	12	12	12	1	20 A	ATM - REC.	2	
	3	FRONT-OF-HOUSE - CASEWORK LTG	20 A	1	12	12	12		1.18	1.2	12	12	1	20 A	ATM - REC.	4	
	5	BACK-OF-HOUSE - LIGHTING	20 A	1	12	12	12	0.33	1.2	12	12	12	1	20 A	ATM - REC.	6	
	7	EXTERIOR LTG	20 A	1	12	12	12		0.08	0.72	12	12	12	1	20 A	RETAIL - CONV. REC.	8	
	9	EXTERIOR SIGNAGE	20 A	1	12	12	12	1.2	0.36		12	12	12	1	20 A	RETAIL - POS REC.	10	
	11	EXTERIOR SIGNAGE	20 A	1	12	12	12		1.2	0.36	12	12	12	1	20 A	RETAIL - POS REC.	12	
	13	EXTERIOR SIGNAGE	20 A	1	12	12	12	1.2	0.36		12	12	12	1	20 A	RETAIL - POS REC.	14	
	15	ILLUMINATED DISPLAY - POWER	20 A	1	12	12	12		0.2	0.36	12	12	12	1	20 A	RETAIL - POS REC.	16	
	17	ILLUMINATED DISPLAY - POWER	20 A	1	12	12	12	0.2	0.36		12	12	12	1	20 A	RETAIL - POS REC.	18	
	19	ILLUMINATED DISPLAY - POWER	20 A	1	12	12	12		0.2	0.54	12	12	12	1	20 A	RETAIL - COUNTER	20	
	21	ILLUMINATED SIGNAGE - POWER	20 A	1	12	12	12	0.2	1.2		12	12	12	1	20 A	RETAIL - NEON SIGNAGE	22	
	23	SHELL SPACE LTG	20 A	1	12	12	12		0.54	0.54	12	12	12	1	20 A	OFFICE 104 - CONV. REC.	24	
--	25	SPARE	20 A	1	--	--	0	0.18		12	12	12	1	20 A	UNISEX 103 - CONV. REC.	26	
--	27	SPARE	20 A	1	--	--	--		0	0.18	12	12	12	1	20 A	BREAK 105 - REFRIG.	28	
--	29	SPARE	20 A	1	--	--	--	0	0.18		12	12	12	1	20 A	BREAK 105 - MICRO.	30	
--	31	SPACE	--	--	--	--	--			0	0.18	12	12	1	20 A	BREAK 105 - DISP.	32	
--	33	SPACE	--	--	--	--	--	0	0.18			12	12	12	1	20 A	BREAK 105 - CONV.	34	
--	35	SPACE	--	--	--	--	--			0	0.72	12	12	12	1	20 A	BREAK 105, FULL. 107, STOR. 108 -...	36	
--	37	SPACE	--	--	--	--	--	0	0			--	--	--	1	20 A	SPARE	38	--
--	39	SPACE	--	--	--	--	--			0	0	--	--	--	1	20 A	SPARE	40	--
--	41	SPACE	--	--	--	--	--	0	0			--	--	--	1	20 A	SPARE	42	--
								Total Load:	9.44 kVA	8.20 kVA								
								Total Amps:	78.67	68.36								

LOAD SUMMARY

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTALS*	
Lighting	3.223 kVA	100.00%	3.223 kVA		
Power	5.6 kVA	100.00%	5.6 kVA	**TOTAL CONNECTED LOAD:**	17.64 kVA
Receptacles	8.82 kVA	100.00%	8.82 kVA	**TOTAL ESTIMATED DEMAND LOAD:**	17.643 kVA
				TOTAL CONNECTED AMPS:	73.51 A
				TOTAL ESTIMATED DEMAND AMPS:	73.5 A

*TOTAL DEMAND CALCS SUBTRACT ANY REDUNDANT LOAD AND THE SMALLER OF ANY NONCOINCIDENT HVAC LOADS. THIS CALC IS DONE AT EACH PANEL.

CIRCUIT KEY NOTES:

SWITCHBOARD SES

ENCLOSURE: NEMA 3R
FED FROM: 0 A/0P @
LOCATION:

SOLID NEUTRAL GROUND BUS

MAIN: 400 A MLO
VOLTS: 120/240 Single
PHASE: 1
WIRE: 3
SCCR: 200 kA
ISC UNKNOWN 0 A

NOTES: **EXISTING SERVICE ENTRANCE SECTION.**

CKT	LOAD DESCRIPTION	LOAD	POLES	FRAME	TRIP	TYPE	ACC.	WIRE AND RACEWAY	CIRCUIT KEY
1	PANEL 'A'	17.64 kVA	2	200 A	200 A			3#3/0 & 1#6 EGC IN 2" C.	
2	PANEL 'B'	21.21 kVA	2	200 A	200 A			3#3/0 & 1#6 EGC IN 2" C.	
3	SPACE	0 kVA	--	--	--	--	--	--	--
4	SPACE	0 kVA	--	--	--	--	--	--	--

LOAD SUMMARY (INCLUDES ALL TUBS IN THIS PANEL)

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTALS*	
HVAC	15.6 kVA	100.00%	15.6 kVA		
Lighting	3.223 kVA	100.00%	3.223 kVA	**TOTAL CONNECTED LOAD:**	38.85 kVA
Motor	0.567 kVA	100.00%	0.567 kVA	**TOTAL ESTIMATED DEMAND LOAD:**	38.85 kVA
Other	4.5 kVA	100.00%	4.5 kVA	**TOTAL CONNECTED AMPS:**	161.88 A
Power	5.6 kVA	100.00%	5.6 kVA	**TOTAL ESTIMATED DEMAND AMPS:**	161.9 A
Receptacles	9.36 kVA	100.00%	9.36 kVA		

*TOTAL DEMAND CALCS SUBTRACT ANY REDUNDANT LOAD AND THE SMALLER OF ANY NONCOINCIDENT HVAC LOADS. THIS CALC IS DONE AT EACH PANEL.

CIRCUIT KEY...

PANEL B

MOUNTING: SURFACE
ENCLOSURE: NEMA PB 1
FED FROM: 200 A/2P @ SES
LOCATION: Space 2

SOLID NEUTRAL GROUND BUS

MAIN: 200 A MLO
VOLTS: 120/240 Single
PHASE: 1
WIRE: 3
SCCR: 35 kA
ISC: 27.11 kA

NOTES:

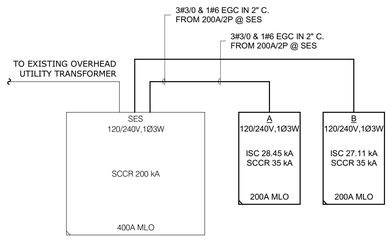
KEY	CKT NO.	LOAD DESCRIPTION	OCPD AMPS	P	WIRE SIZE H	N	G	A	B	WIRE SIZE G	N	H	P	OCPD AMPS	LOAD DESCRIPTION	CKT NO.	KEY
	1	EXISTING RTU-1	35 A	2	8	--	10	2.4	0				--	--	--	1	20 A	SPARE	2	--
--	3	--	--	--	--	--	--			2.4	0		--	--	--	1	20 A	SPARE	4	--
	5	EXISTING RTU-2	35 A	2	8	--	10	2.4	0				--	--	--	1	20 A	SPARE	6	--
--	7	--	--	--	--	--	--			2.4	0		--	--	--	1	20 A	SPARE	8	--
	9	EXISTING RTU-3	45 A	2	6	--	8	3	0				--	--	--	1	20 A	SPARE	10	--
--	11	--	--	--	--	--	--			3	0		--	--	--	1	20 A	SPARE	12	--
	13	WH-1	25 A	2	10	--	10	2.25	0				--	--	--	1	20 A	SPARE	14	--
--	15	--	--	--	--	--	--			2.25	0		--	--	--	1	20 A	SPARE	16	--
	17	EF-1	20 A	1	12	12	12	0.53	0				--	--	--	1	20 A	SPARE	18	--
	19	CP-1	20 A	1	12	12	12			0.04	0		--	--	--	1	20 A	SPARE	20	--
	21	ROOF - RECEPT	20 A	1	12	12	12	0.54	0				--	--	--	1	20 A	SPARE	22	--
--	23	SPARE	20 A	1	--	--	--			0	0		--	--	--	1	20 A	SPARE	24	--
--	25	SPARE	20 A	1	--	--	--	0	0				--	--	--	1	20 A	SPARE	26	--
--	27	SPARE	20 A	1	--	--	--			0	0		--	--	--	1	20 A	SPARE	28	--
--	29	SPARE	20 A	1	--	--	--	0	0				--	--	--	1	20 A	SPARE	30	--
								Total Load:	11.12 kVA	10.09 kVA								
								Total Amps:	92.65	84.08								

LOAD SUMMARY

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTALS*	
HVAC	15.6 kVA	100.00%	15.6 kVA		
Motor	0.567 kVA	100.00%	0.567 kVA	**TOTAL CONNECTED LOAD:**	21.21 kVA
Other	4.5 kVA	100.00%	4.5 kVA	**TOTAL ESTIMATED DEMAND LOAD:**	21.207 kVA
Receptacles	0.54 kVA	100.00%	0.54 kVA	**TOTAL CONNECTED AMPS:**	88.36 A
				TOTAL ESTIMATED DEMAND AMPS:	88.4 A

*TOTAL DEMAND CALCS SUBTRACT ANY REDUNDANT LOAD AND THE SMALLER OF ANY NONCOINCIDENT HVAC LOADS. THIS CALC IS DONE AT EACH PANEL.

CIRCUIT KEY NOTES:



1 **SINGLE LINE DIAGRAM**
NO SCALE

LOCATION	DESCRIPTION	Isc	Isc	Isc
	SRP Service Transformer, 240V, 3φ - 400 AMPS			*ALL ALUMINUM #1/0 AND ABOVE*
	Available Fault Current *28,539 Amps*	*AS DESIGNED*	*ALL COPPER*	
F1	**From EXISTING SERVICE ENTRANCE SECTION 'SES' to PANEL 'A'**			
	50ft with (1) #3/0 CU in STEEL conduit	**28,446** *Amps*	**28,446** *Amps*	**27,886** *Amps*
F2	**From EXISTING SERVICE ENTRANCE SECTION 'SES' to PANEL 'B'**			
	55ft with (1) #3/0 CU in STEEL conduit	**27,105** *Amps*	**27,105** *Amps*	**26,520** *Amps*

ELECTRICAL - RISER DIAGRAM NOTES:

1. THE RISER DIAGRAM IS INTENDED TO CONVEY THE COMPONENTS OF THE ELECTRICAL DISTRIBUTION SYSTEM. REFER TO ELECTRICAL DRAWINGS, DETAILS, DISTRIBUTION / PANEL / EQUIPMENT / EQUIPMENT CONNECTION SCHEDULES, AND SPECIFICATIONS FOR ADDITIONAL INFORMATION.
2. SHORT CIRCUIT CURRENT RATINGS (SCCR) FOR EQUIPMENT ARE MINIMUM REQUIREMENTS FOR BUSS BRACING AND DEVICE RATING. ALL EQUIPMENT SHALL BE FULLY RATED UNLESS SPECIFICALLY NOTED AS SERIES RATED.
3. TRANSFER SWITCHES (SCCR) RATINGS ARE INTENDED AS WITHSTAND AND CLOSE RATINGS (WCR).
4. THE BASIS OF DESIGN: THE CONTRACTOR SHALL BE RESPONSIBLE FOR DERATING AND SIZING CONDUCTORS AND CONDUITS TO EQUAL OR EXCEED AMPACITY OF THE BASIS OF DESIGN CIRCUITS WHEN ALTERNATIVE METHODS OR MATERIALS OTHER THAN THE BASIS OF DESIGN ARE APPLIED.
   a. RACEWAY: EMT UNLESS OTHERWISE NOTED
   b. FEEDER CHARACTERISTICS: ALL CURRENT CARRYING CONDUCTORS SHALL BE COPPER UNLESS NOTED OTHERWISE. CONDUCTOR SIZES ARE BASED ON AMERICAN WIRE GAUGE AWG AND KCMIL THOUSANDS OF CIRCULAR MIL. REFER TO SPECIFICATION SECTION 25 05 13 WIRE AND CABLE FOR ADDITIONAL INFORMATION
   c. GROUNDING AND BONDING CONDUCTORS SHALL BE COPPER.
   d. CONDUCTORS (MOTORS): COPPER
   e. [AL] INDICATES ALUMINUM CONDUCTOR
   f. [BLANK] OR [CU] INDICATES COPPER CONDUCTOR
5. PROVIDE GROUNDING ELECTRODE AND BONDING SYSTEM PER CODE REQUIREMENTS. PROVIDE THE FOLLOWING MINIMUM CONNECTIONS AND COMPONENTS. REFER TO SPECIFICATION SECTION 26 05 26 GROUNDING AND BONDING AND DETAILS WHEN APPLICABLE:
   a. ELECTRICAL GROUND FIELD
   b. CONCRETE ENCASED UFER GROUND
   c. METALLIC WATER MAIN
   d. BUILDING STEEL, EFFECTIVELY GROUNDED
   e. INTERSYSTEM BONDING TERMINAL [IBT]
6. PROVIDE O.Z. GEDNEY OR EQUAL GROUND BUSHING FOR ALL SERVICE AND FEEDER RACEWAYS BONDED TO GROUND BUS WITH CONDUCTOR SIZED TO MAXIMUM FEEDER GROUND CAPACITY.
7. CONDUCTORS AND GROUND SIZES ON THE LINE AND LOAD SIDES OF ALL DISCONNECT SWITCHES SHALL BE IDENTICAL UNLESS NOTED OTHERWISE.
8. REFER TO COVER SHEET FOR ADDITIONAL EQUIPMENT TAG INFORMATION (SPD-#, M-#, ETC).
9. REFER TO GROUNDING ELECTRODE SYSTEM AND BONDING DETAILS
   a. EGC – EQUIPMENT GROUNDING CONDUCTOR
   b. GEC – GROUNDING ELECTRODE CONDUCTOR
   c. SSBJ – SUPPLY SIDE BONDING JUMPER
10. CIRCUIT BREAKER CHARACTERISTICS AND ACCESSORIES:
   a. [CB] INDICATES CIRCUIT BREAKER
   b. [FU] INDICATES FUSED SWITCH
   c. [NF] INDICATES NON-FUSED SWITCH
   d. [MLO] INDICATES MAIN LUG ONLY
   e. [MCB] INDICATES MAIN CIRCUIT BREAKER
   f. [MCCB] INDICATES MOLDED CASE CIRCUIT BREAKER
   g. [GFI] INDICATES GROUND FAULT RELAY
   h. [AER] INDICATES ARC ENERGY REDUCTION SYSTEM
   i. [100% RATED] INDICATES INSULATED CASE BREAKER RATED FOR FULL CONTINUOUS CAPACITY OF CIRCUIT BREAKER NAMEPLATE
   j. [LOCK] INDICATES PADLOCK HASP
   k. [RED] INDICATES RED HANDLE

ELECTRICAL DISTRIBUTION SCHEDULE NOTES:

1. BRANCH PANEL KEY:
   a. *G = GROUND FAULT CIRCUIT INTERRUPT
   b. *P = PADLOCK HASP
   c. *R = RED HANDLE
   d. *NB = NEW BREAKER
   e. *RB = REPLACE EXISTING BREAKER WITH NEW BREAKER
   f. *EB = EXISTING BREAKER

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SINGLE LINE DIAGRAM AND CALCULATIONS

E301

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